

THE ZCASH ARBORETUM

# I

## Math Guide

Mathematical foundations for Halo 2 and Zcash Orchard

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## Abstract

This volume of *The Zcash Arboretum* develops the mathematics underlying the cryptography of Zcash from first principles, assuming only elementary algebra and calculus. Its subjects, in order, are the notation of sets, functions, and relations in which every later claim is stated; the integers and their remainders modulo a prime, where the extended Euclidean algorithm computes inverses and so reduces division to multiplication by an inverse; the abstract structures—groups, rings, and fields—that axiomatise the laws this arithmetic obeys; polynomials over a field, whose bounded number of roots allows two polynomials of large degree to be compared by evaluation at a single random point; the finite fields, their possible sizes, and their cyclic multiplicative structure; the number theory of squares, square roots, and the discrete logarithm; linear algebra and the inner products underlying short commitments to long vectors; the roots of unity that supply evaluation domains and the fast Fourier transform; elliptic curves, whose point groups, when chosen appropriately, admit no known subexponential discrete-logarithm attack; and the probability, asymptotics, and model of computation required to state a claim of 128-bit security with a precise meaning.

Later volumes of the series depend on three parts of this volume: the Pasta curve cycle, constructed in §10; the polynomial machinery on which Halo 2 rests, developed in §5 and §9; and the language of security statements, fixed in §11. A first reading should take the sections in order; a reader who needs one of these parts may begin at its section and follow its references back to the earlier sections on which it depends.

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# 1 Notation: sets, functions, and relations

This section fixes the set-theoretic and logical language of the series. It defines sets, functions and relations; proves that a function has a two-sided inverse exactly when it is bijective (Theorem 1.1), that the classes of an equivalence relation partition the set (Proposition 1.2), and that a partial order and its strict companion determine each other (Proposition 1.3); fixes the conventions of proof; and closes with three tables giving the spoken form of every recurring symbol (§1.5).

## 1.1 Sets

A *set* is a collection of distinct objects, its *elements*; one writes  $x \in A$  for membership and  $x \notin A$  for its negation. A set may be given by listing,  $\{a, b, c\}$ , or by *set-builder* notation  $\{x \in A : P(x)\}$  (“the  $x$  in  $A$  such that  $P(x)$ ”). One writes  $A \subseteq B$  when every element of  $A$  lies in  $B$ , and  $A = B$  exactly when  $A \subseteq B$  and  $B \subseteq A$  — so proving two sets equal means proving two inclusions. The usual operations are union  $A \cup B$ , intersection  $A \cap B$ , difference  $A \setminus B$ , and the empty set  $\emptyset$ . The *Cartesian product* is  $A \times B = \{(a, b) : a \in A, b \in B\}$ , and  $A^n$  is the set of  $n$ -tuples over  $A$ . For a finite set  $A$ ,  $|A|$  denotes its number of elements. The standard number systems are the naturals  $\mathbb{N}$ , the integers  $\mathbb{Z}$ , the rationals  $\mathbb{Q}$ , the reals  $\mathbb{R}$ , and the complex numbers  $\mathbb{C}$ , with  $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$ ; we construct the underlying arithmetic of the prime field  $\mathbb{F}_p$  in §2, certify it as a field—and first call it  $\mathbb{F}_p$ —in §4, and treat general finite fields  $\mathbb{F}_q$  in §6.

## 1.2 Functions

A *function*  $f: A \rightarrow B$  assigns to each  $a \in A$  a unique value  $f(a) \in B$ ;  $A$  is the *domain* and  $B$  the *codomain*. It is *injective* (one-to-one) if  $f(a) = f(a')$  implies  $a = a'$ ; *surjective* (onto) if every  $b \in B$  equals  $f(a)$  for some  $a \in A$ ; and *bijective* if both. The *composition* of  $f: A \rightarrow B$  and  $g: B \rightarrow C$  is  $(g \circ f)(a) = g(f(a))$ ; composition is associative,  $h \circ (g \circ f) = (h \circ g) \circ f$ , both sides sending  $a$  to  $h(g(f(a)))$ . The *identity* on  $A$  is  $\text{id}_A: A \rightarrow A$ ,  $\text{id}_A(a) = a$ .

**Theorem 1.1** (Inverse functions). *A function  $f: A \rightarrow B$  has a two-sided inverse  $f^{-1}: B \rightarrow A$ , meaning  $f^{-1} \circ f = \text{id}_A$  and  $f \circ f^{-1} = \text{id}_B$ , if and only if  $f$  is bijective. When it exists, the inverse is unique.*

*Proof.* Suppose a two-sided inverse  $g$  exists. If  $f(a) = f(a')$ , applying  $g$  gives  $a = g(f(a)) = g(f(a')) = a'$ , so  $f$  is injective; and every  $b \in B$  satisfies  $b = f(g(b))$ , so  $f$  is surjective. Conversely, suppose  $f$  is bijective. Each  $b \in B$  has at least one preimage by surjectivity and at most one by injectivity, so we may define  $g(b)$  to be the unique  $a \in A$  with  $f(a) = b$ ; then  $g(f(a)) = a$  and  $f(g(b)) = b$  hold by construction. Finally, if  $g$  and  $g'$  are both two-sided inverses, associativity of composition gives  $g = g \circ \text{id}_B = g \circ (f \circ g') = (g \circ f) \circ g' = \text{id}_A \circ g' = g'$ ; the inverse is unique, and we write it  $f^{-1}$ .  $\square$

For  $S \subseteq A$  the *image* is  $f(S) = \{f(a) : a \in S\}$ , and for  $T \subseteq B$  the *preimage* is  $f^{-1}(T) = \{a \in A : f(a) \in T\}$ , defined whether or not  $f$  is invertible. The two uses of  $f^{-1}$  agree: when the inverse function exists, the preimage of  $T$  under  $f$  is exactly the image of  $T$  under  $f^{-1}$ .

### 1.3 Relations and equivalences

A *binary relation* on  $A$  is a subset  $R \subseteq A \times A$ ; one writes  $a \sim b$  for  $(a, b) \in R$ . It is an *equivalence relation* if it is *reflexive* ( $a \sim a$ ), *symmetric* ( $a \sim b \Rightarrow b \sim a$ ), and *transitive* ( $a \sim b$  and  $b \sim c$  imply  $a \sim c$ ). The *equivalence class* of  $a$  is  $[a] = \{x \in A : x \sim a\}$ , and the set of distinct classes is the *quotient*  $A/\sim$ .

**Proposition 1.2** (Classes partition the set). *Let  $\sim$  be an equivalence relation on a set  $A$ . The distinct equivalence classes partition  $A$ : they are nonempty, pairwise disjoint, and their union is  $A$ .*

*Proof.* Reflexivity gives  $a \in [a]$  for every  $a \in A$ , so each class is nonempty and every element lies in some class; the union of the classes is therefore  $A$ . For disjointness, suppose the classes  $[a]$  and  $[b]$  share an element  $x$ , and let  $y \in [a]$  be arbitrary. From  $y \sim a$  and  $a \sim x$  (the latter by symmetry from  $x \sim a$ ), transitivity gives  $y \sim x$ ; combining with  $x \sim b$  gives  $y \sim b$ , so  $y \in [b]$ . Thus  $[a] \subseteq [b]$ , and exchanging the roles of  $a$  and  $b$  gives the reverse inclusion; two classes that meet coincide.  $\square$

The volume uses this construction to form the integers modulo  $n$  (§2) and the cosets of a subgroup (§3).

A second kind of relation orders elements rather than identifying them. A *partial order* on  $A$  is a relation  $\leq$  that is reflexive, transitive, and *antisymmetric*:  $a \leq b$  and  $b \leq a$  imply  $a = b$ . Four partial orders recur in the series: the usual  $\leq$  on  $\mathbb{Z}$ ; inclusion  $\subseteq$  on the subsets of a fixed set, whose antisymmetry is the two-inclusions rule of §1.1; divisibility  $|$  on  $\mathbb{N}$  (§2.2); and the *prefix* relation on finite sequences, under which a sequence precedes every sequence that extends it. Two elements are *comparable* if  $a \leq b$  or  $b \leq a$ , and a partial order under which every two elements are comparable is a *total* (or *linear*) order. The order  $\leq$  on  $\mathbb{Z}$  is total; the other three are not:  $\{1\}$  and  $\{2\}$  are incomparable under inclusion, as are 2 and 3 under divisibility and the one-term sequences (1) and (2) under prefix.

Every partial order has a *strict companion*,  $a < b$  meaning  $a \leq b$  and  $a \neq b$ .

**Proposition 1.3** (Strict and non-strict orders). *If  $\leq$  is a partial order on  $A$ , its strict companion  $<$  is irreflexive ( $a < a$  holds for no  $a$ ) and transitive. Conversely, if  $<$  is irreflexive and transitive, then “ $a < b$  or  $a = b$ ” defines a partial order on  $A$  whose strict companion is  $<$ .*

*Proof.* Irreflexivity of the companion is the clause  $a \neq a$ . For transitivity let  $a < b < c$ ; then  $a \leq c$  because  $\leq$  is transitive, and  $a = c$  would give  $a \leq b \leq a$ , hence  $a = b$  by antisymmetry, contradicting  $a < b$ . Conversely, the relation “ $a < b$  or  $a = b$ ” is reflexive by construction and transitive by cases: if either hypothesis is an equality, the conclusion follows from the other by substitution, and otherwise from transitivity of  $<$ . It is antisymmetric, since  $a \neq b$  with both  $a \leq b$  and  $b \leq a$  would mean  $a < b$  and  $b < a$ , whence  $a < a$ , against irreflexivity. Its strict companion is  $<$  again: “( $a < b$  or  $a = b$ ) and  $a \neq b$ ” reduces to  $a < b$ , which already excludes  $a = b$  by irreflexivity.  $\square$

A strict order  $<$  is a *strict total order* when its non-strict companion is total, equivalently when for all  $a, b$  exactly one of  $a < b$ ,  $a = b$ ,  $b < a$  holds (*trichotomy*): at least one holds by totality, and any two together would give  $a < a$ , by substitution or by transitivity. The usual  $<$  on  $\mathbb{Z}$  is the

standard example. A function  $f: A \rightarrow B$  between ordered sets is *monotone* (order-preserving) if  $a \leq a'$  implies  $f(a) \leq f(a')$ , and *strictly increasing* if  $a < a'$  implies  $f(a) < f(a')$ . The volume assumes only one further property of an order, the well-ordering of  $\leq$  on  $\mathbb{N}$ , stated at the start of §2.1.

## 1.4 Logic and proof conventions

We employ the symbols  $\forall$  (“for all”),  $\exists$  (“there exists”),  $\exists!$  (“there exists a unique”), and “:” or “s.t.” for “such that”. Negation swaps the quantifiers:  $\neg(\forall x P(x))$  is  $\exists x \neg P(x)$ , and  $\neg(\exists x P(x))$  is  $\forall x \neg P(x)$ . The order of unlike quantifiers matters:  $\forall x \exists y P(x, y)$  (the  $y$  may depend on  $x$ ) is weaker than  $\exists y \forall x P(x, y)$  (one  $y$  works for all  $x$ ).

Throughout the series we prove an implication  $P \Rightarrow Q$  directly (assume  $P$ , derive  $Q$ ), by contraposition (assume  $\neg Q$ , derive  $\neg P$ ), or by contradiction; we prove a statement  $\forall x P(x)$  by taking an arbitrary  $x$ , and  $\exists x P(x)$  by exhibiting a witness. We prove statements about all natural numbers by induction.

## 1.5 Symbols and their pronunciation

Tables 1–3 list the three alphabets of the series—blackboard-bold letters for the standard number systems and structures, Greek letters such as  $\lambda$  and  $\rho$  for parameters and challenges, and relational and operator symbols—with their English readings: the Greek letters in the International Phonetic Alphabet, the other symbols as the words read aloud. Readings vary between speakers and regions; the alternatives shown are not exhaustive. Each symbol is defined where it is first used; the notes here are reminders, not definitions.

**Table 1.** Blackboard-bold letters. “Blackboard bold” names the double-struck style itself; in speech one usually says only the letter or the system it denotes. Elliptic curves are written italic  $E$  or calligraphic  $\mathcal{E}$ , never blackboard bold. The squared  $\mathbb{P}^2$  of §10 is unrelated: it denotes the projective plane.

| Symbol         | Name         | Said               | Typical role in the series                     |
|----------------|--------------|--------------------|------------------------------------------------|
| $\mathbb{N}$   | blackboard N | “natural numbers”  | the naturals                                   |
| $\mathbb{Z}$   | blackboard Z | “the integers”     | the integers (German <i>Zahlen</i> )           |
| $\mathbb{Q}$   | blackboard Q | “the rationals”    | the rationals ( <i>quotients</i> )             |
| $\mathbb{R}$   | blackboard R | “the reals”        | the real numbers                               |
| $\mathbb{C}$   | blackboard C | “the complexes”    | the complex numbers                            |
| $\mathbb{F}_q$ | blackboard F | “eff-cue”          | the finite field of $q$ elements               |
| $\mathbb{G}$   | blackboard G | “group G”          | an abstract group                              |
| $\mathbb{E}$   | blackboard E | “expectation”      | expectation of a random variable (§11)         |
| $\mathbb{P}$   | blackboard P | “the Pallas curve” | the Pallas group (§10; protocol-spec notation) |

**Table 2.** The 24 letters of the Greek alphabet in lower and upper case. The English names are given in the International Phonetic Alphabet, British (Received Pronunciation) and American (General American), as in the Cambridge, Collins and Oxford dictionaries and Wiktionary, the most widely attested form first; rarer variants are omitted. The last column gives the Modern Greek name in the simplified spelling of the Triantafyllidis dictionary, with its pronunciation; traditional spellings such as  $\gamma\acute{\alpha}\mu\mu\alpha$ ,  $\kappa\acute{\alpha}\pi\pi\alpha$  and  $\mu\upsilon$  are also current. The capital  $\Sigma$  is also the summation sign and names the  $\Sigma$ -protocol family of later volumes; the capital  $\Pi$  is also the product sign and, in those volumes, a conventional name for a protocol or problem.

| Lower                   | Capital | Name    | British                 | American                 | Greek              |
|-------------------------|---------|---------|-------------------------|--------------------------|--------------------|
| $\alpha$                | A       | alpha   | /ˈælfə/                 | /ˈælfə/                  | άλφα [ˈalfa]       |
| $\beta$                 | B       | beta    | /ˈbi:tə/                | /ˈbeɪtə/                 | βήτα [ˈvita]       |
| $\gamma$                | Γ       | gamma   | /ˈgæmə/                 | /ˈgæmə/                  | γάμα [ˈɣama]       |
| $\delta$                | Δ       | delta   | /ˈdeltə/                | /ˈdeltə/                 | δέλτα [ˈdelta]     |
| $\epsilon, \varepsilon$ | E       | epsilon | /ˈepsɪlɒn, ɛpˈsaɪlən/   | /ˈepsələ:n, ˈepsɪlə:n/   | έψιλον [ˈepsilon]  |
| $\zeta$                 | Z       | zeta    | /ˈzi:tə/                | /ˈzeɪtə, ˈzi:tə/         | ζήτα [ˈzita]       |
| $\eta$                  | H       | eta     | /ˈi:tə/                 | /ˈeɪtə, ˈi:tə/           | ήτα [ˈita]         |
| $\theta$                | Θ       | theta   | /ˈθi:tə/                | /ˈθeɪtə, ˈθi:tə/         | θήτα [ˈθita]       |
| $\iota$                 | I       | iota    | /aɪˈəʊtə/               | /aɪˈoʊtə/                | ιώτα [ˈjota]       |
| $\kappa$                | K       | kappa   | /ˈkæpə/                 | /ˈkæpə/                  | κάπα [ˈkapa]       |
| $\lambda$               | Λ       | lambda  | /ˈlæmdə/                | /ˈlæmdə/                 | λάμδα [ˈlamða]     |
| $\mu$                   | M       | mu      | /mju:/                  | /mu:, mju:/              | μι [mi]            |
| $\nu$                   | N       | nu      | /nju:/                  | /nu:, nju:/              | νι [ni]            |
| $\xi$                   | Ξ       | xi      | /saɪ, zaɪ, ksai/        | /saɪ, zaɪ, ksai/         | ξι [ksi]           |
| $\omicron$              | O       | omicron | /əʊˈmaɪkrɒn, ˈɒmɪkrɒn/  | /ˈoʊmɪkra:n, ˈɑ:mɪkra:n/ | όμικρον [ˈomikron] |
| $\pi$                   | Π       | pi      | /paɪ/                   | /paɪ/                    | πι [pi]            |
| $\rho$                  | P       | rho     | /rəʊ/                   | /rou/                    | ρο [ro]            |
| $\sigma$                | Σ       | sigma   | /ˈsɪgmə/                | /ˈsɪgmə/                 | σίγμα [ˈsiɣma]     |
| $\tau$                  | T       | tau     | /taʊ, tɔ:/              | /taʊ, tɔ:/               | ταυ [taf]          |
| $\upsilon$              | Υ       | upsilon | /ju:pˈsaɪlən, ˈʌpsɪlɒn/ | /ˈʌpsələ:n, ˈʊpsɪlə:n/   | ύψιλον [ˈipsilon]  |
| $\phi, \varphi$         | Φ       | phi     | /faɪ/                   | /faɪ, fi:/               | φι [fi]            |
| $\chi$                  | X       | chi     | /kaɪ/                   | /kaɪ, ki:/               | χι [çi]            |
| $\psi$                  | Ψ       | psi     | /psaɪ, saɪ/             | /saɪ, psai, psi:/        | ψι [psi]           |
| $\omega$                | Ω       | omega   | /ˈəʊmɪgə, əʊˈmi:gə/     | /ouˈmeɪgə, ouˈmegə/      | ωμέγα [oˈmɛɣa]     |

**Table 3.** Relational and operator symbols. Every use of  $\oplus$  in the series is the bitwise exclusive or. The encoding  $P^*$  is protocol-spec notation, defined in later volumes.

| Symbol                                | Said                                   | Meaning                                                          |
|---------------------------------------|----------------------------------------|------------------------------------------------------------------|
| $\in, \notin$                         | “in”, “not in”                         | set membership                                                   |
| $\subseteq, \subset$                  | “subset of”                            | (proper) inclusion                                               |
| $\cup, \cap$                          | “union”, “intersect”                   | set union and intersection                                       |
| $A \setminus B$                       | “A minus B”                            | set difference                                                   |
| $\times$                              | “times” / “cross”                      | Cartesian product                                                |
| $ , \nmid$                            | “divides”, “does not divide”           | divisibility                                                     |
| $\preceq, <$                          | “precedes”                             | partial order, strict companion                                  |
| $\equiv$                              | “congruent to”                         | congruence, qualified (mod $n$ )                                 |
| $a \bmod n$                           | “a mod n”                              | remainder of division by $n$                                     |
| $:=, =:$                              | “is defined as”                        | definitional equality                                            |
| $\mapsto$                             | “maps to”                              | the action of a function                                         |
| $g \circ f$                           | “g after f”                            | function composition                                             |
| $\lfloor x \rfloor, \lceil x \rceil$  | “floor”, “ceiling”                     | round down, round up                                             |
| $\log x, \log_2 x, \log_g b$          | “log”, “log base two”, “log base $g$ ” | natural, base-2, and discrete (base $g$ ) logarithm              |
| $\langle a, b \rangle$                | “inner product”                        | inner product; $\langle g \rangle$ the subgroup generated by $g$ |
| $\oplus$                              | “x-or”                                 | bitwise exclusive or                                             |
| $x \parallel y$                       | “x concat y”                           | concatenation of bit or byte strings                             |
| $x \leftarrow D, x \xleftarrow{\$} S$ | “drawn from”                           | sampling ( $\$$ marks uniform); also assignment                  |
| $[k]P$                                | “k times P”                            | scalar multiplication of a point ( $\$10$ )                      |
| $P^*$                                 | “P star”                               | canonical bit-encoding of a point or element                     |
| $\cdot$                               | “dot”                                  | product; argument placeholder, as in $F(\cdot)$                  |
| $ A $                                 | “size of A” / “order of A”             | cardinality                                                      |

## 2 The integers and modular arithmetic

This section develops the integers  $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$  from a small set of assumptions and constructs the integers modulo  $n$ —the set  $\mathbb{Z}/n\mathbb{Z}$  of congruence classes—with its units and the algorithms that compute with them. It proves that division modulo  $n$  is multiplication by a *modular inverse*, that the class of  $a$  has one exactly when  $\gcd(a, n) = 1$  (Theorem 2.24), and that an extension of the algorithm of Euclid’s *Elements* computes it efficiently, as cryptography requires. Every step of a prover or verifier in the proof systems of this series is an addition, a multiplication, or a division of remainders modulo a fixed large prime; addition and multiplication are computed in  $\mathbb{Z}$ , after which multiples of the modulus are discarded. The finite number systems underlying elliptic curves, the scalar arithmetic of Halo 2, and the polynomial commitments of Orchard (short values that bind a prover to entire polynomials, which a verifier then tests by evaluation) all reduce to computations with congruence classes.

### 2.1 Well-ordering and the integers

We take the basic algebraic properties of  $\mathbb{Z}$  for granted: addition and multiplication are associative and commutative, and multiplication distributes over addition; the integers 0 and 1 are identities for addition and multiplication respectively; every integer  $a$  has a negative  $-a$ ; there are no zero divisors (if  $ab = 0$  then  $a = 0$  or  $b = 0$ ); and  $\mathbb{Z}$  carries a total order  $\leq$  compatible with the arithmetic. The one further property assumed as an axiom is the following.

**Theorem 2.1** (Well-ordering principle). *Every nonempty subset  $S \subseteq \mathbb{N} = \{0, 1, 2, \dots\}$  of the natural numbers has a least element: there exists  $m \in S$  such that  $m \leq s$  for all  $s \in S$ .*

The well-ordering principle is logically equivalent to the principle of mathematical induction; we use both freely. The existence results below (division with remainder, termination of the Euclidean algorithm, Bézout’s identity, prime factorisation) all rest on it.

### 2.2 Divisibility

**Definition 2.2.** Let  $a, b \in \mathbb{Z}$ . One says that  $a$  *divides*  $b$ , written  $a \mid b$ , if there exists an integer  $c \in \mathbb{Z}$  with  $b = ac$ . In this case  $a$  is called a *divisor* or *factor* of  $b$ , and  $b$  is a *multiple* of  $a$ . If no such  $c$  exists one writes  $a \nmid b$ .

The following consequences are used throughout, usually without citation.

**Proposition 2.3.** *Let  $a, b, c, d \in \mathbb{Z}$ . Then:*

1.  $a \mid a$  (reflexivity),  $1 \mid a$ , and  $a \mid 0$ .
2. If  $a \mid b$  and  $b \mid c$  then  $a \mid c$  (transitivity).
3. If  $a \mid b$  and  $a \mid c$  then  $a \mid (bx + cy)$  for all  $x, y \in \mathbb{Z}$ ; in particular  $a \mid (b \pm c)$ .
4. If  $a \mid b$  and  $b \neq 0$  then  $|a| \leq |b|$ .

5. If  $a \mid b$  and  $b \mid a$  then  $a = \pm b$ .

6.  $0 \mid b$  if and only if  $b = 0$ .

*Proof.* Part (1) is immediate:  $a = a \cdot 1$ ,  $a = 1 \cdot a$ ,  $0 = a \cdot 0$ . For (2), write  $b = ae$  and  $c = bf$ ; then  $c = a(e f)$ , so  $a \mid c$ . For (3), write  $b = ae$  and  $c = af$ ; then  $bx + cy = a(ex + fy)$ . For (4), write  $b = ac$  with  $b \neq 0$ , so  $c \neq 0$  and hence  $|c| \geq 1$ ; then  $|b| = |a| |c| \geq |a|$ . For (5), if either is 0 then by (6) so is the other and the claim holds; otherwise (4) gives  $|a| \leq |b|$  and  $|b| \leq |a|$ , so  $|a| = |b|$  and  $a = \pm b$ . For (6),  $0 \mid b$  means  $b = 0 \cdot c = 0$ ; conversely  $0 \mid 0$ .  $\square$

Restricted to the natural numbers, where (5) becomes  $a = b$ , the relation  $\mid$  is a partial order in the sense of §1.3; on  $\mathbb{Z}$  it fails to be antisymmetric because of signs. The *units* of  $\mathbb{Z}$ —the integers dividing 1—are exactly  $\pm 1$ , and two integers that divide each other are called *associates*; in  $\mathbb{Z}$  the associates of  $a$  are  $\pm a$ .

## 2.3 The division algorithm

**Theorem 2.4** (Division algorithm). *Let  $a \in \mathbb{Z}$  and  $b \in \mathbb{Z}$  with  $b > 0$ . Then there exist unique integers  $q$  (the quotient) and  $r$  (the remainder) such that*

$$a = bq + r, \quad 0 \leq r < b.$$

*Proof. Existence.* Consider the set

$$S = \{a - bk : k \in \mathbb{Z}, a - bk \geq 0\} \subseteq \mathbb{N}.$$

First,  $S \neq \emptyset$ . If  $a \geq 0$  then  $a = a - b \cdot 0 \in S$ . If  $a < 0$  then, since  $b \geq 1$ , taking  $k = a$  gives  $a - ba = a(1 - b) \geq 0$  (a product of a nonpositive and a nonpositive factor), so  $a - ba \in S$ . Thus  $S$  is a nonempty subset of  $\mathbb{N}$ , and by the well-ordering principle (Theorem 2.1) it has a least element  $r = a - bq$  for some  $q \in \mathbb{Z}$ . By construction  $r \geq 0$ . If  $r \geq b$ , then  $r - b = a - b(q + 1) \geq 0$  would lie in  $S$  and be strictly smaller than  $r$ , contradicting minimality. Hence  $0 \leq r < b$ .

*Uniqueness.* Suppose  $a = bq + r = bq' + r'$  with  $0 \leq r, r' < b$ . Then  $b(q - q') = r' - r$ , so  $b \mid (r' - r)$ . But  $-b < r' - r < b$ , and the only multiple of  $b$  strictly between  $-b$  and  $b$  is 0. Hence  $r' = r$ , and then  $b(q - q') = 0$  with  $b \neq 0$  forces  $q = q'$ .  $\square$

**Remark 2.5.** The hypothesis  $b > 0$  is only for definiteness. For general  $b \neq 0$  one obtains unique  $q, r$  with  $a = bq + r$  and  $0 \leq r < |b|$ , by applying the theorem to  $|b|$  and adjusting the sign of  $q$ . The remainder convention  $0 \leq r < |b|$  differs from the truncation behaviour of many programming languages, where the remainder of a negative dividend may be negative; the results below use the former.

We denote the remainder  $r$  produced by the division algorithm by  $a \bmod b$ , and the quotient  $q$  by  $\lfloor a/b \rfloor$  when  $b > 0$  (the *floor* of the rational number  $a/b$ ). With this notation,  $b \mid a$  holds precisely when  $a \bmod b = 0$ .

## 2.4 The greatest common divisor and the Euclidean algorithm

**Definition 2.6.** A *common divisor* of integers  $a$  and  $b$  is an integer  $d$  with  $d \mid a$  and  $d \mid b$ . The *greatest common divisor* of  $a$  and  $b$ , not both zero, is the largest such  $d$ ; it is denoted  $\gcd(a, b)$ . By convention  $\gcd(0, 0) = 0$ . Integers  $a$  and  $b$  are *coprime* (or *relatively prime*) if  $\gcd(a, b) = 1$ .

The greatest common divisor is well defined when  $(a, b) \neq (0, 0)$ : the set of common divisors is nonempty (it contains 1) and bounded above (every common divisor of, say, the nonzero  $a$  has absolute value at most  $|a|$  by Proposition 2.3(4)), so it has a largest element, which is positive since 1 is a common divisor. The computation of the gcd rests on the following lemma.

**Lemma 2.7.** Let  $a, b \in \mathbb{Z}$ . For any  $q \in \mathbb{Z}$ ,

$$\gcd(a, b) = \gcd(b, a - qb).$$

In particular, if  $a = bq + r$  then  $\gcd(a, b) = \gcd(b, r)$ , and  $\gcd(a, 0) = |a|$ .

*Proof.* It suffices to show the two pairs  $(a, b)$  and  $(b, a - qb)$  have exactly the same common divisors; equal sets of common divisors have equal greatest elements (and the same coprimality status). If  $d \mid a$  and  $d \mid b$ , then  $d \mid (a - qb)$  by Proposition 2.3(3), so  $d$  is a common divisor of  $b$  and  $a - qb$ . Conversely, if  $d \mid b$  and  $d \mid (a - qb)$ , then  $d \mid ((a - qb) + qb) = a$ , so  $d$  is a common divisor of  $a$  and  $b$ . The two sets of common divisors coincide. Finally  $\gcd(a, 0) = |a|$  because the common divisors of  $a$  and 0 are exactly the divisors of  $a$ , the largest of which in absolute value is  $|a|$  (and  $|a| > 0$  when  $a \neq 0$ ; the case  $a = 0$  is the convention  $\gcd(0, 0) = 0$ ).  $\square$

**Theorem 2.8** (Euclidean algorithm). Let  $a, b$  be integers with  $b \neq 0$ . Define a sequence of remainders by  $r_0 = |a|$ ,  $r_1 = |b|$ , and, as long as  $r_i \neq 0$ , let  $r_{i+1} = r_{i-1} \bmod r_i$ , so that

$$r_{i-1} = r_i q_i + r_{i+1}, \quad 0 \leq r_{i+1} < r_i.$$

The sequence  $r_1 > r_2 > \dots$  of positive remainders is strictly decreasing, so some  $r_{n+1} = 0$ ; the last nonzero remainder satisfies  $r_n = \gcd(a, b)$ .

*Proof.* The remainders  $r_1, r_2, \dots$  are nonnegative integers and, while nonzero, strictly decreasing because  $r_{i+1} = r_{i-1} \bmod r_i < r_i$ . A strictly decreasing sequence of nonnegative integers cannot be infinite (again by well-ordering: otherwise the set of values would have no least element), so the algorithm terminates with some  $r_{n+1} = 0$  and  $r_n > 0$ . By Lemma 2.7, each step preserves the gcd:

$$\gcd(a, b) = \gcd(r_0, r_1) = \gcd(r_1, r_2) = \dots = \gcd(r_n, r_{n+1}) = \gcd(r_n, 0) = r_n,$$

using  $\gcd(|a|, |b|) = \gcd(a, b)$  (signs do not affect common divisors).  $\square$

*Example 2.9.* Compute  $\gcd(1071, 462)$ :

$$1071 = 2 \cdot 462 + 147,$$

$$462 = 3 \cdot 147 + 21,$$

$$147 = 7 \cdot 21 + 0.$$

The last nonzero remainder is 21, so  $\gcd(1071, 462) = 21$ . As a check,  $1071 = 21 \cdot 51$  and  $462 = 21 \cdot 22$ , with  $\gcd(51, 22) = 1$ .

*Remark 2.10.* The number of division steps of the Euclidean algorithm for inputs  $a > b > 0$  is  $O(\log b)$ ; the worst case occurs at consecutive Fibonacci numbers (Lamé's theorem), where every quotient  $q_i$  equals 1 except the final one (which equals 2) and the remainders shrink as slowly as possible. With schoolbook arithmetic on  $k$ -bit inputs the whole computation costs  $O(k^2)$  bit operations. This bound makes gcd-based operations such as modular inversion practical at the sizes used in Halo 2 and Orchard.

## 2.5 The Bézout identity and the extended Euclidean algorithm

Back-substitution through the equations of the Euclidean algorithm expresses the gcd as an *integer linear combination* of the inputs.

**Theorem 2.11** (Bézout's identity). *Let  $a, b \in \mathbb{Z}$ , not both zero, and let  $d = \gcd(a, b)$ . Then there exist integers  $u, v \in \mathbb{Z}$  with*

$$au + bv = d.$$

*Moreover  $d$  is the smallest positive integer expressible in the form  $ax + by$  with  $x, y \in \mathbb{Z}$ , and the set of all such combinations is exactly the set of multiples of  $d$ :*

$$\{ax + by : x, y \in \mathbb{Z}\} = d\mathbb{Z} = \{kd : k \in \mathbb{Z}\}.$$

*Proof.* Let  $I = \{ax + by : x, y \in \mathbb{Z}\}$ . This set contains  $a$  and  $b$  and is closed under addition and under multiplication by any integer; concretely, if  $s = ax_1 + by_1$  and  $t = ax_2 + by_2$  lie in  $I$ , then for any  $m, n \in \mathbb{Z}$  one has

$$ms + nt = a(mx_1 + nx_2) + b(my_1 + ny_2) \in I.$$

Since  $a, b$  are not both zero,  $I$  contains a positive element (one of  $\pm a, \pm b$ ), so the set  $I \cap \mathbb{N}_{>0}$  of positive elements of  $I$  is nonempty. By well-ordering it has a least element  $g = au + bv > 0$ .

We claim that  $g$  divides every element of  $I$ . Indeed, let  $c \in I$  and write  $c = gq + r$  with  $0 \leq r < g$  by the division algorithm. Then  $r = c - gq \in I$  because  $I$  is closed under the operations just described. By minimality of  $g$  among positive elements of  $I$ , and  $0 \leq r < g$ , necessarily  $r = 0$ . Hence  $g \mid c$  for every  $c \in I$ ; in particular  $g \mid a$  and  $g \mid b$ , so  $g$  is a common divisor of  $a$  and  $b$ . Conversely, any common divisor  $e$  of  $a$  and  $b$  divides  $au + bv = g$  by Proposition 2.3(3). Therefore  $g$  is a common divisor that is divisible by every common divisor; being positive, it is in particular the *greatest* common divisor, so  $g = d$ . The displayed equation  $au + bv = d$  follows,  $d$  is the least positive element of  $I$ , and  $I = g\mathbb{Z} = d\mathbb{Z}$  because every element of  $I$  is a multiple of  $g$ , as shown, and conversely every multiple  $kg = a(ku) + b(kv)$  lies in  $I$ .  $\square$

*Remark 2.12.* The proof yields a characterisation of the gcd that is independent of the order  $\leq$ :  $d = \gcd(a, b)$  is, up to sign, the unique common divisor of  $a$  and  $b$  that is divisible by every common divisor. Later sections use this formulation outside the integers, where “largest” may make no sense but “divisible by every common divisor” still does. We call a nonempty subset  $I \subseteq$

$\mathbb{Z}$  closed under addition and under multiplication by arbitrary integers an *ideal*; the content of Bézout's identity is that every ideal of  $\mathbb{Z}$  has the form  $d\mathbb{Z}$  for a single generator  $d$ . Ideals recur in Section 4 in a construction that generalises the passage from integers to congruence classes below.

The Bézout coefficients  $u, v$  are not unique. If  $au + bv = d$ , then for every  $k \in \mathbb{Z}$ ,

$$a\left(u + k\frac{b}{d}\right) + b\left(v - k\frac{a}{d}\right) = d,$$

since the added terms  $k\frac{ab}{d}$  and  $-k\frac{ab}{d}$  cancel; and these are all the solutions. To prove completeness, observe first that  $a/d$  and  $b/d$  are coprime: a common divisor  $e$  of both makes  $de$  a common divisor of  $a$  and  $b$ , so  $de \mid au + bv = d$  by Proposition 2.3(3), forcing  $e = \pm 1$ . Now suppose  $au' + bv' = d$ . Subtracting  $au + bv = d$  and dividing by  $d > 0$  gives  $\frac{a}{d}(u' - u) = -\frac{b}{d}(v' - v)$ , so  $b/d$  divides  $\frac{a}{d}(u' - u)$ ; by Proposition 2.15(1) below (whose proof rests only on Bézout's identity, already established),  $b/d$  divides  $u' - u$ , say  $u' = u + k\frac{b}{d}$  with  $k \in \mathbb{Z}$ . Substituting back gives  $b(v' - v + k\frac{a}{d}) = 0$ , so  $v' = v - k\frac{a}{d}$  when  $b \neq 0$ ; and when  $b = 0$  we have  $u' = u$ ,  $a/d = \pm 1$ , and  $k = (v - v')\frac{a}{d}$  places  $(u', v')$  in the family.

**Theorem 2.13** (Extended Euclidean algorithm). *There is an algorithm that, on input  $a, b \in \mathbb{Z}$  not both zero, outputs  $d = \gcd(a, b)$  together with integers  $u, v$  satisfying  $au + bv = d$ , using the same number of division steps as the Euclidean algorithm.*

*Proof.* Alongside the remainder sequence of Theorem 2.8, maintain two auxiliary sequences  $(s_i)$  and  $(t_i)$  with the invariant

$$r_i = as_i + bt_i \quad \text{for all } i. \quad (*)$$

Initialise (taking  $a, b \geq 0$  for clarity; we restore signs at the end)

$$r_0 = a, s_0 = 1, t_0 = 0; \quad r_1 = b, s_1 = 0, t_1 = 1,$$

so  $(*)$  holds for  $i = 0, 1$ . At step  $i \geq 1$ , having  $r_{i-1} = r_i q_i + r_{i+1}$  from the division algorithm, set

$$s_{i+1} = s_{i-1} - q_i s_i, \quad t_{i+1} = t_{i-1} - q_i t_i.$$

Then

$$as_{i+1} + bt_{i+1} = (as_{i-1} + bt_{i-1}) - q_i(as_i + bt_i) = r_{i-1} - q_i r_i = r_{i+1},$$

so induction preserves  $(*)$ . When the algorithm terminates with  $r_{n+1} = 0$  and  $r_n = d = \gcd(a, b)$ , the invariant for  $i = n$  reads  $d = as_n + bt_n$ ; output  $u = s_n, v = t_n$ . If  $b = 0$ , then  $a \neq 0$  and termination is immediate with  $n = 0$ : no division step runs, the last nonzero remainder is  $r_0 = a = \gcd(a, 0)$  (Lemma 2.7), and the invariant at  $i = 0$  already gives the output  $(d, u, v) = (a, 1, 0)$ . If  $b \neq 0$ , termination and the value of  $d$  are exactly as in Theorem 2.8. In either case each step does only a constant amount of extra work. For general inputs, run the algorithm on  $(|a|, |b|)$  to obtain  $|a|u' + |b|v' = d$  (recall  $\gcd(|a|, |b|) = \gcd(a, b)$ ) and output  $u = \operatorname{sgn}(a)u', v = \operatorname{sgn}(b)v'$ , so that  $au + bv = d$ .  $\square$

*Example 2.14.* Run the extended algorithm on  $a = 1071, b = 462$  (continuing Example 2.9). The quotients are  $q_1 = 2, q_2 = 3, q_3 = 7$ .

| $i$ | $r_i$ | $s_i$ | $t_i$ |
|-----|-------|-------|-------|
| 0   | 1071  | 1     | 0     |
| 1   | 462   | 0     | 1     |
| 2   | 147   | 1     | -2    |
| 3   | 21    | -3    | 7     |
| 4   | 0     | 22    | -51   |

For instance, row 2 is row 0 minus  $q_1 = 2$  times row 1:  $s_2 = 1 - 2 \cdot 0 = 1, t_2 = 0 - 2 \cdot 1 = -2$ , and  $1071 \cdot 1 + 462 \cdot (-2) = 1071 - 924 = 147$ . The last nonzero remainder is  $r_3 = 21 = \gcd(1071, 462)$ , with

$$1071 \cdot (-3) + 462 \cdot 7 = -3213 + 3234 = 21.$$

Thus  $(u, v) = (-3, 7)$ . The final row  $(s_4, t_4) = (22, -51) = (b/d, -a/d)$  up to sign records the homogeneous solution:  $1071 \cdot 22 + 462 \cdot (-51) = 0$ .

**Proposition 2.15** (Euclid's lemma). *Let  $a, b, c \in \mathbb{Z}$ .*

1. *If  $c \mid ab$  and  $\gcd(c, a) = 1$ , then  $c \mid b$ .*
2. *If  $\gcd(a, b) = 1, a \mid n$ , and  $b \mid n$ , then  $ab \mid n$ .*
3. *If  $\gcd(a, b) = 1$  and  $\gcd(a, c) = 1$ , then  $\gcd(a, bc) = 1$ .*

*Proof.* (1) By Bézout there are  $u, v$  with  $cu + av = 1$ . Multiply by  $b$ :  $b = cub + avb = c(ub) + (ab)v$ . Since  $c \mid ab$ ,  $c$  divides both terms on the right, hence  $c \mid b$ .

(2) Write  $n = ak$ . Since  $b \mid n = ak$  and  $\gcd(b, a) = 1$ , part (1) gives  $b \mid k$ , say  $k = bm$ . Then  $n = a(bm) = (ab)m$ , so  $ab \mid n$ .

(3) By Bézout,  $ax_1 + by_1 = 1$  and  $ax_2 + cy_2 = 1$  for suitable integers. Multiplying,

$$1 = (ax_1 + by_1)(ax_2 + cy_2) = a \underbrace{(ax_1x_2 + x_1cy_2 + by_1x_2)}_{\in \mathbb{Z}} + bc(y_1y_2),$$

which exhibits 1 as an integer combination of  $a$  and  $bc$ . Hence  $\gcd(a, bc)$  divides 1, so it equals 1.  $\square$

## 2.6 Primes and the fundamental theorem of arithmetic

**Definition 2.16.** An integer  $p > 1$  is *prime* if its only positive divisors are 1 and  $p$ . An integer  $n > 1$  that is not prime is *composite*; thus  $n$  is composite iff  $n = ab$  for some integers  $1 < a, b < n$ . The integer 1 is a *unit* and is by convention neither prime nor composite.

Excluding 1 from the primes is necessary for factorisation into primes to be unique, since otherwise arbitrarily many factors of 1 could be inserted. The property of primes on which uniqueness rests is the prime case of Euclid's lemma.

**Lemma 2.17** (Euclid's lemma for primes). *Let  $p$  be prime and  $a, b \in \mathbb{Z}$ . If  $p \mid ab$  then  $p \mid a$  or  $p \mid b$ . More generally, if  $p \mid a_1 a_2 \cdots a_k$  then  $p \mid a_i$  for some  $i$ .*

*Proof.* The only positive divisors of  $p$  are 1 and  $p$ , so  $\gcd(p, a) \in \{1, p\}$ . If  $\gcd(p, a) = p$  then  $p \mid a$  and the claim holds; otherwise  $\gcd(p, a) = 1$ , and Proposition 2.15(1) gives  $p \mid b$ . The general statement follows by induction on  $k$ : from  $p \mid a_1(a_2 \cdots a_k)$  it follows that  $p \mid a_1$  or  $p \mid a_2 \cdots a_k$ , and the inductive hypothesis handles the latter.  $\square$

**Theorem 2.18** (Fundamental theorem of arithmetic). *Every integer  $n > 1$  can be written as a product of primes,*

$$n = p_1 p_2 \cdots p_k,$$

*and this factorisation is unique up to the order of the factors. Equivalently, there is a unique expression*

$$n = p_1^{e_1} p_2^{e_2} \cdots p_r^{e_r}, \quad p_1 < p_2 < \cdots < p_r \text{ prime, } e_i \geq 1.$$

*Proof. Existence.* Suppose, for contradiction, that some integer  $> 1$  is not a product of primes, and let  $n$  be the least such (well-ordering). Then  $n$  is not prime (a prime is trivially a product of one prime), so  $n = ab$  with  $1 < a, b < n$ . By minimality, each of  $a, b$  is a product of primes; concatenating these factorisations expresses  $n$  as a product of primes, a contradiction. Hence every  $n > 1$  factors into primes.

*Uniqueness.* Suppose

$$p_1 p_2 \cdots p_k = q_1 q_2 \cdots q_\ell$$

are two factorisations of the same integer into primes. The argument proceeds by induction on  $k$ . If  $k = 0$  the product is the empty product 1, forcing  $\ell = 0$  as well (a nonempty product of primes is  $> 1$ ). For the inductive step,  $p_1$  divides the right-hand product, so by Lemma 2.17  $p_1 \mid q_j$  for some  $j$ . Since  $q_j$  is prime, its only divisor  $> 1$  is itself, so  $p_1 = q_j$ . Cancelling this common factor (legitimate as  $\mathbb{Z}$  has no zero divisors) yields

$$p_2 \cdots p_k = q_1 \cdots \widehat{q_j} \cdots q_\ell,$$

where the hat denotes omission. By the inductive hypothesis these two factorisations agree up to order, with  $k - 1 = \ell - 1$ ; restoring  $p_1 = q_j$  shows the original factorisations agree up to order. Collecting equal primes into prime powers gives the canonical form, whose uniqueness is immediate from the uniqueness of the multiset of primes—the factors counted with multiplicity, without regard to order.  $\square$

*Remark 2.19.* Existence rests on the order structure (well-ordering); uniqueness rests on Euclid's lemma, hence on Bézout, hence on the division algorithm. In some number systems with a division-like structure, Euclid's lemma fails and factorisation into irreducibles is *not* unique.

## 2.7 Congruences and residue classes

Congruence formalises the arithmetic of remainders. Fix an integer  $n \geq 1$ , the *modulus*.

**Definition 2.20.** Let  $n \geq 1$  and  $a, b \in \mathbb{Z}$ . The integer  $a$  is *congruent to  $b$  modulo  $n$* , written

$$a \equiv b \pmod{n},$$

if  $n \mid (a - b)$ , equivalently if  $a$  and  $b$  leave the same remainder upon division by  $n$ .

The two formulations agree: if  $a = nq_1 + r_1$  and  $b = nq_2 + r_2$  with  $0 \leq r_1, r_2 < n$ , then  $a - b = n(q_1 - q_2) + (r_1 - r_2)$ , and  $n \mid (a - b)$  iff  $n \mid (r_1 - r_2)$  iff  $r_1 = r_2$  (since  $|r_1 - r_2| < n$ ).

**Proposition 2.21.** Fix  $n \geq 1$ . Congruence modulo  $n$  is an equivalence relation on  $\mathbb{Z}$  that is moreover compatible with addition and multiplication: if  $a \equiv a' \pmod{n}$  and  $b \equiv b' \pmod{n}$ , then

$$a + b \equiv a' + b' \pmod{n}, \quad ab \equiv a'b' \pmod{n}.$$

Consequently  $a^k \equiv (a')^k \pmod{n}$  for all  $k \geq 0$ , and one may substitute congruent values freely in any polynomial expression with integer coefficients.

*Proof.* Reflexivity ( $n \mid 0$ ), symmetry ( $n \mid (a - b) \Rightarrow n \mid (b - a)$ ), and transitivity ( $n \mid (a - b)$  and  $n \mid (b - c)$  give  $n \mid (a - c)$  by Proposition 2.3(3)) are immediate. For compatibility, write  $a - a' = ns$  and  $b - b' = nt$ . Then

$$(a + b) - (a' + b') = n(s + t),$$

and

$$ab - a'b' = ab - a'b + a'b - a'b' = (a - a')b + a'(b - b') = n(sb + a't),$$

so both  $n \mid ((a + b) - (a' + b'))$  and  $n \mid (ab - a'b')$ . The statement about powers and polynomial expressions follows by induction from the multiplicative case.  $\square$

Because congruence is an equivalence relation, it partitions  $\mathbb{Z}$  into disjoint *congruence classes* (also called *residue classes*).

**Definition 2.22.** The *congruence class* (or *residue class*) of  $a$  modulo  $n$  is

$$[a]_n = \{b \in \mathbb{Z} : b \equiv a \pmod{n}\} = \{a + kn : k \in \mathbb{Z}\} = a + n\mathbb{Z}.$$

We write the set of all congruence classes as

$$\mathbb{Z}/n\mathbb{Z} = \{[a]_n : a \in \mathbb{Z}\}.$$

By the division algorithm every integer is congruent modulo  $n$  to exactly one of  $0, 1, \dots, n - 1$ , namely its remainder; hence the  $n$  classes  $[0]_n, [1]_n, \dots, [n - 1]_n$  are distinct and exhaust  $\mathbb{Z}/n\mathbb{Z}$ . The set  $\{0, 1, \dots, n - 1\}$  is the standard *system of representatives*, called the *least nonnegative residues*.

## 2.8 Units modulo $n$ and modular inverses

Division is the inverse of multiplication, so division of remainders modulo  $n$  reduces to determining the  $a$  for which the congruence  $ax \equiv 1 \pmod{n}$  admits a solution.

**Definition 2.23.** Let  $n \geq 1$ . A class  $[a]_n \in \mathbb{Z}/n\mathbb{Z}$  is a *unit* (or is *invertible*) if there exists an integer  $x$  with  $ax \equiv 1 \pmod{n}$ . Any class  $[x]_n$  with this property is the (*modular*) *inverse* of  $[a]_n$ , written  $[a]_n^{-1}$ ; one also calls  $x$  a modular inverse of  $a$  modulo  $n$ . We write the set of units as

$$(\mathbb{Z}/n\mathbb{Z})^\times = \{ [a]_n : [a]_n \text{ is a unit} \}.$$

The definition is well posed on classes: by Proposition 2.21, if  $a' \equiv a$  and  $x' \equiv x \pmod{n}$  then  $a'x' \equiv ax \pmod{n}$ , so whether  $ax \equiv 1 \pmod{n}$  holds depends only on the classes  $[a]_n$  and  $[x]_n$ . The uniqueness part of the next theorem justifies the definite article in “the inverse”.

**Theorem 2.24.** Let  $n \geq 1$  and  $a \in \mathbb{Z}$ . The class  $[a]_n$  is a unit in  $\mathbb{Z}/n\mathbb{Z}$  if and only if  $\gcd(a, n) = 1$ . When  $\gcd(a, n) = 1$ , the inverse is unique, and the extended Euclidean algorithm computes a representative: if  $au + nv = 1$ , then  $[a]_n^{-1} = [u]_n$ .

*Proof.* Suppose  $\gcd(a, n) = 1$ . By Bézout (Theorem 2.11) there are  $u, v$  with  $au + nv = 1$ . Reducing modulo  $n$ ,  $au \equiv 1 \pmod{n}$ , so  $[u]_n$  is an inverse of  $[a]_n$ . The extended Euclidean algorithm (Theorem 2.13) produces such a  $u$  explicitly.

Conversely, if  $[a]_n$  is a unit, say  $ax \equiv 1 \pmod{n}$ , then  $ax - 1 = nk$  for some  $k$ , i.e.  $ax - nk = 1$ . Any common divisor of  $a$  and  $n$  divides the left side, hence divides 1; thus  $\gcd(a, n) = 1$ .

Uniqueness: if  $ax \equiv 1$  and  $ax' \equiv 1 \pmod{n}$ , then  $x \equiv x(ax') \equiv (xa)x' \equiv x' \pmod{n}$ —a computation using only the associativity and commutativity of integer multiplication together with Proposition 2.21—so  $[x]_n = [x']_n$ .  $\square$

*Example 2.25.* Consider the computation of  $[17]_{43}^{-1}$ . The extended Euclidean algorithm on  $(43, 17)$ :

$$\begin{aligned} 43 &= 2 \cdot 17 + 9, \\ 17 &= 1 \cdot 9 + 8, \\ 9 &= 1 \cdot 8 + 1, \\ 8 &= 8 \cdot 1 + 0, \end{aligned}$$

so  $\gcd(43, 17) = 1$ . Back-substituting:

$$1 = 9 - 8 = 9 - (17 - 9) = 2 \cdot 9 - 17 = 2(43 - 2 \cdot 17) - 17 = 2 \cdot 43 - 5 \cdot 17.$$

Hence  $-5 \cdot 17 \equiv 1 \pmod{43}$ , and  $[17]_{43}^{-1} = [-5]_{43} = [38]_{43}$ . Check:  $17 \cdot 38 = 646 = 15 \cdot 43 + 1$ , so  $17 \cdot 38 \equiv 1 \pmod{43}$ .

Later sections use the number of units.

**Definition 2.26** (Euler totient). *Euler’s totient function*  $\varphi(n)$  is the number of integers  $a$  with  $1 \leq a \leq n$  and  $\gcd(a, n) = 1$ . Equivalently,  $\varphi(n) = |(\mathbb{Z}/n\mathbb{Z})^\times|$ .

The two counts agree: the classes of  $1, 2, \dots, n$  run exactly once through all of  $\mathbb{Z}/n\mathbb{Z}$  (the class of  $n$  being  $[0]_n$ ), and by Theorem 2.24 the class  $[a]_n$  is a unit precisely when  $\gcd(a, n) = 1$ .

*Example 2.27.* We have  $\varphi(1) = 1$ , and  $\varphi(p) = p - 1$  for prime  $p$ : for  $1 \leq a < p$  the positive divisor  $\gcd(a, p)$  of  $p$  is 1 or  $p$ , and  $p \nmid a$  rules out  $p$ , so every one of the  $p - 1$  nonzero classes is a unit. By direct enumeration,  $\varphi(12) = |\{1, 5, 7, 11\}| = 4$ .

Division by  $a$  modulo  $n$  is therefore multiplication by the inverse class  $[a]_n^{-1}$ . By Theorem 2.24 that inverse exists precisely when  $\gcd(a, n) = 1$ , hence for every nonzero class when the modulus is prime, and the extended Euclidean algorithm computes it in  $O(\log n)$  division steps (Remark 2.10). The laws used in this section—associativity, commutativity, distributivity, identities, negatives, and inverses—are the defining laws of the structures that Sections 3 and 4 introduce, with  $\mathbb{Z}/n\mathbb{Z}$  as an instance.

## 3 Groups

This section defines the *group*, a set with one associative operation that has an identity and inverses, and proves the results on groups that the series uses: Lagrange's theorem (Theorem 3.29), with the corollary on which the rest of the volume relies, Euler's theorem and Fermat's little theorem (Corollary 3.32); the subgroups and the classification of cyclic groups (Theorems 3.24 and 3.43); the quotient of an abelian group by a subgroup; and the basic theory of homomorphisms and isomorphisms. Almost every object met later in this series is a group: the integers modulo a prime under addition, the nonzero residues under multiplication, the points of an elliptic curve, the symmetries of a finite field. Key agreement over a public channel needs this structure: associativity makes repeated combination of a public element with itself independent of bracketing, and identity and inverses make repetition counts obey an arithmetic.

### 3.1 The group axioms

**Definition 3.1** (Binary operation). Let  $S$  be a set. A *binary operation* on  $S$  is a function

$$*: S \times S \rightarrow S, \quad (a, b) \mapsto a * b.$$

Closure, the property that  $a * b$  lies in  $S$ , is part of the statement that  $*$  maps *into*  $S$ . It is named separately because  $*$  restricts to a binary operation on a subset  $H \subseteq S$  exactly when  $H$  is *closed* under  $*$ , the condition on which subgroups (§3.4) rest.

**Definition 3.2** (Group). A *group* is a pair  $(G, *)$  consisting of a set  $G$  and a binary operation  $*$  on  $G$  satisfying three axioms.

(G1) *Associativity*. For all  $a, b, c \in G$ ,  $(a * b) * c = a * (b * c)$ .

(G2) *Identity*. There exists  $e \in G$  such that  $e * a = a * e = a$  for all  $a \in G$ .

(G3) *Inverses*. For each  $a \in G$  there exists  $b \in G$  such that  $a * b = b * a = e$ , where  $e$  is an identity as in (G2).

A set with an operation satisfying only (G1) is a *semigroup*; if it also satisfies (G2) it is a *monoid*. A group is thus a monoid in which every element is invertible.

The number of elements of  $G$ , written  $|G|$ , is the *order* of the group; if  $|G|$  is finite,  $G$  is a *finite group*, and otherwise an *infinite group*. When the operation is understood, one writes “the group  $G$ ”.

The axioms assert only the *existence* of an identity and of inverses; their uniqueness, proved next, justifies the notations  $e$  and  $a^{-1}$ .

**Proposition 3.3.** *Let  $(G, *)$  be a group.*

1. *The identity element is unique.*
2. *Each  $a \in G$  has a unique inverse, denoted  $a^{-1}$ .*
3. (Cancellation.) *For all  $a, x, y \in G$ : if  $a * x = a * y$  then  $x = y$ , and if  $x * a = y * a$  then  $x = y$ .*
4. *For all  $a, b \in G$ ,  $(a^{-1})^{-1} = a$  and  $(a * b)^{-1} = b^{-1} * a^{-1}$ .*

*Proof.* (1) Suppose  $e$  and  $e'$  both satisfy (G2). Using that  $e'$  is an identity,  $e = e * e'$ ; using that  $e$  is an identity,  $e * e' = e'$ . Hence  $e = e'$ .

(2) Suppose  $b$  and  $b'$  are both inverses of  $a$ . Then

$$b = b * e = b * (a * b') = (b * a) * b' = e * b' = b',$$

using (G2), the inverse property of  $b'$ , associativity, and the inverse property of  $b$ .

(3) Suppose  $a * x = a * y$ . Multiplying on the left by  $a^{-1}$  and reassociating,

$$x = (a^{-1} * a) * x = a^{-1} * (a * x) = a^{-1} * (a * y) = (a^{-1} * a) * y = y.$$

The right-cancellation law is proved symmetrically, multiplying on the right by  $a^{-1}$ .

(4) Since  $a * a^{-1} = a^{-1} * a = e$ , the element  $a$  is an inverse of  $a^{-1}$ ; by (2) it is *the* inverse, so  $(a^{-1})^{-1} = a$ . For the product formula, compute

$$(a * b) * (b^{-1} * a^{-1}) = a * (b * b^{-1}) * a^{-1} = a * e * a^{-1} = e,$$

and symmetrically  $(b^{-1} * a^{-1}) * (a * b) = e$ ; by (2),  $(a * b)^{-1} = b^{-1} * a^{-1}$ . The reversal of order is known as the “socks–shoes” rule.  $\square$

Two notational conventions are standard. *Multiplicative* notation writes the operation as juxtaposition  $ab$  (or  $a \cdot b$ ), the identity as 1, the inverse as  $a^{-1}$ , and powers as  $a^n = \underbrace{a \cdots a}_{n \text{ factors}}$  for  $n \geq 1$ , with  $a^0 = 1$  and  $a^{-n} = (a^{-1})^n$ . *Additive* notation, reserved almost exclusively for commutative groups because the symbol  $+$  connotes commutativity, writes  $a + b$ , the identity as 0, the inverse as  $-a$ , and  $na$  for the  $n$ -fold sum. The exponent laws

$$a^m a^n = a^{m+n}, \quad (a^m)^n = a^{mn}$$

hold for all integers  $m, n$ , proved by induction together with Proposition 3.3. Henceforth  $ab$  denotes  $a * b$ , and the symbol  $*$  is omitted unless needed for clarity.

The power notation presupposes that an  $n$ -fold product does not depend on its bracketing, whereas axiom (G1) concerns only three factors. The next proposition proves this; thereafter products and powers are written without brackets.

**Proposition 3.4** (Generalised associativity). *Let  $a_1, \dots, a_n$ ,  $n \geq 1$ , be elements of a group—or of any set with an associative operation. Every way of bracketing the product  $a_1 a_2 \dots a_n$  yields the same element, namely the left-normed product  $L_n = (\dots ((a_1 a_2) a_3) \dots) a_n$ . In particular  $a^n$  is a single well-defined element, however the  $n$  factors are combined.*

*Proof.* By strong induction on  $n$ . For  $n \leq 2$  there is only one bracketing, and for  $n = 3$  the two bracketings agree by (G1). Let  $n \geq 3$  and assume the claim for all shorter products. Any bracketed product  $P$  of  $a_1, \dots, a_n$  has an outermost multiplication, splitting it as  $P = QR$  with  $Q$  a bracketed product of  $a_1, \dots, a_k$  and  $R$  one of  $a_{k+1}, \dots, a_n$ , for some  $1 \leq k < n$ . The induction hypothesis evaluates both halves:  $Q = L_k$ , and  $R$  is the left-normed product of  $a_{k+1}, \dots, a_n$ . If  $k = n - 1$ , then  $R = a_n$  and  $P = L_{n-1} a_n = L_n$  directly. If  $k < n - 1$ , write  $R = R' a_n$  with  $R'$  the left-normed product of  $a_{k+1}, \dots, a_{n-1}$ ; then (G1) and the induction hypothesis, applied to the bracketed product  $L_k R'$  of the  $n - 1$  elements  $a_1, \dots, a_{n-1}$ , give

$$P = L_k(R' a_n) = (L_k R') a_n = L_{n-1} a_n = L_n.$$

Every bracketing therefore equals  $L_n$ . □

**Definition 3.5** (Abelian group). A group  $(G, *)$  is *abelian* (or *commutative*) if  $a * b = b * a$  for all  $a, b \in G$ ; otherwise it is *non-abelian*.

The term honours Niels Henrik Abel. In an abelian group the factors of a product may be reordered arbitrarily, and the socks–shoes formula becomes  $(ab)^{-1} = b^{-1} a^{-1} = a^{-1} b^{-1}$ .

## 3.2 First examples

*Example 3.6.* The pair  $(\mathbb{Z}, +)$  is an infinite abelian group: addition is associative and commutative, the identity is 0, and the inverse of  $n$  is  $-n$ . Likewise  $(\mathbb{Q}, +)$ ,  $(\mathbb{R}, +)$ , and  $(\mathbb{C}, +)$  are abelian groups. By contrast  $(\mathbb{Z}, \cdot)$  is *not* a group: the integer 2 has no multiplicative inverse in  $\mathbb{Z}$ .

*Example 3.7.* The nonzero rationals  $\mathbb{Q}^\times = \mathbb{Q} \setminus \{0\}$  form an abelian group under multiplication, with identity 1 and inverse  $1/a$ ; likewise  $\mathbb{R}^\times$  and  $\mathbb{C}^\times$ . Zero must be excluded because it has no multiplicative inverse. More generally, for any field  $F$  (for example  $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ ; the abstract notion is defined in Section 4) the nonzero elements  $F^\times = F \setminus \{0\}$  form an abelian group under multiplication. This property is part of the definition of a field.

Most finite examples below are built from congruence classes. On the set  $\mathbb{Z}/n\mathbb{Z}$  of residue classes  $[a]_n$  of Section 2, define operations through representatives,

$$[a]_n + [b]_n = [a + b]_n, \quad [a]_n [b]_n = [ab]_n.$$

By Proposition 2.21 the right-hand sides do not depend on the representatives chosen, so both operations are well defined, and each law of integer arithmetic—associativity, commutativity, the identities  $[0]_n$  and  $[1]_n$ , negatives—passes to classes by computing on representatives. (The full structural statement, that these operations make  $\mathbb{Z}/n\mathbb{Z}$  a commutative ring, is Theorem 4.6 in Section 4; the present section uses only the laws listed above.) Under addition, all of  $\mathbb{Z}/n\mathbb{Z}$  is a group, as Example 3.9 below records. For  $n > 1$ , multiplication does not make it a group, since  $[0]_n$  has no inverse; Section 2 identified the invertible classes as the *units*, the classes  $[a]_n$  with  $\gcd(a, n) = 1$  (Theorem 2.24). The units form a group.

**Proposition 3.8.** *Under multiplication of classes, the set  $(\mathbb{Z}/n\mathbb{Z})^\times$  of units of  $\mathbb{Z}/n\mathbb{Z}$  is an abelian group. Its order is  $|\mathbb{Z}/n\mathbb{Z}|^\times = \varphi(n)$ , Euler's totient of Definition 2.26: the number of integers  $a$  with  $1 \leq a \leq n$  and  $\gcd(a, n) = 1$ .*

*Proof.* Multiplication of classes is associative and commutative with identity  $[1]_n$ , as derived above from Proposition 2.21 by computing on representatives, and  $[1]_n$  is a unit (it is its own inverse). The operation is closed on units: if  $[a]_n$  and  $[b]_n$  are units with inverses  $[a]_n^{-1}$  and  $[b]_n^{-1}$ , then  $[ab]_n$  has inverse  $[b]_n^{-1}[a]_n^{-1}$ , since  $(ab)(b^{-1}a^{-1}) \equiv 1 \pmod{n}$  for any representatives of the inverse classes. Every unit has an inverse by definition, and that inverse is itself a unit (it is invertible, with inverse the original class). Hence  $(\mathbb{Z}/n\mathbb{Z})^\times$  is an abelian group. By Theorem 2.24 its elements are exactly the classes  $[a]_n$  with  $\gcd(a, n) = 1$  and  $1 \leq a \leq n$ , of which there are  $\varphi(n)$  by Definition 2.26.  $\square$

*Example 3.9.* The pair  $(\mathbb{Z}/n\mathbb{Z}, +)$  is an abelian group of order  $n$ : the discussion above derived associativity, commutativity, the identity  $[0]_n$ , and negatives from Proposition 2.21, and the classes number exactly  $n$  (the least nonnegative residues, Definition 2.22). The invertible classes form the abelian group  $(\mathbb{Z}/n\mathbb{Z})^\times$  of order  $\varphi(n)$  under multiplication (Theorem 2.24 and Proposition 3.8). When  $n = p$  is prime, every nonzero class is a unit (Example 2.27), so  $(\mathbb{Z}/p\mathbb{Z})^\times$  has order  $p - 1$ —this is Example 3.7 specialised to  $\mathbb{Z}/p\mathbb{Z}$ , which Section 4 shows to be a field and denotes  $\mathbb{F}_p$ .

*Example 3.10.* The one-element set  $\{e\}$ , with its only possible operation  $e * e = e$ , is the *trivial group*. The group  $\mathbb{Z}/2\mathbb{Z} = \{0, 1\}$  is the smallest nontrivial group. A further example of small order is the *Klein four-group*  $V = \{e, a, b, c\}$ , in which every element composed with itself gives  $e$  and the product of any two distinct nonidentity elements is the third; the verification that this table satisfies the axioms is routine. The group  $V$  is abelian of order 4.

The groups  $V$  and  $\mathbb{Z}/4\mathbb{Z}$  both have order 4 but are not isomorphic, that is, structurally different (§3.8): in  $V$  every element combined with itself gives the identity, so  $V$  has no element of order 4 in the sense defined next, whereas in  $\mathbb{Z}/4\mathbb{Z}$  the class 1 must be added to itself four times to give 0.

### 3.3 The order of an element

**Definition 3.11** (Order of an element). Let  $G$  be a group and  $a \in G$ . The *order* of  $a$ , written  $\text{ord}(a)$ , is the least positive integer  $n$  with  $a^n = e$ , if such an  $n$  exists;  $a$  then has *finite order*. If no positive power of  $a$  equals  $e$ , then  $a$  has *infinite order* and one writes  $\text{ord}(a) = \infty$ .

The order  $\text{ord}(a)$  of an element shares its name with the order  $|G|$  of a group because it equals the order of the smallest subgroup of  $G$  containing  $a$  (Proposition 3.21).

**Proposition 3.12.** *Let  $a$  have finite order  $d = \text{ord}(a)$ , and let  $m, k \in \mathbb{Z}$ .*

1.  $a^m = e$  if and only if  $d \mid m$ .
2.  $a^m = a^k$  if and only if  $m \equiv k \pmod{d}$ .
3. The distinct powers of  $a$  are exactly  $e = a^0, a^1, \dots, a^{d-1}$ .

*Proof.* (1) If  $d \mid m$ , write  $m = dq$ ; then  $a^m = (a^d)^q = e^q = e$ . Conversely, suppose  $a^m = e$  and divide with remainder (Theorem 2.4):  $m = dq + r$  with  $0 \leq r < d$ . Then

$$e = a^m = (a^d)^q a^r = a^r,$$

so  $r$  is a nonnegative integer smaller than  $d$  with  $a^r = e$ . Minimality of  $d$  among *positive* exponents forces  $r = 0$ , whence  $d \mid m$ .

(2) We have  $a^m = a^k$  if and only if  $a^{m-k} = e$  (multiply by  $a^{-k}$  and use the exponent laws), which by (1) holds if and only if  $d \mid (m - k)$ , i.e.  $m \equiv k \pmod{d}$  (Definition 2.20).

(3) By (2) the powers  $a^0, a^1, \dots, a^{d-1}$  are pairwise distinct, their exponents being pairwise incongruent modulo  $d$ ; and every power  $a^m$  equals  $a^{m \bmod d}$ , again by (2).  $\square$

The final step of part (1) is the *division-minimality argument*: to show that every witness of some property is a multiple of the least positive witness  $m$ , divide an arbitrary witness  $t = mq + r$  with  $0 \leq r < m$ , show that  $r$  is itself a witness, and conclude  $r = 0$  by minimality. It has the form of the proof of Bézout's identity (Theorem 2.11), carried over to a group, and recurs for the least positive power of  $g$  lying in a subgroup (Theorem 3.24) and for the least positive element of a subgroup of  $\mathbb{Z}$  (Proposition 3.26).

*Example 3.13.* In  $(\mathbb{Z}/n\mathbb{Z}, +)$ , with the powers of Definition 3.11 read additively, the  $k$ -fold sum of the class 1 is the class  $k$ , which is 0 exactly when  $n \mid k$ , so  $\text{ord}(1) = n$ . More generally  $\text{ord}(a) = n / \gcd(a, n)$  for any class  $a$ , by Corollary 3.23 below read additively. In  $\mathbb{C}^\times$  the element  $i$  has order 4: its successive powers are

$$i^1 = i, \quad i^2 = -1, \quad i^3 = -i, \quad i^4 = 1.$$

In  $(\mathbb{Z}, +)$  every nonzero element has infinite order: no nonzero multiple of a nonzero integer is 0.

### 3.4 Subgroups

**Definition 3.14** (Subgroup). A subset  $H \subseteq G$  of a group  $(G, *)$  is a *subgroup*, written  $H \leq G$ , if  $H$  is itself a group under the restriction of  $*$  to  $H \times H$ : explicitly,  $H$  is nonempty, closed under  $*$ , contains the identity  $e$  of  $G$ , and contains  $a^{-1}$  whenever it contains  $a$ . Every group has the *trivial* subgroup  $\{e\}$  and  $G$  itself; a subgroup other than  $G$  is *proper*.

The definition is unambiguous. The identity of a subgroup coincides with the identity of  $G$ : if  $e_H \in H$  satisfies  $e_H * e_H = e_H$ , then cancelling  $e_H$  in  $G$  against  $e_H * e = e_H$  (Proposition 3.3(3)) gives  $e_H = e$ . Inverses computed in  $H$  agree with those computed in  $G$ , by uniqueness of inverses in  $G$ .

Since a subset inherits associativity from  $G$ , a single condition suffices.

**Proposition 3.15** (One-step subgroup test). *Let  $G$  be a group and  $H \subseteq G$ . Then  $H \leq G$  if and only if*

1.  $H$  is nonempty, and
2. for all  $a, b \in H$ , the element  $ab^{-1}$  lies in  $H$ .

*Proof.* If  $H \leq G$ , then  $H$  is nonempty and, for  $a, b \in H$ , it contains  $b^{-1}$  and hence  $ab^{-1}$ ; both conditions hold.

Conversely, suppose (1) and (2) hold. Pick  $a \in H$  by (1); taking  $b = a$  in (2) gives  $e = aa^{-1} \in H$ . For any  $b \in H$ , taking the pair  $(e, b)$  in (2) gives  $b^{-1} = eb^{-1} \in H$ , so  $H$  is closed under inverses. Finally, for  $a, b \in H$  the previous step gives  $b^{-1} \in H$ , and applying (2) to the pair  $(a, b^{-1})$  gives  $a(b^{-1})^{-1} = ab \in H$ , so  $H$  is closed under the operation. Associativity is inherited from  $G$ , so  $H$  is a group.  $\square$

*Remark 3.16.* For a *finite* nonempty subset  $H$ , closure under the operation alone forces  $H \leq G$ . Given  $a \in H$ , closure puts all the powers  $a, a^2, a^3, \dots$  in  $H$ ; since  $H$  is finite they cannot all be distinct, so  $a^i = a^j$  for some  $i < j$ , giving  $a^{j-i} = e \in H$ . Then  $a^{-1} = a^{j-i-1} \in H$ : this is a positive power of  $a$  when  $j - i \geq 2$ , and when  $j - i = 1$  we have  $a = e$ , whose inverse  $e$  already lies in  $H$ .

*Example 3.17.* The even integers  $2\mathbb{Z} = \{2k : k \in \mathbb{Z}\}$  form a subgroup of  $(\mathbb{Z}, +)$ : the set is nonempty, and  $2k - 2l = 2(k - l)$  is again even (the subgroup test, written additively). More generally  $n\mathbb{Z} = \{nk : k \in \mathbb{Z}\} \leq \mathbb{Z}$  for every  $n \geq 0$ , by the same computation—and these are *all* the subgroups of  $\mathbb{Z}$ , as Proposition 3.26 will show.

**Proposition 3.18.** *The intersection  $\bigcap_{i \in I} H_i$  of any family of subgroups  $H_i \leq G$  is a subgroup of  $G$ .*

*Proof.* The identity  $e$  lies in every  $H_i$ , so the intersection is nonempty. If  $a$  and  $b$  lie in the intersection, then for each  $i$  both lie in  $H_i$ , so  $ab^{-1} \in H_i$  by Proposition 3.15; hence  $ab^{-1}$  lies in the intersection, and the one-step test applies.  $\square$

**Definition 3.19** (Generated subgroup). For a subset  $S \subseteq G$ , the *subgroup generated by  $S$* , written  $\langle S \rangle$ , is the intersection of all subgroups of  $G$  containing  $S$ . By Proposition 3.18 it is a subgroup, and by construction it is the smallest subgroup of  $G$  containing  $S$ .

Concretely,  $\langle S \rangle$  consists of all finite products  $s_1^{\varepsilon_1} s_2^{\varepsilon_2} \dots s_k^{\varepsilon_k}$  with  $s_i \in S$  and  $\varepsilon_i \in \{+1, -1\}$ , the empty product being  $e$ . The set of such products contains  $S$ , contains  $e$ , and is closed under multiplication (concatenate) and under inversion (reverse the factors and negate the exponents, by the socks–shoes rule), so it is a subgroup containing  $S$ ; conversely, every subgroup containing  $S$  is closed under these operations and so contains every such product. Hence the two descriptions coincide.

### 3.5 Cyclic groups

**Definition 3.20** (Cyclic group). A group  $G$  is *cyclic* if there exists  $g \in G$  with

$$G = \langle g \rangle = \{g^k : k \in \mathbb{Z}\};$$

such an element  $g$  is a *generator* of  $G$ .

The displayed equality agrees with Definition 3.19: the set  $\{g^k : k \in \mathbb{Z}\}$  contains  $g$  and is closed under products and inverses by the exponent laws, so it is a subgroup containing  $g$ ; conversely, any subgroup containing  $g$  contains all its powers. Every cyclic group is abelian, since  $g^m g^n = g^{m+n} = g^n g^m$ .

**Proposition 3.21.** Let  $G = \langle g \rangle$  be cyclic.

1. If  $g$  has infinite order, the powers  $g^k$  ( $k \in \mathbb{Z}$ ) are pairwise distinct, so  $G$  is infinite.
2. If  $g$  has finite order  $d$ , then  $G = \{e, g, g^2, \dots, g^{d-1}\}$  has exactly  $d$  elements.

In both cases  $|\langle g \rangle| = \text{ord}(g)$ .

*Proof.* (1) Suppose  $g^i = g^j$  with  $i \neq j$ ; without loss of generality  $i > j$ . Then  $g^{i-j} = e$  with  $i-j > 0$ , contradicting infinite order. So the powers are pairwise distinct and  $G$  is infinite.

(2) This is Proposition 3.12(3): the distinct powers of  $g$  are exactly  $g^0, g^1, \dots, g^{d-1}$ , and these exhaust  $G = \{g^k : k \in \mathbb{Z}\}$ .  $\square$

*Example 3.22.* The group  $(\mathbb{Z}, +)$  is infinite cyclic, generated by 1 and also by  $-1$ . The group  $(\mathbb{Z}/n\mathbb{Z}, +)$  is cyclic of order  $n$ , generated by 1 (Example 3.13). The unit group  $(\mathbb{Z}/8\mathbb{Z})^\times = \{1, 3, 5, 7\}$  is *not* cyclic: each of its four elements squares to 1 ( $3^2 = 9 \equiv 1$ ,  $5^2 = 25 \equiv 1$ ,  $7^2 = 49 \equiv 1 \pmod{8}$ ), so no element has order 4. By contrast  $(\mathbb{Z}/5\mathbb{Z})^\times = \{1, 2, 3, 4\}$  is cyclic, generated by 2: its powers run

$$2^1 \equiv 2, \quad 2^2 \equiv 4, \quad 2^3 \equiv 3, \quad 2^4 \equiv 1 \pmod{5},$$

exhausting the group.

**Corollary 3.23.** Let  $g$  have finite order  $n$  and let  $k \in \mathbb{Z}$ . Then

$$\text{ord}(g^k) = \frac{n}{\gcd(n, k)}.$$

In particular  $g^k$  generates  $\langle g \rangle$  if and only if  $\gcd(n, k) = 1$ , so a cyclic group of order  $n$  has exactly  $\phi(n)$  generators.

*Proof.* Set  $d = \gcd(n, k)$  and write  $k = dk'$ ,  $n = dn'$  with  $\gcd(n', k') = 1$ . For  $m \geq 1$ ,

$$(g^k)^m = g^{km} = e \iff n \mid km \iff dn' \mid dk'm \iff n' \mid k'm \iff n' \mid m,$$

using Proposition 3.12(1) for the first equivalence and Euclid's lemma (Proposition 2.15(1), with  $\gcd(n', k') = 1$ ) for the last. The least positive such  $m$  is  $n' = n/d$ , which is therefore  $\text{ord}(g^k)$ .

For the second claim,  $\langle g^k \rangle \subseteq \langle g \rangle$  always, and by Proposition 3.21 the subgroup  $\langle g^k \rangle$  has  $\text{ord}(g^k) = n / \gcd(n, k)$  elements; it equals the  $n$ -element set  $\langle g \rangle$  exactly when  $\gcd(n, k) = 1$ . Since the powers  $g^1, g^2, \dots, g^n$  run through each element of  $\langle g \rangle$  exactly once (Proposition 3.12(3)), the generators correspond to the  $k \in \{1, \dots, n\}$  with  $\gcd(n, k) = 1$ , of which there are  $\varphi(n)$  by Definition 2.26.  $\square$

**Theorem 3.24** (Subgroups of cyclic groups). *Let  $G = \langle g \rangle$  be a cyclic group.*

1. *Every subgroup of  $G$  is cyclic.*
2. *If  $G$  is infinite, its subgroups are exactly  $\langle g^m \rangle$  for  $m \geq 0$ ; distinct  $m \geq 0$  give distinct subgroups, and for  $m \geq 1$  the subgroup  $\langle g^m \rangle$  is isomorphic to  $(\mathbb{Z}, +)$ .*
3. *If  $|G| = n$ , then for each positive divisor  $d \mid n$  there is exactly one subgroup of  $G$  of order  $d$ , namely  $\langle g^{n/d} \rangle$ , and these are all the subgroups of  $G$ . Subgroups of  $G$  thus correspond bijectively to positive divisors of  $n$ .*

*Proof.* (1) Let  $H \leq G$ . If  $H = \{e\}$ , then  $H = \langle e \rangle$  is cyclic. Otherwise  $H$  contains some  $g^t \neq e$ , so  $t \neq 0$ ; since  $H$  also contains  $(g^t)^{-1} = g^{-t}$ , it contains a power of  $g$  with positive exponent. Let  $m$  be the least positive integer with  $g^m \in H$  (well-ordering, Theorem 2.1). We claim  $H = \langle g^m \rangle$ . The inclusion  $\supseteq$  holds by closure. For  $\subseteq$ , take any element of  $H$ ; it is some  $g^t$ , since  $H \subseteq G = \langle g \rangle$ . Divide  $t = mq + r$  with  $0 \leq r < m$ ; then

$$g^r = g^{t-mq} = g^t (g^m)^{-q} \in H,$$

and minimality of  $m$  forces  $r = 0$ , by the division-minimality argument. Hence  $g^t = (g^m)^q \in \langle g^m \rangle$ .

(2) Let  $G$  be infinite. Then  $g$  has infinite order, for otherwise  $G$  would be finite by Proposition 3.21(2). By part (1) every subgroup is  $\{e\} = \langle g^0 \rangle$  or  $\langle g^m \rangle$  with  $m \geq 1$  least positive, and conversely each  $\langle g^m \rangle$  is a subgroup. For distinctness, observe that for  $m \geq 1$  the positive exponents  $t$  with  $g^t \in \langle g^m \rangle$  are exactly the positive multiples of  $m$ : the elements of  $\langle g^m \rangle$  are the powers  $g^{mk}$ , and  $g^t = g^{mk}$  forces  $t = mk$  because distinct exponents give distinct powers (Proposition 3.21(1)). The least such exponent recovers  $m$ , so distinct  $m \geq 1$  give distinct subgroups, and each differs from  $\{e\}$  because  $g^m \neq e$ . Finally, for  $m \geq 1$  the element  $g^m$  has infinite order (if  $(g^m)^t = g^{mt} = e$  then  $mt = 0$ , so  $t = 0$ ), so  $\langle g^m \rangle$  is an infinite cyclic group and is therefore isomorphic to  $(\mathbb{Z}, +)$  by Theorem 3.43(1) below. No circularity arises: the proof of that theorem uses only Propositions 3.12 and 3.21.

(3) Let  $|G| = n$ , so  $\text{ord}(g) = n$  by Proposition 3.21.

*Existence.* Let  $d \mid n$  be a positive divisor. Since  $n/d$  divides  $n$ , we have  $\gcd(n, n/d) = n/d$ , so Corollary 3.23 gives

$$\text{ord}(g^{n/d}) = \frac{n}{\gcd(n, n/d)} = \frac{n}{n/d} = d,$$

and  $\langle g^{n/d} \rangle$  is a subgroup of order  $d$  by Proposition 3.21.

*Uniqueness.* Let  $H \leq G$  with  $|H| = d$ . If  $d = 1$  then  $H = \{e\} = \langle g^n \rangle = \langle g^{n/1} \rangle$ , since  $g^n = e$ . Otherwise, as in part (1),  $H = \langle g^m \rangle$  for the least positive  $m$  with  $g^m \in H$ , and Propositions 3.12 and 3.21 together with Corollary 3.23 give

$$d = |H| = \text{ord}(g^m) = \frac{n}{\gcd(n, m)}, \quad \text{so} \quad \gcd(n, m) = \frac{n}{d}.$$

In particular  $n/d$  divides  $m$ , say  $m = (n/d)m'$ ; then  $g^m = (g^{n/d})^{m'}$ , so  $H = \langle g^m \rangle \subseteq \langle g^{n/d} \rangle$ . Both sides have exactly  $d$  elements, so  $H = \langle g^{n/d} \rangle$ .

*All subgroups arise.* Any  $H \leq G$  equals some  $\langle g^m \rangle$  (or  $\{e\}$ ), whose order  $n/\gcd(n, m)$  is a positive divisor  $d$  of  $n$ ; by uniqueness  $H = \langle g^{n/d} \rangle$ . The assignment  $d \mapsto \langle g^{n/d} \rangle$  is therefore surjective onto the set of subgroups, and it is injective because subgroups of different orders are distinct. This is the asserted bijection.  $\square$

*Remark 3.25.* Specialised to  $G = \mathbb{Z}/n\mathbb{Z}$  with generator 1 (written additively), Theorem 3.24(3) reads: for each divisor  $d \mid n$  there is exactly one subgroup of order  $d$ , namely

$$\langle n/d \rangle = \{0, n/d, 2n/d, \dots, (d-1)n/d\},$$

itself cyclic of order  $d$ , hence isomorphic to  $\mathbb{Z}/d\mathbb{Z}$  by the classification below. For  $n = 12$  the six subgroups correspond to the divisors 1, 2, 3, 4, 6, 12, and inclusion between subgroups mirrors divisibility between the corresponding divisors: the subgroup lattice of  $\mathbb{Z}/12\mathbb{Z}$  is the divisor lattice of 12, with chains such as  $1 \mid 2 \mid 4, 1 \mid 2 \mid 6$ , and  $3 \mid 6$ .

The classification of the subgroups of  $\mathbb{Z}$  announced in Example 3.17 is Theorem 3.24(2) read additively for  $G = (\mathbb{Z}, +) = \langle 1 \rangle$ ; a direct proof follows.

**Proposition 3.26.** *Every subgroup of  $(\mathbb{Z}, +)$  has the form  $n\mathbb{Z}$  for a unique integer  $n \geq 0$ .*

*Proof.* Let  $H \leq \mathbb{Z}$ . If  $H = \{0\}$ , then  $H = 0\mathbb{Z}$ . Otherwise  $H$  contains a nonzero integer and, being closed under negation, contains a positive one; let  $n$  be its least positive element (Theorem 2.1). Closure under addition and negation gives  $n\mathbb{Z} \subseteq H$ . Conversely, for any  $h \in H$ , divide  $h = nq + r$  with  $0 \leq r < n$ ; then  $r = h - nq \in H$ , and minimality of  $n$  forces  $r = 0$ , so  $h \in n\mathbb{Z}$ . Hence  $H = n\mathbb{Z}$ . For uniqueness, the integer  $n$  is recovered from  $H$  as its least positive element (and  $n = 0$  exactly for  $H = \{0\}$ ), so distinct  $n \geq 0$  give distinct subgroups.  $\square$

### 3.6 Cosets and Lagrange's theorem

The cosets of a subgroup  $H \leq G$ , defined next, partition  $G$  into blocks of size  $|H|$ ; counting the blocks gives Lagrange's theorem.

**Definition 3.27** (Cosets and index). Let  $H \leq G$  and  $a \in G$ . The *left coset* of  $H$  by  $a$  is

$$aH = \{ah : h \in H\},$$

and the *right coset* is  $Ha = \{ha : h \in H\}$ . In additive notation one writes  $a + H = \{a + h : h \in H\}$ . The set of left cosets is denoted  $G/H$ , and the *index* of  $H$  in  $G$  is  $[G : H] = |G/H|$ , the number of left cosets.

**Lemma 3.28** (Cosets partition the group). *Let  $H \leq G$ .*

1. *Each  $a \in G$  lies in  $aH$ ; the left cosets cover  $G$ .*
2. *Two left cosets are either equal or disjoint:  $aH = bH$  if and only if  $a^{-1}b \in H$ , and otherwise  $aH \cap bH = \emptyset$ .*
3. *Every left coset has the same cardinality as  $H$ : the map  $h \mapsto ah$  is a bijection from  $H$  onto  $aH$ .*

*The same statements hold for right cosets, with  $ba^{-1} \in H$  as the criterion in (2).*

*Proof.* (1) Since  $e \in H$ , we have  $a = ae \in aH$ .

(2) First suppose the cosets meet:  $ab_1 = bb_2$  for some  $b_1, b_2 \in H$ . Then  $a^{-1}b = b_1b_2^{-1} \in H$ . Conversely, suppose  $a^{-1}b = h \in H$ , so  $b = ah$ . Then every element  $bb' \in bH$  equals  $a(hb') \in aH$ , giving  $bH \subseteq aH$ ; and since  $b^{-1}a = h^{-1} \in H$ , the same argument with the roles of  $a$  and  $b$  exchanged gives  $aH \subseteq bH$ . Hence  $aH = bH$ . Combining the two directions: if two cosets share even one element they coincide, so distinct cosets are disjoint.

(3) The map  $\lambda_a: H \rightarrow aH, h \mapsto ah$ , is surjective by the definition of  $aH$  and injective by left cancellation (Proposition 3.3(3)):  $ah = ah'$  implies  $h = h'$ . Hence  $|aH| = |H|$ .

For right cosets, the same three arguments apply with sides exchanged; in (2) the criterion becomes  $ba^{-1} \in H$ .  $\square$

**Theorem 3.29** (Lagrange's theorem). *Let  $G$  be a finite group and  $H \leq G$ . Then  $|H|$  divides  $|G|$ , and*

$$|G| = [G : H] |H|.$$

*Proof.* By Lemma 3.28, the distinct left cosets of  $H$  partition  $G$ : they cover  $G$  by part (1) and are pairwise disjoint by part (2). Since  $G$  is finite there are finitely many of them, say  $[G : H] = k$ , and each has exactly  $|H|$  elements by part (3). Summing the sizes of the blocks of the partition,

$$|G| = \sum_{\text{distinct cosets } aH} |aH| = k |H| = [G : H] |H|. \quad \square$$

**Corollary 3.30.** *In a finite group  $G$ , the order of every element divides  $|G|$ .*

*Proof.* For  $a \in G$ , the cyclic subgroup  $\langle a \rangle$  has order  $\text{ord}(a)$  by Proposition 3.21 (the order is finite, since  $\langle a \rangle \subseteq G$  is finite), and Theorem 3.29 makes  $|\langle a \rangle|$  a divisor of  $|G|$ .  $\square$

**Corollary 3.31.** *If  $G$  is a finite group of order  $N$ , then  $a^N = e$  for every  $a \in G$ .*

*Proof.* Let  $d = \text{ord}(a)$ . By Corollary 3.30,  $d \mid N$ , say  $N = dm$ ; then  $a^N = (a^d)^m = e^m = e$ .  $\square$

**Corollary 3.32** (Euler's theorem; Fermat's little theorem). *Let  $n \geq 1$  and  $a \in \mathbb{Z}$  with  $\gcd(a, n) = 1$ . Then*

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

*In particular, for a prime  $p$  and  $a$  not divisible by  $p$ ,*

$$a^{p-1} \equiv 1 \pmod{p},$$

*and  $a^p \equiv a \pmod{p}$  holds for all integers  $a$ .*

*Proof.* Apply Corollary 3.31 to the group  $(\mathbb{Z}/n\mathbb{Z})^\times$ , which has order  $\varphi(n)$  (Proposition 3.8). For  $\gcd(a, n) = 1$  the class  $[a]_n$  is a unit (Theorem 2.24), so  $[a]_n^{\varphi(n)} = [1]_n$ , which is the first congruence. For  $n = p$  prime,  $\varphi(p) = p - 1$  (Example 2.27), and  $p \nmid a$  gives  $\gcd(a, p) = 1$ , yielding the second congruence. Multiplying it by  $a$  gives  $a^p \equiv a \pmod{p}$ ; and when  $p \mid a$  both sides are congruent to 0, so the last congruence holds for all  $a$ .  $\square$

*Remark 3.33.* The congruences of Corollary 3.32 are proved only here; Section 7 restates them under the heading *Fermat's little theorem and Euler's theorem*, cites this proof, and develops their cryptographic role.

**Corollary 3.34.** *Every group  $G$  of prime order  $p$  is cyclic, hence isomorphic to  $\mathbb{Z}/p\mathbb{Z}$ , and has no proper nontrivial subgroups.*

*Proof.* Pick  $a \in G$  with  $a \neq e$  (possible since  $p \geq 2$ ). The subgroup  $\langle a \rangle$  has at least two elements, and by Theorem 3.29 its order divides  $p$ ; a divisor of  $p$  exceeding 1 equals  $p$ , so  $\langle a \rangle = G$  and  $G$  is cyclic. The isomorphism with  $\mathbb{Z}/p\mathbb{Z}$  is Theorem 3.43 below. Finally, any subgroup has order 1 or  $p$  by Lagrange, i.e. is  $\{e\}$  or  $G$ .  $\square$

### 3.7 Quotients of abelian groups

The quotient of an abelian group by a subgroup underlies the quotient rings of Section 4 and, through them,  $\mathbb{Z}/n\mathbb{Z}$  and the polynomial quotients used throughout the series.

**Proposition 3.35** (Quotient of an abelian group). *Let  $A$  be an abelian group, written additively, and let  $H \leq A$ . The operation*

$$(a + H) + (b + H) = (a + b) + H$$

*on the coset set  $A/H$  is well defined and makes  $A/H$  an abelian group—the quotient group of  $A$  by  $H$ —with identity  $H = 0 + H$  and inverse  $-(a + H) = (-a) + H$ . If  $A$  is finite, then*

$$|A/H| = [A : H] = \frac{|A|}{|H|}.$$

*Proof.* The substantive claim is that the operation is well defined: its output is stated in terms of chosen representatives  $a$  and  $b$ , and replacing them by other representatives of the same cosets

must leave the result unchanged. Suppose then that  $a + H = a' + H$  and  $b + H = b' + H$ . By Lemma 3.28(2), written additively, this means  $a' - a \in H$  and  $b' - b \in H$ . Then

$$(a' + b') - (a + b) = (a' - a) + (b' - b) \in H,$$

since  $H$  is closed under addition; applying Lemma 3.28(2) once more,  $(a' + b') + H = (a + b) + H$ . The operation is well defined. This representative-independence check is the template for every coset-wise construction in the volume; it recurs, in the same words, for quotient-ring multiplication in Section 4.

The axioms then pass coset-wise from  $A$ : for associativity,

$$\begin{aligned} ((a + H) + (b + H)) + (c + H) &= ((a + b) + c) + H = (a + (b + c)) + H \\ &= (a + H) + ((b + H) + (c + H)); \end{aligned}$$

commutativity follows the same way from commutativity in  $A$ ; the coset  $0 + H = H$  is an identity; and  $(-a) + H$  is an inverse of  $a + H$ . Hence  $A/H$  is an abelian group. When  $A$  is finite, the number of cosets is  $[A : H] = |A|/|H|$  by Theorem 3.29—the same partition-into-equal-blocks count as before.  $\square$

For  $A = (\mathbb{Z}, +)$  and  $H = n\mathbb{Z}$ , the cosets are the residue classes  $[a]_n = a + n\mathbb{Z}$  of Definition 2.22 and coset addition is addition of classes. The notation  $\mathbb{Z}/n\mathbb{Z}$ , introduced in Section 2 as a name for the set of classes, is thus an instance of the quotient notation  $A/H$ , and  $(\mathbb{Z}/n\mathbb{Z}, +)$  is the quotient group of  $\mathbb{Z}$  by  $n\mathbb{Z}$ .

*Remark 3.36.* For a non-abelian group  $G$  the same construction succeeds exactly for those subgroups  $N \leq G$  whose left and right cosets coincide,  $gN = Ng$  for all  $g \in G$ ; these are the *normal* subgroups. Commutativity makes every subgroup of an abelian group normal, which is why no extra hypothesis was needed in Proposition 3.35. The present volume needs only the abelian case and does not develop the general theory.

### 3.8 Homomorphisms and isomorphisms

Groups are compared through the maps that respect their operations.

**Definition 3.37** (Group homomorphism). Let  $(G, *)$  and  $(G', \circ)$  be groups. A function  $\phi: G \rightarrow G'$  is a *group homomorphism* if

$$\phi(a * b) = \phi(a) \circ \phi(b) \quad \text{for all } a, b \in G.$$

The set of all homomorphisms from  $G$  to  $G'$  is denoted  $\text{Hom}(G, G')$ .

**Proposition 3.38.** Let  $\phi: G \rightarrow G'$  be a homomorphism, and write  $e, e'$  for the identities of  $G$  and  $G'$ .

1.  $\phi(e) = e'$ .
2.  $\phi(a^{-1}) = \phi(a)^{-1}$  for all  $a \in G$ .

3.  $\phi(a^n) = \phi(a)^n$  for all  $a \in G$  and  $n \in \mathbb{Z}$ .
4. The image  $\phi(G)$  is a subgroup of  $G'$ .

*Proof.* (1) From  $e = e * e$  we get  $\phi(e) = \phi(e) \circ \phi(e)$ ; cancelling  $\phi(e)$  in  $G'$  (Proposition 3.3(3)) gives  $e' = \phi(e)$ .

(2) Applying  $\phi$  to  $a * a^{-1} = e = a^{-1} * a$  and using (1),

$$\phi(a) \circ \phi(a^{-1}) = e' = \phi(a^{-1}) \circ \phi(a),$$

so  $\phi(a^{-1})$  is an inverse of  $\phi(a)$ ; by uniqueness of inverses,  $\phi(a^{-1}) = \phi(a)^{-1}$ .

(3) For  $n \geq 0$ , induct: the case  $n = 0$  is (1), and  $\phi(a^{n+1}) = \phi(a^n * a) = \phi(a)^n \circ \phi(a) = \phi(a)^{n+1}$ . For  $n < 0$ , write  $n = -m$  with  $m > 0$  and combine with (2):  $\phi(a^{-m}) = \phi((a^{-1})^m) = \phi(a^{-1})^m = (\phi(a)^{-1})^m = \phi(a)^{-m}$ .

(4) The image is nonempty, containing  $e' = \phi(e)$ . For  $x = \phi(a)$  and  $y = \phi(b)$  in  $\phi(G)$ ,

$$x \circ y^{-1} = \phi(a) \circ \phi(b)^{-1} = \phi(a * b^{-1}) \in \phi(G),$$

using (2); the one-step subgroup test (Proposition 3.15) concludes.  $\square$

**Definition 3.39** (Kernel and image). The *kernel* and *image* of a homomorphism  $\phi: G \rightarrow G'$  are

$$\ker(\phi) = \{a \in G : \phi(a) = e'\} = \phi^{-1}(\{e'\}), \quad \text{im}(\phi) = \phi(G) = \{\phi(a) : a \in G\}.$$

**Proposition 3.40.** For any homomorphism  $\phi: G \rightarrow G'$ , the kernel is a subgroup,  $\ker(\phi) \leq G$ , and  $\phi$  is injective if and only if  $\ker(\phi) = \{e\}$ .

*Proof.* The kernel contains  $e$  by Proposition 3.38(1), so it is nonempty; and for  $a, b \in \ker(\phi)$ ,

$$\phi(ab^{-1}) = \phi(a) \circ \phi(b)^{-1} = e' \circ (e')^{-1} = e',$$

so  $ab^{-1} \in \ker(\phi)$ ; the one-step test gives  $\ker(\phi) \leq G$ .

If  $\phi$  is injective and  $\phi(a) = e' = \phi(e)$ , then  $a = e$ ; the kernel is trivial. Conversely, suppose  $\ker(\phi) = \{e\}$  and  $\phi(a) = \phi(b)$ . Then  $\phi(ab^{-1}) = \phi(a) \circ \phi(b)^{-1} = e'$ , so  $ab^{-1} \in \ker(\phi) = \{e\}$ , forcing  $ab^{-1} = e$ , i.e.  $a = b$ .  $\square$

Kernels are exactly the subgroups by which one may quotient: for abelian  $G$ , Proposition 3.35 forms  $G/\ker(\phi)$  directly, and in general the kernel satisfies the normality condition of Remark 3.36—for  $n \in \ker(\phi)$  and  $g \in G$ , one computes  $\phi(gng^{-1}) = \phi(g) \circ e' \circ \phi(g)^{-1} = e'$ , so conjugating the kernel by  $g$  lands back in the kernel, whence  $g\ker(\phi) = \ker(\phi)g$ .

**Definition 3.41** (Isomorphism). An *isomorphism* is a bijective homomorphism  $\phi: G \rightarrow G'$ ; when one exists, the groups are *isomorphic*, written  $G \cong G'$ . An isomorphism from  $G$  to itself is an *automorphism*. An injective homomorphism is a *monomorphism*; a surjective one is an *epimorphism*.

**Proposition 3.42.** *The set-theoretic inverse of an isomorphism is an isomorphism, and  $\cong$  is an equivalence relation on any collection of groups.*

*Proof.* Let  $\phi: G \rightarrow G'$  be an isomorphism, with inverse function  $\phi^{-1}$  (which exists since  $\phi$  is a bijection, Theorem 1.1). For  $x, y \in G'$ , write  $x = \phi(a)$  and  $y = \phi(b)$ ; then

$$\phi^{-1}(x \circ y) = \phi^{-1}(\phi(a * b)) = a * b = \phi^{-1}(x) * \phi^{-1}(y),$$

so  $\phi^{-1}$  is a homomorphism, and it is a bijection; hence an isomorphism. For the equivalence relation: reflexivity is witnessed by the identity automorphism  $\text{id}_G$ ; symmetry by the inverse just constructed; and transitivity because a composite of homomorphisms is a homomorphism (apply the defining property twice) and a composite of bijections is a bijection.  $\square$

Any property expressible through the operation alone, such as the orders of elements, the lattice of subgroups, or commutativity, transfers across an isomorphism, so groups are classified *up to isomorphism*; the next theorem does so for cyclic groups.

**Theorem 3.43** (Classification of cyclic groups). *Every cyclic group is isomorphic to exactly one of the following:*

1. *the infinite cyclic group  $(\mathbb{Z}, +)$ , if it is infinite;*
2. *the finite cyclic group  $(\mathbb{Z}/n\mathbb{Z}, +)$ , if it has order  $n$ .*

*In particular, two cyclic groups are isomorphic if and only if they have the same order.*

*Proof.* Let  $G = \langle g \rangle$  and define

$$\phi: \mathbb{Z} \rightarrow G, \quad \phi(k) = g^k.$$

The exponent law  $g^{k+l} = g^k g^l$  says exactly that  $\phi$  is a homomorphism from  $(\mathbb{Z}, +)$  to  $G$ , and  $\phi$  is surjective because every element of  $G$  is a power of  $g$ .

*Case 1:  $g$  has infinite order.* Distinct exponents give distinct powers (Proposition 3.21(1)), so  $\phi$  is injective, hence a bijective homomorphism, and  $\mathbb{Z} \cong G$ .

*Case 2:  $\text{ord}(g) = n$ .* By Proposition 3.12(1),

$$\ker(\phi) = \{k \in \mathbb{Z} : g^k = e\} = n\mathbb{Z}.$$

Define the induced map

$$\bar{\phi}: \mathbb{Z}/n\mathbb{Z} \rightarrow G, \quad \bar{\phi}([k]_n) = g^k.$$

It is well defined: if  $[k]_n = [l]_n$  then  $n \mid (k - l)$ , so  $g^k = g^l$  by Proposition 3.12(2). It is a homomorphism:  $\bar{\phi}([k]_n + [l]_n) = \bar{\phi}([k + l]_n) = g^{k+l} = g^k g^l = \bar{\phi}([k]_n) \bar{\phi}([l]_n)$ . It is surjective because  $\phi$  is. And it is injective by Proposition 3.40: if  $\bar{\phi}([k]_n) = g^k = e$ , then  $n \mid k$  (Proposition 3.12(1)), so  $[k]_n = [0]_n$ ; the kernel of  $\bar{\phi}$  is trivial. Hence  $\mathbb{Z}/n\mathbb{Z} \cong G$ .

*Distinctness.* Isomorphic groups have equal order, an isomorphism being in particular a bijection. The group  $\mathbb{Z}$  is infinite and the groups  $\mathbb{Z}/n\mathbb{Z}$  for  $n = 1, 2, 3, \dots$  have the pairwise distinct finite orders  $n$ , so no two of the listed groups are isomorphic, and a cyclic group is isomorphic to exactly

one of them—the one matching its order. The final claim follows: cyclic groups of equal order are isomorphic to the same listed group, hence to each other (Proposition 3.42), and isomorphic groups have equal order.  $\square$

In key agreement over a public channel, two parties fix a finite group  $G$  and a public element  $g \in G$ , of order  $n = \text{ord}(g)$ ; each combines  $g$  with itself a secret number of times,  $a$  and  $b$  respectively, publishes the result, and applies its own count to the other's. By generalised associativity (Proposition 3.4),  $a$  repetitions of  $g$  give the element  $g^a$  however bracketed, and the exponent laws give

$$(g^a)^b = g^{ab} = (g^b)^a,$$

so the two parties obtain a common secret element, while an eavesdropper sees only  $g$  and the two published values. By the classification of cyclic groups,  $\langle g \rangle$  is isomorphic to  $\mathbb{Z}/n\mathbb{Z}$ , so exponents add, negate, and matter only modulo  $n$  (Proposition 3.12), in the modular arithmetic of Section 2. By Lagrange's theorem,  $n$  divides  $|G|$ . That publishing  $g^a$  hides  $a$  is not a theorem of algebra but a statement about computation: it fails in some groups, such as  $(\mathbb{Z}/n\mathbb{Z}, +)$ , where recovering  $a$  from  $ag$  is a single modular division, and appears to hold in others. The sections that follow and later volumes make this hiding property precise and construct the groups for which it is credible.

## 4 Rings and fields

An arithmetic circuit, the object a verifier checks in a proof system, is a network of gates, each adding or multiplying two previously computed values; inverting a random challenge, or cancelling a common factor from both sides of an identity, requires *division* as well. This section defines the *ring*, a set with two operations obeying the arithmetic laws of  $\mathbb{Z}$ , and the *field*, a ring in which division by every nonzero element is possible. It shows that the residue classes  $\mathbb{Z}/n\mathbb{Z}$  of §2, in which this monograph computes, form a ring (Theorem 4.6), the quotient of  $\mathbb{Z}$  by the ideal  $n\mathbb{Z}$ , and a field exactly when  $n$  is prime (Corollary 4.26); it proves that every finite integral domain is a field (Theorem 4.29) and that the characteristic of a domain is 0 or prime (Theorem 4.32). Polynomials, linear algebra, roots of unity, and elliptic curves are developed over fields in the sections that follow, and the finite fields  $\mathbb{F}_p$  and  $\mathbb{F}_q$  underlie essentially all the elliptic-curve and proof-system constructions of the monograph. From §3 we assume only the theory of *abelian* groups; additive structure is written additively with identity 0, multiplicative structure multiplicatively with identity 1.

### 4.1 Rings

**Definition 4.1** (Ring). A *ring* is a set  $R$  equipped with two binary operations  $+$  and  $\cdot$  such that:

1.  $(R, +)$  is an abelian group: addition is associative and commutative, there is an element 0 with  $a + 0 = a$  for all  $a$ , and every  $a$  has an *additive inverse*  $-a$  with  $a + (-a) = 0$ ;
2. multiplication is associative:  $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ ;

3. there is a *multiplicative identity* 1 with  $1 \cdot a = a \cdot 1 = a$  for all  $a$ ;
4. multiplication distributes over addition on both sides:  $a \cdot (b + c) = a \cdot b + a \cdot c$  and  $(a + b) \cdot c = a \cdot c + b \cdot c$ .

The ring is *commutative* if moreover  $a \cdot b = b \cdot a$  for all  $a, b \in R$ . We write  $ab$  for  $a \cdot b$ , let multiplication bind tighter than addition, and call 0 the *zero* and 1 the *unity* of the ring.

*Remark 4.2.* Some authors omit requirement (3), saying “ring with unity” for the notion above and “rng” for the weaker one. In this monograph *every ring has a 1*, and the structure-preserving maps between rings—the ring homomorphisms of Definition 4.21 below—are required to preserve it. In the examples used here “ring” usually means “commutative ring”; commutativity is always stated explicitly when a result depends on it.

The distributive law, the only axiom coupling the two operations, yields the following identities, which are used without comment hereafter.

**Proposition 4.3** (Elementary ring arithmetic). *Let  $R$  be a ring and  $a, b \in R$ . Then:*

1.  $0 \cdot a = a \cdot 0 = 0$ ;
2.  $(-a)b = a(-b) = -(ab)$ ;
3.  $(-a)(-b) = ab$ ;
4. for all integers  $m, n \in \mathbb{Z}$ ,  $(ma)(nb) = (mn)(ab)$ , where  $na$  denotes the  $n$ -fold additive multiple ( $0a = 0$ ,  $(n+1)a = na + a$ , and  $(-n)a = -(na)$ , as for any abelian group, §3).

*Proof.* (1) From  $0 = 0 + 0$  and distributivity,  $0 \cdot a = (0 + 0) \cdot a = 0 \cdot a + 0 \cdot a$ ; adding  $-(0 \cdot a)$  to both sides leaves  $0 = 0 \cdot a$ . The computation for  $a \cdot 0$  is symmetric.

(2) By distributivity and (1),  $ab + (-a)b = (a + (-a))b = 0 \cdot b = 0$ , so  $(-a)b$  is the additive inverse of  $ab$ , that is,  $(-a)b = -(ab)$ ; likewise  $ab + a(-b) = a(b + (-b)) = 0$  gives  $a(-b) = -(ab)$ .

(3) Applying (2) twice,  $(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$ , the last step because negation is an involution in the abelian group  $(R, +)$ .

(4) First fix  $a, b$  and show  $a(nb) = n(ab)$  for  $n \geq 0$  by induction: the case  $n = 0$  is (1), and  $a((n+1)b) = a(nb + b) = a(nb) + ab = n(ab) + ab = (n+1)(ab)$  by distributivity and the inductive hypothesis. For  $n < 0$  write  $n = -m$  with  $m > 0$ ; then  $a(nb) = a(-mb) = -(a(mb)) = -(m(ab)) = n(ab)$  by (2). The same argument in the first factor gives  $(ma)c = m(ac)$  for all  $c$ , and combining,

$$(ma)(nb) = m(a(nb)) = m(n(ab)) = (mn)(ab),$$

the final equality being the usual bookkeeping for iterated multiples in an abelian group, which we omit.  $\square$

*Remark 4.4* (The zero ring). Nothing in Definition 4.1 forces  $1 \neq 0$ . If  $1 = 0$  in a ring  $R$ , then every  $a \in R$  satisfies  $a = 1 \cdot a = 0 \cdot a = 0$  by Proposition 4.3(1), so  $R = \{0\}$ : the *zero ring* (or *trivial ring*), a ring whose two operations are the only possible ones. Whenever the zero ring must

be excluded we impose the assumption  $1 \neq 0$  explicitly; the definition of a field below builds  $1 \neq 0$  in as a standing assumption.

*Example 4.5* (Basic rings). 1. The integers  $\mathbb{Z}$  form the prototype commutative ring; much of ring theory consists of determining which properties of  $\mathbb{Z}$  hold in general.

2. The rationals  $\mathbb{Q}$ , the reals  $\mathbb{R}$ , and the complexes  $\mathbb{C}$  are commutative rings, and moreover fields in the sense of §4.4.

3. The residue classes  $\mathbb{Z}/n\mathbb{Z} = \{\overline{0}, \overline{1}, \dots, \overline{n-1}\}$  for  $n \geq 1$ —in this section’s examples we abbreviate the class  $[a]_n$  of Definition 2.22 to  $\overline{a}$  when the modulus is clear—under the induced operations form a commutative ring whose additive group is the cyclic group of order  $n$  (§3); this is Theorem 4.6 below. We write  $\mathbb{Z}/n\mathbb{Z}$ , or  $\mathbb{Z}_n$  when no ambiguity arises. The case  $n = 1$  gives the zero ring of Remark 4.4.

4. The *polynomial ring*  $R[x]$  over a commutative ring  $R$  consists of the polynomials  $a_0 + a_1x + \dots + a_dx^d$  with coefficients  $a_i \in R$ , added coefficient-wise and multiplied by expanding and collecting powers of  $x$ ; it is a commutative ring with the same 0 and 1 as  $R$  (the constant polynomials). The theory of polynomials over a field is developed in its own section later in the volume.

5. The *matrix ring*  $M_n(R)$  of  $n \times n$  matrices over a commutative ring  $R$ ,  $n \geq 1$ , is a ring under matrix addition and multiplication. It is *not* commutative for  $n \geq 2$  and  $R \neq \{0\}$ , even when  $R$  is commutative: in  $M_2(\mathbb{Z})$ ,

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \text{but} \quad \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

It is the basic noncommutative example.

6. The *product ring*  $R \times S$  of two rings carries the componentwise operations, with zero  $(0, 0)$  and identity  $(1, 1)$ ; likewise any finite product  $R_1 \times \dots \times R_k$ .

7. The zero ring  $\{0\}$ .

**Theorem 4.6.** *Let  $n \geq 1$ . The operations*

$$[a]_n + [b]_n := [a + b]_n, \quad [a]_n \cdot [b]_n := [ab]_n$$

*are well defined and make  $\mathbb{Z}/n\mathbb{Z}$  a commutative ring with additive identity  $[0]_n$  and multiplicative identity  $[1]_n$ . It has exactly  $n$  elements.*

*Proof.* The right-hand sides are defined in terms of chosen representatives  $a, b$ , so we must check that different choices give the same class. This is exactly the compatibility half of Proposition 2.21: if  $a' \equiv a$  and  $b' \equiv b \pmod{n}$ , then  $a' + b' \equiv a + b$  and  $a'b' \equiv ab \pmod{n}$ , so  $[a' + b']_n = [a + b]_n$  and  $[a'b']_n = [ab]_n$ . The operations are therefore well defined on classes.

Every ring axiom is now inherited from the corresponding law of  $\mathbb{Z}$  by passing to classes. For instance, distributivity:

$$\begin{aligned} [a]_n([b]_n + [c]_n) &= [a]_n[b + c]_n = [a(b + c)]_n = [ab + ac]_n \\ &= [ab]_n + [ac]_n = [a]_n[b]_n + [a]_n[c]_n, \end{aligned}$$

using only the definitions and the distributive law of the integers. Associativity and commutativity of both operations follow by the identical pattern. The class  $[0]_n$  is an additive identity and  $[1]_n$  a multiplicative one, since  $[a + 0]_n = [a]_n$  and  $[1 \cdot a]_n = [a]_n$ ; and  $-[a]_n = [-a]_n$  because  $[a]_n + [-a]_n = [a + (-a)]_n = [0]_n$ .

Finally, by the division algorithm every integer is congruent modulo  $n$  to exactly one of  $0, 1, \dots, n-1$  (the least nonnegative residues, Definition 2.22 and the discussion following it), so the classes  $[0]_n, \dots, [n-1]_n$  are distinct and exhaust  $\mathbb{Z}/n\mathbb{Z}$ : the ring has exactly  $n$  elements.  $\square$

## 4.2 Units, zero divisors, and integral domains

Units are the elements by which one can divide, and zero divisors are the obstruction to *cancellation*, the removal of a nonzero common factor from both sides of an equation. The integers admit cancellation, since a product of nonzero integers is nonzero, but only  $\pm 1$  divide 1.

**Definition 4.7** (Unit). An element  $u$  of a ring  $R$  is a *unit* (or is *invertible*) if there exists  $v \in R$  with  $uv = vu = 1$ . Such a  $v$  is unique: if  $uv = vu = 1$  and  $uw' = v'u = 1$ , then  $v = v \cdot 1 = v(uv') = (vu)v' = 1 \cdot v' = v'$ . It is called the *inverse* of  $u$  and written  $u^{-1}$ . The set of units of  $R$  is denoted  $R^\times$ .

**Proposition 4.8.** For any ring  $R$ , the set  $R^\times$  of units is a group under the multiplication of  $R$ , called the group of units (or multiplicative group) of  $R$ .

*Proof.* The identity 1 is a unit (it is its own inverse), so  $1 \in R^\times$ . For closure, let  $u, w \in R^\times$ ; then

$$(uw)(w^{-1}u^{-1}) = u(ww^{-1})u^{-1} = u \cdot 1 \cdot u^{-1} = 1,$$

and symmetrically  $(w^{-1}u^{-1})(uw) = 1$ , so  $uw$  is a unit with  $(uw)^{-1} = w^{-1}u^{-1}$ . Multiplication in  $R^\times$  is associative because it is associative in  $R$ . Finally each  $u^{-1}$  is itself a unit, its inverse being  $u$ ; so every element of  $R^\times$  has an inverse in  $R^\times$ , and the group axioms are verified.  $\square$

**Example 4.9** (Units). 1. The units of the integers are  $\mathbb{Z}^\times = \{1, -1\}$ : if  $uv = 1$  in  $\mathbb{Z}$  then  $|u| |v| = 1$  forces  $|u| = 1$ .

2. In  $\mathbb{Q}$ ,  $\mathbb{R}$ , and  $\mathbb{C}$  every nonzero element is a unit:  $\mathbb{Q}^\times = \mathbb{Q} \setminus \{0\}$ ,  $\mathbb{R}^\times = \mathbb{R} \setminus \{0\}$ ,  $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$ . Commutative rings with  $1 \neq 0$  in which *every* nonzero element is a unit are precisely the fields of §4.4; this is their defining feature. Both qualifiers are needed: the zero ring of Remark 4.4 satisfies the unit condition vacuously, and noncommutative rings satisfying it—the *division rings*—exist but are not fields.

3. In  $\mathbb{Z}/n\mathbb{Z}$ , the class  $\bar{a}$  is a unit if and only if  $\gcd(a, n) = 1$ , by Theorem 2.24; thus  $(\mathbb{Z}/n\mathbb{Z})^\times$  consists of the residue classes coprime to the modulus, and its order is Euler's totient  $\varphi(n)$  (Definition 2.26).

4. In the matrix ring,  $M_n(R)^\times$  is the group of invertible  $n \times n$  matrices. Over a field it is the *general linear group*  $GL_n$ .

**Definition 4.10** (Zero divisor). A nonzero element  $a$  of a ring  $R$  is a *left zero divisor* if  $ab = 0$  for some nonzero  $b \in R$ , and a *right zero divisor* if  $ba = 0$  for some nonzero  $b$ . In a commutative ring the two notions coincide and we say *zero divisor*. A nonzero element that is neither is called *regular* (or *cancellable*).

The term “cancellable” is justified by the following equivalence.

**Proposition 4.11.** A nonzero element  $a$  of a ring  $R$  is not a left zero divisor if and only if it is left-cancellable:  $ab = ac$  implies  $b = c$  for all  $b, c \in R$ .

*Proof.* Suppose  $a$  is not a left zero divisor and  $ab = ac$ . Then  $a(b - c) = ab - ac = 0$ , using Proposition 4.3(2) to expand  $a(b + (-c))$ ; since  $a$  is nonzero and not a left zero divisor, the factor  $b - c$  must be 0, so  $b = c$ . Conversely, suppose  $a$  is a left zero divisor, say  $ab = 0$  with  $b \neq 0$ . Then  $ab = 0 = a \cdot 0$  while  $b \neq 0$ : cancellation fails.  $\square$

**Remark 4.12.** A unit is never a zero divisor: if  $u \in R^\times$  and  $ub = 0$ , then  $b = 1 \cdot b = (u^{-1}u)b = u^{-1}(ub) = u^{-1} \cdot 0 = 0$ . The converse fails in general—in  $\mathbb{Z}$  every nonzero element is a non-zero-divisor, yet only  $\pm 1$  are units—but every nonzero non-zero-divisor is a unit in a finite commutative ring, by the injectivity-implies-surjectivity argument that proves Theorem 4.29 below.

**Example 4.13** (Zero divisors). 1. In  $\mathbb{Z}/6\mathbb{Z}$ ,  $\bar{2} \cdot \bar{3} = \bar{6} = \bar{0}$ , so  $\bar{2}$  and  $\bar{3}$  are zero divisors. In general a nonzero class  $\bar{a} \in \mathbb{Z}/n\mathbb{Z}$  is a zero divisor precisely when  $d := \gcd(a, n) > 1$ : in that case  $a \cdot (n/d) = (a/d) \cdot n \equiv 0 \pmod{n}$  while  $n/d \neq 0$  (as  $0 < n/d < n$ ); and when  $d = 1$  the class is a unit (Example 4.9(3)), hence no zero divisor by Remark 4.12. Thus in  $\mathbb{Z}/n\mathbb{Z}$  every nonzero element is either a unit or a zero divisor, a dichotomy revisited structurally by Theorem 4.29.

2. In a product ring  $R \times S$  with both factors nonzero,  $(1, 0)(0, 1) = (0, 0)$ : product rings always have zero divisors.
3. In  $M_2(\mathbb{R})$ , the diagonal matrices  $\text{diag}(1, 0)$  and  $\text{diag}(0, 1)$  multiply to the zero matrix; more generally every nonzero *singular* matrix—the standard name for a square matrix that is not invertible—is a zero divisor there.

**Definition 4.14** (Integral domain). An *integral domain* (or simply *domain*) is a commutative ring  $R$  with  $1 \neq 0$  and no zero divisors:  $ab = 0$  implies  $a = 0$  or  $b = 0$ . Equivalently, by Proposition 4.11, a domain is a nonzero commutative ring in which every nonzero element is cancellable.

**Proposition 4.15.** Let  $n \geq 2$ . The ring  $\mathbb{Z}/n\mathbb{Z}$  has no zero divisors if and only if  $n$  is prime; that is,  $\mathbb{Z}/n\mathbb{Z}$  is an integral domain if and only if  $n$  is prime.

*Proof.* If  $n = ab$  is composite with  $1 < a, b < n$ , then  $[a]_n[b]_n = [n]_n = [0]_n$  with both factors nonzero (their representatives lie strictly between 0 and  $n$ ), so  $\mathbb{Z}/n\mathbb{Z}$  has zero divisors. Conversely let  $n = p$  be prime and suppose  $[a]_p[b]_p = [0]_p$ , that is,  $p \mid ab$ . Euclid's lemma for primes (Lemma 2.17) gives  $p \mid a$  or  $p \mid b$ , that is,  $[a]_p = [0]_p$  or  $[b]_p = [0]_p$ . Since  $p \geq 2$  we also have  $[1]_p \neq [0]_p$ , so  $\mathbb{Z}/p\mathbb{Z}$  is an integral domain.  $\square$

*Example 4.16* (Integral domains). The integers  $\mathbb{Z}$  form an integral domain, the example from which the name derives. Every field is a domain (Proposition 4.28 below). The ring  $\mathbb{Z}/n\mathbb{Z}$  is a domain exactly when  $n$  is prime (Proposition 4.15; the composite case is witnessed concretely by Example 4.13(1)). If  $R$  is a domain, so is the polynomial ring  $R[x]$ : let  $f$  and  $g$  be nonzero of degrees  $m$  and  $n$ —the largest indices carrying nonzero coefficients  $f_m$  and  $g_n$ , the *leading coefficients*. In  $fg$  the coefficient of  $x^{m+n}$  is  $\sum_{i+j=m+n} f_i g_j$ ; every term with  $i > m$  or  $j > n$  has a vanishing factor, and  $i \leq m, j \leq n, i + j = m + n$  force  $i = m, j = n$ , so the coefficient equals  $f_m g_n$ , nonzero because  $R$  is a domain. Hence  $fg \neq 0$ . Non-domains include  $\mathbb{Z}/6\mathbb{Z}$ , product rings with two nonzero factors, and the matrix rings  $M_n(R)$  for  $n \geq 2$  (Example 4.13).

### 4.3 Ideals and quotient rings

The *ideals* of a ring are the subsets by which one can form a quotient preserving *both* operations, just as any subgroup of an abelian group yields a quotient group (Proposition 3.35). They are needed here to construct  $\mathbb{Z}/n\mathbb{Z}$  as a quotient and to indicate the origin of the finite fields  $\mathbb{F}_q$ ; the general theory belongs to commutative algebra and is omitted.

**Definition 4.17** (Ideal). Let  $R$  be a commutative ring. An *ideal* of  $R$  is a subset  $I \subseteq R$  such that

1.  $I$  is a subgroup of the additive group  $(R, +)$ :  $0 \in I$ , and  $a - b \in I$  whenever  $a, b \in I$ ; and
2.  $I$  *absorbs* multiplication by arbitrary ring elements:  $ra \in I$  whenever  $r \in R$  and  $a \in I$ .

In noncommutative rings one distinguishes left, right, and two-sided ideals according to the side on which absorption holds; commutative rings suffice for this volume, and there the three notions coincide.

No ideal other than  $R$  contains a unit, so none is a subring: if an ideal  $I$  contains a unit  $u$ , then  $1 = u^{-1}u \in I$  by absorption, so  $r = r \cdot 1 \in I$  for every  $r \in R$  and  $I = R$ . Consequently a field has no ideals other than the two trivial ones, and hence no quotients other than a copy of itself and the zero ring.

*Example 4.18* (Ideals). 1. In any commutative ring  $R$ , the subsets  $\{0\}$  and  $R$  are ideals—the *trivial* and *improper* ideals respectively.

2. For  $a \in R$ , the *principal ideal generated by  $a$*  is

$$(a) := aR = \{ar : r \in R\},$$

the set of multiples of  $a$ ; absorption holds because  $r'(ar) = a(r'r)$ . In  $\mathbb{Z}$ , the principal ideal  $(n) = n\mathbb{Z}$  is the set of multiples of  $n$ . Moreover *every* ideal of  $\mathbb{Z}$  is principal: an ideal

$I \neq \{0\}$  contains a nonzero element and hence (closing under negation) a positive one, so it contains a least positive element  $n$  by well-ordering; for any  $a \in I$  the division algorithm gives  $a = qn + r$  with  $0 \leq r < n$ , whence  $r = a - qn \in I$  by absorption and subgroup closure, forcing  $r = 0$  by minimality; so  $I = n\mathbb{Z}$ . This sharpens the observation of Remark 2.12, where ideals of  $\mathbb{Z}$  first appeared as the sets of Bézout combinations.

3. For  $a_1, \dots, a_k \in R$ , the *ideal generated by*  $a_1, \dots, a_k$  is  $(a_1, \dots, a_k) = \{r_1 a_1 + \dots + r_k a_k : r_i \in R\}$ , the smallest ideal containing them all.

An integral domain in which every ideal is principal, as in (2), is called a *principal ideal domain* (PID).

**Definition 4.19** (Quotient ring operations). Let  $I$  be an ideal of a commutative ring  $R$ . Since  $I$  is a subgroup of the abelian group  $(R, +)$ , the additive cosets  $a + I$  form the quotient group  $R/I$  under coset addition  $(a + I) + (b + I) = (a + b) + I$  (Proposition 3.35). Define a multiplication of cosets by

$$(a + I)(b + I) := ab + I.$$

**Proposition 4.20.** *With the operations of Definition 4.19, the set  $R/I$  is a commutative ring—the quotient ring of  $R$  by  $I$ —with zero  $0 + I = I$  and identity  $1 + I$ . In particular the coset multiplication is well defined.*

*Proof.* Well-definedness is where absorption is needed. Let  $a' = a + s$  and  $b' = b + t$  be other representatives of the same cosets, with  $s, t \in I$ . Then

$$a'b' = (a + s)(b + t) = ab + at + sb + st,$$

and each of  $at, sb, st$  lies in  $I$  by the absorption property; hence  $a'b' - ab \in I$  and  $a'b' + I = ab + I$ . The product coset is therefore independent of the representatives chosen.

The ring axioms now pass to cosets exactly as in the proof of Theorem 4.6. For instance, distributivity reads

$$\begin{aligned} (a + I)((b + I) + (c + I)) &= (a + I)((b + c) + I) = a(b + c) + I \\ &= (ab + ac) + I = (a + I)(b + I) + (a + I)(c + I), \end{aligned}$$

and associativity and commutativity of both operations follow by the same pattern from the corresponding laws in  $R$ . The coset  $I = 0 + I$  is the additive identity,  $1 + I$  the multiplicative identity, and  $-(a + I) = (-a) + I$ .  $\square$

Ring homomorphisms, the maps preserving ring structure, make precise the sense in which the quotient construction generalises the passage from  $\mathbb{Z}$  to  $\mathbb{Z}/n\mathbb{Z}$ .

**Definition 4.21** (Ring homomorphism). A *ring homomorphism* between rings  $R$  and  $S$  is a map  $\phi: R \rightarrow S$  with

$$\phi(a + b) = \phi(a) + \phi(b), \quad \phi(ab) = \phi(a)\phi(b), \quad \phi(1) = 1$$

for all  $a, b \in R$ . Its *kernel* is  $\ker \phi = \{a \in R : \phi(a) = 0\}$ .

The kernel of a ring homomorphism out of a commutative ring is always an ideal: it is a subgroup of  $(R, +)$  because  $\phi$  is in particular a homomorphism of additive groups, and it absorbs because  $a \in \ker \phi$  gives  $\phi(ra) = \phi(r)\phi(a) = \phi(r) \cdot 0 = 0$ .

*Example 4.22.* Take  $R = \mathbb{Z}$  and  $I = n\mathbb{Z}$ . The cosets  $a + n\mathbb{Z}$  are literally the residue classes  $[a]_n$  of Definition 2.22, and the coset operations of Definition 4.19 are exactly residue arithmetic: the quotient construction recovers the ring  $\mathbb{Z}/n\mathbb{Z}$  of Example 4.5(3), so modular arithmetic is an instance of it. The same construction with a different ring produces the general finite fields: quotients  $\mathbb{F}_p[x]/(f)$  of a polynomial ring by the principal ideal of an *irreducible* polynomial  $f$  of degree  $k$ —irreducibility, the polynomial analogue of primality, is defined in §5—furnish the finite fields  $\mathbb{F}_q$  with  $q = p^k$  elements. The construction is recorded here and developed in §6; the curves of Halo 2 and Orchard are defined over *prime* fields, so the prime case  $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$  is the main case in what follows.

*Remark 4.23.* The quotient map

$$\pi: \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}, \quad \pi(a) = [a]_n,$$

is a surjective ring homomorphism with kernel  $n\mathbb{Z}$ . Every quotient construction in the volume’s algebra follows this pattern: one chooses a ring and an ideal of elements to be “declared zero”, and computes with cosets. On notation: we abbreviate  $[a]_n$  to  $\bar{a}$  (as in the examples above) or drop the decoration altogether, writing plain  $a$ , whenever the modulus is clear from context;  $\mathbb{Z}/n\mathbb{Z}$  and  $\mathbb{Z}_n$  are used interchangeably; and  $\mathbb{Z}_n$  always means this ring, never the  $n$ -adic integers (which do not appear in this series).

## 4.4 Fields

**Definition 4.24** (Field). A *field* is a commutative ring  $F$  with  $1 \neq 0$  in which every nonzero element is a unit. Equivalently, a field is a set  $F$  with two operations  $+$  and  $\cdot$  such that  $(F, +)$  is an abelian group with identity 0, the nonzero elements  $(F \setminus \{0\}, \cdot)$  form an abelian group with identity 1, and multiplication distributes over addition. Concretely, in a field one may add, subtract, multiply, and divide by any nonzero element, with  $a/b := ab^{-1}$ . The group  $F^\times = (F \setminus \{0\}, \cdot)$  is the *multiplicative group* of the field. (Two steps of the equivalence require an argument. Passing from the first formulation to the second, one must show  $F \setminus \{0\}$  is closed under multiplication; that is Proposition 4.28 below. Passing back, the ring axioms must be checked for products involving 0: distributivity forces  $0 \cdot a = a \cdot 0 = 0$  by the computation of Proposition 4.3(1), whereupon every product with a zero factor vanishes and the associativity, commutativity, and identity laws extend from  $F \setminus \{0\}$  to all of  $F$ ; and  $1 \neq 0$  holds because the group  $F \setminus \{0\}$  contains its identity 1.)

**Definition 4.25** (Subfield and extension field). A subset  $F \subseteq K$  of a field  $K$  is a *subfield* when it contains 0 and 1 and is itself a field under the operations of  $K$  restricted to  $F$ . When  $F$  is a subfield of  $K$ , the larger field  $K$  is called an *extension field* (or simply an *extension*) of  $F$ ; the two phrases name the same relation. More generally, an injective ring homomorphism  $F \hookrightarrow$

$K$  between fields exhibits  $K$  as an extension of the copy of  $F$  inside it; by the usual abuse of language we identify  $F$  with that copy.

The unit criterion of §2 identifies the residue rings that are fields.

**Corollary 4.26.** *If  $p$  is prime, then every nonzero class in  $\mathbb{Z}/p\mathbb{Z}$  is a unit, so  $\mathbb{Z}/p\mathbb{Z}$  is a field; it is denoted  $\mathbb{F}_p$ . More generally,  $\mathbb{Z}/n\mathbb{Z}$  is a field if and only if  $n$  is prime.*

*Proof.* Let  $p$  be prime and  $[a]_p \neq [0]_p$ , so that  $p \nmid a$ . The only positive divisors of  $p$  are 1 and  $p$ , so  $\gcd(a, p) = 1$ , and Theorem 2.24 makes  $[a]_p$  a unit. Since  $[1]_p \neq [0]_p$  for  $p \geq 2$ , the ring  $\mathbb{Z}/p\mathbb{Z}$  is a field. Conversely, if  $n = ab$  is composite with  $1 < a, b < n$ , then  $[a]_n \neq [0]_n$  but  $\gcd(a, n) = a > 1$ , so  $[a]_n$  is not a unit (Theorem 2.24 again) and the ring is not a field; and  $n = 1$  gives the zero ring, which is not a field because a field has  $1 \neq 0$  by definition.  $\square$

**Example 4.27 (Fields).** 1. The rationals  $\mathbb{Q}$ , the reals  $\mathbb{R}$ , and the complexes  $\mathbb{C}$  are fields. The integers are not: 2 has no inverse in  $\mathbb{Z}$ .

2. For  $p$  prime,  $\mathbb{F}_p := \mathbb{Z}/p\mathbb{Z}$  is a field (Corollary 4.26; Theorem 4.29 below recovers the same fact abstractly). It is called the *prime field* of characteristic  $p$ —Definition 4.30 names the invariant—and underlies every later cryptographic construction in the series. Its finite extensions  $\mathbb{F}_q$  with  $q = p^k$  elements exist for every prime power, arising from the quotient construction of Example 4.22, and are developed in §6.

3. The subset  $\mathbb{Q}(i) = \{a + bi : a, b \in \mathbb{Q}\} \subseteq \mathbb{C}$  is a field: the inverse of a nonzero  $a + bi$  is  $(a - bi)/(a^2 + b^2)$ , which again has rational coordinates. More generally the *number fields*, obtained by adjoining to  $\mathbb{Q}$  roots of polynomial equations with rational coefficients, are fields.

4. For any field  $F$ , the *rational functions*  $F(x) = \{f/g : f, g \in F[x], g \neq 0\}$  form a field under the usual arithmetic of fractions—the field of fractions of the polynomial ring  $F[x]$ .

**Proposition 4.28.** *Every field is an integral domain.*

*Proof.* Let  $F$  be a field. Commutativity and  $1 \neq 0$  hold by definition. Suppose  $ab = 0$  with  $a \neq 0$ . Then  $a$  is a unit, and multiplying by its inverse,

$$b = 1 \cdot b = (a^{-1}a)b = a^{-1}(ab) = a^{-1} \cdot 0 = 0.$$

Hence  $F$  has no zero divisors.  $\square$

The converse fails:  $\mathbb{Z}$  is a domain but not a field (Example 4.16). For finite domains the converse holds, by a counting argument that recurs, the *injectivity-implies-surjectivity argument*: to invert an element  $a$  of a finite structure, one shows that multiplication by  $a$  is injective, hence onto by finiteness, and reads off a preimage of 1. The same argument proves the claim of Remark 4.12.

**Theorem 4.29.** *Every finite integral domain is a field.*

*Proof.* Let  $R$  be a finite integral domain and  $a \in R$  nonzero. Consider the multiplication map

$$\mu_a: R \rightarrow R, \quad \mu_a(x) = ax.$$

It is injective: if  $ax = ay$ , then  $a(x - y) = 0$ , and since  $R$  is a domain and  $a \neq 0$  this forces  $x - y = 0$ , that is,  $x = y$  (equivalently,  $a$  is cancellable by Proposition 4.11). An injective map from a finite set to itself is surjective: the image  $\mu_a(R)$  has  $|R|$  distinct elements and sits inside  $R$ , so it is all of  $R$ . In particular 1 lies in the image, so  $ab = 1$  for some  $b \in R$ ; commutativity gives  $ba = 1$  as well, so  $b = a^{-1}$  and  $a$  is a unit. Since  $R$  is a domain,  $1 \neq 0$ , and every nonzero element has been shown invertible:  $R$  is a field.  $\square$

With Proposition 4.28, the theorem shows that a nonzero finite commutative ring is a field exactly when it has no zero divisors. Applied to  $\mathbb{Z}/p\mathbb{Z}$ , a finite domain by Proposition 4.15, it recovers Corollary 4.26, but establishes only that each inverse *exists*; the extended Euclidean algorithm of §2 *computes* it, in  $O(\log p)$  division steps. Cryptography uses the existence statement in proofs and the algorithm in computation.

## 4.5 The characteristic

Every ring receives a canonical map from the integers; the characteristic, an invariant of the ring, describes that map and determines the image of  $\mathbb{Z}$  inside the ring.

**Definition 4.30** (Characteristic). Let  $R$  be a ring, and for  $n \in \mathbb{Z}$  let  $n \cdot 1$  denote the  $n$ -fold additive multiple of 1 (so  $n \cdot 1 = 1 + \dots + 1$  with  $n$  summands for  $n > 0$ ,  $0 \cdot 1 = 0$ , and  $(-n) \cdot 1 = -(n \cdot 1)$ ). The *characteristic*  $\text{char}(R)$  is the least positive integer  $n$  with  $n \cdot 1 = 0$ , if such an  $n$  exists, and 0 otherwise.

The map

$$\iota: \mathbb{Z} \rightarrow R, \quad \iota(n) = n \cdot 1,$$

is a ring homomorphism: additivity is the bookkeeping of iterated multiples in the abelian group  $(R, +)$ , multiplicativity is Proposition 4.3(4) with  $a = b = 1$ —namely  $\iota(m)\iota(n) = (m \cdot 1)(n \cdot 1) = (mn)(1 \cdot 1) = \iota(mn)$ —and  $\iota(1) = 1$ . Its kernel is an ideal of  $\mathbb{Z}$ , hence of the form  $n\mathbb{Z}$  for a unique  $n \geq 0$  by Example 4.18(2), and  $\text{char}(R)$  is exactly this nonnegative generator:  $\text{char}(R) = 0$  precisely when  $\iota$  is injective, so that a faithful copy of  $\mathbb{Z}$  sits inside  $R$ , and  $\text{char}(R) = n > 0$  when the kernel is  $n\mathbb{Z}$ .

**Example 4.31** (Characteristics). The rings  $\mathbb{Z}$ ,  $\mathbb{Q}$ ,  $\mathbb{R}$ , and  $\mathbb{C}$  all have characteristic 0: no positive multiple of 1 vanishes. For the residue rings,  $\text{char}(\mathbb{Z}/n\mathbb{Z}) = n$ : the multiple  $n \cdot \bar{1} = \bar{n} = \bar{0}$  vanishes, and no smaller positive multiple does, since  $k \cdot \bar{1} = \bar{k} \neq \bar{0}$  for  $0 < k < n$ . In particular  $\text{char}(\mathbb{F}_p) = p$ .

**Theorem 4.32.** *The characteristic of an integral domain—in particular, of a field—is either 0 or a prime number.*

*Proof.* Let  $R$  be an integral domain with  $\text{char}(R) = n > 0$ ; we show  $n$  is prime. First  $n \neq 1$ : otherwise  $1 \cdot 1 = 1 = 0$ , making  $R$  the zero ring (Remark 4.4) and contradicting  $1 \neq 0$  in a domain. Suppose  $n = ab$  with  $1 < a, b < n$ . By Proposition 4.3(4),

$$0 = n \cdot 1 = (ab)(1 \cdot 1) = (a \cdot 1)(b \cdot 1),$$

so  $a \cdot 1 = 0$  or  $b \cdot 1 = 0$  since  $R$  is a domain. Either case exhibits a positive integer smaller than  $n$  whose multiple of 1 vanishes, contradicting the minimality of  $n = \text{char}(R)$ . Hence  $n$  admits no such factorisation, and being greater than 1, it is prime.  $\square$

*Remark 4.33* (The prime subfield). Let  $F$  be a field. If  $\text{char}(F) = 0$ , the homomorphism  $\iota$  embeds a copy of  $\mathbb{Z}$  in  $F$ , and dividing—possible in a field—a copy of  $\mathbb{Q}$  as well. If  $\text{char}(F) = p$ , the image of  $\iota$  is a copy of  $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ . The smallest subfield so obtained is called the *prime subfield* of  $F$ ; every field is thus an extension of  $\mathbb{Q}$  or of some  $\mathbb{F}_p$ . The fields  $\mathbb{F}_q$  underlying the cryptography treated in this series all have prime characteristic  $p$ . In characteristic  $p$  the “freshman’s dream” identity  $(a + b)^p = a^p + b^p$  holds and gives rise to the Frobenius endomorphism; both are developed in the finite-fields section (Lemma 6.13).

## 5 Polynomials over a field

Throughout this section  $F$  denotes a field. The section constructs the ring  $F[X]$  with its division, gcd and factorisation theory, and proves that a nonzero polynomial of degree  $d$  over a field has at most  $d$  roots. From this bound it derives Lagrange interpolation (§5.7) and the Schwartz–Zippel lemma (§5.10), which justifies the polynomial identity testing of the later volumes: two polynomials encoding a computation are compared by evaluating both at one random  $\alpha$ , not coefficient by coefficient at the cost of reading the computation. Halo 2 and Zcash Orchard use these results in their polynomial commitment schemes and their *arithmetisation*, the encoding of a computation as identities between polynomials; polynomials also underlie finite fields and error-correcting codes.

### 5.1 The polynomial ring

A polynomial is a formal expression, a finite list of coefficients, and *not a function*: evaluating it yields a function, which over a finite field carries less information than the polynomial (§5.9).

**Definition 5.1** (Polynomial). A *polynomial* over  $F$  in the *indeterminate*  $X$  is a formal expression

$$f = a_0 + a_1X + a_2X^2 + \dots + a_nX^n = \sum_{i=0}^n a_iX^i,$$

where  $n \in \mathbb{N} = \{0, 1, 2, \dots\}$  and the *coefficients*  $a_0, \dots, a_n$  lie in  $F$ . Formally,  $f$  is identified with its coefficient sequence  $(a_0, a_1, a_2, \dots)$ , in which only finitely many entries are nonzero. Two polynomials are equal precisely when their coefficient sequences agree term by term:

$$\sum_i a_iX^i = \sum_i b_iX^i \iff a_i = b_i \text{ for every } i \in \mathbb{N}.$$

The symbol  $X$  is an *indeterminate* (a formal variable): it is not an element of  $F$  and denotes no particular value; it is a placeholder organising the coefficients. The set of all polynomials over  $F$  in  $X$  is denoted  $F[X]$ .

**Remark 5.2** (The sequence model). In the rigorous form of Definition 5.1,  $F[X]$  is the set of all functions  $a: \mathbb{N} \rightarrow F$  that are *eventually zero*: there is some  $N$  with  $a(i) = 0$  for all  $i > N$ . The expression  $\sum_i a_i X^i$  is notation for the sequence  $(a_0, a_1, \dots)$ , and the two viewpoints are used interchangeably: the arithmetic of sequences defined next is the arithmetic the notation suggests.

**Definition 5.3** (Ring operations on  $F[X]$ ). Let  $f = \sum_i a_i X^i$  and  $g = \sum_i b_i X^i$  lie in  $F[X]$ , with the shorter coefficient sequence zero-padded so that both run over the same index set. Their *sum* and *product* are

$$f + g = \sum_{i \geq 0} (a_i + b_i) X^i, \quad fg = \sum_{k \geq 0} c_k X^k \quad \text{where} \quad c_k = \sum_{i+j=k} a_i b_j = \sum_{i=0}^k a_i b_{k-i}.$$

The coefficient  $c_k$  is the *convolution* of the two coefficient sequences. Both defining sums are finite—only finitely many  $a_i$  and  $b_j$  are nonzero, hence only finitely many  $c_k$  are nonzero—so the sum and the product are again polynomials.

The convolution formula results from multiplying out the product  $(\sum_i a_i X^i)(\sum_j b_j X^j)$  and collecting like powers of  $X$  using  $X^i \cdot X^j = X^{i+j}$ ; as a convolution it involves no substitution of a value for  $X$ . The proof of the ring axioms (Definition 4.1) likewise involves no evaluation: it writes the coefficient of  $X^m$  on each side of an identity as a finite sum over index decompositions, then reindexes or regroups that sum, as in the associativity step below.

**Theorem 5.4** ( $F[X]$  is a commutative ring). *With the operations of Definition 5.3, the set  $F[X]$  is a commutative ring whose additive identity is the zero polynomial  $0 = (0, 0, 0, \dots)$  and whose multiplicative identity is the constant polynomial  $1 = (1, 0, 0, \dots)$ . Moreover the map  $c \mapsto (c, 0, 0, \dots)$  embeds  $F$  into  $F[X]$  as a subring, the constant polynomials; in this sense  $F \subseteq F[X]$ .*

*Proof. Additive structure.* Addition is coordinatewise, so its commutativity and associativity are inherited directly from the corresponding laws of the additive group of  $F$ , the sequence  $(0, 0, \dots)$  is an additive identity, and  $(-a_0, -a_1, \dots)$  is an additive inverse of  $(a_0, a_1, \dots)$ . Hence  $(F[X], +)$  is an abelian group.

*Commutativity of multiplication.* The coefficient of  $X^k$  in  $fg$  is  $\sum_{i+j=k} a_i b_j$ ; reindexing the sum by  $(i, j) \mapsto (j, i)$  and using commutativity of multiplication in  $F$ ,

$$\sum_{i+j=k} a_i b_j = \sum_{i+j=k} b_i a_j,$$

which is the coefficient of  $X^k$  in  $gf$ .

*Associativity of multiplication.* Write  $h = \sum_l c_l X^l$ . The coefficient of  $X^m$  in  $(fg)h$  is

$$\sum_{k+l=m} \left( \sum_{i+j=k} a_i b_j \right) c_l = \sum_{i+j+l=m} a_i b_j c_l,$$

and expanding  $f(gh)$  instead produces  $\sum_{i+j+l=m} a_i(b_j c_l)$ , the same symmetric triple sum by associativity in  $F$ . Hence  $(fg)b = f(gh)$  coefficient by coefficient.

*Distributivity.* The coefficient of  $X^k$  in  $f(g + h)$  is  $\sum_{i+j=k} a_i(b_j + c_j)$ , which distributivity in  $F$  splits as  $\sum_{i+j=k} a_i b_j + \sum_{i+j=k} a_i c_j$ : the coefficient of  $X^k$  in  $fg + fh$ . The other distributive law follows from this one and commutativity.

*Identity.* With  $1 = (1, 0, 0, \dots)$ , the  $k$ -th coefficient of  $1 \cdot f$  is  $\sum_{i+j=k} \mathbf{1}[i=0] a_j = a_k$ , where  $\mathbf{1}[i=0]$  denotes the coefficient sequence of 1 (equal to 1 at index 0 and 0 elsewhere); so  $1 \cdot f = f$ .

*Embedding.* Let  $\iota(c) = (c, 0, 0, \dots)$ . The map  $\iota$  preserves sums coordinatewise, and  $\iota(c)\iota(d)$  has  $k$ -th coefficient  $cd$  for  $k = 0$  and 0 otherwise, so  $\iota(c)\iota(d) = \iota(cd)$ ; also  $\iota(1) = 1$ . Distinct constants give distinct sequences, so  $\iota$  is an injective ring homomorphism (Definition 4.21) and its image is a subring of  $F[X]$  isomorphic to  $F$ .  $\square$

*Remark 5.5* (Scalar multiplication). Multiplying  $f = \sum_i a_i X^i$  by a constant  $c \in F$ , viewed as the constant polynomial  $\iota(c)$ , gives the *scalar multiple*  $cf = \sum_i (ca_i) X^i$ . With addition, this makes  $F[X]$  a *vector space* over  $F$ : for the present purpose, a set closed under addition and under scaling by field elements, treated in the linear-algebra section later in the volume. The monomials  $1, X, X^2, X^3, \dots$  form an infinite coordinate system (a *basis*). Later arguments use this linear structure, for instance to count polynomials of bounded degree.

## 5.2 Degree

**Definition 5.6** (Degree, leading coefficient, monic). Let  $f = \sum_i a_i X^i \in F[X]$  be nonzero. The *degree*  $\deg f$  is the largest index  $n$  with  $a_n \neq 0$ ; the coefficient  $a_n$  is the *leading coefficient*  $\text{lc}(f)$ , the term  $a_n X^n$  is the *leading term*, and  $a_0$  is the *constant term*. A nonzero polynomial is *monic* if its leading coefficient is 1. The constant polynomials are the elements of  $F$  embedded by Theorem 5.4; the nonzero ones have degree 0. The zero polynomial is assigned  $\deg 0 = -\infty$ , with the conventions  $-\infty < n$ ,  $-\infty + n = -\infty$ , and  $-\infty + (-\infty) = -\infty$  for every  $n \in \mathbb{Z}$ . Polynomials of degree 1, 2, and 3 are called *linear*, *quadratic*, and *cubic* respectively.

The convention  $\deg 0 = -\infty$  makes the degree formulas below hold without exception, so later inductions on degree need no special case for the zero polynomial.

**Proposition 5.7** (Degree of sums and products). For all  $f, g \in F[X]$ ,

$$\deg(f + g) \leq \max\{\deg f, \deg g\}, \quad (1)$$

with equality whenever  $\deg f \neq \deg g$ , and

$$\deg(fg) = \deg f + \deg g. \quad (2)$$

*Proof.* If either polynomial is zero, both statements reduce to the  $-\infty$  conventions of Definition 5.6. So let  $f$  and  $g$  be nonzero, of degrees  $m$  and  $n$  with leading coefficients  $a_m$  and  $b_n$ .

*Sum.* The coefficient of  $X^k$  in  $f + g$  is  $a_k + b_k$ , which vanishes for  $k > \max\{m, n\}$ ; this is (1). If  $m > n$ , the coefficient of  $X^m$  is  $a_m + 0 = a_m \neq 0$ , giving equality (symmetrically for  $n > m$ ). When

$m = n$ , cancellation  $a_m + b_m = 0$  can lower the degree, which is why the sum bound is only an inequality.

*Product.* The coefficient  $c_{m+n} = \sum_{i+j=m+n} a_i b_j$  has only the surviving term  $i = m, j = n$ : terms with  $i > m$  have  $a_i = 0$ , and terms with  $i < m$  force  $j > n$ , so  $b_j = 0$ . Hence  $c_{m+n} = a_m b_n$ , and the same argument shows  $c_k = 0$  for  $k > m + n$ . Since  $F$  is a field it has no zero divisors (Proposition 4.28), so  $a_m b_n \neq 0$ ; therefore  $\deg(fg) = m + n$  and  $\text{lc}(fg) = \text{lc}(f)\text{lc}(g)$ .  $\square$

The product formula (2) is the first result to use the field hypothesis, through the absence of zero divisors in  $F$ ; the next three results and the rest of the section depend on it.

**Corollary 5.8** ( $F[X]$  is an integral domain). *The ring  $F[X]$  has no zero divisors: if  $fg = 0$ , then  $f = 0$  or  $g = 0$ .*

*Proof.* If  $f$  and  $g$  are both nonzero, then  $\deg(fg) = \deg f + \deg g \geq 0$  by (2), so  $fg \neq 0$ . Contrapositively,  $fg = 0$  forces  $f = 0$  or  $g = 0$ .  $\square$

**Corollary 5.9** (Cancellation). *Let  $f, g, h \in F[X]$  with  $h \neq 0$ . If  $fh = gh$ , then  $f = g$ .*

*Proof.* From  $fh = gh$  we get  $(f - g)h = 0$ ; since  $h \neq 0$ , Corollary 5.8 forces  $f - g = 0$ .  $\square$

**Proposition 5.10** (Units of  $F[X]$ ). *A polynomial  $f \in F[X]$  is a unit—has a multiplicative inverse in  $F[X]$ —if and only if it is a nonzero constant. Thus  $F[X]^\times = F^\times$ .*

*Proof.* A nonzero constant  $c$  has the inverse  $c^{-1} \in F \subseteq F[X]$ . Conversely,  $fg = 1$  gives  $\deg f + \deg g = \deg 1 = 0$  by (2); since degrees of nonzero polynomials are nonnegative,  $\deg f = \deg g = 0$ , and  $f$  is a nonzero constant.  $\square$

**Definition 5.11** (Associates). Two polynomials  $f, g \in F[X]$  are *associates* if  $f = ug$  for some unit  $u \in F^\times$ , that is, if they differ only by a nonzero scalar factor.

Every nonzero polynomial  $f$  has exactly one monic associate,  $\text{lc}(f)^{-1}f$ . Choosing the monic representative, as below, makes *the* greatest common divisor and *the* irreducible factor unique.

### 5.3 Division with remainder

Division with remainder, with *degree* in the role of absolute value on  $\mathbb{Z}$ , carries the arithmetic of  $\mathbb{Z}$  (§2) over to  $F[X]$ : greatest common divisors, the Euclidean algorithm, Bézout's identity, irreducibles in place of primes, and unique factorisation.

**Theorem 5.12** (Division algorithm). *Let  $f, g \in F[X]$  with  $g \neq 0$ . There exist unique polynomials  $q, r \in F[X]$  such that*

$$f = qg + r \quad \text{and} \quad \deg r < \deg g.$$

*We call  $q$  the quotient and  $r$  the remainder, and write  $q = f \text{ div } g$  and  $r = f \bmod g$ .*

*Proof. Existence.* Fix  $g$ , of degree  $n$  with leading coefficient  $b_n$ , and induct on  $\deg f$ . If  $\deg f < n$ —including  $f = 0$  with  $\deg f = -\infty$ —take  $q = 0$  and  $r = f$ . Otherwise let  $m = \deg f \geq n$ , with leading coefficient  $a_m$ , and form

$$f_1 = f - \frac{a_m}{b_n} X^{m-n} g.$$

The subtracted term has degree  $m$  and leading coefficient  $(a_m/b_n) b_n = a_m$ , so it cancels the leading term of  $f$ : either  $f_1 = 0$  or  $\deg f_1 < m$ . By the induction hypothesis  $f_1 = q_1 g + r$  with  $\deg r < n$ , and then  $f = (q_1 + (a_m/b_n) X^{m-n})g + r$ . The field hypothesis enters at this step: division by the leading coefficient  $b_n$  requires  $b_n^{-1}$  to exist.

*Uniqueness.* Suppose  $f = qg + r = q'g + r'$  with both remainders of degree  $< n$ . Then  $(q - q')g = r' - r$ . If  $q \neq q'$ , then  $\deg((q - q')g) = \deg(q - q') + n \geq n$  by (2), whereas  $\deg(r' - r) \leq \max\{\deg r', \deg r\} < n$  by (1)—a contradiction. Hence  $q = q'$ , and then  $r = f - qg = r'$ .  $\square$

**Remark 5.13** (Polynomial long division). The existence proof is constructive: it is *polynomial long division*, which repeatedly eliminates the current leading term by subtracting a monomial multiple of  $g$ . Each step strictly lowers the degree of the running remainder, so the process terminates after at most  $\deg f - \deg g + 1$  steps when  $\deg f \geq \deg g$ , and takes no steps when  $f = 0$  or  $\deg f < \deg g$ . Later inductions on degree use the same leading-term elimination.

For example, let  $f = X^3 + 2X + 1$  and  $g = X^2 + 1$  over  $\mathbb{Q}$ . Subtracting  $X \cdot g = X^3 + X$  leaves  $X + 1$ , whose degree 1 is below  $\deg g = 2$ ; hence  $q = X$  and  $r = X + 1$ :

$$X^3 + 2X + 1 = X(X^2 + 1) + (X + 1).$$

**Definition 5.14** (Divisibility). For  $f, g \in F[X]$ , we say  $g$  *divides*  $f$ , written  $g \mid f$ , if  $f = qg$  for some  $q \in F[X]$ ; equivalently, for  $g \neq 0$ , if the remainder  $f \bmod g$  is 0. In that case  $g$  is a *divisor* (or *factor*) of  $f$ .

## 5.4 Roots and the factor theorem

This subsection defines evaluation, where a polynomial first gives rise to a function, and proves the root bound (Theorem 5.18). Every later construction of the volume on polynomials rests on that bound, including the cyclicity of the multiplicative group of a finite field.

**Definition 5.15** (Evaluation and roots). Let  $f = \sum_{i=0}^n a_i X^i \in F[X]$  and  $\alpha \in F$ . The *evaluation* of  $f$  at  $\alpha$  is the field element

$$f(\alpha) = \sum_{i=0}^n a_i \alpha^i \in F,$$

obtained by substituting  $\alpha$  for the indeterminate  $X$ . An element  $\alpha \in F$  is a *root* (or *zero*) of  $f$  if  $f(\alpha) = 0$ .

**Proposition 5.16** (Evaluation is a ring homomorphism). Fix  $\alpha \in F$ . The evaluation map  $\text{ev}_\alpha: F[X] \rightarrow F$ ,  $f \mapsto f(\alpha)$ , satisfies

$$(f + g)(\alpha) = f(\alpha) + g(\alpha), \quad (fg)(\alpha) = f(\alpha)g(\alpha), \quad 1(\alpha) = 1$$

for all  $f, g \in F[X]$ ; that is,  $\text{ev}_\alpha$  is a ring homomorphism.

*Proof.* Additivity is immediate from the coordinatewise definition of addition. For multiplicativity, with  $c_k = \sum_{i+j=k} a_i b_j$  as in Definition 5.3,

$$(fg)(\alpha) = \sum_k c_k \alpha^k = \sum_k \sum_{i+j=k} a_i b_j \alpha^{i+j} = \left( \sum_i a_i \alpha^i \right) \left( \sum_j b_j \alpha^j \right) = f(\alpha) g(\alpha),$$

the middle equality because expanding the product of the two finite sums and grouping terms by  $k = i + j$  produces exactly the double sum. The formal reason evaluation respects the algebra is that the convolution defining polynomial multiplication collapses to ordinary multiplication in  $F$  via  $\alpha^i \alpha^j = \alpha^{i+j}$ . Finally  $1(\alpha) = 1$  by direct substitution.  $\square$

**Theorem 5.17** (Remainder theorem; factor theorem). *Let  $f \in F[X]$  and  $\alpha \in F$ .*

1. (Remainder theorem.) *The remainder of  $f$  upon division by the linear polynomial  $X - \alpha$  is the constant  $f(\alpha)$ : there is  $q \in F[X]$  with  $f = q(X - \alpha) + f(\alpha)$ .*
2. (Factor theorem.) *The element  $\alpha$  is a root of  $f$  if and only if  $(X - \alpha) \mid f$ .*

*Proof.* (1) Divide  $f$  by  $X - \alpha$  (Theorem 5.12): since  $\deg(X - \alpha) = 1$ , the remainder has degree  $< 1$ , hence is a constant  $r \in F$ . Evaluating  $f = q(X - \alpha) + r$  at  $\alpha$  via Proposition 5.16 gives  $f(\alpha) = q(\alpha)(\alpha - \alpha) + r = r$ .

(2) By (1) and the uniqueness of the remainder,  $(X - \alpha) \mid f$  holds exactly when the remainder  $f(\alpha)$  is 0, that is, when  $\alpha$  is a root of  $f$ .  $\square$

**Theorem 5.18** (At most  $d$  roots). *A nonzero polynomial  $f \in F[X]$  of degree  $d$  has at most  $d$  distinct roots in  $F$ .*

*Proof.* We induct on  $d = \deg f$ . If  $d = 0$ , then  $f$  is a nonzero constant, which evaluates to itself everywhere and so has  $0 \leq d$  roots. Let  $d \geq 1$ . If  $f$  has no root in  $F$  the claim holds; otherwise let  $\alpha$  be a root. By the factor theorem,  $f = (X - \alpha)q$ , and  $\deg q = d - 1$  by (2). Let  $\beta \neq \alpha$  be any other root of  $f$ . Evaluating with Proposition 5.16,

$$0 = f(\beta) = (\beta - \alpha)q(\beta);$$

since  $\beta - \alpha \neq 0$  and  $F$  has no zero divisors (Proposition 4.28), we get  $q(\beta) = 0$ . Every root of  $f$  other than  $\alpha$  is therefore a root of  $q$ , and by the induction hypothesis  $q$  has at most  $d - 1$  distinct roots. Hence  $f$  has at most  $(d - 1) + 1 = d$ .  $\square$

*Remark 5.19* (Why a field is needed). Both ingredients of the proof, the absence of zero divisors and the degree bookkeeping, are necessary. Over a ring with zero divisors the bound fails: in  $\mathbb{Z}/8\mathbb{Z}$  the degree-2 polynomial  $X^2 - 1$  has the four roots  $\bar{1}, \bar{3}, \bar{5}, \bar{7}$ , since  $1^2 = 1$ ,  $3^2 = 9 \equiv 1$ ,  $5^2 = 25 \equiv 1$ , and  $7^2 = 49 \equiv 1 \pmod{8}$ . The polynomial methods of Halo 2 therefore require a field, specifically a finite field  $\mathbb{F}_p$  or  $\mathbb{F}_q$ .

**Corollary 5.20** (Few points determine a low-degree polynomial). *Let  $f, g \in F[X]$  both have degree at most  $d$ . If  $f(\alpha) = g(\alpha)$  for  $d + 1$  distinct values  $\alpha \in F$ , then  $f = g$  as polynomials.*

*Proof.* The difference  $h = f - g$  has  $\deg h \leq \max\{\deg f, \deg g\} \leq d$  by (1) and vanishes at  $d + 1$  distinct points. Were  $h$  nonzero, it would be a polynomial of degree at most  $d$  with  $d + 1$  distinct roots, contradicting Theorem 5.18. Hence  $h = 0$ , that is,  $f = g$ .  $\square$

## 5.5 Greatest common divisors

**Definition 5.21** (Greatest common divisor). Let  $f, g \in F[X]$ , not both zero. A *greatest common divisor* of  $f$  and  $g$  is a polynomial  $d$  such that

1.  $d \mid f$  and  $d \mid g$ ; and
2. every common divisor  $c$  of  $f$  and  $g$  satisfies  $c \mid d$ .

A greatest common divisor is determined only up to multiplication by a unit; the unique *monic* one (Theorem 5.23) is singled out and denoted  $\gcd(f, g)$ . When  $\gcd(f, g) = 1$ , the polynomials  $f$  and  $g$  are called *coprime* (or *relatively prime*). Since every polynomial divides 0, we have  $\gcd(f, 0) = \text{lc}(f)^{-1}f$  for  $f \neq 0$ .

**Lemma 5.22** (Euclidean step). *Let  $f, g \in F[X]$  with  $g \neq 0$ , and let  $f = qg + r$  be the division of Theorem 5.12. Then the pairs  $(f, g)$  and  $(g, r)$  have exactly the same common divisors; in particular, their greatest common divisors coincide.*

*Proof.* If  $c \mid f$  and  $c \mid g$ , then  $c \mid (f - qg) = r$ ; so  $c$  is a common divisor of  $(g, r)$ . Conversely, if  $c \mid g$  and  $c \mid r$ , then  $c \mid (qg + r) = f$ . The two sets of common divisors are therefore equal, hence so are their maximal elements under divisibility.  $\square$

**Theorem 5.23** (Euclidean algorithm; Bézout's identity). *Let  $f, g \in F[X]$ , not both zero. Then  $\gcd(f, g)$  exists and is unique as a monic polynomial, and there exist  $s, t \in F[X]$  with*

$$\gcd(f, g) = sf + tg. \quad (3)$$

*Concretely: set  $r_0 = f$  and  $r_1 = g$ , and while  $r_i \neq 0$  divide*

$$r_{i-1} = q_i r_i + r_{i+1}, \quad \deg r_{i+1} < \deg r_i.$$

*The degrees strictly decrease, so some  $r_{N+1} = 0$ ; the last nonzero remainder  $r_N$ , made monic, is  $\gcd(f, g)$ .*

*Proof. Termination.* If  $r_1 = 0$ , the algorithm stops at once with  $N = 0$  and  $r_N = f \neq 0$ . Otherwise the degrees  $\deg r_1 > \deg r_2 > \dots$  form a strictly decreasing sequence of nonnegative integers, which cannot continue forever; so some  $r_{N+1} = 0$  with  $r_N \neq 0$ .

*The gcd property.* Lemma 5.22, applied at each division, preserves the set of common divisors:

$$\begin{aligned}\{\text{common divisors of } (f, g)\} &= \{\text{common divisors of } (r_1, r_2)\} \\ &= \dots = \{\text{common divisors of } (r_N, 0)\},\end{aligned}$$

and the last set is exactly the set of divisors of  $r_N$ , since every polynomial divides 0. Thus  $r_N$  divides both  $f$  and  $g$ , and every common divisor of  $f$  and  $g$  divides  $r_N$ : the monic associate  $\text{lc}(r_N)^{-1}r_N$  is a monic greatest common divisor.

*Uniqueness.* Two monic greatest common divisors divide each other, so they differ by a unit  $u \in F^\times$  (Proposition 5.10); comparing leading coefficients of monic polynomials forces  $u = 1$ .

*Bézout.* We show by induction on  $i$  that each remainder  $r_i$  is an  $F[X]$ -linear combination of  $f$  and  $g$ . The bases  $r_0 = 1 \cdot f + 0 \cdot g$  and  $r_1 = 0 \cdot f + 1 \cdot g$  are trivial, and the recurrence  $r_{i+1} = r_{i-1} - q_i r_i$  expresses  $r_{i+1}$  as a combination once  $r_{i-1}$  and  $r_i$  are. Hence  $r_N = s_0 f + t_0 g$  for some  $s_0, t_0 \in F[X]$ ; dividing through by  $\text{lc}(r_N)$  yields (3).  $\square$

*Remark 5.24* (Extended Euclidean algorithm). Carrying the linear-combination coefficients along with the remainders, as in the integer case (Theorem 2.13), produces  $s$  and  $t$  alongside the gcd. The standard application is inversion in a quotient  $F[X]/(m)$  for irreducible  $m$ : if  $\gcd(f, m) = 1$ , then  $sf + tm = 1$ , so  $s$  is the inverse of  $f$  modulo  $m$ . Division in the extension fields  $\mathbb{F}_q$  constructed in Section 6 is performed this way, and the algorithm also computes gcds of the polynomials a proof system manipulates. The curves of Halo 2 and Orchard, however, are defined over *prime* fields  $\mathbb{F}_p$ , not extension fields, and field inversion there uses the integer extended Euclidean algorithm or Fermat's little theorem, not the polynomial algorithm in  $F[X]/(m)$ .

**Corollary 5.25** (Euclid's lemma in  $F[X]$ ). *Let  $p, f, g \in F[X]$ .*

1. *If  $\gcd(p, f) = 1$  and  $p \mid fg$ , then  $p \mid g$ .*
2. *If  $p$  is irreducible and  $p \mid fg$ , then  $p \mid f$  or  $p \mid g$ .*

*Here irreducible is the polynomial analogue of prime, made precise in Definition 5.26 below.*

*Proof.* (1) Multiplying the Bézout identity  $sp + tf = 1$  by  $g$  gives  $spg + tfg = g$ . The polynomial  $p$  divides both terms on the left—the second because  $p \mid fg$ —hence divides  $g$ .

(2) Suppose  $p \nmid f$ . The gcd  $\gcd(p, f)$  is a monic divisor of  $p$  that is not the monic associate of  $p$  (otherwise  $p \mid f$ ). By irreducibility the only monic divisors of  $p$  are 1 and its monic associate, so  $\gcd(p, f) = 1$  and part (1) applies.  $\square$

## 5.6 Unique factorisation

**Definition 5.26** (Irreducible polynomial). A polynomial  $p \in F[X]$  is *irreducible over  $F$*  if  $\deg p \geq 1$  and  $p$  admits no factorisation  $p = gh$  with both  $\deg g \geq 1$  and  $\deg h \geq 1$ ; equivalently,  $p$  is nonconstant and its only factorisations are the trivial ones, (unit)  $\times$  (associate of  $p$ ). A nonconstant polynomial that is not irreducible is *reducible*.

*Remark 5.27* (Irreducibility is relative to the base field). The polynomial  $X^2 + 1$  is irreducible over  $\mathbb{R}$  (it has no real root, and a real quadratic factors precisely when it has a root, by the test below), yet splits as  $(X - i)(X + i)$  over  $\mathbb{C}$ , and factors as  $(X + 1)^2$  over  $\mathbb{F}_2$ , by direct expansion in characteristic 2. Degree-1 polynomials are irreducible over every field. The *rootlessness test* states that a polynomial of degree 2 or 3 is irreducible over  $F$  if and only if it has no root in  $F$ , because any nontrivial factorisation of a quadratic or cubic must include a linear factor, which by the factor theorem (Theorem 5.17) supplies a root. The test fails from degree 4 onward: a product of two irreducible quadratics has no roots yet is reducible.

**Theorem 5.28** ( $F[X]$  is a unique factorisation domain). *Every nonconstant  $f \in F[X]$  factors as*

$$f = c p_1 p_2 \cdots p_k$$

*with  $c = \text{lc}(f) \in F^\times$  a constant and each  $p_i$  monic irreducible. The factorisation is unique up to order: if also  $f = c' q_1 \cdots q_l$  with  $c' \in F^\times$  and each  $q_j$  monic irreducible, then  $c = c'$ ,  $k = l$ , and the  $q_j$  are a reordering of the  $p_i$ .*

*Proof.* The proof is modelled on the fundamental theorem of arithmetic (Theorem 2.18), with degree in place of absolute value.

*Existence*, by induction on  $\deg f \geq 1$ . If  $f$  is irreducible, then  $f = \text{lc}(f) \cdot p_1$  with  $p_1$  its monic associate. If  $f = gh$  is reducible with  $1 \leq \deg g, \deg h < \deg f$ , the induction hypothesis factors  $g$  and  $h$ , and multiplying the two factorisations—collecting the two constants into one—factors  $f$ .

*Uniqueness*. Comparing leading coefficients in the two factorisations gives  $c = c' = \text{lc}(f)$ , since monic factors contribute leading coefficient 1 by (2). Cancelling the constants (Corollary 5.9) leaves  $p_1 \cdots p_k = q_1 \cdots q_l$ ; we induct on  $k$ . For  $k = 1$  the left side is irreducible, so  $l = 1$  and  $p_1 = q_1$ . For  $k \geq 2$ :  $p_1$  divides  $q_1 \cdots q_l$ , so iterating Euclid's lemma (Corollary 5.25(2)) gives  $p_1 \mid q_j$  for some  $j$ . A monic irreducible dividing a monic irreducible equals it:  $q_j = p_1 h$  forces  $\deg h = 0$  by irreducibility of  $q_j$ , and comparing leading coefficients gives  $h = 1$ . Cancel  $p_1 = q_j$  from both sides (Corollary 5.9). If  $l = 1$ , the cancellation leaves  $p_2 \cdots p_k = 1$ , which is impossible: the left side is a product of  $k - 1 \geq 1$  nonconstant polynomials and so has degree at least 1 by (2), while  $\deg 1 = 0$ . Hence  $l \geq 2$ , and the induction hypothesis, applied to  $p_2 \cdots p_k = \prod_{j' \neq j} q_{j'}$ , concludes  $k = l$  and matches the remaining factors.  $\square$

*Remark 5.29* (The ring  $F[X]$  is a principal ideal domain). The degree function and Theorem 5.12 make  $F[X]$  a *Euclidean domain*—an integral domain with division with remainder relative to a size measure—and every Euclidean domain is a *principal ideal domain* (PID): each ideal (Definition 4.17) of  $F[X]$  is  $(g) = \{qg : q \in F[X]\}$  for a single generator  $g$ , namely any nonzero element of least degree in a nonzero ideal (the zero ideal is generated by 0), by the same division-minimality argument that proved every ideal of  $\mathbb{Z}$  principal (Example 4.18(2)). The generator of the ideal  $\{sf + tg : s, t \in F[X]\}$  is a gcd of  $f$  and  $g$ , and Bézout's identity (3) is the explicit form of this principality. The results above follow the standard chain

$$\text{Euclidean} \implies \text{PID} \implies \text{UFD}$$

for  $F[X]$ , ending at Theorem 5.28.

## 5.7 Lagrange interpolation

Lagrange interpolation, the constructive converse of Corollary 5.20, underlies secret sharing, Reed–Solomon error-correcting codes, and the polynomial encodings in Halo 2.

**Theorem 5.30** (Lagrange interpolation). *Let  $x_0, \dots, x_d \in F$  be distinct (the nodes) and let  $y_0, \dots, y_d \in F$  be arbitrary targets. There exists a unique  $f \in F[X]$  with  $\deg f \leq d$  and  $f(x_i) = y_i$  for all  $i$ , given explicitly by*

$$f(X) = \sum_{i=0}^d y_i \ell_i(X), \quad \text{where} \quad \ell_i(X) = \prod_{j \neq i} \frac{X - x_j}{x_i - x_j}$$

are the Lagrange basis polynomials for the nodes.

*Proof.* The  $\ell_i$  are polynomials. Each denominator  $\prod_{j \neq i} (x_i - x_j)$  is a product of nonzero field elements, the nodes being distinct, hence is an invertible constant; so  $\ell_i$  is a polynomial, of degree exactly  $d$  (a product of  $d$  linear factors divided by a nonzero constant).

*The delta property.* Writing  $\delta_{im}$  for the Kronecker symbol (equal to 1 if  $i = m$  and 0 otherwise),

$$\ell_i(x_m) = \delta_{im} \quad (0 \leq i, m \leq d) : \quad (4)$$

at  $m = i$  every factor  $(x_i - x_j)/(x_i - x_j)$  is 1, while at  $m \neq i$  the factor with  $j = m$  has numerator  $x_m - x_m = 0$ . Consequently  $f(x_m) = \sum_i y_i \ell_i(x_m) = y_m$  for each  $m$ , and  $\deg f \leq d$  by (1), each summand having degree at most  $d$ .

*Uniqueness.* Two interpolants of degree at most  $d$  agree at the  $d + 1$  distinct nodes, hence are equal by Corollary 5.20.  $\square$

*Example 5.31* (A worked interpolation over  $\mathbb{Q}$ ). This example computes the polynomial of degree at most 2 with  $f(0) = 1$ ,  $f(1) = 3$ , and  $f(2) = 7$ . The basis polynomials for the nodes 0, 1, 2 are

$$\begin{aligned} \ell_0 &= \frac{(X-1)(X-2)}{(0-1)(0-2)} = \frac{(X-1)(X-2)}{2}, \\ \ell_1 &= \frac{(X-0)(X-2)}{(1-0)(1-2)} = -X(X-2), \\ \ell_2 &= \frac{(X-0)(X-1)}{(2-0)(2-1)} = \frac{X(X-1)}{2}. \end{aligned}$$

Hence

$$f = 1 \cdot \ell_0 + 3 \cdot \ell_1 + 7 \cdot \ell_2 = \frac{1}{2}(X^2 - 3X + 2) + 3(2X - X^2) + \frac{7}{2}(X^2 - X) = X^2 + X + 1,$$

which satisfies  $f(0) = 1$ ,  $f(1) = 3$ ,  $f(2) = 7$  as required.

*Remark 5.32* (The evaluation isomorphism). Fix  $d + 1$  distinct nodes and let  $V = \{f \in F[X] : \deg f \leq d\}$ . In the language of the linear-algebra section later in the volume,  $V$  is a  $(d + 1)$ -dimensional vector space over  $F$  with basis  $1, X, \dots, X^d$ , and the evaluation map

$$\text{ev}: V \rightarrow F^{d+1}, \quad f \mapsto (f(x_0), \dots, f(x_d)),$$

is  $F$ -linear. Its surjectivity is the existence half of Theorem 5.30, and its injectivity the uniqueness half; so  $\text{ev}$  is a linear isomorphism  $V \cong F^{d+1}$ . By the delta property (4), the Lagrange basis  $\{\ell_i\}$  is the basis of  $V$  matched to the standard coordinates of  $F^{d+1}$ : the polynomial with prescribed values  $(\gamma_0, \dots, \gamma_d)$  has coordinates  $(\gamma_0, \dots, \gamma_d)$  in the basis  $\{\ell_i\}$ . Conversion between a polynomial's *coefficient representation* and its *evaluation representation* underlies fast polynomial arithmetic via the fast Fourier transform, treated with the roots of unity later in the volume, and the evaluation-domain encodings used throughout Halo 2.

**Remark 5.33** (The vanishing polynomial). Alongside the Lagrange basis, the node set  $\{x_0, \dots, x_d\}$  determines its *vanishing polynomial*

$$Z(X) = \prod_{i=0}^d (X - x_i),$$

the unique monic polynomial of degree  $d + 1$  vanishing at every node. A polynomial  $g$  vanishes on the whole node set if and only if  $Z \mid g$ . One direction is immediate from Proposition 5.16. For the other, apply the factor theorem repeatedly:  $g(x_0) = 0$  gives  $g = (X - x_0)g_1$ ; evaluating at  $x_1 \neq x_0$  gives  $0 = (x_1 - x_0)g_1(x_1)$ , so  $g_1(x_1) = 0$  since  $F$  has no zero divisors, and  $g_1 = (X - x_1)g_2$ ; continuing through the remaining nodes strips off one linear factor per node. The repetition requires the nodes to be distinct, so that the linear factors  $X - x_i$  are pairwise coprime. A proof system uses the Lagrange basis and the vanishing polynomial to argue that a committed polynomial agrees with prescribed values on an evaluation domain.

## 5.8 Multiplicity and the formal derivative

A linear factor may divide a polynomial more than once, as in  $X^2 - 2X + 1 = (X - 1)^2$ . This subsection defines the multiplicity of a root, bounds the number of roots counted with multiplicity, and detects repeated roots with a derivative defined without limits.

**Definition 5.34** (Multiplicity of a root). Let  $f \in F[X]$  be nonzero and let  $\alpha \in F$  be a root of  $f$ . The *multiplicity*  $\text{mult}_\alpha(f)$  is the largest integer  $m \geq 1$  with  $(X - \alpha)^m \mid f$ . A root of multiplicity 1 is *simple*; a root of multiplicity at least 2 is *repeated* (or *multiple*).

The largest such  $m$  exists: the powers  $(X - \alpha)^m$  have strictly increasing degrees, so only finitely many can divide the fixed  $f$ , and at least one does by the factor theorem. In an equivalent characterisation,  $m = \text{mult}_\alpha(f)$  holds exactly when  $f = (X - \alpha)^m g$  with  $g(\alpha) \neq 0$ . Writing  $f = (X - \alpha)^m g$  with  $m$  maximal forces  $g(\alpha) \neq 0$ , else the factor theorem would split a further factor  $X - \alpha$  off  $g$ ; conversely, if  $f = (X - \alpha)^m g$  with  $g(\alpha) \neq 0$ , then  $(X - \alpha)^{m+1} \nmid f$  would give  $(X - \alpha)^m g = (X - \alpha)^{m+1} h$ , and cancelling (Corollary 5.9)  $g = (X - \alpha)h$ , making  $\alpha$  a root of  $g$ .

The number of roots counted *with* multiplicity is also bounded by the degree; the proof uses the following lemma.

**Lemma 5.35** (Coprime factors multiply). Let  $f_1, \dots, f_r \in F[X]$  be pairwise coprime ( $\gcd(f_i, f_j) = 1$  for  $i \neq j$ ) and suppose each  $f_i$  divides  $f$ . Then  $f_1 f_2 \dots f_r \mid f$ .

*Proof.* First let  $r = 2$ , and write  $f = f_1 u = f_2 v$ . Bézout (3) gives  $s f_1 + t f_2 = 1$ , so

$$f = s f_1 f + t f_2 f = s f_1 (f_2 v) + t f_2 (f_1 u) = f_1 f_2 (s v + t u),$$

and  $f_1 f_2 \mid f$ . For  $r > 2$ , induct: by hypothesis  $f_1 \cdots f_{r-1} \mid f$ , and the product  $f_1 \cdots f_{r-1}$  is coprime to  $f_r$ —a nonconstant common divisor would have a monic irreducible factor (Theorem 5.28), which by Euclid's lemma (Corollary 5.25(2)) would divide some  $f_i$  with  $i < r$  as well as  $f_r$ , contradicting  $\gcd(f_i, f_r) = 1$ . The case  $r = 2$  then applies.  $\square$

**Proposition 5.36** (Counting roots with multiplicity). *Let  $f \in F[X]$  be nonzero of degree  $d$ , with all its distinct roots  $\alpha_1, \dots, \alpha_r \in F$  of multiplicities  $m_1, \dots, m_r$ . Then*

$$f = (X - \alpha_1)^{m_1} \cdots (X - \alpha_r)^{m_r} g$$

*for some  $g \in F[X]$  with no roots in  $F$ , and  $\sum_i m_i \leq d$ . The number of roots counted with multiplicity is thus at most  $d$ , with equality if and only if  $f$  splits into linear factors over  $F$ .*

*Proof.* We first check that the powers  $(X - \alpha_i)^{m_i}$  are pairwise coprime. Each  $X - \alpha_i$  is monic irreducible (degree 1), and by unique factorisation the monic divisors of  $(X - \alpha_i)^{m_i}$  are exactly the powers  $(X - \alpha_i)^e$  with  $0 \leq e \leq m_i$ . For  $i \neq j$  the only monic polynomial that is simultaneously a power of  $X - \alpha_i$  and of  $X - \alpha_j$  is 1, since  $X - \alpha_i = X - \alpha_j$  would force  $\alpha_i = \alpha_j$ ; so  $\gcd((X - \alpha_i)^{m_i}, (X - \alpha_j)^{m_j}) = 1$ .

Each  $(X - \alpha_i)^{m_i}$  divides  $f$  by the definition of multiplicity, so Lemma 5.35 gives  $f = (X - \alpha_1)^{m_1} \cdots (X - \alpha_r)^{m_r} g$ . Maximality of each  $m_i$  gives  $g(\alpha_i) \neq 0$ : a further factor  $X - \alpha_i$  inside  $g$  would raise the multiplicity. Nor has  $g$  any other root: a root  $\beta$  of  $g$  is a root of  $f$ , hence one of the  $\alpha_i$ . Taking degrees with (2),

$$m_1 + \cdots + m_r + \deg g = d,$$

so  $\sum_i m_i \leq d$ , with equality precisely when  $\deg g = 0$ , that is, when  $f$  is a constant times a product of linear factors.  $\square$

**Definition 5.37** (Formal derivative). The *formal derivative* is the map  $D: F[X] \rightarrow F[X]$  defined on  $f = \sum_{i=0}^n a_i X^i$  by

$$f' = Df = \sum_{i=1}^n i a_i X^{i-1} = a_1 + 2a_2 X + \cdots + n a_n X^{n-1},$$

where the coefficient  $i a_i$  means  $a_i$  added to itself  $i$  times in  $F$ —that is,  $a_i$  multiplied by the image of the integer  $i$  under the canonical map  $\mathbb{Z} \rightarrow F$  (Definition 4.30). No limits are involved: the definition is formal and algebraic, valid over any field, and is introduced for one purpose, to detect repeated roots without appeal to analysis.

**Proposition 5.38** (Rules of formal differentiation). *For all  $f, g \in F[X]$ ,  $c \in F$ ,  $\alpha \in F$ , and  $m \geq 1$ :*

1. (Linearity.)  $(f + g)' = f' + g'$  and  $(cf)' = cf'$ .
2. (Product rule.)  $(fg)' = f'g + fg'$ .
3. (Power rule.)  $((X - \alpha)^m)' = m(X - \alpha)^{m-1}$ .

*Proof.* (1) Linearity is immediate from the coordinatewise definition of  $D$ , which is linear in the coefficients.

(2) Both sides are *bilinear* in the pair  $(f, g)$ : by part (1), each is linear in  $f$  with  $g$  fixed and linear in  $g$  with  $f$  fixed. It therefore suffices to verify the identity on monomials  $f = X^a, g = X^b$ , since every polynomial is a linear combination of monomials and the general case follows by expanding  $f = \sum_a u_a X^a, g = \sum_b v_b X^b$  and applying the monomial case termwise. On monomials,

$$(X^a X^b)' = (X^{a+b})' = (a+b) X^{a+b-1} = a X^{a-1} X^b + X^a b X^{b-1} = (X^a)' X^b + X^a (X^b)',$$

with the convention that a term carrying the integer coefficient 0 (the case  $a = 0$  or  $b = 0$ ) is the zero polynomial.

(3) Induct on  $m$ . The base is  $(X - \alpha)' = 1$ . For the step, the product rule gives

$$\begin{aligned} ((X - \alpha)^m)' &= ((X - \alpha)(X - \alpha)^{m-1})' \\ &= (X - \alpha)^{m-1} + (X - \alpha)(m-1)(X - \alpha)^{m-2} \\ &= m(X - \alpha)^{m-1}. \end{aligned} \quad \square$$

The proof of part (3) of the next theorem needs a repeated root in an extension field not given in advance. The following lemma constructs that extension by applying the quotient construction of Example 4.22 to polynomial rings; its proof transposes the argument that made  $\mathbb{Z}/p\mathbb{Z}$  a field (Corollary 4.26) from  $\mathbb{Z}$  to  $F[X]$ .

**Lemma 5.39** (Adjoining a root). *Let  $h \in F[X]$  be irreducible. Then the quotient ring  $K = F[X]/(h)$  is a field; the map  $a \mapsto a + (h)$  embeds  $F$  into  $K$  as a subfield, so that  $K$  is an extension of  $F$ ; and the class  $\alpha = X + (h)$  is a root of  $h$  in  $K$ .*

*Proof.* The principal ideal  $(h) = hF[X]$  is an ideal of the commutative ring  $F[X]$  (Example 4.18(2)), so  $K$  is a commutative ring under the coset operations (Proposition 4.20). Its identity is not zero:  $1 + (h) = 0 + (h)$  would say  $h \mid 1$ , impossible since a factorisation  $1 = hk$  has degree  $\deg h + \deg k = 0$  by (2), while  $\deg h \geq 1$ .

Every nonzero class is a unit. Let  $f + (h) \neq 0 + (h)$ , i.e.  $h \nmid f$ . The gcd  $\gcd(h, f)$  is then a monic divisor of the irreducible  $h$  other than its monic associate, hence equals 1, exactly as in the proof of Corollary 5.25(2); Bézout's identity (3) supplies  $s, t \in F[X]$  with  $sf + th = 1$ , and reducing modulo  $(h)$  gives  $(s + (h))(f + (h)) = 1 + (h)$ . So  $K$  is a field.

The map  $\iota(a) = a + (h)$  on constants preserves sums, products, and 1 directly from the coset operations of Definition 4.19, and it is injective:  $\iota(a) = 0$  with  $a \neq 0$  would make  $h$  divide a nonzero constant, again impossible by degrees. We identify  $F$  with its image, making  $K$  an extension field of  $F$ .

Finally, write  $h = \sum_i c_i X^i$  and compute in  $K$ :

$$h(\alpha) = \sum_i \iota(c_i) (X + (h))^i = \left( \sum_i c_i X^i \right) + (h) = h + (h) = 0 + (h),$$

so  $\alpha$  is a root of  $h$  in  $K$ . □

**Theorem 5.40** (Derivative test for repeated roots; separability). *Let  $f \in F[X]$  be nonzero and  $\alpha \in F$ .*

1. *The element  $\alpha$  is a repeated root of  $f$ —a root of multiplicity at least 2—if and only if  $f(\alpha) = 0$  and  $f'(\alpha) = 0$ .*
2. *If  $f$  has a repeated root in  $F$ , then  $\gcd(f, f') \neq 1$ .*
3. *Call  $f$  separable if it has no repeated root in any field containing  $F$ . Then  $f$  is separable if and only if  $\gcd(f, f') = 1$  in  $F[X]$ .*

*Proof.* (1) Let  $\alpha$  be a root of multiplicity  $m \geq 1$  and write  $f = (X - \alpha)^m g$  with  $g(\alpha) \neq 0$ . The product and power rules give

$$f' = m(X - \alpha)^{m-1}g + (X - \alpha)^m g' = (X - \alpha)^{m-1} [mg + (X - \alpha)g'].$$

If  $m = 1$ , the first factor is 1 and evaluating at  $\alpha$  gives  $f'(\alpha) = 1 \cdot g(\alpha) = g(\alpha) \neq 0$ : a simple root of  $f$  is not a root of  $f'$ . If  $m \geq 2$ , then  $(X - \alpha)^{m-1}$  divides  $f'$  and vanishes at  $\alpha$ , so  $f'(\alpha) = 0$ . Together with the factor theorem for  $f(\alpha) = 0$ , this proves the equivalence.

(2) A repeated root  $\alpha \in F$  is a common root of  $f$  and  $f'$  by (1), so the factor theorem puts the common factor  $X - \alpha$  into both, and  $(X - \alpha) \mid \gcd(f, f')$ .

(3) The argument rests on the fact that *gcds are stable under field extension*: for an extension field  $K$  of  $F$  (Definition 4.25), the Euclidean algorithm of Theorem 5.23 run on  $f, f' \in F[X]$  uses only field operations on their coefficients, which lie in  $F$ ; run inside  $K[X]$  on the same inputs it produces the identical quotients and remainders. Hence the gcd computed in  $K[X]$  equals the gcd computed in  $F[X]$ .

Suppose  $\gcd(f, f') = 1$  in  $F[X]$ , and let  $K \supseteq F$  be any extension. By stability,  $\gcd(f, f') = 1$  in  $K[X]$  as well, so by (2) applied over  $K$ , the polynomial  $f$  has no repeated root in  $K$ :  $f$  is separable.

Conversely, suppose  $d = \gcd(f, f')$  is nonconstant; we exhibit a repeated root in some extension. Let  $b$  be a monic irreducible factor of  $d$  (Theorem 5.28), and let  $K = F[X]/(b)$  be the extension field supplied by Lemma 5.39, in which  $b$  has a root  $\alpha$ . The polynomial  $b$  divides  $d$ , which divides both  $f$  and  $f'$ ; evaluating the factorisations at  $\alpha$  (Proposition 5.16) gives  $f(\alpha) = f'(\alpha) = 0$ . By (1), applied over  $K$ , the element  $\alpha$  is a repeated root of  $f$  in  $K$ :  $f$  is not separable.  $\square$

**Remark 5.41** (Formal derivatives in positive characteristic). The integer coefficient  $i$  in  $f' = \sum_i i a_i X^{i-1}$  is interpreted in  $F$ , and may vanish there. Over a field of characteristic  $p$ —where  $p = 0$  in  $F$ ,  $p$  prime, the setting of  $\mathbb{F}_p$  and  $\mathbb{F}_q$  (Definition 4.30)—the derivative of  $X^p$  is  $p X^{p-1} = 0$ . Thus  $f = X^p - 1$  over  $\mathbb{F}_p$  has  $f' = 0$  and  $\gcd(f, f') = \gcd(f, 0) = f \neq 1$ ; correspondingly,

$$X^p - 1 = (X - 1)^p \quad \text{in } \mathbb{F}_p[X],$$

a factorisation the finite-fields section establishes through the freshman's-dream identity  $(a + b)^p = a^p + b^p$  of characteristic  $p$  (Lemma 6.13, applied to  $(X + (-1))^p$  together with  $(-1)^p = -1$  in every  $\mathbb{F}_p$ ): a single root 1 of multiplicity  $p$ . Arguments over the finite fields of Orchard must account for this characteristic- $p$  behaviour.

## 5.9 Polynomials versus polynomial functions

This subsection determines when a polynomial may be identified with the function it induces.

**Definition 5.42** (Polynomial function). Every  $f \in F[X]$  induces a *polynomial function*

$$\tilde{f}: F \rightarrow F, \quad \alpha \mapsto f(\alpha).$$

The assignment  $f \mapsto \tilde{f}$  is a ring homomorphism from  $F[X]$  to the ring  $F^F$  of *all* functions  $F \rightarrow F$  under pointwise operations ( $(u + v)(\alpha) = u(\alpha) + v(\alpha)$  and  $(uv)(\alpha) = u(\alpha)v(\alpha)$ ), by Proposition 5.16 applied at every point.

**Proposition 5.43** (When formal equals functional). *Let  $f, g \in F[X]$ .*

1. *If  $F$  is infinite, then  $\tilde{f} = \tilde{g}$  as functions implies  $f = g$  as polynomials. Equivalently,  $f \mapsto \tilde{f}$  is injective, and polynomials over an infinite field may be identified with polynomial functions.*
2. *If  $F$  is finite, the map is not injective. Over  $\mathbb{F}_p$ , the nonzero polynomial  $X^p - X$  induces the zero function, since Fermat's little theorem (Corollary 3.32) gives  $\alpha^p = \alpha$  for every  $\alpha \in \mathbb{F}_p$ .*

*Proof.* (1) Let  $h = f - g$ , so that  $\tilde{h}$  is the zero function:  $h$  vanishes at every point of the infinite set  $F$ . Were  $h$  nonzero of degree  $d$ , it would have at most  $d$  roots (Theorem 5.18), yet it has infinitely many—a contradiction. Hence  $h = 0$ .

(2) Let  $F$  be finite, with  $q = |F|$  elements, and consider the *vanishing-polynomial witness*

$$w(X) = \prod_{\alpha \in F} (X - \alpha),$$

a monic polynomial of degree  $q$ —in particular nonzero as a formal object—which vanishes at every point of  $F$  by construction. Thus  $\tilde{w} = \tilde{0}$  while  $w \neq 0$ : the map  $f \mapsto \tilde{f}$  identifies distinct polynomials. Over  $F = \mathbb{F}_p$  the witness specialises to  $X^p - X$ : by Fermat's little theorem every element of  $\mathbb{F}_p$  is a root of the monic degree- $p$  polynomial  $X^p - X$ , so Proposition 5.36 factors it as  $\prod_{\alpha \in \mathbb{F}_p} (X - \alpha)$ —the  $p$  distinct linear factors exhaust the degree—and  $X^p - X$  induces the zero function despite having degree  $p \geq 1$ .  $\square$

**Remark 5.44** (Why polynomials are defined formally). Part (2) of Proposition 5.43 is the reason polynomials are defined as coefficient sequences and not as functions. Cryptography works over finite fields, where the functional view loses information: the polynomials  $X$  and  $X^p$  induce the same function on  $\mathbb{F}_p$ , yet have degrees 1 and  $p$ . The security of polynomial commitment schemes rests on the *degree*, a property of the formal object invisible to the induced function over a finite field. A prover commits to a low-degree *formal* polynomial, and the verifier tests it by *evaluation*. The Schwartz–Zippel lemma quantifies the gap between agreement at the tested points and identity to the claimed formal polynomial.

### 5.10 The Schwartz–Zippel lemma

To draw  $\alpha$  *uniformly at random* from a finite nonempty set  $S$  means to select each element with equal likelihood  $1/|S|$ ; the *probability* of an event is then the fraction of elements of  $S$  producing it, so that  $\Pr[f(\alpha) = 0] = |\{\alpha \in S : f(\alpha) = 0\}|/|S|$ . The theorem needs only this counting case; probability is developed later in the volume.

**Theorem 5.45** (Univariate Schwartz–Zippel). *Let  $f \in F[X]$  be nonzero of degree at most  $d$ , let  $S \subseteq F$  be finite and nonempty, and let  $\alpha$  be drawn uniformly at random from  $S$ . Then*

$$\Pr[f(\alpha) = 0] \leq \frac{d}{|S|}. \quad (5)$$

*More generally, if  $f, g \in F[X]$  are distinct, each of degree at most  $d$ , then  $\Pr[f(\alpha) = g(\alpha)] \leq d/|S|$ .*

*Proof.* By Theorem 5.18, the nonzero  $f$  has at most  $d$  roots in  $F$ , hence at most  $d$  in the subset  $S$ . Under the uniform draw,

$$\Pr[f(\alpha) = 0] = \frac{|\{\alpha \in S : f(\alpha) = 0\}|}{|S|} \leq \frac{d}{|S|}.$$

For the general form, apply this to  $h = f - g$ , which is nonzero and has degree at most  $d$  by (1), noting that  $f(\alpha) = g(\alpha)$  exactly when  $h(\alpha) = 0$ .  $\square$

**Remark 5.46** (Polynomial identity testing). Inequality (5) is the probabilistic refinement of Corollary 5.20: a *single* random evaluation point detects unequal polynomials of degree at most  $d$ , except with probability at most  $d/|S|$ . With  $|S|$  exponentially large ( $S = \mathbb{F}_q$  with  $q \approx 2^{254}$ , the 255-bit fields underlying Orchard) and  $d$  polynomially bounded, the failure probability is exponentially small, hence *negligible*, a term given its precise asymptotic meaning later in the volume. The bound is the soundness argument for polynomial identity testing, and hence for the polynomial commitment schemes and arithmetisation checks of Halo 2, which verify a purported polynomial identity by evaluating both sides at a random challenge and comparing the results.

**Remark 5.47** (Multivariate generalisation). The full Schwartz–Zippel lemma extends to polynomials in several indeterminates  $f \in F[X_1, \dots, X_n]$ : a nonzero polynomial of total degree  $d$ —the largest sum of exponents  $i_1 + \dots + i_n$  over the monomials with nonzero coefficient—vanishes at a uniformly random point of  $S^n$  with probability at most  $d/|S|$ . The univariate case proved here is the base case of the multivariate inductive proof and the only version the later volumes invoke directly. Where an identity in several random challenges arises, for instance in a permutation argument drawing a pair of challenges  $(\beta, \gamma)$ , the multivariate bound applies and follows from the univariate lemma applied once per variable.

Over a 255-bit field  $\mathbb{F}_q$ , distinct polynomials  $f$  and  $g$  of degree at most  $d$  agree on a uniformly random challenge  $\alpha \in \mathbb{F}_q$  with probability at most  $d/q \approx d \cdot 2^{-254}$ , astronomically small even with  $d$  in the millions.

## 6 Finite fields

This section studies the finite fields, the fields of §4 with finitely many elements. A verifier must divide by every nonzero value, which the arithmetic of a 64-bit processor word, modulo  $2^{64}$ , does not permit: in  $\mathbb{Z}/2^{64}\mathbb{Z}$  no even class has an inverse (Theorem 2.24). The section gives two inversion algorithms for  $\mathbb{F}_p$  and proves that every finite field has  $p^k$  elements for a prime  $p$  (Theorem 6.9), so no field has 6, 10, or 12 elements. Fields of size  $2^k$  exist for every positive  $k$ , but the field with  $2^{64}$  elements is not  $\mathbb{Z}/2^{64}\mathbb{Z}$ : it is the polynomial quotient of Theorem 6.10, with different addition and multiplication. The section then defines the Frobenius map  $x \mapsto x^p$  and proves that the multiplicative group  $\mathbb{F}_q^\times$  is *cyclic*, the powers of a single element running through every nonzero value (Theorem 6.17). Cyclicity gives discrete logarithms their meaning and guarantees the roots of unity on which the polynomial constructions of later sections depend.

### 6.1 The prime field and two inversion routes

Most constructions of the series are carried out over the field  $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$  of Corollary 4.26, for a prime  $p$  of roughly 255 bits. Its addition, subtraction, and multiplication are the residue arithmetic of §2; inversion requires an algorithm, since a proof that an inverse exists (§4) does not compute it.

**Construction 6.1** (Inversion by the extended Euclidean algorithm). Given  $a$  with  $0 < a < p$ , run the extended Euclidean algorithm (Theorem 2.13) on the pair  $(a, p)$ . Since  $p$  is prime and  $p \nmid a$ , the gcd is 1, and the algorithm outputs integers  $u, v$  with  $au + pv = 1$ . Reducing modulo  $p$  gives  $au \equiv 1 \pmod{p}$ , that is,

$$a^{-1} = [u]_p \quad \text{in } \mathbb{F}_p.$$

The cost is  $O(\log p)$  division steps (Theorem 2.13). This is the standard general-purpose method; a worked instance is Example 2.25, which computed  $[17]_{43}^{-1} = [38]_{43}$  by exactly this route.

The second method rests on Fermat's little theorem and on the following fast exponentiation.

**Construction 6.2** (Square-and-multiply). Let  $a$  be an element of any structure with an associative multiplication, and let  $e \geq 1$  have binary expansion  $e = \sum_i e_i 2^i$  with digits  $e_i \in \{0, 1\}$ . Squaring repeatedly produces the values

$$a^{2^0} = a, \quad a^{2^{i+1}} = (a^{2^i})^2,$$

and multiplying together the factors indexed by the nonzero digits gives

$$a^e = \prod_{i: e_i=1} a^{2^i}.$$

The computation uses  $\lceil \log_2 e \rceil$  squarings and at most  $\lceil \log_2 e \rceil$  further multiplications— $O(\log e)$  multiplications in all, against the  $e - 1$  multiplications of the naive product  $a \cdot a \cdots a$ . For the 255-bit exponents of later sections the former is a few hundred multiplications and the latter a count beyond the reach of any computer.

*Remark 6.3.* Written additively, the same algorithm computes the multiple  $e \cdot P$  of a group element  $P$  by repeated doubling and selective adding. Under the name *double-and-add* it is the algorithm for scalar multiplication on elliptic curves (Definition 10.20).

**Construction 6.4** (Inversion by Fermat's little theorem). Let  $a \in \mathbb{F}_p$  be nonzero. Fermat's little theorem (Corollary 3.32) gives  $a^{p-1} = 1$ , hence  $a \cdot a^{p-2} = 1$ , and so

$$a^{-1} = a^{p-2},$$

computed by square-and-multiply (Construction 6.2) with exponent  $p - 2$ , at a cost of  $O(\log p)$  modular multiplications when  $p > 2$ . For  $p = 2$ , the only nonzero element is 1, whose inverse is 1; the exponent is zero and no multiplication is needed.

*Remark 6.5* (Constant-time inversion). The two methods have comparable asymptotic cost. In Construction 6.4 the sequence of operations performed—which steps square, which multiply—depends only on the binary digits of the *public* exponent  $p - 2$ , never on the secret input  $a$ ; the method is *constant-time-friendly*. The extended Euclidean method, by contrast, performs a sequence of divisions whose lengths and quotients depend on  $a$  itself, so its running time and memory-access pattern can leak information about  $a$  to an observer timing the computation. Side-channel-resistant implementations therefore often prefer the Fermat method despite its slightly larger constant factor.

## 6.2 Characteristic, prime subfield, and prime-power order

This subsection determines the possible cardinalities of a finite field through the characteristic of Section 4.5, defined by the canonical homomorphism

$$\iota: \mathbb{Z} \rightarrow F, \quad \iota(n) = n \cdot 1_F,$$

whose kernel is an ideal  $n\mathbb{Z}$  of  $\mathbb{Z}$ ; the characteristic  $\text{char}(F)$  is its nonnegative generator (Definition 4.30).

**Theorem 6.6.** *Every finite field  $F$  has prime characteristic:  $\text{char}(F) = p$  for some prime  $p$ . In particular  $\text{char}(\mathbb{F}_p) = p$ .*

*Proof.* The map  $\iota$  cannot be injective, since its domain  $\mathbb{Z}$  is infinite and its codomain  $F$  is finite; two distinct integers  $m < n$  must satisfy  $\iota(m) = \iota(n)$ , and then  $n - m$  is a nonzero element of  $\ker \iota$ . Hence  $\ker \iota = n\mathbb{Z}$  with  $n > 0$ , that is,  $\text{char}(F) > 0$ ; and the characteristic of a field, being nonzero, is prime by Theorem 4.32. For  $F = \mathbb{F}_p$  itself, Example 4.31 computed  $\text{char}(\mathbb{Z}/p\mathbb{Z}) = p$  directly.  $\square$

Every field of characteristic  $p$  contains a canonical copy of  $\mathbb{F}_p$  (Remark 4.33), defined as follows.

**Definition 6.7** (Subfield; prime subfield). A *subfield* of a field  $F$  is a subset  $K \subseteq F$  containing 0 and 1 and closed under addition, negation, multiplication, and inversion of nonzero elements; with the restricted operations,  $K$  is itself a field. The *prime subfield* of  $F$  is the intersection of

all subfields of  $F$ —equivalently, the smallest subfield of  $F$ , and equivalently again the subfield generated by  $1_F$ .

The intersection of any family of subfields is again a subfield, since each defining closure property is preserved under intersection; the intersection of *all* subfields is contained in every subfield, hence is the smallest one. That this smallest subfield is exactly what  $1_F$  generates—the closure of  $\{1_F\}$  under the field operations—follows from the classification below, whose proof exhibits the prime subfield inside that closure.

**Proposition 6.8** (Classification of prime subfields). *Let  $F$  be a field. If  $\text{char}(F) = p > 0$ , the prime subfield of  $F$  is isomorphic to  $\mathbb{F}_p$ . If  $\text{char}(F) = 0$ , it is isomorphic to  $\mathbb{Q}$ .*

*Proof.* Suppose first that  $\text{char}(F) = p > 0$ , so that  $\ker \iota = p\mathbb{Z}$ . Define

$$\bar{\iota}: \mathbb{F}_p = \mathbb{Z}/p\mathbb{Z} \rightarrow F, \quad \bar{\iota}([a]_p) = a \cdot 1_F.$$

A single equivalence shows the map well defined and injective:  $a \cdot 1_F = b \cdot 1_F$  holds if and only if  $(a - b) \cdot 1_F = 0$ , i.e.  $a - b \in \ker \iota = p\mathbb{Z}$ , i.e.  $[a]_p = [b]_p$ . It is a ring homomorphism because  $\iota$  is one (Section 4.5) and  $\bar{\iota}$  evaluates  $\iota$  on representatives of residue classes. Its image  $K = \{a \cdot 1_F : a \in \mathbb{Z}\}$  is therefore a subring of  $F$  isomorphic to the field  $\mathbb{F}_p$ , and  $K$  is a subfield: for  $[a]_p \neq [0]_p$  the computation  $\bar{\iota}([a]_p) \bar{\iota}([a]_p^{-1}) = \bar{\iota}([1]_p) = 1_F$  exhibits the inverse of each nonzero element of  $K$  inside  $K$ . Finally,  $K$  lies in every subfield: a subfield contains  $1_F$ , hence all its additive multiples  $a \cdot 1_F$  by closure under addition and negation. So  $K$  is contained in the intersection of all subfields and is itself a subfield, so  $K$  equals that intersection: the prime subfield is  $K \cong \mathbb{F}_p$ .

Now suppose  $\text{char}(F) = 0$ , so that  $\iota$  is injective and  $b \cdot 1_F \neq 0$  for every nonzero integer  $b$ . Define

$$j: \mathbb{Q} \rightarrow F, \quad j\left(\frac{a}{b}\right) = (a \cdot 1_F)(b \cdot 1_F)^{-1}.$$

The map is well defined: if  $a/b = c/d$  with  $b, d \neq 0$ , then  $ad = bc$  in  $\mathbb{Z}$ , so  $(a \cdot 1_F)(d \cdot 1_F) = (b \cdot 1_F)(c \cdot 1_F)$  by applying  $\iota$ , and multiplying both sides by  $(b \cdot 1_F)^{-1}(d \cdot 1_F)^{-1}$  gives  $(a \cdot 1_F)(b \cdot 1_F)^{-1} = (c \cdot 1_F)(d \cdot 1_F)^{-1}$ . That  $j$  preserves sums, products, and 1 is the usual arithmetic of fractions, transported by the homomorphism  $\iota$  and commutativity. The map is injective:  $j(a/b) = 0$  forces  $a \cdot 1_F = 0$ , hence  $a = 0$ . The image of  $j$  is thus a subfield of  $F$  isomorphic to  $\mathbb{Q}$ ; and it lies in every subfield  $K$ , since  $K$  contains  $1_F$ , hence every  $a \cdot 1_F$ , hence—being closed under inversion and multiplication—every  $(a \cdot 1_F)(b \cdot 1_F)^{-1}$ . As before the image must equal the intersection of all subfields, and the prime subfield is a copy of  $\mathbb{Q}$ .  $\square$

The size constraint follows by counting over the prime subfield. Every field  $F$  of characteristic  $p$  is a vector space over its prime subfield  $\mathbb{F}_p$ : vectors are the elements of  $F$ , vector addition is the field addition, and scalar multiplication  $c \cdot x$  for  $c \in \mathbb{F}_p \subseteq F$  is the field multiplication, so the vector-space axioms are instances of the ring axioms of  $F$ . The proof uses two facts, proved in Section 8 independently of the present theorem: a space with a finite spanning set has a basis, and coordinates with respect to a basis are unique.

**Theorem 6.9** (Finite fields have prime-power order). *Every finite field  $F$  has cardinality  $|F| = p^k$  for some prime  $p$  and integer  $k \geq 1$ , where  $p = \text{char}(F)$ .*

*Proof.* By Theorem 6.6 the characteristic of  $F$  is a prime  $p$ , and by Proposition 6.8 the prime subfield of  $F$  is a copy of  $\mathbb{F}_p$ , with which we identify it. As explained above,  $F$  is then a vector space over  $\mathbb{F}_p$ . It is spanned by the finite set  $F$  itself, so it has a finite basis  $e_1, \dots, e_k$ ; and  $k \geq 1$  because  $1 \neq 0$  makes  $F$  nonzero. Every element of  $F$  is thus

$$c_1 e_1 + c_2 e_2 + \dots + c_k e_k$$

for a *unique* tuple  $(c_1, \dots, c_k)$  with each  $c_i \in \mathbb{F}_p$ : existence because the  $e_i$  span, uniqueness because coordinates with respect to a basis are unique (Section 8). The assignment  $(c_1, \dots, c_k) \mapsto \sum_i c_i e_i$  is therefore a bijection from  $\mathbb{F}_p^k$  to  $F$ , and counting tuples gives  $|F| = p \cdot p \cdots p = p^k$ .  $\square$

**Theorem 6.10** (Existence and uniqueness of  $\mathbb{F}_q$ ). *For every prime power  $q = p^k$  there exists a field with exactly  $q$  elements, and any two finite fields of the same cardinality are isomorphic. One constructs such a field as the quotient  $\mathbb{F}_p[x]/(f)$  of the polynomial ring by the ideal generated by an irreducible polynomial  $f$  of degree  $k$ —the construction recorded in Example 4.22.*

*Remark 6.11.* The fields on the critical path of Halo 2 and Orchard are *prime* fields, the case  $q = p$  with  $k = 1$ , where the theorem is proved: existence is Corollary 4.26, and uniqueness holds because a field with  $p$  elements has order  $p^k = p$  with  $k = 1$  by Theorem 6.9, so it equals its own prime subfield, a copy of  $\mathbb{F}_p$  (Proposition 6.8). The general case  $k \geq 2$  would require polynomial factorisation over field extensions; no later result depends on it, so it is stated without proof. *Convention, fixed henceforth:* the letter  $q = p^k$  denotes a prime power, and  $\mathbb{F}_q$  a field with  $q$  elements, of characteristic  $p$ ; results are stated for general  $\mathbb{F}_q$  whenever the proof is identical to the prime case, and for the applications to Halo 2 one may take  $q = p$  throughout.

### 6.3 The Frobenius endomorphism

The identity  $(a + b)^p = a^p + b^p$ , a common error over  $\mathbb{Q}$  and  $\mathbb{R}$ , where it is false for every  $p \geq 2$ , is a theorem in characteristic  $p$ . Its proof expands  $(a + b)^p$  by the binomial theorem.

**Lemma 6.12** (Binomial theorem; the coefficients count subsets). *Let  $R$  be a commutative ring. For all  $a, b \in R$  and every integer  $n \geq 0$ ,*

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k},$$

*where the integer coefficient  $\binom{n}{k}$  acts as the additive multiple  $\binom{n}{k} \cdot x = x + \dots + x$ , equals the number of  $k$ -element subsets of an  $n$ -element set, and satisfies*

$$\binom{n}{k} = \frac{n!}{k! (n-k)!}.$$

*Proof.* Expand the product  $(a + b)^n = (a + b) \cdots (a + b)$  by distributivity without collecting terms: the result is a sum of  $2^n$  products, one for each way of choosing, from every factor, either its  $a$  or its  $b$ . By commutativity such a product equals  $a^k b^{n-k}$  exactly when  $a$  was chosen from  $k$  of the  $n$  factors, and it is determined by the set of those  $k$  factors; so the coefficient of  $a^k b^{n-k}$  is the number of  $k$ -element subsets of the  $n$  factors, and collecting the terms gives the displayed sum.

For the formula, count the sequences of  $k$  distinct elements of an  $n$ -element set in two ways. Choosing the entries in turn, there are  $n$  choices for the first,  $n - 1$  for the second, and  $n - k + 1$  for the  $k$ -th:  $n(n - 1) \cdots (n - k + 1) = n! / (n - k)!$  sequences. Alternatively, each such sequence is a  $k$ -element subset together with an ordering of it, and a  $k$ -element set has  $k!$  orderings — the same count with  $k$  in place of  $n$  — so there are  $k!$  sequences per subset, and the number of subsets is  $n! / (k! (n - k)!)$ .  $\square$

**Lemma 6.13** (Freshman's dream). *Let  $R$  be a commutative ring of prime characteristic  $p$ . Then for all  $a, b \in R$ ,*

$$(a + b)^p = a^p + b^p,$$

*and more generally  $(a + b)^{p^n} = a^{p^n} + b^{p^n}$  for every  $n \geq 1$ .*

*Proof.* By the binomial theorem (Lemma 6.12),

$$(a + b)^p = \sum_{i=0}^p \binom{p}{i} a^i b^{p-i},$$

where the integer coefficient  $\binom{p}{i}$  acts as the additive multiple  $\binom{p}{i} \cdot x = x + \cdots + x$ . We claim that  $p \mid \binom{p}{i}$  for  $0 < i < p$ . From the factorial identity of the same lemma

$$\binom{p}{i} i! (p - i)! = p!$$

the prime  $p$  divides the right-hand side. It does not divide  $i! (p - i)!$ : that product is a product of integers all smaller than  $p$ , and if  $p$  divided the product it would divide one of the factors by Euclid's lemma for primes (Lemma 2.17), which is impossible. Applying Euclid's lemma to the left-hand side instead,  $p$  must divide  $\binom{p}{i}$ , say  $\binom{p}{i} = pm$ .

Now let  $0 < i < p$  and consider the corresponding term. Writing  $\iota(n) = n \cdot 1_R$  for the canonical homomorphism of Section 4.5, the coefficient acts as multiplication by  $\iota(\binom{p}{i}) = \iota(pm) = \iota(p) \iota(m) = 0 \cdot \iota(m) = 0$ , since  $\text{char}(R) = p$  means exactly  $\iota(p) = p \cdot 1_R = 0$ . Every middle term of the binomial expansion therefore vanishes, and the terms  $i = p$  and  $i = 0$  leave  $(a + b)^p = a^p + b^p$ .

The general case follows by induction on  $n$ , the base case  $n = 1$  being the identity just proved. For  $n \geq 2$ , apply the base case to the elements  $a^{p^{n-1}}$  and  $b^{p^{n-1}}$ :

$$(a + b)^{p^n} = ((a + b)^{p^{n-1}})^p = (a^{p^{n-1}} + b^{p^{n-1}})^p = a^{p^n} + b^{p^n},$$

using the inductive hypothesis in the middle step.  $\square$

A homomorphism from a structure to itself is an *endomorphism*. In characteristic  $p$  the  $p$ -th power map is one: it preserves addition by the lemma, and multiplication by commutativity.

**Definition 6.14** (Frobenius endomorphism). Let  $F$  be a field of characteristic  $p > 0$ . The *Frobenius map* of  $F$  is

$$\text{Frob}: F \rightarrow F, \quad \text{Frob}(x) = x^p.$$

**Proposition 6.15.** *Let  $F$  be a field of characteristic  $p$ . The Frobenius map is a ring homomorphism  $F \rightarrow F$  fixing the prime subfield pointwise. If  $F$  is finite, Frob is an automorphism of  $F$ —a bijective ring homomorphism of  $F$  onto itself.*

*Proof.* Multiplicativity follows from commutativity and associativity alone:  $(xy)^p = x^p y^p$ . Additivity is the freshman's dream (Lemma 6.13), and  $1^p = 1$ ; so Frob is a ring homomorphism.

Injectivity is an instance of a general fact used again later: *every ring homomorphism between fields is injective*. If  $\phi: F \rightarrow F'$  had  $\phi(x) = 0$  for some  $x \neq 0$ , then

$$1 = \phi(1) = \phi(x^{-1}x) = \phi(x^{-1})\phi(x) = \phi(x^{-1}) \cdot 0 = 0$$

in  $F'$ , contradicting  $1 \neq 0$  in a field. In particular Frob is injective.

The prime subfield is fixed pointwise. By Proposition 6.8 its elements are the values  $\bar{i}([a]_p) = a \cdot 1_F$ , and since  $\bar{i}$  is a ring homomorphism,

$$\text{Frob}(\bar{i}([a]_p)) = \bar{i}([a]_p)^p = \bar{i}([a]_p^p) = \bar{i}([a^p]_p) = \bar{i}([a]_p),$$

the last step because  $a^p \equiv a \pmod{p}$  for all integers  $a$  (Corollary 3.32).

Finally, let  $F$  be finite. An injective map from a finite set to itself is surjective—the injectivity-implies-surjectivity argument of Section 4.4—so Frob is a bijective ring homomorphism of  $F$  onto itself, an automorphism.  $\square$

The iterates  $\text{Frob}^n(x) = x^{p^n}$  are again homomorphisms, by the general case of the freshman's dream. On  $\mathbb{F}_p$  the Frobenius map is the identity; on larger fields of characteristic  $p$  it is not, and Remark 6.24 records its role in cryptography.

## 6.4 Cyclicity of the multiplicative group

The multiplicative group  $\mathbb{F}_q^\times$  is a finite abelian group of order  $q - 1$ , since every nonzero element of a field is invertible. Finite abelian groups need not be cyclic: the unit group  $(\mathbb{Z}/8\mathbb{Z})^\times$  of Example 3.22 has order 4 and no element of order 4. The proof that  $\mathbb{F}_q^\times$  is cyclic is a counting argument of Gauss, resting on an identity for the totient function  $\varphi$  and using only its *counting definition* (Definition 2.26):  $\varphi(n)$  is the number of integers  $a$  with  $1 \leq a \leq n$  and  $\gcd(a, n) = 1$ .

**Lemma 6.16** (Gauss's divisor-sum identity). *For every integer  $n \geq 1$ ,*

$$\sum_{d|n} \varphi(d) = n,$$

*the sum running over the positive divisors  $d$  of  $n$ .*

*Proof.* Partition the set  $\{1, 2, \dots, n\}$  according to the value of  $\gcd(a, n)$ . For each  $a$  the  $\gcd$  is a positive divisor  $d$  of  $n$ , so the classes

$$C_d = \{a : 1 \leq a \leq n, \gcd(a, n) = d\}, \quad d \mid n,$$

are disjoint and cover  $\{1, \dots, n\}$ ; hence  $n = \sum_{d \mid n} |C_d|$ .

We claim  $|C_d| = \varphi(n/d)$ . A member of  $C_d$  is divisible by  $d$ , so it has the form  $a = db$  with  $1 \leq b \leq n/d$ ; we show that, for such  $a$ ,

$$\gcd(db, n) = d \iff \gcd(b, n/d) = 1.$$

For the forward direction, suppose  $e = \gcd(b, n/d) > 1$ ; then  $de$  divides both  $db = a$  and  $d \cdot (n/d) = n$ , so  $de$  is a common divisor exceeding  $d$  and  $\gcd(a, n) \geq de > d$ . For the converse, suppose  $\gcd(b, n/d) = 1$ . The integer  $d$  is a common divisor of  $a$  and  $n$ , so  $d$  divides  $g = \gcd(a, n)$  by the strongest-common-divisor property (Remark 2.12); write  $g = dc$  with  $c \geq 1$ . From  $dc \mid db$  we get  $c \mid b$ , and from  $dc \mid n = d(n/d)$  we get  $c \mid n/d$  (cancel  $d$  in each divisibility); so  $c$  divides  $\gcd(b, n/d) = 1$ , forcing  $c = 1$  and  $g = d$ .

The members of  $C_d$  are therefore exactly the integers  $db$  with  $1 \leq b \leq n/d$  and  $\gcd(b, n/d) = 1$ , and by the counting definition of the totient (Definition 2.26) there are precisely  $\varphi(n/d)$  of them. Summing,

$$n = \sum_{d \mid n} \varphi\left(\frac{n}{d}\right) = \sum_{d \mid n} \varphi(d),$$

the second equality because  $d \mapsto n/d$  is a bijection of the set of positive divisors of  $n$  onto itself.  $\square$

**Theorem 6.17** (Finite subgroups of  $F^\times$  are cyclic). *Let  $F$  be any field and let  $G \leq F^\times$  be a finite subgroup of the multiplicative group, of order  $N$ . Then  $G$  is cyclic. In particular, for a finite field  $\mathbb{F}_q$  the whole multiplicative group  $\mathbb{F}_q^\times$  is cyclic of order  $q - 1$ .*

*Proof.* For each divisor  $d \mid N$ , let

$$\psi(d) = |\{a \in G : \text{ord}(a) = d\}|$$

count the elements of  $G$  of order exactly  $d$ . Every element of the finite group  $G$  has finite order dividing  $N = |G|$  (Corollary 3.30), so the sets counted by the  $\psi(d)$  partition  $G$  and

$$\sum_{d \mid N} \psi(d) = N. \tag{6}$$

We claim that for each  $d \mid N$ , either  $\psi(d) = 0$  or  $\psi(d) = \varphi(d)$ . Suppose  $\psi(d) > 0$  and choose  $a \in G$  of order  $d$ . The powers  $a^0, a^1, \dots, a^{d-1}$  are  $d$  distinct elements of  $G$  (Proposition 3.12(3)), and each satisfies

$$(a^j)^d = (a^d)^j = 1^j = 1,$$

so each is a root in  $F$  of the polynomial  $X^d - 1 \in F[X]$ . That polynomial has degree  $d$ , so it has at most  $d$  roots in  $F$  (Theorem 5.18); the  $d$  distinct powers of  $a$  therefore account for all of its roots.

Now take any  $b \in G$  of order  $d$ . Then  $b^d = 1$ , so  $b$  is a root of  $X^d - 1$  and hence  $b = a^j$  for some  $0 \leq j < d$ . By Corollary 3.23,

$$\text{ord}(a^j) = \frac{d}{\gcd(d, j)},$$

which equals  $d$  exactly when  $\gcd(j, d) = 1$ . The elements of order  $d$  in  $G$  are thus precisely the powers  $a^j$  with  $0 \leq j < d$  and  $\gcd(j, d) = 1$ , and by the counting definition of the totient there are  $\varphi(d)$  of them:  $\psi(d) = \varphi(d)$ . This proves the claim, and with it the inequality  $\psi(d) \leq \varphi(d)$  for every  $d \mid N$ .

Summing the inequality over the divisors of  $N$  and comparing (6) with Gauss's identity (Lemma 6.16),

$$N = \sum_{d \mid N} \psi(d) \leq \sum_{d \mid N} \varphi(d) = N.$$

The two ends are equal, so the inequality is an equality term by term:  $\psi(d) = \varphi(d)$  for every divisor  $d \mid N$ . In particular

$$\psi(N) = \varphi(N) \geq 1,$$

since  $a = 1$  always witnesses  $\gcd(1, N) = 1$  in the totient count. Hence  $G$  contains an element  $g$  of order  $N$ ; its powers form a subgroup of  $G$  with  $N$  distinct elements (Proposition 3.12(3)), which must be all of  $G$ . So  $G = \langle g \rangle$  is cyclic.

For the final statement, let  $F = \mathbb{F}_q$  be finite. The multiplicative group  $\mathbb{F}_q^\times$  has  $q - 1$  elements—every nonzero element of a field is a unit—and is a finite subgroup of itself, so it is cyclic of order  $q - 1$ .  $\square$

**Corollary 6.18** (Element-order census in cyclic groups). *A cyclic group  $G$  of order  $N$  contains, for each divisor  $d \mid N$ , exactly  $\varphi(d)$  elements of order  $d$ , and no elements of any other order. In particular  $G$  has exactly  $\varphi(N)$  generators.*

*Proof.* Write  $G = \langle g \rangle$  with  $\text{ord}(g) = N$ ; the elements of  $G$  are the powers  $g^j$ ,  $0 \leq j < N$ , each appearing once (Proposition 3.12(3)). By Corollary 3.23,  $\text{ord}(g^j) = N / \gcd(N, j)$ , which is always a divisor of  $N$ ; and  $\text{ord}(g^j) = d$  if and only if  $\gcd(j, N) = N/d$ . By the gcd computation in the proof of Lemma 6.16 (applied with  $n = N$  and divisor  $N/d$ ), the exponents  $j \in \{1, \dots, N\}$  with  $\gcd(j, N) = N/d$  are exactly  $j = (N/d)t$  with  $1 \leq t \leq d$  and  $\gcd(t, d) = 1$ , of which there are  $\varphi(d)$  by Definition 2.26. The generators are the elements of order  $N$ , numbering  $\varphi(N)$ .  $\square$

**Definition 6.19** (Primitive root). A generator of the cyclic group  $\mathbb{F}_q^\times$  is called a *primitive root* (or *primitive element*) of  $\mathbb{F}_q$ . Equivalently, an element  $g \in \mathbb{F}_q^\times$  is primitive if and only if  $\text{ord}(g) = q - 1$ .

Primitive roots exist by Theorem 6.17; by Corollary 6.18 there are exactly  $\varphi(q - 1)$  of them.

*Remark 6.20.* The group-theoretic term “generator of  $\mathbb{F}_q^\times$ ” and the polynomial term “primitive  $(q - 1)$ -th root of unity” (Definition 9.3) denote the same elements; the identification is restated once roots of unity have been defined in general.

*Example 6.21* (Primitive roots of  $\mathbb{F}_7$ ). The group  $\mathbb{F}_7^\times$  has order 6. Testing  $g = 3$ , the successive powers modulo 7 are

$$3^1 = 3, \quad 3^2 = 2, \quad 3^3 = 6, \quad 3^4 = 4, \quad 3^5 = 5, \quad 3^6 = 1,$$

which run through all of  $\{1, \dots, 6\}$ : the element 3 is a primitive root of  $\mathbb{F}_7$ . By contrast 2 is not: its powers are 2, 4, 1, so  $\text{ord}(2) = 3 < 6$ . The complete set of primitive roots consists of the powers  $3^j$  with  $\gcd(j, 6) = 1$  (Corollary 3.23), namely  $j = 1$  and  $j = 5$ : the primitive roots of  $\mathbb{F}_7$  are  $3^1 = 3$  and  $3^5 = 5$ , and there are  $\phi(6) = 2$  of them, as the census predicts.

Cyclicity also determines the number of solutions of the equation  $x^n = 1$  in a finite field, on which the evaluation domains of later sections depend.

**Corollary 6.22** (Roots of unity in  $\mathbb{F}_q$ ). *Let  $\mathbb{F}_q$  be a finite field and  $n \geq 1$ . The solutions of  $x^n = 1$  in  $\mathbb{F}_q^\times$  number exactly  $\gcd(n, q - 1)$ , and they form the cyclic subgroup of  $\mathbb{F}_q^\times$  of that order. In particular, when  $n \mid q - 1$  the equation has exactly  $n$  solutions.*

*Proof.* Fix a primitive root  $g$  (Theorem 6.17) and write each element of  $\mathbb{F}_q^\times$  uniquely as  $x = g^j$  with  $0 \leq j \leq q - 2$  (Proposition 3.12(3)). Then  $x^n = g^{jn}$ , and  $g^{jn} = 1$  holds if and only if  $(q - 1) \mid jn$  (Proposition 3.12(1)). Set  $d = \gcd(n, q - 1)$  and write  $q - 1 = dm$ ,  $n = dn'$  with  $\gcd(m, n') = 1$ . Then

$$(q - 1) \mid jn \iff dm \mid jdn' \iff m \mid jn' \iff m \mid j,$$

the last step by Euclid's lemma (Proposition 2.15(1), with  $\gcd(m, n') = 1$ ). The qualifying exponents in  $\{0, \dots, q - 2\}$  are therefore the multiples of  $m = (q - 1)/d$ , namely  $j = tm$  for  $t = 0, 1, \dots, d - 1$ : exactly  $d$  of them. The corresponding solutions  $x = (g^m)^t$  are exactly the powers of  $g^m$ , which by Corollary 3.23 has order  $(q - 1)/\gcd(q - 1, m) = (q - 1)/m = d$ ; so the solution set is the cyclic subgroup  $\langle g^m \rangle$  of order  $d = \gcd(n, q - 1)$ . When  $n \mid q - 1$  we have  $d = n$ , giving exactly  $n$  solutions.  $\square$

*Remark 6.23* (Two-adicity and proof-system fields). The corollary makes a curve such as Pallas or Vesta suitable for Halo 2: the field size  $q$  is chosen so that  $q - 1$  is divisible by a large power of 2, and the corollary then *guarantees* a multiplicative subgroup of 2-power order. That subgroup is the evaluation domain of the number-theoretic transform, on which efficient polynomial arithmetic rests. The construction of these domains and the transform itself are treated in Section 9; the curves are introduced in Section 10.

*Remark 6.24* (Cyclicity and Frobenius in cryptography). The cyclicity of  $\mathbb{F}_q^\times$  (Theorem 6.17) underlies the discrete-logarithm problem—every nonzero element has the form  $g^j$ , the problem is to recover  $j$ , and the presumed hardness of doing so is a foundational assumption of later volumes—and, through Corollary 6.22, guarantees the existence of the roots of unity used in polynomial commitment schemes. The Frobenius endomorphism (Definition 6.14) induces, on an elliptic curve over  $\mathbb{F}_p$ , an endomorphism of the curve whose characteristic polynomial  $T^2 - tT + p$  encodes the number of curve points as  $N = p + 1 - t$ , where  $t$  is the Hasse trace; the order of the cryptographic group is thus determined by the field-theoretic Frobenius. Both connections are taken up in Section 10.

## 7 Number theory for cryptography

This section treats the number theory behind two requirements of elliptic-curve cryptography. A point  $(x, y)$  of a curve  $y^2 = x^3 + ax + b$  (Section 10) is determined by  $x$  up to the sign of  $y$ , so wire formats send  $x$  with one bit selecting the sign, halving the transmission; decompression must decide whether  $x^3 + ax + b$  is a square modulo  $p$  and, if it is, *compute* a square root, in the time of a few modular exponentiations on numbers hundreds of bits long. Elliptic-curve cryptography also publishes powers  $g^x$  of a known group element while the exponent  $x$  is a secret key, so exponentiation must be efficient and its inversion infeasible. The section proves the Chinese Remainder Theorem and Euler's criterion, gives square-root algorithms for  $p \equiv 3 \pmod{4}$  (Proposition 7.17) and in general (Theorem 7.19), and proves the Pohlig–Hellman reduction of a discrete logarithm to discrete logarithms in subgroups of prime order, whence the prime group orders of Halo 2 and Orchard (Corollary 7.24).

Throughout,  $n$  and  $m$  denote positive integers and  $p$  denotes a prime. Statements proved in Section 2, Section 3 and Section 6 (the cyclicity theorem) reappear in arithmetic form, with proofs by citation.

### 7.1 Euler's totient function and the Chinese Remainder Theorem

A class  $[a]_n \in \mathbb{Z}/n\mathbb{Z}$  is a unit if and only if  $\gcd(a, n) = 1$  (Theorem 2.24), the units form a finite abelian group  $(\mathbb{Z}/n\mathbb{Z})^\times$  (Proposition 3.8), and Euler's totient  $\varphi(n)$  counts them:  $\varphi(n) = |(\mathbb{Z}/n\mathbb{Z})^\times|$  (Definition 2.26). In particular  $\varphi(p) = p - 1$  for a prime  $p$  (Example 2.27), so  $\mathbb{F}_p^\times$  has order  $p - 1$ . Through the orders of elliptic-curve groups, the totient enters the parameter choices of Halo 2 and Orchard. Since Section 2 computes  $\varphi(n)$  only by enumeration, this subsection derives a formula from its values on prime powers and its multiplicativity on coprime factors.

**Proposition 7.1** (Totient of a prime power). *For a prime  $p$  and an integer  $k \geq 1$ ,*

$$\varphi(p^k) = p^k - p^{k-1} = p^k \left(1 - \frac{1}{p}\right).$$

*Proof.* Among  $1, \dots, p^k$ , we claim the integers *not* coprime to  $p^k$  are exactly the multiples of  $p$ . If  $p \mid a$  then  $p \mid \gcd(a, p^k)$ , so the gcd exceeds 1. Conversely, if  $g = \gcd(a, p^k) > 1$ , then  $g$  has a prime factor  $\ell$ ; from  $\ell \mid g \mid p^k$  and Euclid's lemma for primes (Lemma 2.17) we get  $\ell = p$ , and  $\ell \mid g \mid a$  gives  $p \mid a$ . The multiples of  $p$  in the range are  $p, 2p, \dots, p^{k-1} \cdot p$ , of which there are  $p^{k-1}$ ; subtracting them from the  $p^k$  integers in the range leaves  $\varphi(p^k) = p^k - p^{k-1}$ .  $\square$

Multiplicativity follows from the Chinese Remainder Theorem, which the discrete-logarithm subsection uses again. The theorem is stated in the language of rings (Section 4), for the product ring  $R \times S$  with componentwise operations (Example 4.5); a bijective ring homomorphism (Definition 4.21) is a *ring isomorphism*, as for groups (Definition 3.41).

**Theorem 7.2** (Chinese Remainder Theorem). *If  $\gcd(m, n) = 1$ , then the map*

$$\mathbb{Z}/mn\mathbb{Z} \longrightarrow \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}, \quad [a]_{mn} \longmapsto ([a]_m, [a]_n)$$

is a ring isomorphism. It restricts to a group isomorphism

$$(\mathbb{Z}/mn\mathbb{Z})^\times \cong (\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/n\mathbb{Z})^\times.$$

*Proof.* The map is well defined: if  $[a]_{mn} = [a']_{mn}$  then  $mn \mid a - a'$ , hence  $m \mid a - a'$  and  $n \mid a - a'$ , so the two components agree. It is a ring homomorphism because each component is: reduction modulo  $m$  (and modulo  $n$ ) sends sums to sums, products to products, and 1 to 1 (Theorem 4.6), and the product ring's operations are componentwise.

For injectivity it suffices, since the map is in particular a homomorphism of additive groups, to check that the kernel is trivial (Proposition 3.40). Suppose  $[a]_m = [0]_m$  and  $[a]_n = [0]_n$ , that is,  $m \mid a$  and  $n \mid a$ . Since  $\gcd(m, n) = 1$ , Euclid's lemma (Proposition 2.15(2)) gives  $mn \mid a$ , i.e.  $[a]_{mn} = [0]_{mn}$ .

Both sides have exactly  $mn$  elements (Theorem 4.6 for each factor), and an injective map between finite sets of equal size is bijective: the  $mn$  distinct inputs have  $mn$  distinct images, which must exhaust the codomain. Hence the map is a ring isomorphism.

For the multiplicative statement, a ring isomorphism  $\phi$  matches units with units: if  $uv = 1$  then  $\phi(u)\phi(v) = \phi(1) = 1$ , and the same applies to  $\phi^{-1}$ . The units of the product ring are exactly the pairs of units,  $(u, v)^{-1} = (u^{-1}, v^{-1})$  existing precisely when both components are invertible. Restricting  $\phi$  to unit groups therefore gives a bijection  $(\mathbb{Z}/mn\mathbb{Z})^\times \rightarrow (\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/n\mathbb{Z})^\times$  that respects multiplication: a group isomorphism.  $\square$

**Corollary 7.3** (Multiplicativity and the product formula). *If  $\gcd(m, n) = 1$  then  $\phi(mn) = \phi(m)\phi(n)$ . Hence if  $n = \prod_i p_i^{k_i}$  is the prime factorisation of  $n$ , then*

$$\phi(n) = \prod_i (p_i^{k_i} - p_i^{k_i-1}) = n \prod_{p|n} \left(1 - \frac{1}{p}\right),$$

*the last product running over the distinct primes dividing  $n$ .*

*Proof.* The unit-group isomorphism of Theorem 7.2 is in particular a bijection, and the cardinality of a product of two sets is the product of the cardinalities, so  $\phi(mn) = \phi(m)\phi(n)$ .

For the product formula, the prime powers  $p_i^{k_i}$  are pairwise coprime: a common divisor exceeding 1 of two of them would have a prime factor dividing both, which by Euclid's lemma for primes (Lemma 2.17) would equal both  $p_i$  and  $p_j$ . Moreover each  $p_i^{k_i}$  is coprime to the product of the others, by repeated application of Proposition 2.15(3). Induction on the number of factors therefore extends multiplicativity to  $\phi(n) = \prod_i \phi(p_i^{k_i})$ , and Proposition 7.1 evaluates each factor:

$$\phi(n) = \prod_i (p_i^{k_i} - p_i^{k_i-1}) = \prod_i p_i^{k_i} \left(1 - \frac{1}{p_i}\right) = n \prod_{p|n} \left(1 - \frac{1}{p}\right). \quad \square$$

For example, the product formula gives  $\phi(12) = 12\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right) = 4$ , in agreement with the direct enumeration  $\{1, 5, 7, 11\}$  of Example 2.27. Gauss's divisor-sum identity  $\sum_{d|n} \phi(d) = n$  is Lemma 6.16.

## 7.2 Fermat's little theorem and Euler's theorem

Both congruences are Corollary 3.32, proved from Lagrange's theorem.

**Theorem 7.4** (Euler's theorem). *If  $\gcd(a, n) = 1$ , then*

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

*Proof.* The hypothesis says that  $[a]_n$  is a unit (Theorem 2.24), i.e. an element of the finite group  $(\mathbb{Z}/n\mathbb{Z})^\times$ , whose order is  $\varphi(n)$  (Definition 2.26). By Corollary 3.31—every element  $g$  of a finite group of order  $N$  satisfies  $g^N = e$ , a consequence of Lagrange's theorem—we get  $[a]_n^{\varphi(n)} = [1]_n$ , which is the stated congruence. This is Corollary 3.32 verbatim.  $\square$

**Corollary 7.5** (Fermat's little theorem). *Let  $p$  be prime. If  $p \nmid a$ , then  $a^{p-1} \equiv 1 \pmod{p}$ . Moreover  $a^p \equiv a \pmod{p}$  for every integer  $a$ .*

*Proof.* For  $p \nmid a$  we have  $\gcd(a, p) = 1$ , so Theorem 7.4 with  $n = p$  and  $\varphi(p) = p - 1$  (Example 2.27) gives the first congruence; multiplying it by  $a$  gives  $a^p \equiv a \pmod{p}$ . When  $p \mid a$  both  $a^p$  and  $a$  are congruent to 0, so the second congruence holds for all  $a$ .  $\square$

**Remark 7.6** (Cryptographic use of Euler and Fermat). Euler's theorem underlies RSA: if  $ed \equiv 1 \pmod{\varphi(n)}$ , say  $ed = 1 + k\varphi(n)$ , then

$$(a^e)^d = a^{1+k\varphi(n)} = a \cdot (a^{\varphi(n)})^k \equiv a \pmod{n}$$

whenever  $\gcd(a, n) = 1$ , so  $e$  can serve as a public encryption exponent and  $d$  as its private inverse. Orchard's elliptic-curve cryptography instead uses Fermat's little theorem in the field  $\mathbb{F}_p$ : every nonzero  $x$  satisfies  $x^{p-1} = 1$ , so inversion is computable as  $x^{-1} = x^{p-2}$  (Construction 6.4).

## 7.3 Multiplicative order and primitive roots

**Definition 7.7** (Multiplicative order). Let  $\gcd(a, n) = 1$ . The *multiplicative order* of  $a$  modulo  $n$ , written  $\text{ord}_n(a)$ , is the least positive integer  $k$  with  $a^k \equiv 1 \pmod{n}$ . Such a  $k$  exists because  $a^{\varphi(n)} \equiv 1$  by Theorem 7.4.

The quantity  $\text{ord}_n(a)$  is exactly the order of  $[a]_n$  as an element of the group  $(\mathbb{Z}/n\mathbb{Z})^\times$ , so the order theory of Section 3 applies verbatim:  $a^m \equiv 1 \pmod{n}$  if and only if  $\text{ord}_n(a) \mid m$  (Proposition 3.12(1))—in particular  $\text{ord}_n(a) \mid \varphi(n)$ , by Euler's theorem—and

$$\text{ord}_n(a^j) = \frac{\text{ord}_n(a)}{\gcd(j, \text{ord}_n(a))}$$

(Corollary 3.23).

**Definition 7.8** (Primitive root). A *primitive root* modulo  $n$  is an element  $g$  with  $\text{ord}_n(g) = \varphi(n)$ ; equivalently,  $g$  generates  $(\mathbb{Z}/n\mathbb{Z})^\times$  as a cyclic group, so that  $\{g^0, g^1, \dots, g^{\varphi(n)-1}\}$  lists all the units.

Primitive roots need not exist for every modulus. The unit group  $(\mathbb{Z}/8\mathbb{Z})^\times = \{1, 3, 5, 7\}$  has order  $\varphi(8) = 4$ , yet every one of its elements squares to 1 (Example 3.22): the element orders are 1 for the identity and 2 for each of 3, 5, 7, so no element has order 4 and no primitive root modulo 8 exists. The unit group is isomorphic to the Klein four-group of Example 3.10: every element is its own inverse, and the product of two distinct non-identity elements  $a$  and  $b$  is neither  $e$  (which would force  $b = a^{-1} = a$ ) nor  $a$  nor  $b$  (which would force  $b = e$  or  $a = e$ ), so it is the third. The case that cryptography requires is  $n = p$  prime, where the multiplicative group is always cyclic.

**Corollary 7.9** ( $\mathbb{F}_p^\times$  is cyclic). For every prime  $p$ , the group  $\mathbb{F}_p^\times = (\mathbb{Z}/p\mathbb{Z})^\times$  is cyclic of order  $p - 1$ . In particular primitive roots modulo  $p$  exist, and there are exactly  $\varphi(p - 1)$  of them; Example 6.21 exhibits the two primitive roots 3 and 5 of  $\mathbb{F}_7$ .

*Proof.* The group  $\mathbb{F}_p^\times$  is a finite subgroup, of order  $p - 1$ , of the multiplicative group of the field  $\mathbb{F}_p$ , hence cyclic by Theorem 6.17. A generator  $g$  has order  $p - 1$ , and the powers  $g^k$  that again generate are those with  $\gcd(k, p - 1) = 1$  (Corollary 3.23), numbering  $\varphi(p - 1)$ .  $\square$

**Remark 7.10** (Cyclicity and prime-order groups). Cyclicity is the reason the prime-order groups of elliptic-curve cryptography behave exactly like  $\mathbb{Z}/q\mathbb{Z}$  under addition once a generator is fixed: a cyclic group of order  $q$  is isomorphic to  $(\mathbb{Z}/q\mathbb{Z}, +)$  via  $g^k \leftrightarrow k$  (Theorem 3.43). The full classification, stated without proof and not used later, is that primitive roots modulo  $n$  exist exactly when  $n \in \{1, 2, 4, p^k, 2p^k\}$  for an odd prime  $p$ .

## 7.4 Quadratic residues and the Euler criterion

The squares modulo a prime decide whether a curve point with a given  $x$ -coordinate exists, the first decompression task.

**Definition 7.11** (Quadratic residue). Let  $p$  be an odd prime and  $a$  an integer with  $p \nmid a$ . We say  $a$  is a *quadratic residue* modulo  $p$  (a QR) if the congruence  $x^2 \equiv a \pmod{p}$  has a solution, and a *quadratic non-residue* (a QNR) otherwise.

**Proposition 7.12** (Exactly half are residues). For an odd prime  $p$ , exactly  $\frac{p-1}{2}$  of the nonzero residues modulo  $p$  are quadratic residues and  $\frac{p-1}{2}$  are non-residues. The squaring map  $x \mapsto x^2$  is exactly two-to-one on  $\mathbb{F}_p^\times$ .

*Proof.* Consider  $\sigma: \mathbb{F}_p^\times \rightarrow \mathbb{F}_p^\times$ ,  $\sigma(x) = x^2$ ; its image is exactly the set of quadratic residues. For  $x, y \in \mathbb{F}_p^\times$ ,

$$\sigma(x) = \sigma(y) \iff x^2 - y^2 = 0 \iff (x - y)(x + y) = 0 \iff y = \pm x,$$

the middle equivalence because a field has no zero divisors (Proposition 4.28). Since  $p$  is odd,  $x \neq -x$  for  $x \neq 0$  (otherwise  $2x = 0$  with 2 and  $x$  nonzero), so every fibre of  $\sigma$  over its image has exactly

two elements. Counting,  $|\text{im } \sigma| = \frac{p-1}{2}$ , and the complement—the non-residues—has the same size.  $\square$

**Theorem 7.13** (Euler’s criterion). *Let  $p$  be an odd prime and  $p \nmid a$ . Then*

$$a^{\frac{p-1}{2}} \equiv \begin{cases} 1 & (\text{mod } p) \quad \text{if } a \text{ is a quadratic residue,} \\ -1 & (\text{mod } p) \quad \text{if } a \text{ is a quadratic non-residue.} \end{cases}$$

*Proof.* First, the value is always  $\pm 1$ . Fermat’s little theorem (Corollary 7.5) gives  $(a^{(p-1)/2})^2 = a^{p-1} \equiv 1$ , and the only square roots of 1 modulo a prime are  $\pm 1$ : from  $x^2 \equiv 1$  we get  $p \mid (x-1)(x+1)$ , so  $p \mid x-1$  or  $p \mid x+1$  by Euclid’s lemma for primes (Lemma 2.17), i.e.  $x \equiv \pm 1$ . Hence  $a^{(p-1)/2} \equiv \pm 1 \pmod{p}$ .

If  $a$  is a residue, say  $a \equiv x^2$  with  $p \nmid x$ , then  $a^{(p-1)/2} \equiv x^{p-1} \equiv 1$ , again by Fermat. Thus every one of the  $\frac{p-1}{2}$  quadratic residues (Proposition 7.12) is a root of the polynomial  $t^{(p-1)/2} - 1$  over  $\mathbb{F}_p$ . That polynomial has degree  $\frac{p-1}{2}$ , so it has *at most*  $\frac{p-1}{2}$  roots (Theorem 5.18); the residues therefore account for all of them. A non-residue  $a$  is consequently not a root, yet still satisfies  $a^{(p-1)/2} \equiv \pm 1$ ; the remaining possibility is  $a^{(p-1)/2} \equiv -1$ .  $\square$

**Definition 7.14** (Legendre symbol). For an odd prime  $p$  and an integer  $a$ , the *Legendre symbol* is

$$\left(\frac{a}{p}\right) = \begin{cases} 0 & \text{if } p \mid a, \\ 1 & \text{if } a \text{ is a quadratic residue modulo } p, \\ -1 & \text{if } a \text{ is a quadratic non-residue modulo } p. \end{cases}$$

Euler’s criterion (Theorem 7.13) then reads, uniformly,

$$\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p},$$

so the symbol is computable by a single modular exponentiation, at a cost of  $O(\log p)$  modular multiplications by square-and-multiply (Construction 6.2). The criterion also shows that the symbol is *completely multiplicative* in  $a$ :

$$\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right) \left(\frac{b}{p}\right), \quad \text{since} \quad (ab)^{\frac{p-1}{2}} = a^{\frac{p-1}{2}} b^{\frac{p-1}{2}}.$$

**Example 7.15** (13 is a non-square in both Pasta fields). Orchard uses the Pasta hash-to-curve construction, a procedure, treated in later volumes, that maps arbitrary byte strings to curve points and requires a fixed non-square constant in the base field. The construction fixes  $Z = -13$  as a verifiably non-square constant in both base fields. Let  $p$  and  $q$  be the Pasta primes of Section 10; both satisfy  $p \equiv q \equiv 1 \pmod{4}$ . The verification below works modulo  $p$ ; the computation modulo  $q$  is identical.

First,  $-1$  is a square modulo any prime  $\equiv 1 \pmod{4}$ : the exponent  $\frac{p-1}{2}$  is even, so Euler's criterion gives

$$\left(\frac{-1}{p}\right) \equiv (-1)^{\frac{p-1}{2}} = 1.$$

By complete multiplicativity,

$$\left(\frac{-13}{p}\right) = \left(\frac{-1}{p}\right) \left(\frac{13}{p}\right) = \left(\frac{13}{p}\right),$$

so  $-13$  is a non-square precisely when  $13$  is. Whether  $13$  is a square is settled by one Euler-criterion exponentiation in each field, a computation any computer-algebra system reproduces in milliseconds:

$$13^{\frac{p-1}{2}} \equiv p-1 \equiv -1 \pmod{p}, \quad 13^{\frac{q-1}{2}} \equiv q-1 \equiv -1 \pmod{q},$$

whence  $\left(\frac{13}{p}\right) = \left(\frac{13}{q}\right) = -1$ . Thus  $13$ , and with it  $Z = -13$ , is a quadratic non-residue modulo both Pasta primes, as the hash-to-curve construction requires.

*Example 7.16* ( $5$  is a non-square in the Pallas base field). The same test decides a second constant on which later volumes rely, in this case the constant term of the curve equation. Pallas is the curve  $y^2 = x^3 + 5$  over  $\mathbb{F}_p$  (Section 10), and the arithmetic circuits of Halo 2 encode its point at infinity as the coordinate pair  $(0, 0)$  (Construction 10.13). The encoding is unambiguous only if no genuine point has  $x$ -coordinate  $0$ , i.e. only if  $y^2 = 5$  has no solution in  $\mathbb{F}_p$ : the question is whether  $5$  is a square in the *base* field  $\mathbb{F}_p$  of Pallas, not in its scalar field  $\mathbb{F}_q$ . Euler's criterion decides it in one exponentiation:

$$5^{\frac{p-1}{2}} \equiv p-1 \equiv -1 \pmod{p},$$

so  $\left(\frac{5}{p}\right) = -1$ . The constant  $5$  is a quadratic non-residue modulo the Pallas base-field prime, and no point of Pallas has  $x = 0$ . (The same computation modulo  $q$  gives  $5^{(q-1)/2} \equiv -1 \pmod{q}$ , so Vesta has no such point either.) That no point has  $y = 0$  is a different fact— $5$  is not a cube in  $\mathbb{F}_p$ —which Remark 10.26 reads off from the oddness of the group order; the two together are the hypotheses of Proposition 10.14.

## 7.5 Square roots modulo a prime

Decompression must also *produce* an  $x$  with  $x^2 \equiv a$  for a residue  $a$ , recovering  $y$  from  $y^2$ ; the cost depends on  $p \pmod{4}$ .

**Proposition 7.17** (The  $p \equiv 3 \pmod{4}$  shortcut). *Let  $p \equiv 3 \pmod{4}$  be prime and let  $a$  be a quadratic residue modulo  $p$  with  $p \nmid a$ . Then*

$$x \equiv a^{\frac{p+1}{4}} \pmod{p}$$

*is a square root of  $a$ , and the two square roots of  $a$  are  $\pm x$ .*

*Proof.* Since  $p \equiv 3 \pmod{4}$ , the quantity  $\frac{p+1}{4}$  is a positive integer. Compute

$$x^2 = a^{\frac{p+1}{2}} = a^{\frac{p-1}{2}} \cdot a \equiv \left(\frac{a}{p}\right) \cdot a = a \pmod{p},$$

using Euler's criterion (Theorem 7.13) and the hypothesis that  $a$  is a residue, so  $\left(\frac{a}{p}\right) = 1$ . The roots come in the pair  $\pm x$  by the two-to-one property of squaring (Proposition 7.12).  $\square$

*Remark 7.18* (Cost and the Pasta primes). Primes  $p \equiv 3 \pmod{4}$  are often favoured for this single-exponentiation shortcut. For  $p \equiv 5 \pmod{8}$  a closed-form variant (Atkin's formula) exists. In general the cost of square-root extraction grows with the 2-adic valuation  $s$  of  $p-1$ : writing  $p-1 = 2^s d$  with  $d$  odd, the Tonelli–Shanks algorithm below performs up to  $s$  corrective iterations. The Pasta primes have  $s = 32$ , chosen for FFT-friendliness (Remark 9.10), so their implementations use Tonelli–Shanks.

For  $p \equiv 1 \pmod{4}$ , write  $p-1 = 2^s d$  with  $d$  odd and  $s \geq 2$ . The algorithm works in the subgroup of  $\mathbb{F}_p^\times$  of order  $2^s$ , which contains the obstruction to the single-exponentiation shortcut, and removes that obstruction one power of two at a time. The subgroup is unique since  $\mathbb{F}_p^\times$  is cyclic (Corollary 7.9 and Theorem 3.24(3)).

**Theorem 7.19** (Tonelli–Shanks). *Let  $p$  be an odd prime,  $p-1 = 2^s d$  with  $d$  odd, and let  $a$  be a quadratic residue with  $p \nmid a$ . Given any quadratic non-residue  $z$  modulo  $p$ , one can compute a square root of  $a$  modulo  $p$  using  $O(s^2 + \log p)$  modular multiplications.*

*Proof.* Write  $H \leq \mathbb{F}_p^\times$  for the unique cyclic subgroup of order  $2^s$ . Set  $c = z^d$ . We claim  $c$  generates  $H$ . By Euler's criterion (Theorem 7.13)  $z^{(p-1)/2} \equiv -1$ , so

$$c^{2^{s-1}} = z^{2^{s-1}d} = z^{\frac{p-1}{2}} \equiv -1 \neq 1, \quad c^{2^s} = z^{p-1} \equiv 1$$

by Fermat (Corollary 7.5); hence  $\text{ord}_p(c)$  divides  $2^s$  but not  $2^{s-1}$ , forcing  $\text{ord}_p(c) = 2^s$  and  $\langle c \rangle = H$ .

Initialise

$$x = a^{\frac{d+1}{2}}, \quad t = a^d, \quad m = s$$

(the exponent  $\frac{d+1}{2}$  is an integer because  $d$  is odd). We maintain three invariants:

1.  $x^2 = at$ ;
2. the current  $c$  has order exactly  $2^m$ ;
3. the order of  $t$  divides  $2^{m-1}$ .

Initially (1) holds since  $x^2 = a^{d+1} = a \cdot a^d$ ; (2) was proved above; and (3) holds because  $t^{2^{s-1}} = a^{(p-1)/2} \equiv 1$  by Euler's criterion (Theorem 7.13),  $a$  being a residue.

Each iteration proceeds as follows. If  $t = 1$ , then  $x^2 = a$  by (1) and the algorithm terminates. Otherwise let  $2^i = \text{ord}_p(t)$ , found as the least  $i \geq 1$  with  $t^{2^i} = 1$  by repeatedly squaring  $t$ ; invariant (3) gives  $1 \leq i \leq m-1$ . Set

$$b = c^{2^{m-i-1}},$$

and update

$$x \leftarrow xb, \quad t \leftarrow tb^2, \quad c \leftarrow b^2, \quad m \leftarrow i.$$

The invariants are preserved. For (1),  $(xb)^2 = x^2 b^2 = at b^2$ , which is  $a$  times the new  $t$ . For (2), the old  $c$  has order  $2^m$ , so  $\text{ord}_p(b) = 2^m / \gcd(2^m, 2^{m-i-1}) = 2^{i+1}$  (Corollary 3.23), and the new

$c = b^2$  has order  $2^i$ —exactly 2 to the new  $m$ . For (3), both the old  $t$  and  $b^2$  have order exactly  $2^i$ , so  $t^{2^{i-1}}$  and  $(b^2)^{2^{i-1}}$  each have order 2; the only element of order 2 in  $\mathbb{F}_p^\times$  is  $-1$ , because the square roots of 1 are  $\pm 1$  (as in the proof of Theorem 7.13); hence

$$(t b^2)^{2^{i-1}} = t^{2^{i-1}} (b^2)^{2^{i-1}} = (-1)(-1) = 1,$$

so the new  $t$  has order dividing  $2^{i-1}$ , which is  $2^{m-1}$  for the new  $m$ .

Each iteration strictly decreases  $m$ , from  $s$  through a descending chain of nonnegative integers, so after at most  $s$  iterations the order of  $t$  reaches  $2^0 = 1$ , i.e.  $t = 1$ , and  $x$  is the desired root. For the cost: locating  $i$  and computing  $b$  each take at most  $m \leq s$  squarings, so the loop costs  $O(s)$  multiplications per iteration and  $O(s^2)$  in total; the three initial exponentiations, with exponents smaller than  $p$ , cost  $O(\log p)$  multiplications each by square-and-multiply (Construction 6.2).  $\square$

Finding the non-residue  $z$  is the only randomised step in this section. A uniformly random element of  $\mathbb{F}_p^\times$  is a non-residue with probability exactly  $\frac{1}{2}$  (Proposition 7.12), and each candidate is checked by one Euler-criterion exponentiation (Definition 7.14), i.e.  $O(\log p)$  modular multiplications, so a few random trials find  $z$  with high probability. Implementations hard-code one non-residue per field, verified once, as Example 7.15 verifies  $Z = -13$ . This volume uses no deterministic polynomial-time method for finding a non-residue.

*Remark 7.20* (Which root, and detecting failure). The square roots of a residue come in a pair  $\pm x$  (Proposition 7.12), so a *canonical* root is fixed by convention—for instance “choose the root whose least nonnegative representative is even”, as elliptic-curve point encodings do; the compressed point’s extra bit selects between the two. Both algorithms detect a non-residue  $a$ . The shortcut of Proposition 7.17 still outputs a candidate  $a^{(p+1)/4}$ , which fails the check  $x^2 \equiv a$  at the cost of one squaring. The variable-time Tonelli–Shanks algorithm cannot complete its update: Euler’s criterion gives  $a^{(p-1)/2} \equiv -1$ , so  $t = a^d$  satisfies  $t^{2^{i-1}} \equiv -1$  and has order exactly  $2^i$ , and the first search for the least  $i \geq 1$  with  $t^{2^i} = 1$  already returns  $i = m$ , which invariant (3) of the proof of Theorem 7.19 forbids and for which the update exponent  $2^{m-i-1}$  is undefined. The algorithm can therefore report “not a residue” as soon as  $i = m$  occurs, or verify  $a^{(p-1)/2} \equiv 1$  up front, at the same  $O(\log p)$  cost as the initial exponentiations.

## 7.6 The discrete logarithm problem

The statements below concern a cyclic group written multiplicatively. In Orchard the group is the group of points of an elliptic curve, of large prime order and written additively; every statement transfers by renaming the operation.

**Definition 7.21** (Discrete logarithm problem). Let  $G = \langle g \rangle$  be a finite cyclic group of order  $N$  with generator  $g$ . Given  $h \in G$ , the *discrete logarithm* of  $h$  to base  $g$ , written  $\log_g h$ , is the unique residue  $x \in \mathbb{Z}/N\mathbb{Z}$  with  $g^x = h$ . The *discrete logarithm problem* (DLP) is to compute  $x$  from  $(G, g, h)$ .

Existence and uniqueness of  $\log_g h$  are the content of the classification of cyclic groups: the map  $x \mapsto g^x$  is a group isomorphism  $(\mathbb{Z}/N\mathbb{Z}, +) \xrightarrow{\sim} G$  (Theorem 3.43, whose map  $\tilde{\phi}$  this is), so it

has a well-defined inverse, and  $\log_g$  is that inverse. Computing  $g^x$  from  $x$  takes  $O(\log N)$  group operations by square-and-multiply (Construction 6.2); computing  $x$  from  $g^x$  is the DLP, for which no efficient algorithm is known in well-chosen groups. This asymmetry is the one-wayness on which elliptic-curve cryptography rests.

*Remark 7.22* (Generic complexity and group choice). Two *generic* algorithms solve the DLP in any group of order  $N$ , using only the group operation. *Baby-step giant-step* is deterministic: set  $m = \lceil \sqrt{N} \rceil$  and write the unknown as  $x = im + j$  with  $0 \leq i, j < m$ ; tabulate the “baby steps”  $g^j$  for all  $j$ , then walk the “giant steps”  $h(g^{-m})^i$  for  $i = 0, 1, 2, \dots$  until one equals a tabulated  $g^j$ , at which point  $hg^{-im} = g^j$  gives  $x = im + j$ . Both the table and the walk are  $O(\sqrt{N})$ , so the cost is  $O(\sqrt{N})$  time and space. *Pollard’s rho* (not developed here) is a randomised alternative with the same  $O(\sqrt{N})$  expected running time—on a heuristic analysis, made explicit in Section 10—but only  $O(1)$  space, and is the algorithm used in practical attacks. Conversely, Shoup’s theorem shows that  $\Omega(\sqrt{N})$  group operations are *necessary* for any generic algorithm when  $N$  is prime, so for generic attacks the square-root cost is exact. Reaching a  $\lambda$ -bit security level—no attack cheaper than  $2^\lambda$  operations—therefore requires  $N \approx 2^{2\lambda}$ . This is why the Pallas and Vesta scalar fields have prime order roughly 255 bits long, targeting roughly 128-bit security (Section 10). These costs hold for *generic* algorithms: in  $(\mathbb{Z}/p\mathbb{Z})^\times$ , sub-exponential index-calculus attacks exploit the ring structure of the integers behind the group. No such attack is known for elliptic-curve groups, which therefore permit much smaller parameters than multiplicative groups of comparable security.

The next theorem makes “well-chosen” precise.

**Theorem 7.23** (Pohlig–Hellman reduction). *Let  $G = \langle g \rangle$  be cyclic of order  $N = \prod_{i=1}^r q_i^{e_i}$ , the  $q_i$  distinct primes. The discrete logarithm in  $G$  is computable by solving discrete logarithms in subgroups of prime order  $q_i$ — $e_i$  of them for each  $i$ —and combining the answers with the Chinese Remainder Theorem, at a total cost of*

$$O\left(\sum_{i=1}^r e_i (\log N + \sqrt{q_i})\right)$$

*group operations. When  $N$  has a large prime factor  $q_{\max}$  the  $\sqrt{q_{\max}}$  term dominates; when  $N$  is smooth—a product of primes bounded polynomially in  $\log N$ —every term is polynomial in  $\log N$ , and hence so is the cost of the DLP. Corollary 7.24 draws the design consequence.*

*Proof.* Let  $h = g^x$  with  $x$  the unknown. The reduction has two stages.

*Stage (a): splitting across coprime factors.* For each  $i$  set  $N_i = N/q_i^{e_i}$  and project:

$$g_i = g^{N_i}, \quad h_i = h^{N_i}.$$

By Corollary 3.23,  $\text{ord}(g_i) = N / \gcd(N, N_i) = N/N_i = q_i^{e_i}$ , so  $g_i$  generates the unique subgroup of  $G$  of order  $q_i^{e_i}$  (Theorem 3.24(3)). Moreover  $h_i = g^{xN_i} = g_i^x$ , and since  $g_i$  has order  $q_i^{e_i}$ , the exponent matters only modulo  $q_i^{e_i}$  (Proposition 3.12(2)): writing  $x_i = x \bmod q_i^{e_i}$ ,

$$\log_{g_i} h_i = x_i.$$

Solving the  $r$  smaller DLPs yields  $x \bmod q_i^{e_i}$  for every  $i$ ; since the prime powers  $q_i^{e_i}$  are pairwise coprime, the Chinese Remainder Theorem (Theorem 7.2, iterated across the factors as in the proof of Corollary 7.3) determines  $x$  modulo  $N$  uniquely. Each projection costs  $O(\log N)$  operations by square-and-multiply.

*Stage (b): lifting a prime power digit by digit.* Fix one prime power  $q^e$  (dropping the subscript  $i$ ), with  $\gamma = g_i$  of order  $q^e$  and  $\eta = b_i = \gamma^{x'}$ ,  $x' = x_i$ . Write  $x'$  in base  $q$ :

$$x' \equiv x_0 + x_1 q + \dots + x_{e-1} q^{e-1} \pmod{q^e}, \quad 0 \leq x_j < q.$$

The element  $\delta = \gamma^{q^{e-1}}$  has order  $q$  (Corollary 3.23 again). Raise  $\eta$  to the power  $q^{e-1}$ : every digit beyond the first is multiplied by a multiple of  $q^e$  in the exponent and vanishes, leaving

$$\eta^{q^{e-1}} = \gamma^{x' q^{e-1}} = \delta^{x_0},$$

a single DLP in the group  $\langle \delta \rangle$  of prime order  $q$ , which baby-step giant-step (Remark 7.22) solves in  $O(\sqrt{q})$  operations. Once  $x_0, \dots, x_{j-1}$  are known, they are removed and the step is repeated: the element  $\eta_j = \gamma^{-(x_0 + x_1 q + \dots + x_{j-1} q^{j-1})}$  equals  $\gamma$  raised to  $x_j q^j$  (multiples of  $q^{j+1}$ ), so

$$\eta_j^{q^{e-1-j}} = \delta^{x_j},$$

and digit  $x_j$  is again one order- $q$  DLP. Each of the  $e$  digits costs one such DLP plus  $O(\log N)$  operations for the exponentiations.

Summing over the digits of all the prime-power factors gives  $\sum_i e_i$  order- $q_i$  DLPs at  $O(\sqrt{q_i})$  apiece plus  $O(\log N)$  bookkeeping for each, which is the stated bound.  $\square$

**Corollary 7.24** (Why prime order matters). *If  $N$  has a small prime factor  $q$ , an attacker can recover the discrete logarithm modulo  $q$  in  $O(\log N + \sqrt{q})$  group operations— $O(\log N)$  to project into the order- $q$  subgroup and  $O(\sqrt{q})$  for the small DLP there—leaking part of the secret  $x$ . Security therefore requires  $N$  to have a large prime factor; the safest and simplest choice is to make  $N$  itself a large prime, so that  $G$  has no proper nontrivial subgroups and the Pohlig–Hellman reduction yields no smaller problem. For this reason the elliptic-curve groups used in Halo 2 and Orchard have large prime order.*

*Proof.* For the first claim, apply stage (a) of Theorem 7.23 to the single factor  $q$ : the projections  $g^{N/q}, b^{N/q}$  cost  $O(\log N)$  operations, and the resulting DLP in a group of order  $q$  costs  $O(\sqrt{q})$  by baby-step giant-step—yielding  $x \bmod q$ . For the second, if  $N$  is prime then  $G$  has no subgroups besides  $\{e\}$  and  $G$  (Corollary 3.34), so the reduction’s only output is the original problem, and the generic  $\Omega(\sqrt{N})$  lower bound of Remark 7.22 applies in full.  $\square$

**Remark 7.25** (Cofactors and small-subgroup attacks). In practice an elliptic curve over  $\mathbb{F}_p$  has group order  $h \cdot q$ , where  $q$  is a large prime and  $h$  is a small *cofactor*, and cryptographic operations stay in the subgroup of prime order  $q$ . A protocol that does not check that incoming points lie in that subgroup (or clear the cofactor) admits a practical *small-subgroup attack*: an adversary supplies a point of low order and runs Pohlig–Hellman in the small factor to extract information about the secret modulo  $h$ . The Pasta curves of Halo 2 have cofactor  $h = 1$ : their group orders are prime (the primes recorded in Section 10), which eliminates this class of attacks in the group itself rather than by protocol-level checks.

A group of large prime order  $q$  has the field  $\mathbb{Z}/q\mathbb{Z}$  as its scalars. The curves, and the primes  $p$  and  $q$  taken as given here, are the subject of Section 10.

## 8 Linear algebra over a field

This section develops linear algebra from the axioms, assuming only the notion of a field (Section 4): *vector spaces*, whose elements can be added and scaled by elements of a field; bases and dimension (Theorem 8.17); linear maps and rank–nullity (Theorem 8.19); bilinear forms and the standard inner product on the coordinate space  $F^n$ , in practice over a finite field  $\mathbb{F}_q$ ; and the mixed inner product of a vector of scalars with a vector of group elements. For fixed public elements  $G_1, \dots, G_n$  of a suitable group, the mixed inner product  $[a_1]G_1 + \dots + [a_n]G_n$  is the Pedersen-style *commitment* of Halo 2 and Orchard to a list  $(a_1, \dots, a_n)$  of secrets, such as the *wire values* (intermediate values) of an *arithmetic circuit*, with  $n$  possibly in the millions: a single group element, independent of  $n$ , that computationally determines the list, can later be opened or reasoned about, and is additively homomorphic. Remark 8.29 derives the homomorphic property and the collisions that this compression necessarily introduces. Commitments of other kinds, such as hash-based ones, need not be linear. The next section applies these results to the polynomial spaces  $F[X]_{<n}$  over structured evaluation domains.

The section also discharges two deferrals. Remark 5.5 left to it the linear structure of  $F[X]$ , with the monomials as an infinite coordinate system. The proof of Theorem 6.9 used two facts without proof: a space with a finite spanning set has a basis, proved in Theorem 8.17, and coordinates in a basis are unique, which follows from linear independence (Definition 8.14). Neither argument uses finite fields, so no circularity arises. Throughout,  $F$  denotes an arbitrary field with identities 0 and 1; the field axioms of Definition 4.24 are used without citation.

### 8.1 Vector spaces and subspaces

**Definition 8.1** (Vector space). Let  $F$  be a field. A *vector space over  $F$*  (an  *$F$ -vector space*) is a set  $V$  equipped with two operations, *vector addition*  $+: V \times V \rightarrow V$  and *scalar multiplication*  $\cdot: F \times V \rightarrow V$ , such that for all  $u, v, w \in V$  and all  $a, b \in F$ :

1.  $(V, +)$  is an abelian group: addition is associative and commutative, there is a *zero vector*  $\mathbf{0} \in V$  with  $v + \mathbf{0} = v$ , and every  $v$  has an *additive inverse*  $-v$  with  $v + (-v) = \mathbf{0}$ ;
2. scalar multiplication is compatible with the multiplication of  $F$ :  $a \cdot (b \cdot v) = (ab) \cdot v$ ;
3. the identity of  $F$  acts as the identity:  $1 \cdot v = v$ ;
4. scalar multiplication distributes over vector addition:  $a \cdot (u + v) = a \cdot u + a \cdot v$ ;
5. scalar multiplication distributes over scalar addition:  $(a + b) \cdot v = a \cdot v + b \cdot v$ .

The elements of  $V$  are called *vectors*, the elements of  $F$  *scalars*. We write  $av$  for  $a \cdot v$ , and we allow the scalar  $0 \in F$  and the vector  $\mathbf{0} \in V$  to share a symbol whenever context distinguishes them.

The following consequences of the axioms are used hereafter without citation.

**Proposition 8.2.** *Let  $V$  be an  $F$ -vector space,  $v \in V$ , and  $a \in F$ . Then:*

1.  $0 \cdot v = \mathbf{0}$ ;
2.  $a \cdot \mathbf{0} = \mathbf{0}$ ;
3.  $(-1) \cdot v = -v$ ;
4. if  $av = \mathbf{0}$ , then  $a = 0$  or  $v = \mathbf{0}$ .

*Proof.* (1) By distributivity over scalar addition,  $0 \cdot v = (0 + 0) \cdot v = 0 \cdot v + 0 \cdot v$ ; adding  $-(0 \cdot v)$  to both sides leaves  $\mathbf{0} = 0 \cdot v$ .

(2) Likewise  $a \cdot \mathbf{0} = a \cdot (\mathbf{0} + \mathbf{0}) = a \cdot \mathbf{0} + a \cdot \mathbf{0}$ , and cancelling in the abelian group  $(V, +)$  gives  $a \cdot \mathbf{0} = \mathbf{0}$ .

(3) Using the identity axiom, distributivity, and (1),

$$v + (-1) \cdot v = 1 \cdot v + (-1) \cdot v = (1 + (-1)) \cdot v = 0 \cdot v = \mathbf{0},$$

so  $(-1) \cdot v$  is the additive inverse of  $v$ ; by uniqueness of inverses in a group,  $(-1) \cdot v = -v$ .

(4) Suppose  $av = \mathbf{0}$  and  $a \neq 0$ . Since  $F$  is a field,  $a^{-1}$  exists, and by the identity and compatibility axioms

$$v = 1 \cdot v = (a^{-1}a) \cdot v = a^{-1} \cdot (av) = a^{-1} \cdot \mathbf{0} = \mathbf{0},$$

the last step by (2). □

*Example 8.3* (The coordinate space  $F^n$ ). The principal example is the *coordinate space*

$$F^n = \{ (x_1, \dots, x_n) : x_i \in F \},$$

the set of  $n$ -tuples of scalars, under componentwise operations:

$$(x_1, \dots, x_n) + (y_1, \dots, y_n) = (x_1 + y_1, \dots, x_n + y_n), \quad a \cdot (x_1, \dots, x_n) = (ax_1, \dots, ax_n).$$

The zero vector is  $(0, \dots, 0)$ , and every axiom of Definition 8.1 follows from the corresponding field axiom applied in each coordinate. The degenerate case  $n = 0$  is the *trivial* (or *zero*) vector space  $\{\mathbf{0}\}$ , consisting of the empty tuple alone. As a standing convention we write vectors of  $F^n$  in boldface and regard them as columns,  $\mathbf{x} = (x_1, \dots, x_n)^T$ .

*Example 8.4* (Further examples). 1. *Polynomials*. The polynomials  $F[X]$  form an  $F$ -vector space under addition and the scalar multiplication of Remark 5.5. So does the subset

$$F[X]_{<n} = \{ f \in F[X] : f = 0 \text{ or } \deg f < n \}$$

of polynomials of degree less than  $n$ : sums and scalar multiples of such polynomials again have degree less than  $n$  (or are zero), by Proposition 5.7, so the operations restrict and the axioms are inherited. Polynomial commitment schemes operate on this space.

2. *Functions.* For any set  $S$ , the set  $F^S$  of all functions  $S \rightarrow F$  is an  $F$ -vector space under point-wise operations:  $(f + g)(s) = f(s) + g(s)$  and  $(af)(s) = a f(s)$ . Example 8.3 is the special case  $S = \{1, \dots, n\}$ , a tuple being a function of its index.
3. *Matrices.* The set  $F^{m \times n}$  of  $m \times n$  arrays of scalars is an  $F$ -vector space under entrywise addition and scaling—the case of (2) in which  $S$  is the grid of index pairs  $(i, j)$ .
4. *Field extensions.* If  $F$  is a subfield of a larger field  $K$  (§4), then  $K$  is an  $F$ -vector space: vector addition is the addition of  $K$ , and scalar multiplication is the multiplication of  $K$  restricted to pairs in  $F \times K$ , so the axioms are instances of the field axioms of  $K$ . The dimension of this vector space—in the sense of Theorem 8.17 below—is called the *degree* of the extension, written  $[K : F]$ . For instance  $\mathbb{C}$  is an  $\mathbb{R}$ -vector space of dimension 2, with basis  $1, i$ , so  $[\mathbb{C} : \mathbb{R}] = 2$ ; and the finite field  $\mathbb{F}_q$  with  $q = p^k$  is a  $k$ -dimensional  $\mathbb{F}_p$ -vector space,  $[\mathbb{F}_q : \mathbb{F}_p] = k$ , which is the structure used in the proof of Theorem 6.9.

**Definition 8.5** (Subspace). A subset  $W \subseteq V$  of an  $F$ -vector space  $V$  is a (*linear*) *subspace* if  $W$  is itself a vector space under the operations inherited from  $V$ ; equivalently—as the next proposition makes precise—if and only if  $W$  is nonempty and closed under the vector-space operations.

**Proposition 8.6** (Subspace criterion). *A subset  $W \subseteq V$  is a subspace if and only if*

1.  $\mathbf{0} \in W$ ;
2.  $W$  is closed under addition:  $u, v \in W$  implies  $u + v \in W$ ; and
3.  $W$  is closed under scalar multiplication:  $a \in F$  and  $v \in W$  imply  $av \in W$ .

*Equivalently,  $W$  is a subspace if and only if  $W \neq \emptyset$  and  $au + bv \in W$  for all  $a, b \in F$  and  $u, v \in W$ .*

*Proof.* If  $W$  is a subspace, then (1)–(3) hold: closure (2) and (3) is what it means for the inherited operations to be operations *on*  $W$ , and the zero vector of  $W$  is the zero vector of  $V$ , since a zero vector  $\mathbf{0}_W$  of  $W$  satisfies  $\mathbf{0}_W + \mathbf{0}_W = \mathbf{0}_W$ , and cancelling in  $V$  gives  $\mathbf{0}_W = \mathbf{0}$ .

Conversely, suppose (1)–(3) hold. Closure makes the two operations well defined on  $W$ . The additive inverse of any  $v \in W$  is  $(-1)v \in W$ , by (3) and Proposition 8.2(3). Every remaining axiom of Definition 8.1 is a universally quantified identity—associativity, commutativity, compatibility, the two distributive laws, the action of 1—and each holds for all elements of  $V$ , in particular for all elements of  $W$ . Hence  $W$  is a vector space.

For the two-scalar form: if  $W$  is a subspace, then  $au + bv \in W$  by (2) and (3) combined, and  $W \ni \mathbf{0}$  is nonempty. Conversely, choosing  $(a, b) = (1, 1)$  recovers (2), choosing  $(a, b) = (a, 0)$  recovers (3) (using Proposition 8.2(1) to identify  $0v$ ), and  $\mathbf{0} = 0v \in W$  for any  $v$  in the nonempty  $W$ , which is (1).  $\square$

**Example 8.7.** In  $F^3$  the set  $W = \{(x_1, x_2, x_3) : x_1 + x_2 + x_3 = 0\}$  is a subspace: it contains  $\mathbf{0}$ , and if the coordinates of  $\mathbf{x}$  and  $\mathbf{y}$  each sum to 0, then those of  $a\mathbf{x} + b\mathbf{y}$  sum to  $a \cdot 0 + b \cdot 0 = 0$ . By contrast  $\{(x_1, x_2, x_3) : x_1 + x_2 + x_3 = 1\}$  is *not* a subspace: it does not contain  $\mathbf{0}$ . The same computation

shows that the solution set of any single homogeneous linear equation  $c_1x_1 + \cdots + c_nx_n = 0$  (the  $c_i$  fixed scalars) is a subspace of  $F^n$ , and the solution set of a finite *system* of such equations, being an intersection of subspaces, is a subspace by the next proposition. These solution sets are the kernels of linear maps (Definition 8.18).

**Proposition 8.8** (Intersections and sums). *Let  $V$  be an  $F$ -vector space.*

1. *The intersection  $\bigcap_{i \in I} W_i$  of an arbitrary family of subspaces of  $V$  is a subspace.*
2. *The sum  $W_1 + \cdots + W_k = \{w_1 + \cdots + w_k : w_j \in W_j\}$  of finitely many subspaces is a subspace, and it is the smallest subspace of  $V$  containing every  $W_j$ .*

*The union of two subspaces is in general not a subspace.*

*Proof.* (1) Each  $W_i$  contains  $\mathbf{0}$ , so the intersection does. If  $u, v$  lie in every  $W_i$  and  $a, b \in F$ , then  $au + bv \in W_i$  for every  $i$  by Proposition 8.6, hence  $au + bv$  lies in the intersection, which is therefore a subspace by the same criterion.

(2) The sum contains  $\mathbf{0} = \mathbf{0} + \cdots + \mathbf{0}$ , and it contains each  $W_j$  (take every other summand  $\mathbf{0}$ ). For closure, take two elements  $\sum_j w_j$  and  $\sum_j w'_j$  with  $w_j, w'_j \in W_j$ , and scalars  $a, b$ ; regrouping,

$$a \sum_j w_j + b \sum_j w'_j = \sum_j (aw_j + bw'_j),$$

and  $aw_j + bw'_j \in W_j$  since  $W_j$  is a subspace, so the combination lies in the sum. Finally, any subspace  $U$  containing every  $W_j$  contains each summand  $w_j$  of any element of the sum, hence contains the element itself by closure under addition; so the sum is contained in  $U$ , and is the smallest such subspace.

For the union, take  $V = F^2$  (any field has  $|F| > 1$ , since  $1 \neq 0$ ) and the two axes  $\{(x, 0) : x \in F\}$  and  $\{(0, y) : y \in F\}$ —subspaces by the criterion. Their union contains  $(1, 0)$  and  $(0, 1)$  but not the sum  $(1, 1) = (1, 0) + (0, 1)$ , since  $(1, 1)$  lies on neither axis; closure under addition fails.  $\square$

## 8.2 Linear combinations, span, and linear independence

Besides being cut out by equations, a subspace can be generated by a set of vectors, as the set of all vectors obtained from them by addition and scaling.

**Definition 8.9** (Linear combination and span). Let  $S \subseteq V$ . A *linear combination* of vectors in  $S$  is any finite sum

$$a_1v_1 + a_2v_2 + \cdots + a_kv_k, \quad a_i \in F, v_i \in S.$$

The *span* of  $S$ , written  $\text{span}(S)$  or  $\langle S \rangle$ , is the set of all linear combinations of finitely many vectors of  $S$ ; by convention  $\text{span}(\emptyset) = \{\mathbf{0}\}$ . The set  $S$  *spans* (or *generates*)  $V$  if  $\text{span}(S) = V$ , and  $V$  is *finitely generated* (synonymously, *finite-dimensional*) if some finite set spans it.

**Proposition 8.10.** *For any  $S \subseteq V$ , the span  $\text{span}(S)$  is the smallest subspace of  $V$  containing  $S$ : it is a subspace, it contains  $S$ , and it is contained in every subspace of  $V$  that contains  $S$ .*

*Proof.* The zero vector is the empty linear combination (or, for nonempty  $S$ , the combination  $0v$ ), and a sum of two linear combinations of vectors in  $S$ , or a scalar multiple of one, is again such a linear combination—concatenate the lists and rescale the coefficients. By Proposition 8.6,  $\text{span}(S)$  is a subspace, and it contains each  $v \in S$  as the combination  $1 \cdot v$ . If  $W$  is any subspace containing  $S$ , then closure under addition and scaling forces every linear combination of elements of  $S$  into  $W$ , by induction on the number of summands; hence  $\text{span}(S) \subseteq W$ .  $\square$

**Definition 8.11** (Linear independence). A finite list of vectors  $v_1, \dots, v_k \in V$  is *linearly independent* if the only way to express the zero vector as a linear combination of the list is with all coefficients zero:

$$a_1 v_1 + \dots + a_k v_k = \mathbf{0} \implies a_1 = \dots = a_k = 0.$$

Otherwise the list is *linearly dependent*, and a relation  $\sum_i a_i v_i = \mathbf{0}$  with some  $a_i \neq 0$  is called a *dependence relation*. An arbitrary—possibly infinite—set  $S \subseteq V$  is linearly independent if every finite list of distinct vectors from  $S$  is linearly independent.

*Remark 8.12.* Linear independence expresses *non-redundancy*: a list is dependent exactly when some vector in it lies in the span of the others, which is the content of the next lemma. A single vector  $v$  is independent if and only if  $v \neq \mathbf{0}$  (Proposition 8.2(4)); two vectors are dependent if and only if one is a scalar multiple of the other.

**Lemma 8.13** (Dependence and redundancy). *A list  $v_1, \dots, v_k$  with  $k \geq 1$  is linearly dependent if and only if some  $v_j$  is a linear combination of the others. Moreover, if  $v_1, \dots, v_k$  is independent but  $v_1, \dots, v_k, w$  is dependent, then  $w \in \text{span}(v_1, \dots, v_k)$ , and the coefficients expressing  $w$  are unique.*

*Proof.* Suppose first that  $v_j = \sum_{i \neq j} a_i v_i$ . Then

$$\sum_{i \neq j} a_i v_i + (-1) v_j = \mathbf{0}$$

is a dependence relation, its coefficient  $-1$  on  $v_j$  being nonzero (as  $1 \neq 0$  in a field). Conversely, given a dependence relation  $\sum_i a_i v_i = \mathbf{0}$  with  $a_j \neq 0$ , the field supplies  $a_j^{-1}$ , and solving for  $v_j$  gives  $v_j = -a_j^{-1} \sum_{i \neq j} a_i v_i$ , a linear combination of the others.

For the second statement, take a dependence relation  $\sum_i a_i v_i + bw = \mathbf{0}$  for the extended list. The coefficient  $b$  cannot be 0: otherwise the relation would be a nontrivial relation among  $v_1, \dots, v_k$  alone, contradicting their independence. Hence  $w = -b^{-1} \sum_i a_i v_i \in \text{span}(v_1, \dots, v_k)$ . For uniqueness, suppose  $w = \sum_i c_i v_i = \sum_i c'_i v_i$ ; subtracting,  $\sum_i (c_i - c'_i) v_i = \mathbf{0}$ , and independence forces  $c_i = c'_i$  for every  $i$ .  $\square$

### 8.3 Bases, dimension, and linear maps

**Definition 8.14** (Basis). A *basis* of  $V$  is a linearly independent set  $B$  that spans  $V$ . Every  $v \in V$  is then a *unique finite* linear combination of elements of  $B$ : existence because  $B$  spans, uniqueness because two expressions subtract to a relation among finitely many distinct elements of the independent set (as in Lemma 8.13). For a finite basis  $B = (b_1, \dots, b_n)$ , write

$$v = c_1 b_1 + \dots + c_n b_n, \quad [v]_B = (c_1, \dots, c_n),$$

and call the scalars  $c_i$  the *coordinates* of  $v$  in  $B$ .

*Example 8.15* (The standard and monomial bases). The *standard basis* of  $F^n$  is  $e_1, \dots, e_n$ , where  $e_i$  has a 1 in position  $i$  and 0 elsewhere. Both properties reduce to reading off coordinates: the combination  $\sum_i c_i e_i$  is the tuple  $(c_1, \dots, c_n)$ , so every vector is such a combination (spanning) and only the zero combination gives  $\mathbf{0}$  (independence). The *monomial basis* of  $F[X]_{<n}$  is  $1, X, \dots, X^{n-1}$ : a polynomial of degree less than  $n$  is, by construction, a linear combination of these monomials, and independence is the fact that polynomials are equal exactly when their coefficient sequences agree (Definition 5.1).

That all bases of a space have the same length rests on the following lemma, whose proof replaces the vectors of a spanning list, one at a time, by the vectors of an independent list.

**Lemma 8.16** (Steinitz exchange). *If  $v_1, \dots, v_m$  are linearly independent in  $V$  and  $w_1, \dots, w_n$  span  $V$ , then  $m \leq n$ .*

*Proof.* We prove by induction on  $k = 0, 1, \dots, m$  that, after a suitable renumbering of the  $w_j$ , the hybrid list

$$v_1, \dots, v_k, w_{k+1}, \dots, w_n$$

spans  $V$ . The case  $k = 0$  is the hypothesis on the  $w_j$ .

For the induction step, suppose the hybrid list at stage  $k < m$  spans  $V$ . In particular it expresses  $v_{k+1}$ :

$$v_{k+1} = \sum_{i \leq k} a_i v_i + \sum_{j > k} c_j w_j.$$

Some coefficient  $c_j$  with  $j > k$  must be nonzero: were they all zero,  $v_{k+1}$  would be a linear combination of  $v_1, \dots, v_k$ , and Lemma 8.13 would make the list  $v_1, \dots, v_{k+1}$  dependent, contradicting the independence of the  $v_i$ . (This step requires a slot  $j > k$  to exist, i.e.  $k < n$ ; if instead  $k = n$ , the hybrid list is  $v_1, \dots, v_n$  and the displayed equation already yields the same contradiction.) Renumber so that  $c_{k+1} \neq 0$ , and solve for the displaced vector:

$$w_{k+1} = c_{k+1}^{-1} \left( v_{k+1} - \sum_{i \leq k} a_i v_i - \sum_{j > k+1} c_j w_j \right).$$

Every vector of the stage- $k$  list therefore lies in the span of the stage- $(k+1)$  list  $v_1, \dots, v_{k+1}, w_{k+2}, \dots, w_n$ —the  $v_i$  and the remaining  $w_j$  trivially,  $w_{k+1}$  by the display—so by Proposition 8.10 the stage- $(k+1)$  list spans  $V$ , completing the induction.

Each step replaces one  $w$  by one  $v$ , so the induction reaches  $k = m$  only if there are at least  $m$  slots to replace:  $m \leq n$ .  $\square$

**Theorem 8.17** (Dimension). *Let  $V$  be a finitely generated  $F$ -vector space. Then  $V$  has a basis, and any two bases of  $V$  have the same number of elements. This common number is the dimension  $\dim_F V$  (written  $\dim V$  when the field is clear). In particular  $\dim_F F^n = n$  and  $\dim_F F[X]_{<n} = n$ .*

*Proof. Existence.* Let  $w_1, \dots, w_n$  be a finite spanning list. While the list is dependent, Lemma 8.13 exhibits some  $w_j$  in the span of the others; discarding it leaves the span unchanged, since any combination using  $w_j$  can be rewritten without it (substitute and regroup, Proposition 8.10). Each discard shortens the list, so after at most  $n$  discards the process halts at an independent spanning list—a basis. (If  $V$  is the zero space, the process discards everything and halts at the empty list, a basis of  $\{\mathbf{0}\}$  with  $\text{span}(\emptyset) = \{\mathbf{0}\}$ ; the dimension is 0.)

*Invariance.* Every basis is finite: if a basis contained more than the  $n$  elements of a finite spanning list, any  $n + 1$  of its elements would contradict Lemma 8.16. Let  $b_1, \dots, b_m$  and  $b'_1, \dots, b'_n$  be bases. Applying Lemma 8.16 with the first list as the independent one and the second as the spanning one gives  $m \leq n$ ; exchanging the roles gives  $n \leq m$ . Hence  $m = n$ .

The two named spaces have the explicit bases of Example 8.15, each of length  $n$ .  $\square$

Linear maps are the functions between vector spaces that preserve the linear structure; the Pedersen-style commitments of the series are such maps.

**Definition 8.18** (Linear map, kernel, image). A function  $T: V \rightarrow W$  between  $F$ -vector spaces is *linear* (an  *$F$ -linear map*) if

$$T(au + bv) = aT(u) + bT(v) \quad \text{for all } a, b \in F \text{ and } u, v \in V.$$

Its *kernel* and *image* are

$$\ker T = \{v \in V : T(v) = \mathbf{0}\} \subseteq V, \quad \text{im } T = \{T(v) : v \in V\} \subseteq W.$$

A bijective linear map is an *isomorphism*.

Three basic facts follow from the subspace criterion and linearity. First, taking  $a = b = 0$  gives  $T(\mathbf{0}) = \mathbf{0}$ , so  $\mathbf{0} \in \ker T$  and  $\mathbf{0} \in \text{im } T$ . Second, both kernel and image are subspaces: if  $u, v \in \ker T$  then  $T(au + bv) = aT(u) + bT(v) = \mathbf{0}$ , and if  $T(u), T(v) \in \text{im } T$  then  $aT(u) + bT(v) = T(au + bv) \in \text{im } T$ ; Proposition 8.6 applies to each. Third,  $T$  is injective if and only if  $\ker T = \{\mathbf{0}\}$ : if  $T(u) = T(v)$ , then  $T(u - v) = \mathbf{0}$  by linearity, so a trivial kernel forces  $u = v$ ; conversely an injective  $T$  sends only  $\mathbf{0}$  to  $T(\mathbf{0}) = \mathbf{0}$ . Moreover, for fixed  $V$  and  $W$  the set  $\text{Hom}(V, W)$  of all linear maps  $V \rightarrow W$  is itself an  $F$ -vector space under the pointwise operations of Example 8.4(2)—a sum or scalar multiple of linear maps is linear, so  $\text{Hom}(V, W)$  is a subspace of the space of all functions  $V \rightarrow W$ .

**Theorem 8.19** (Rank–nullity). *Let  $T: V \rightarrow W$  be a linear map with  $V$  finite-dimensional. Then*

$$\dim V = \dim \ker T + \dim \text{im } T.$$

*Proof.* The kernel is a subspace of the finite-dimensional  $V$ , and is itself finite-dimensional: an independent list in  $\ker T$  has length at most  $\dim V$  by Lemma 8.16 (tested against a basis of  $V$ ,

which spans), so a longest independent list  $u_1, \dots, u_k$  in  $\ker T$  exists, and it spans  $\ker T$ —any  $u \in \ker T$  outside its span would extend it to a longer independent list by Lemma 8.13. Thus  $u_1, \dots, u_k$  is a basis of  $\ker T$ .

Extend it to a basis of  $V$ : while the current independent list fails to span  $V$ , adjoin any vector outside its span—independence is preserved by Lemma 8.13—and the process terminates, again because independent lists cannot exceed  $\dim V$  in length (Lemma 8.16 against any basis of  $V$ ). Write the resulting basis as  $u_1, \dots, u_k, v_1, \dots, v_r$ , so that  $\dim V = k + r$ .

We claim  $T(v_1), \dots, T(v_r)$  is a basis of  $\operatorname{im} T$ ; the theorem follows, since then  $\dim \operatorname{im} T = r$  and  $\dim V = k + r = \dim \ker T + \dim \operatorname{im} T$ .

*Spanning.* Any element of  $\operatorname{im} T$  is  $T(v)$  for some  $v = \sum_i a_i u_i + \sum_j c_j v_j$ , and linearity with  $T(u_i) = \mathbf{0}$  gives  $T(v) = \sum_j c_j T(v_j)$ .

*Independence.* Suppose  $\sum_j c_j T(v_j) = \mathbf{0}$ . By linearity  $T(\sum_j c_j v_j) = \mathbf{0}$ , so  $\sum_j c_j v_j \in \ker T$ , and expressing this kernel element in the kernel basis gives  $\sum_j c_j v_j = \sum_i a_i u_i$  for some scalars  $a_i$ . Rearranged,  $\sum_i a_i u_i - \sum_j c_j v_j = \mathbf{0}$  is a linear relation among the combined basis of  $V$ , whose independence forces every coefficient—in particular every  $c_j$ —to vanish.  $\square$

Rank–nullity implies that a linear map from a higher-dimensional space into a lower-dimensional one has a nontrivial kernel—the image has dimension at most that of the target, so the kernel has dimension at least the difference—and therefore that distinct inputs collide. This collision is what makes vector commitments compressing and only *computationally* binding (Remark 8.29).

## 8.4 Bilinear forms and the inner product on $F^n$

Commitment and proof systems use expressions linear in *each* of two arguments, chiefly the inner product of two vectors. Over a finite field such a product has no length, angle or positivity, and the applications use only its algebraic properties, which the *bilinear form* captures.

**Definition 8.20** (Bilinear form). A *bilinear form* on an  $F$ -vector space  $V$  is a function  $\beta: V \times V \rightarrow F$  that is linear in each argument separately:

$$\beta(au + a'u', v) = a\beta(u, v) + a'\beta(u', v), \quad \beta(u, bv + b'v') = b\beta(u, v) + b'\beta(u, v'),$$

for all  $u, u', v, v' \in V$  and  $a, a', b, b' \in F$ . The form is *symmetric* if  $\beta(u, v) = \beta(v, u)$  for all  $u, v$ , and *nondegenerate* if  $\beta(u, v) = 0$  for all  $v$  implies  $u = \mathbf{0}$ .

Relative to a fixed basis  $b_1, \dots, b_n$  of  $V$ , a bilinear form is determined by finitely many scalars: expanding both arguments in the basis and applying bilinearity termwise,  $\beta(u, v) = \sum_{i,j} \beta(b_i, b_j) [u]_i [v]_j$ , so  $\beta$  is represented by the unique  $n \times n$  array  $M$  with entries  $M_{ij} = \beta(b_i, b_j)$ , via  $\beta(u, v) = [u]_B^T M [v]_B$  in matrix shorthand. The representation is not used further; only the special case with  $M$  the identity array is needed below.

**Definition 8.21** (Standard inner product on  $F^n$ ). The *standard inner product* (or *dot product*)

on  $F^n$  is

$$\langle \mathbf{a}, \mathbf{b} \rangle = \sum_{i=1}^n a_i b_i = \mathbf{a}^T \mathbf{b}, \quad \mathbf{a} = (a_1, \dots, a_n), \mathbf{b} = (b_1, \dots, b_n).$$

Bilinearity holds because each term  $a_i b_i$  is linear in each factor, and symmetry because multiplication in  $F$  commutes; the representing array in the standard basis is the identity ( $\langle e_i, e_j \rangle$  is 1 if  $i = j$  and 0 otherwise). The form is nondegenerate: if  $\langle \mathbf{a}, \mathbf{b} \rangle = 0$  for every  $\mathbf{b}$ , then in particular  $a_i = \langle \mathbf{a}, e_i \rangle = 0$  for each  $i$ , so  $\mathbf{a} = \mathbf{0}$ .

*Remark 8.22* (No positivity over finite fields). Over  $\mathbb{R}$  the standard inner product is *positive definite*:  $\langle \mathbf{a}, \mathbf{a} \rangle = \sum_i a_i^2 \geq 0$ , with equality only at  $\mathbf{0}$ . Positivity is what yields the Euclidean norm  $\|\mathbf{a}\| = \sqrt{\langle \mathbf{a}, \mathbf{a} \rangle}$  and the Cauchy–Schwarz inequality  $|\langle \mathbf{a}, \mathbf{b} \rangle| \leq \|\mathbf{a}\| \|\mathbf{b}\|$ —the geometric structure of real inner products. Over a finite field  $\mathbb{F}_q$  there is no ordering compatible with the arithmetic, so “ $\geq 0$ ” is meaningless, and this structure is lost: there exist nonzero *isotropic* vectors  $\mathbf{a} \neq \mathbf{0}$  with  $\langle \mathbf{a}, \mathbf{a} \rangle = 0$ . Over  $\mathbb{F}_5$ , for instance, the vector  $\mathbf{a} = (1, 2)$  has

$$\langle \mathbf{a}, \mathbf{a} \rangle = 1^2 + 2^2 = 1 + 4 = 5 = 0$$

in  $\mathbb{F}_5$ . The arguments of proof systems use only the algebraic properties that remain—bilinearity, symmetry, and nondegeneracy—so the absence of positivity does not affect them.

Nondegeneracy identifies vectors with the scalar-valued linear maps on them.

**Proposition 8.23** (Functionals via the inner product). *For each  $\mathbf{a} \in F^n$ , the map  $\mathbf{b} \mapsto \langle \mathbf{a}, \mathbf{b} \rangle$  is a linear map  $F^n \rightarrow F$  (a linear functional). The assignment*

$$F^n \longrightarrow \text{Hom}(F^n, F), \quad \mathbf{a} \longmapsto \langle \mathbf{a}, \cdot \rangle,$$

*is an isomorphism of  $F$ -vector spaces onto the dual space  $\text{Hom}(F^n, F)$  of all linear functionals on  $F^n$ . In particular both spaces have dimension  $n$ .*

*Proof.* Each  $\langle \mathbf{a}, \cdot \rangle$  is linear in  $\mathbf{b}$  by bilinearity, and the assignment itself is linear in  $\mathbf{a}$  for the same reason, mapping into the vector space  $\text{Hom}(F^n, F)$  of Definition 8.18. It is injective: if  $\langle \mathbf{a}, \cdot \rangle$  is the zero functional, then  $\langle \mathbf{a}, \mathbf{b} \rangle = 0$  for every  $\mathbf{b}$ , and nondegeneracy (Definition 8.21) gives  $\mathbf{a} = \mathbf{0}$ , so the kernel is trivial. It is surjective: given a linear functional  $\varphi$ , set  $a_i = \varphi(e_i)$  and  $\mathbf{a} = (a_1, \dots, a_n)$ ; then for every  $\mathbf{b} = \sum_i b_i e_i$ , linearity of  $\varphi$  gives

$$\varphi(\mathbf{b}) = \sum_i b_i \varphi(e_i) = \sum_i a_i b_i = \langle \mathbf{a}, \mathbf{b} \rangle,$$

so  $\varphi = \langle \mathbf{a}, \cdot \rangle$ . A bijective linear map is an isomorphism, and  $\dim \text{Hom}(F^n, F) = \dim F^n = n$  follows since an isomorphism carries a basis to a basis (its inverse is linear, and both directions preserve spanning and independence).  $\square$

*Remark 8.24* (Polynomial evaluation as an inner product). By Proposition 8.23, the notions “a linear functional on  $F^n$ ” and “a vector in  $F^n$ ” are interchangeable. For a fixed point  $z \in F$ , the map sending a polynomial to its value at  $z$  is linear in the coefficients, hence an inner product against a fixed vector: for  $p \in F[X]_{<n}$  with coefficient vector  $\mathbf{c} = (c_0, c_1, \dots, c_{n-1})$ ,

$$p(z) = c_0 + c_1 z + \dots + c_{n-1} z^{n-1} = \langle \mathbf{c}, (1, z, z^2, \dots, z^{n-1}) \rangle.$$

Evaluating a *committed* polynomial at a challenge point is therefore an inner-product claim about the committed coefficient vector, which is the form of statement an inner-product argument proves (Remark 8.32).

## 8.5 The mixed inner product with group elements

The commitments of Halo 2 and Orchard pair a vector of *scalars* with a vector of *group elements*. This subsection develops their algebra over an abstract abelian group; the group used in practice, the points of an elliptic curve, is constructed in the elliptic-curve section later in the volume, and the results apply to it unchanged.

**Construction 8.25** (Scalar multiplication by  $\mathbb{F}_r$  on a prime-order group). Let  $(G, +)$  be a finite abelian group of prime order  $r$ , written additively with identity  $\mathcal{O}$ . For an integer  $a \in \mathbb{Z}$  and  $G \in G$ , write  $[a]G$  for the  $a$ -fold multiple of  $G$ —the additive reading of the powers of §3:  $[0]G = \mathcal{O}$ ,  $[a]G = G + \dots + G$  with  $a$  summands for  $a > 0$ , and  $[-a]G = -([a]G)$ . The brackets separate the scalar notationally from the group element. The multiple  $[a]G$  depends only on the residue of  $a$  modulo  $r$ : the order of  $G$  divides  $|G| = r$  (Corollary 3.30), and  $[a]G = [b]G$  whenever  $a \equiv b \pmod{\text{ord}(G)}$  (Proposition 3.12(2), read additively), so congruence modulo  $r$  suffices. The operation therefore *descends* to scalars in the field  $\mathbb{F}_r = \mathbb{Z}/r\mathbb{Z}$  (Corollary 4.26): for  $a \in \mathbb{F}_r$  and  $G \in G$ , define  $[a]G$  using any integer representative of  $a$ . The rest of the section uses this descended operation, which scales group elements by elements of a *field*.

**Proposition 8.26** ( $G$  as an  $\mathbb{F}_r$ -vector space). *Let  $G$  be an abelian group of prime order  $r$ , with the scalar multiplication of Construction 8.25. Then  $G$  is an  $\mathbb{F}_r$ -vector space of dimension 1; any element  $G \neq \mathcal{O}$  is a basis, and every element of  $G$  is  $[a]G$  for a unique  $a \in \mathbb{F}_r$ .*

*Proof.* The vector-space axioms are the standard laws of multiples. For integers  $a, b$ , the identities  $[a + b]G = [a]G + [b]G$  and  $[a]([b]G) = [ab]G$  are the exponent laws of §3 in additive notation,  $[1]G = G$  is the definition, and  $[a](G + H) = [a]G + [a]H$  holds in any *abelian* group: for  $a > 0$  it is the rearrangement of  $a$  summands  $G + H$  into  $a$  summands  $G$  followed by  $a$  summands  $H$ , valid by commutativity and associativity (formally, induction on  $a$ ), and the cases  $a \leq 0$  follow by negating. Each law respects reduction modulo  $r$  by Construction 8.25, so the four scalar axioms of Definition 8.1 hold with scalars in  $\mathbb{F}_r$ , and  $(G, +)$  is an abelian group by hypothesis:  $G$  is an  $\mathbb{F}_r$ -vector space.

Now let  $G \neq \mathcal{O}$ . By Lagrange's theorem (Theorem 3.29, via Corollary 3.30) the order of  $G$  divides the prime  $r$ , and it exceeds 1 since  $G$  is not the identity; hence  $\text{ord}(G) = r$ . The  $r$  multiples  $[0]G, [1]G, \dots, [r-1]G$  are then pairwise distinct (Proposition 3.12(2), additively:  $[a]G = [b]G$  forces  $a \equiv b \pmod{r}$ ), and  $G$  has exactly  $r$  elements, so the multiples exhaust  $G$ . Thus  $\{G\}$  spans; and it is independent, since  $[a]G = \mathcal{O}$  forces  $a = 0$  in  $\mathbb{F}_r$ , again because  $\text{ord}(G) = r$ . A one-element basis gives  $\dim_{\mathbb{F}_r} G = 1$ , and the uniqueness of  $a$  in  $[a]G$  is the uniqueness of coordinates in a basis (Definition 8.14).  $\square$

**Definition 8.27** (Mixed inner product, multiscalar multiplication). Let  $\mathbf{G} = (G_1, \dots, G_n) \in G^n$  be a fixed tuple of group elements and  $\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{F}_r^n$  a vector of scalars. Their *mixed in-*

*ner product*—in computational contexts a *multiscalar multiplication* (MSM), in cryptographic ones a *Pedersen-style inner product*—is the group element

$$\langle \mathbf{a}, \mathbf{G} \rangle = \sum_{i=1}^n [a_i] G_i \in \mathbb{G}.$$

The angle-bracket notation of Definition 8.21 is reused; the type of the second argument—scalar vector or vector of group elements—determines which product is meant.

**Proposition 8.28** (Bilinearity of the mixed product). *Fix  $\mathbf{G} \in \mathbb{G}^n$ . The map  $\mathbf{a} \mapsto \langle \mathbf{a}, \mathbf{G} \rangle$  is  $\mathbb{F}_r$ -linear from  $\mathbb{F}_r^n$  to  $\mathbb{G}$ :*

$$\langle c\mathbf{a} + c'\mathbf{a}', \mathbf{G} \rangle = [c]\langle \mathbf{a}, \mathbf{G} \rangle + [c']\langle \mathbf{a}', \mathbf{G} \rangle \quad \text{for all } c, c' \in \mathbb{F}_r \text{ and } \mathbf{a}, \mathbf{a}' \in \mathbb{F}_r^n.$$

*Symmetrically, for fixed  $\mathbf{a} \in \mathbb{F}_r^n$  the map  $\mathbf{G} \mapsto \langle \mathbf{a}, \mathbf{G} \rangle$  is a group homomorphism  $\mathbb{G}^n \rightarrow \mathbb{G}$ , and is  $\mathbb{F}_r$ -linear when  $\mathbb{G}^n$  is viewed as an  $\mathbb{F}_r$ -vector space (componentwise, as in Example 8.3 with  $\mathbb{G}$  in place of  $F$ ). Thus  $\langle \cdot, \cdot \rangle: \mathbb{F}_r^n \times \mathbb{G}^n \rightarrow \mathbb{G}$  is bilinear over  $\mathbb{F}_r$ .*

*Proof.* Both statements are the vector-space laws of  $\mathbb{G}$  (Proposition 8.26) applied componentwise. For the first,

$$\langle c\mathbf{a} + c'\mathbf{a}', \mathbf{G} \rangle = \sum_i [ca_i + c'a'_i] G_i = \sum_i ([c][a_i]G_i + [c'][a'_i]G_i) = [c] \sum_i [a_i]G_i + [c'] \sum_i [a'_i]G_i,$$

using  $[a + b]G = [a]G + [b]G$  and  $[ab]G = [a]([b]G)$  in each component and then regrouping the sum in the abelian group  $\mathbb{G}$ . For the second, linearity in  $\mathbf{G}$  follows by the symmetric computation on the other slot of the defining sum, using  $[a](G + H) = [a]G + [a]H$  componentwise.  $\square$

**Remark 8.29** (The mixed product as a commitment). The element  $\langle \mathbf{a}, \mathbf{G} \rangle$  is a *commitment* to the scalar vector  $\mathbf{a}$  under the public “basis”  $\mathbf{G}$ , as at the opening of the section. Proposition 8.28 is its *homomorphic* property: the commitment to  $c\mathbf{a} + c'\mathbf{a}'$  is the  $[c]$ -,  $[c']$ -scaled combination of the individual commitments, computable by anyone holding those commitments and the scalars  $c, c'$ , without knowledge of  $\mathbf{a}$  or  $\mathbf{a}'$ . Provided the  $G_i$  are chosen so that no party can exhibit a nontrivial relation  $\sum_i [a_i]G_i = \mathcal{O}$ —a computational hypothesis, the *discrete-logarithm hardness assumption*, made precise in later volumes of this series—the commitment is *binding*: the map  $\mathbf{a} \mapsto \langle \mathbf{a}, \mathbf{G} \rangle$  is computationally injective, in the sense that no feasible computation produces two vectors with the same image. The map is not injective once  $n \geq 2$ : it is linear from the  $n$ -dimensional  $\mathbb{F}_r^n$  into the one-dimensional  $\mathbb{G}$  (Proposition 8.26), so by rank-nullity (Theorem 8.19) its kernel has dimension at least  $n - 1$ . Collisions therefore exist in large number; binding asserts only that they cannot be *found*, a computational rather than information-theoretic statement. The classical Pedersen commitment to a single value  $a$  with randomness  $\rho$  is the special case  $n = 2$ , with  $H \neq \mathcal{O}$ ,

$$[a]G + [\rho]H = \langle (a, \rho), (G, H) \rangle,$$

which is moreover *perfectly hiding*: for fixed  $a$ , as  $\rho$  ranges uniformly over  $\mathbb{F}_r$ , the term  $[\rho]H$  ranges uniformly over  $\mathbb{G}$  (it is a bijection of  $\mathbb{G}$ , by unique representation in the basis  $\{H\}$ ), so the commitment’s distribution is uniform and carries no information about  $a$ .

## 8.6 Linear algebra in commitment and proof systems

The following remarks locate the preceding results in commitment and proof systems. They are descriptive; later volumes develop each point rigorously.

*Remark 8.30* (Witnesses, constraints, and assignments). An arithmetic circuit over  $\mathbb{F}_q$  becomes, after *arithmetisation*—the encoding of a computation as a system of polynomial constraints on the wire values—a collection of equations that the vector  $\mathbf{z} \in \mathbb{F}_q^N$  of all wire values must satisfy. The *linear* part of such a system—the wiring (or “copy”) constraints of the PLONK family, which assert that certain wires carry equal values—consists of homogeneous linear equations in the entries of  $\mathbf{z}$ , organised in practice as arrays of coefficients (one row per equation) acting on the assignment vector. Satisfying the linear constraints confines  $\mathbf{z}$  to a subspace of  $\mathbb{F}_q^N$  (Example 8.7), or, when public inputs fix some entries, to a translate  $\mathbf{z}_0 + W$  of one—an *affine* subspace. The multiplicative constraints are handled separately, by the polynomial methods of the surrounding sections. Rank–nullity (Theorem 8.19) quantifies the degrees of freedom that remain after the linear constraints are imposed: the dimension of the witness space.

*Remark 8.31* (Commitments as linear maps). A Pedersen vector commitment  $\mathbf{a} \mapsto \langle \mathbf{a}, \mathbf{G} \rangle$  is a linear map  $\mathbb{F}_r^n \rightarrow \mathbf{G}$  (Proposition 8.28), and every algebraic manipulation a verifier performs rests on this linearity: taking known linear combinations of commitments, folding two commitment keys  $\mathbf{G}, \mathbf{G}'$  into  $\mathbf{G} + [x]\mathbf{G}'$  under a challenge  $x$ , and checking claimed openings by re-evaluating the same linear map. A polynomial commitment is the same map applied to a coefficient vector, and an *evaluation* of the committed polynomial at a point  $z$  is the scalar inner product  $\langle \mathbf{c}, (1, z, \dots, z^{n-1}) \rangle$  (Remark 8.24).

*Remark 8.32* (The inner-product relation). The core statement an inner-product argument proves has the form: “I know vectors  $\mathbf{a}, \mathbf{b} \in \mathbb{F}_r^n$  such that

$$P = \langle \mathbf{a}, \mathbf{G} \rangle + \langle \mathbf{b}, \mathbf{H} \rangle + [\langle \mathbf{a}, \mathbf{b} \rangle]U$$

for public  $\mathbf{G}, \mathbf{H} \in \mathbf{G}^n$  and  $U \in \mathbf{G}$ .” The equation combines three of this section’s constructions: two mixed inner products  $\langle \mathbf{a}, \mathbf{G} \rangle$  and  $\langle \mathbf{b}, \mathbf{H} \rangle$ , committing to the witness vectors; one scalar inner product  $\langle \mathbf{a}, \mathbf{b} \rangle$ , the value being argued; and the bilinearity of all three, which is what permits prover and verifier to *fold* a length- $n$  instance into a length- $n/2$  one and recurse. The soundness of the argument reduces to the nondegeneracy of these forms together with the hardness of finding kernel elements of the commitment map.

## 9 Roots of unity and evaluation domains

This section constructs the evaluation domains on which polynomial proof systems of the kind underlying Halo 2 and Zcash Orchard encode computations, and the fast Fourier transform on them. Such a proof system records an  $n$ -step computation as a table over a finite field, one column per register and one row per step, and convinces a verifier that every row follows from its predecessor without the verifier replaying the computation: by Lagrange interpolation (Theorem 5.30) each column is the value table on  $n$  fixed points of a polynomial of degree less than  $n$ , so conditions on the table become polynomial identities, checkable at a single random point by the Schwartz–Zippel lemma (Theorem 5.45). The powers of a primitive  $n$ -th root of unity form a point set

on which conditions on every row, shifts to the next row and the polynomials attached to the set are cheap to evaluate, and on which conversion between values and coefficients, and polynomial multiplication, cost far fewer than the  $n^2$  field operations of the schoolbook methods. Such a root exists in  $\mathbb{F}_q$  exactly when  $n \mid q - 1$  (Theorem 9.7); its powers form the evaluation domain  $H$ , whose vanishing polynomial is  $X^n - 1$  (Theorem 9.13) and whose Lagrange basis has a closed form (Theorem 9.22); evaluation on  $H$  is the number-theoretic transform, which turns polynomial multiplication into entrywise multiplication of value vectors (Theorem 9.32); and the fast Fourier transform computes it in  $O(n \log n)$  operations (Theorem 9.37).

Throughout,  $\mathbb{F}$  denotes a field. When  $\mathbb{F}$  is to be finite, the notation  $\mathbb{F}_q$  denotes the field with  $q$  elements (so  $q$  is a prime power, Theorem 6.9); when we intend a prime field specifically we write  $\mathbb{F}_p$ . The multiplicative group of nonzero elements is  $\mathbb{F}^\times = \mathbb{F} \setminus \{0\}$  (Example 3.7). We reserve the letter  $n$  for the order of the domain constructed below, with  $n \geq 1$ .

## 9.1 Roots of unity

**Definition 9.1** (Root of unity). Let  $\mathbb{F}$  be a field and  $n \in \mathbb{N}_{>0}$ . An element  $\omega \in \mathbb{F}$  is an  $n$ -th root of unity if  $\omega^n = 1$ . We write

$$\mu_n(\mathbb{F}) = \{ \omega \in \mathbb{F} : \omega^n = 1 \}$$

for the set of all  $n$ -th roots of unity in  $\mathbb{F}$ .

Since  $0^n = 0 \neq 1$ , we have  $0 \notin \mu_n(\mathbb{F})$ ; every  $n$ -th root of unity is therefore a unit, i.e. lies in  $\mathbb{F}^\times$ .

**Proposition 9.2.** The set  $\mu_n(\mathbb{F})$  is a finite subgroup of  $\mathbb{F}^\times$  of order at most  $n$ .

*Proof.* The identity 1 satisfies  $1^n = 1$ , so  $1 \in \mu_n(\mathbb{F})$ . If  $\omega, \zeta \in \mu_n(\mathbb{F})$ , then  $(\omega\zeta)^n = \omega^n \zeta^n = 1 \cdot 1 = 1$  because  $\mathbb{F}^\times$  is abelian, so  $\omega\zeta \in \mu_n(\mathbb{F})$ ; and  $(\omega^{-1})^n = (\omega^n)^{-1} = 1^{-1} = 1$ , so  $\omega^{-1} \in \mu_n(\mathbb{F})$ . Hence  $\mu_n(\mathbb{F})$  is a subgroup of  $\mathbb{F}^\times$  (Definition 3.14). For the order bound, every element of  $\mu_n(\mathbb{F})$  is a root of the polynomial  $X^n - 1 \in \mathbb{F}[X]$ , which is nonzero of degree  $n$  and so has at most  $n$  roots in  $\mathbb{F}$  (Theorem 5.18). Thus  $|\mu_n(\mathbb{F})| \leq n$ .  $\square$

The bound  $|\mu_n(\mathbb{F})| \leq n$  need not be attained; it is attained exactly when  $\mathbb{F}$  contains a *primitive*  $n$ -th root of unity.

**Definition 9.3** (Primitive root of unity). An element  $\omega \in \mathbb{F}$  is a *primitive  $n$ -th root of unity* if  $\omega^n = 1$  and  $\omega^k \neq 1$  for every integer  $k$  with  $1 \leq k < n$ . Equivalently,  $\omega$  is a primitive  $n$ -th root of unity if its multiplicative order  $\text{ord}(\omega)$  (Definition 3.11) equals  $n$ .

The order  $\text{ord}(\omega)$  is the least positive integer  $d$  with  $\omega^d = 1$ ; it exists for any  $\omega \in \mu_n(\mathbb{F})$  because  $n$  itself is such an exponent and the positive integers are well-ordered. The two formulations in Definition 9.3 agree, since both state that  $n$  is the least positive exponent  $d$  with  $\omega^d = 1$ .

**Remark 9.4** (Primitive roots as roots of unity). The group-theoretic term “generator of  $\mathbb{F}_q^\times$ ” and the polynomial term “primitive  $(q - 1)$ -th root of unity” name the same elements, as announced in Remark 6.20: a primitive root of a finite field  $\mathbb{F}_q$ , that is, a generator of the cyclic group  $\mathbb{F}_q^\times$  in the sense of Definition 6.19, is an element of multiplicative order  $q - 1$ , which by Definition 9.3 is precisely a primitive  $(q - 1)$ -th root of unity.

**Lemma 9.5** (Order divides exponent). *Let  $\omega \in \mathbb{F}^\times$  have finite order  $d = \text{ord}(\omega)$ . Then for any integer  $m$  we have  $\omega^m = 1$  if and only if  $d \mid m$ . In particular,  $\omega$  is an  $n$ -th root of unity if and only if  $d \mid n$ , and  $\omega$  is a primitive  $n$ -th root of unity if and only if  $d = n$ .*

*Proof.* The first claim is Proposition 3.12(1), read in the abelian group  $\mathbb{F}^\times$ : if  $d \mid m$ , write  $m = dk$ , so  $\omega^m = (\omega^d)^k = 1$ ; conversely, division with remainder  $m = qd + r$ ,  $0 \leq r < d$ , gives  $1 = \omega^m = \omega^r$ , and minimality of  $d$  forces  $r = 0$ . The remaining clauses instantiate it:  $\omega^n = 1 \iff d \mid n$  is the case  $m = n$ , and primitivity is the case  $d = n$  by Definition 9.3.  $\square$

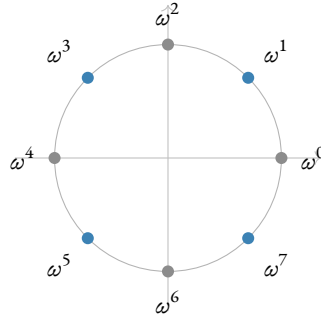
**Proposition 9.6.** *If  $\omega \in \mathbb{F}$  is a primitive  $n$ -th root of unity, then*

$$\mu_n(\mathbb{F}) = \langle \omega \rangle = \{1, \omega, \omega^2, \dots, \omega^{n-1}\}$$

*is a cyclic group of order exactly  $n$ , and these  $n$  powers are pairwise distinct.*

*Proof.* The powers  $1, \omega, \dots, \omega^{n-1}$  are pairwise distinct: if  $\omega^i = \omega^j$  with  $0 \leq i \leq j \leq n-1$ , then  $\omega^{j-i} = 1$  with  $0 \leq j-i < n$ , so primitivity (Lemma 9.5, with  $d = n$ ) forces  $j-i = 0$ . Thus  $\langle \omega \rangle$  has exactly  $n$  elements, all of which lie in  $\mu_n(\mathbb{F})$  since  $(\omega^k)^n = (\omega^n)^k = 1$ . But  $|\mu_n(\mathbb{F})| \leq n$  by Proposition 9.2, so the inclusion  $\langle \omega \rangle \subseteq \mu_n(\mathbb{F})$  is an equality of  $n$ -element sets.  $\square$

Over the complex numbers a primitive  $n$ -th root of unity always exists, namely  $\omega = e^{2\pi i/n}$ ; the  $n$ -th roots of unity are  $n$  equally spaced points on the unit circle, and the primitive ones lie at the angles  $2\pi k/n$  with  $\gcd(k, n) = 1$ . Over a finite field, the case used in cryptography, one exists only for the  $n$  determined in the next subsection.



**Figure 1.** The eight 8-th roots of unity in  $\mathbb{C}$ , equally spaced on the unit circle at angles  $2\pi k/8$ , with  $\omega = e^{2\pi i/8}$ . The four primitive ones ( $k = 1, 3, 5, 7$ , exactly the  $k$  coprime to 8; marked in colour) each generate the whole cycle; the grey points  $\omega^0 = 1, \omega^2 = i, \omega^4 = -1, \omega^6 = -i$  have orders 1, 4, 2, 4 and generate proper subgroups. Over a finite field the cyclic group structure persists without the geometry of the circle.

## 9.2 Existence over finite fields: the condition $n \mid q - 1$

The multiplicative group  $\mathbb{F}_q^\times$  of a finite field is cyclic of order  $q - 1$  (Theorem 6.17); the existence theorem for roots of unity rests on this fact.

**Theorem 9.7** (Existence and count of primitive  $n$ -th roots of unity). *Let  $\mathbb{F}_q$  be a finite field and  $n \in \mathbb{N}_{>0}$  with  $\gcd(n, q) = 1$  (automatic when  $n < p$ , where  $p = \text{char } \mathbb{F}_q$ ; equivalently, in characteristic  $p$ , exactly the condition  $p \nmid n$ ). Then:*

1. *The field  $\mathbb{F}_q$  contains a primitive  $n$ -th root of unity if and only if  $n \mid q - 1$ .*
2. *When  $n \mid q - 1$ , the group  $\mu_n(\mathbb{F}_q)$  is cyclic of order exactly  $n$ , and the number of primitive  $n$ -th roots of unity in  $\mathbb{F}_q$  is  $\varphi(n)$ , where  $\varphi$  is Euler's totient function (Definition 2.26).*

*Proof.* Let  $g$  be a generator of  $\mathbb{F}_q^\times$ , which is cyclic of order  $q - 1$  by Theorem 6.17.

( $\Rightarrow$ ) Suppose  $\omega$  is a primitive  $n$ -th root of unity in  $\mathbb{F}_q$ . Then  $\omega \in \mathbb{F}_q^\times$  has order  $n$  by Definition 9.3, and by Lagrange's theorem the order of an element divides the group order  $q - 1$  (Corollary 3.30). Hence  $n \mid q - 1$ .

( $\Leftarrow$ ) Suppose  $n \mid q - 1$ , say  $q - 1 = nm$ . Set  $\omega = g^m$ . Then  $\omega^n = g^{mn} = g^{q-1} = 1$ , so  $\omega \in \mu_n(\mathbb{F}_q)$ . Its order is

$$\text{ord}(\omega) = \text{ord}(g^m) = \frac{\text{ord}(g)}{\gcd(m, \text{ord}(g))} = \frac{q-1}{\gcd(m, q-1)} = \frac{q-1}{m} = n,$$

using the order-of-a-power formula (Corollary 3.23) together with  $m \mid q - 1$  (so  $\gcd(m, q - 1) = m$ ). Thus  $\omega$  is a primitive  $n$ -th root of unity, and by Proposition 9.6,  $\mu_n(\mathbb{F}_q) = \langle \omega \rangle$  is cyclic of order exactly  $n$ .

*Count.* In the cyclic group  $\mu_n(\mathbb{F}_q) = \langle \omega \rangle$  of order  $n$ , the element  $\omega^k$  has order  $n / \gcd(k, n)$ , again by Corollary 3.23. This equals  $n$  precisely when  $\gcd(k, n) = 1$ , and the number of  $k \in \{0, 1, \dots, n-1\}$  with  $\gcd(k, n) = 1$  is by definition  $\varphi(n)$  (Definition 2.26). Hence there are exactly  $\varphi(n)$  primitive  $n$ -th roots of unity.

(On the coprimality hypothesis. The condition  $\gcd(n, q) = 1$  is what guarantees that  $X^n - 1$  is separable— $\gcd(X^n - 1, nX^{n-1}) = 1$  when  $n \neq 0$  in  $\mathbb{F}_q$ —hence has  $n$  distinct roots in a splitting field over  $\mathbb{F}_q$  (Theorem 5.40); that all  $n$  of them lie in  $\mathbb{F}_q$  itself is exactly the condition  $n \mid q - 1$ . If  $p = \text{char } \mathbb{F}_q$  divided  $n$ , then  $X^n - 1$  would have repeated roots and  $\mu_n(\mathbb{F}_q)$  would be strictly smaller. But  $n \mid q - 1$  already forces  $\gcd(n, q) = 1$ , since  $\gcd(q - 1, q) = 1$ , so the hypothesis is subsumed.)  $\square$

*Remark 9.8.* The proof shows that for a finite field the statement reduces to  $\mathbb{F}_q$  has a primitive  $n$ -th root of unity  $\iff n \mid q - 1$ , with no separate coprimality condition, because  $n \mid q - 1$  makes  $n$  coprime to  $q$ ; the theorem states the condition  $\gcd(n, q) = 1$  only to exhibit the role of the characteristic.

*Example 9.9.* Take  $\mathbb{F}_q = \mathbb{F}_p$  with  $p = 7$ . Here  $q - 1 = 6$ , and 3 is a generator of  $\mathbb{F}_7^\times$ : its successive powers are 3, 2, 6, 4, 5, 1, all six nonzero residues (Example 6.21). For  $n = 3 \mid 6$ , set  $m = 6/3 = 2$  and  $\omega = 3^2 = 2$ . Since  $2^1 = 2$ ,  $2^2 = 4$  and  $2^3 = 8 \equiv 1$ , the element  $\omega = 2$  has order 3 and  $\mu_3(\mathbb{F}_7) = \{1, 2, 4\}$ . There are  $\varphi(3) = 2$  primitive cube roots of unity, namely 2 and 4. By contrast  $4 \nmid 6$ , so  $\mathbb{F}_7$  has no primitive 4-th root of unity:  $X^4 - 1 = (X^2 - 1)(X^2 + 1)$ , and  $X^2 + 1$  is irreducible over  $\mathbb{F}_7$  because it has no root there,  $-1$  not being a square modulo 7.

*Remark 9.10* (Two-adicity and FFT-friendly primes). For the fast algorithms below,  $n$  should be a power of two, or at least *smooth*—a product of small primes, ideally with many factors of two. By Theorem 9.7, a primitive  $2^k$ -th root of unity exists exactly when  $2^k \mid q - 1$ . The largest  $k$  with  $2^k \mid q - 1$  is called the *two-adicity* of the field. Designers choose primes  $p$  of high two-adicity,  $p - 1 = 2^k \cdot c$  for large  $k$ , as scalar fields of proof systems so that large smooth evaluation domains exist. Both Pasta primes—those underlying the Pallas and Vesta curves of Section 10—satisfy  $2^{32} \mid p - 1$ , with two-adicity exactly 32 (the primes are specified in §5.4.9.6, “Pallas and Vesta”, of the Zcash protocol specification), so each Pasta field contains primitive  $2^k$ -th roots of unity for every  $k \leq 32$ , hence power-of-two evaluation domains of every size up to  $2^{32}$ ; every construction of this section therefore applies over the fields used by Halo 2 and Orchard.

### 9.3 The evaluation domain $H$ and its vanishing polynomial

**Definition 9.11** (Evaluation domain). Let  $\mathbb{F}$  be a field containing a primitive  $n$ -th root of unity  $\omega$ . The *evaluation domain* of order  $n$  is the multiplicative subgroup

$$H = \langle \omega \rangle = \{1, \omega, \omega^2, \dots, \omega^{n-1}\} = \mu_n(\mathbb{F}) \subseteq \mathbb{F}^\times,$$

a cyclic group of order  $n$  by Proposition 9.6. We index its elements as  $\omega^0, \omega^1, \dots, \omega^{n-1}$ , and abbreviate  $\omega^i$  by  $b_i$  when convenient. Both  $\mathbb{F}$  and  $\omega$  remain fixed for the rest of the section.

**Definition 9.12** (Vanishing polynomial). The *vanishing polynomial* (or *zerofier*) of  $H$  is

$$Z_H(X) = \prod_{b \in H} (X - b) \in \mathbb{F}[X].$$

On a generic  $n$ -point set the vanishing polynomial (Remark 5.33) is a product of  $n$  linear factors with no simpler form; on the group  $H$  it reduces to a binomial.

**Theorem 9.13** (Closed form and factorisation of the vanishing polynomial). *With  $H = \mu_n(\mathbb{F})$  of order  $n$  generated by a primitive  $n$ -th root of unity  $\omega$ ,*

$$Z_H(X) = X^n - 1 = \prod_{i=0}^{n-1} (X - \omega^i).$$

*Moreover  $Z_H$  is separable: it has  $n$  distinct roots and  $\gcd(Z_H, Z'_H) = 1$ .*

*Proof.* Every  $b \in H$  satisfies  $b^n = 1$ , hence is a root of  $X^n - 1$ . The  $n$  elements of  $H$  are distinct (Proposition 9.6), so  $X^n - 1$  has  $n$  distinct roots, namely all of  $H$ , and factors as  $X^n - 1 = c \prod_{b \in H} (X - b)$  for some constant  $c \in \mathbb{F}^\times$  by iterating the factor theorem (Proposition 5.36, with the degree leaving no room for further factors). Comparing leading coefficients—both  $X^n - 1$  and the product are monic—gives  $c = 1$ , whence  $X^n - 1 = \prod_{b \in H} (X - b) = Z_H(X)$ .

For separability, the formal derivative is  $Z'_H(X) = nX^{n-1}$  (Definition 5.37). The existence of the  $n$  distinct roots already forces  $\text{char } \mathbb{F} \nmid n$ : if  $\text{char } \mathbb{F} = \ell$  divided  $n$ , say  $n = \ell m$ , the freshman’s dream (Lemma 6.13) would give  $X^n - 1 = (X^m - 1)^\ell$ , a polynomial with at most  $m < n$  distinct

roots—a contradiction. Hence  $n \neq 0$  in  $\mathbb{F}$ , so  $nX^{n-1}$  is nonzero at every element of  $H$ , since each is nonzero. Thus it has no root in common with  $X^n - 1$ , so  $\gcd(Z_H, Z'_H) = 1$  and  $Z_H$  is separable (Theorem 5.40).  $\square$

The proof shows that  $\text{char } \mathbb{F} \nmid n$ , so that division by  $n$  is defined in  $\mathbb{F}$ . By the theorem, a polynomial  $f$  vanishes on all of  $H$  if and only if  $X^n - 1$  divides it, and  $Z_H$  is evaluated at any point by a single exponentiation  $X^n$ , not  $n$  multiplications.

**Proposition 9.14** (Cosets and shifts). *Let  $\gamma \in \mathbb{F}^\times$  and let  $\gamma H = \{\gamma h : h \in H\}$  be the corresponding coset (Definition 3.27). Then the polynomial vanishing on  $\gamma H$  is*

$$Z_{\gamma H}(X) = \prod_{b \in H} (X - \gamma b) = X^n - \gamma^n.$$

*In particular, distinct cosets of  $H$  in  $\mathbb{F}^\times$  have vanishing polynomials  $X^n - c$  for distinct constants  $c = \gamma^n$ , and the cosets of  $H$  partition  $\mathbb{F}^\times$  (Lemma 3.28) into translates of  $H$ , on each of which  $X^n$  is constant.*

*Proof.* Substitute  $Y = X/\gamma$ :

$$\prod_{b \in H} (X - \gamma b) = \gamma^n \prod_{b \in H} (Y - b) = \gamma^n (Y^n - 1) = \gamma^n ((X/\gamma)^n - 1) = X^n - \gamma^n,$$

using Theorem 9.13 for the middle equality. The constant is  $c = \gamma^n$ , and

$$\gamma_1 H = \gamma_2 H \iff \gamma_1/\gamma_2 \in H \iff (\gamma_1/\gamma_2)^n = 1 \iff \gamma_1^n = \gamma_2^n,$$

so distinct cosets give distinct constants  $c$ .  $\square$

**Lemma 9.15** (Finite geometric series). *Let  $\mathbb{F}$  be a field,  $\zeta \in \mathbb{F}$  with  $\zeta \neq 1$ , and  $n \geq 1$ . Then*

$$\sum_{i=0}^{n-1} \zeta^i = \frac{\zeta^n - 1}{\zeta - 1}.$$

*Proof.* Write  $S = 1 + \zeta + \zeta^2 + \dots + \zeta^{n-1}$ . Multiplying by  $\zeta$  raises the exponent of every term by one:

$$\zeta S = \zeta + \zeta^2 + \dots + \zeta^{n-1} + \zeta^n.$$

In the difference  $\zeta S - S$ , the terms  $\zeta, \zeta^2, \dots, \zeta^{n-1}$  appear in both lines and cancel; only the two extreme terms remain:

$$(\zeta - 1)S = \zeta S - S = \zeta^n - 1.$$

Since  $\zeta \neq 1$ , the factor  $\zeta - 1$  is invertible, and dividing by it yields the claim.  $\square$

**Proposition 9.16** (Sum of roots of unity). *If  $n > 1$  then  $\sum_{i=0}^{n-1} \omega^i = 0$ , and more generally for any integer  $k$ ,*

$$\sum_{i=0}^{n-1} \omega^{ki} = \begin{cases} n & \text{if } n \mid k, \\ 0 & \text{if } n \nmid k. \end{cases}$$

*Proof.* If  $n \mid k$ , then  $\omega^{ki} = (\omega^n)^{ki/n} = 1$  for each  $i$ , so the sum is  $n$ . If  $n \nmid k$ , set  $\zeta = \omega^k \neq 1$  (since  $\omega$  has order  $n$ , we have  $\omega^k = 1 \iff n \mid k$  by Lemma 9.5), and Lemma 9.15 gives

$$\sum_{i=0}^{n-1} \zeta^i = \frac{\zeta^n - 1}{\zeta - 1} = \frac{(\omega^n)^k - 1}{\zeta - 1} = \frac{1 - 1}{\zeta - 1} = 0,$$

the denominator being nonzero because  $\zeta \neq 1$ . The first claim is the case  $k = 1$ , valid for  $n > 1$  since then  $n \nmid 1$ .  $\square$

*Remark 9.17* (Vanishing by rotation invariance). For  $n \nmid k$  the vanishing also follows without the geometric-series formula. The  $n$  terms of  $\sum_i \omega^{ki}$  run through the subgroup  $\langle \zeta \rangle$  generated by  $\zeta = \omega^k$ , each element appearing equally often; multiplying the sum by  $\zeta$  permutes the terms cyclically, so the sum satisfies  $\zeta S = S$ , and  $(\zeta - 1)S = 0$  with  $\zeta \neq 1$  forces  $S = 0$ . Figure 1 shows the complex case, in which rotation permutes the equally spaced points.

*Remark 9.18* (Why “orthogonality”). Call two vectors in  $\mathbb{F}^n$  *orthogonal* when their standard inner product (Definition 8.21) vanishes. For each frequency  $k$  let  $v_k = (1, \omega^k, \omega^{2k}, \dots, \omega^{(n-1)k})$  be its vector of powers, and let  $\bar{v}_k = (1, \omega^{-k}, \omega^{-2k}, \dots, \omega^{-(n-1)k})$  be the same construction with  $\omega^{-1}$  in place of  $\omega$ : the *mirror* of  $v_k$ , which traverses the cycle in the opposite direction and takes the role of the complex conjugate in the classical setting (on the unit circle  $\bar{z} = z^{-1}$ ). Then  $\langle \bar{v}_k, v_{k'} \rangle = \sum_i \omega^{i(k'-k)}$ , which Proposition 9.16 evaluates to 0 for  $k \neq k'$  and to  $n$  for  $k = k'$ : distinct frequency vectors are orthogonal, and each pairs with its own mirror to  $n$ . (Which factor carries the mirror is a convention: with the roles exchanged the exponent is  $k - k'$ , and the criterion  $n \mid k' - k$  is symmetric; the order chosen matches the product  $\bar{V}V$  computed in Theorem 9.29.) This is the finite-field counterpart of the orthogonality of the exponentials  $\theta \mapsto e^{2\pi i k \theta}$  in classical Fourier analysis, with the sum over the cycle in place of the integral over the circle. The proof of the inverse transform (Theorem 9.29) rests on it, the  $v_k$  being the columns of the transform matrix.

## 9.4 The Lagrange basis on $H$

Since  $H$  has exactly  $n$  points, values on  $H$  determine polynomials of degree less than  $n$  (Corollary 5.20); the basis adapted to values on  $H$  is the Lagrange basis of §5.7.

**Definition 9.19.** Let  $\mathbb{F}[X]_{<n}$  denote the  $\mathbb{F}$ -vector space of polynomials of degree strictly less than  $n$ , together with the zero polynomial. It has the monomial basis  $\{1, X, \dots, X^{n-1}\}$  (Definition 8.14), so  $\dim_{\mathbb{F}} \mathbb{F}[X]_{<n} = n$  (Theorem 8.17).

**Definition 9.20.** For each  $i \in \{0, 1, \dots, n-1\}$ , the  $i$ -th *Lagrange basis polynomial* for  $H = \{\omega^0, \dots, \omega^{n-1}\}$  is

$$L_i(X) = \prod_{\substack{j=0 \\ j \neq i}}^{n-1} \frac{X - \omega^j}{\omega^i - \omega^j} \in \mathbb{F}[X]_{<n},$$

the basis polynomial  $\ell_i$  of Theorem 5.30 for the nodes  $\omega^0, \dots, \omega^{n-1}$ .

**Proposition 9.21** (Interpolation property). *The polynomials  $L_0, \dots, L_{n-1}$  satisfy the Kronecker-delta conditions*

$$L_i(\omega^j) = \delta_{ij} = \begin{cases} 1 & i = j, \\ 0 & i \neq j, \end{cases}$$

*they form a basis of  $\mathbb{F}[X]_{<n}$  and they give the unique interpolant: for any data  $(v_0, \dots, v_{n-1}) \in \mathbb{F}^n$ , the polynomial*

$$f(X) = \sum_{i=0}^{n-1} v_i L_i(X)$$

*is the unique element of  $\mathbb{F}[X]_{<n}$  with  $f(\omega^i) = v_i$  for all  $i$ .*

*Proof.* Each  $L_i$  is well defined and of degree  $n - 1$  because the nodes  $\omega^j$  are distinct (Proposition 9.6), so the denominators  $\omega^i - \omega^j$  with  $j \neq i$  are nonzero. The delta conditions, the interpolation  $f(\omega^j) = v_j$ , and the uniqueness are Theorem 5.30 (with (4)) applied to these nodes with  $d = n - 1$ ; uniqueness rests, as there, on the root count: two interpolants in  $\mathbb{F}[X]_{<n}$  differ by a polynomial of degree  $< n$  with  $n$  distinct roots, which is the zero polynomial (Theorem 5.18). It remains to see that the  $L_i$  form a basis of  $\mathbb{F}[X]_{<n}$ . They are linearly independent: if  $\sum_i c_i L_i = 0$ , evaluation at  $\omega^j$  yields  $c_j = 0$  for every  $j$  (Definition 8.11). Being  $n$  independent vectors in the  $n$ -dimensional space  $\mathbb{F}[X]_{<n}$  (Definition 9.19), they form a basis (Theorem 8.17).  $\square$

The root-count step recurs and is called the *root-count device*: two polynomials of degree  $< n$  that agree on all  $n$  points of  $H$  are equal. It gives the uniqueness above, Proposition 9.24, and Example 9.25 below.

On a generic point set the Lagrange polynomials admit no closed form simpler than the defining product; on the *group*  $H$  they are expressed through the vanishing polynomial.

**Theorem 9.22** (Closed form of the Lagrange basis on  $H$ ). *For  $H = \mu_n(\mathbb{F})$  with primitive root  $\omega$ , and for each  $i$ ,*

$$L_i(X) = \frac{\omega^i}{n} \cdot \frac{X^n - 1}{X - \omega^i} = \frac{\omega^i (X^n - 1)}{n (X - \omega^i)}.$$

*In particular, dividing out one linear factor and scaling yields all  $n$  Lagrange polynomials from the single polynomial  $Z_H(X) = X^n - 1$ .*

*Proof.* Fix  $i$ . From Theorem 9.13,  $Z_H(X) = \prod_{j=0}^{n-1} (X - \omega^j)$ , so

$$\frac{Z_H(X)}{X - \omega^i} = \prod_{\substack{j=0 \\ j \neq i}}^{n-1} (X - \omega^j),$$

which is exactly the numerator of  $L_i$  in Definition 9.20. It remains to evaluate the denominator  $\prod_{j \neq i} (\omega^i - \omega^j)$ . Writing  $Z_H = (X - \omega^i) g$  with  $g = \prod_{j \neq i} (X - \omega^j)$ , the product rule (Proposition 5.38) gives  $Z'_H = g + (X - \omega^i) g'$ , so

$$Z'_H(\omega^i) = g(\omega^i) = \prod_{\substack{j=0 \\ j \neq i}}^{n-1} (\omega^i - \omega^j)$$

—the derivative at a simple root equals the product of the differences to the other roots, the computation already made for simple roots in the proof of Theorem 5.40. Concretely  $Z'_H(X) = nX^{n-1}$ , so

$$Z'_H(\omega^i) = n(\omega^i)^{n-1} = n\omega^{in}\omega^{-i} = n \cdot 1 \cdot \omega^{-i} = \frac{n}{\omega^i},$$

using  $\omega^{in} = (\omega^n)^i = 1$ . Therefore

$$L_i(X) = \frac{Z_H(X)/(X - \omega^i)}{Z'_H(\omega^i)} = \frac{(X^n - 1)/(X - \omega^i)}{n/\omega^i} = \frac{\omega^i(X^n - 1)}{n(X - \omega^i)},$$

as claimed.  $\square$

*Remark 9.23* (Barycentric form). The closed form gives the *barycentric* interpolation formula on  $H$  a compact form. Writing  $f(X) = \sum_i v_i L_i(X)$  and substituting Theorem 9.22,

$$f(X) = \frac{X^n - 1}{n} \sum_{i=0}^{n-1} \frac{\omega^i v_i}{X - \omega^i}.$$

Evaluating  $f$  at a point  $z \notin H$  thus costs one evaluation of  $z^n - 1$  and  $n$  terms of the sum, with no per-point polynomial reconstruction. A verifier can use this formula when the values  $v_i$  are known, as for public-input polynomials; a commitment to hidden values does not by itself permit this evaluation.

**Proposition 9.24** (Lagrange polynomials sum to one). *The Lagrange basis polynomials for  $H$  satisfy  $\sum_{i=0}^{n-1} L_i(X) = 1$  identically.*

*Proof.* The polynomial  $g(X) = \sum_i L_i(X) - 1$  lies in  $\mathbb{F}[X]_{<n}$  and vanishes at every  $\omega^j$ , because  $\sum_i L_i(\omega^j) = \sum_i \delta_{ij} = 1$  (Proposition 9.21). By the root-count device—a polynomial of degree  $< n$  with  $n$  distinct roots is zero (Theorem 5.18)— $g \equiv 0$ .  $\square$

*Example 9.25* (The all-ones value vector). The constant polynomial 1 interpolates the data  $v_i = 1$  for all  $i$ ; Proposition 9.24 is the statement  $\sum_i 1 \cdot L_i = 1$ . Likewise the values of  $X$  on  $H$  are  $v_i = \omega^i$ , and, for  $n \geq 2$ , Proposition 9.21 gives  $\sum_i \omega^i L_i(X) = X$ : both sides have degree  $< n$  and agree on all of  $H$ , so the root-count device applies. These identities are used to rewrite monomials in the Lagrange basis.

## 9.5 Evaluation and interpolation as mutually inverse linear maps

An element of  $\mathbb{F}[X]_{<n}$  is described by its  $n$  coefficients or by its  $n$  values on  $H$ ; passage between these two coordinate systems is a linear isomorphism, the specialisation to  $H$  of the evaluation isomorphism for general nodes in Remark 5.32.

**Definition 9.26** (Evaluation map). The *evaluation map* on  $H$  is the  $\mathbb{F}$ -linear map (Definition 8.18)

$$\text{ev}_H: \mathbb{F}[X]_{<n} \longrightarrow \mathbb{F}^n, \quad \text{ev}_H(f) = (f(\omega^0), f(\omega^1), \dots, f(\omega^{n-1})).$$

Linearity is immediate from  $(af + bg)(\omega^i) = af(\omega^i) + bg(\omega^i)$ , an instance of Proposition 5.16.

**Theorem 9.27** (Evaluation and interpolation are mutually inverse). *The evaluation map  $\text{ev}_H$  is a vector-space isomorphism. Its inverse is the interpolation map*

$$\text{int}_H: \mathbb{F}^n \longrightarrow \mathbb{F}[X]_{<n}, \quad \text{int}_H(v_0, \dots, v_{n-1}) = \sum_{i=0}^{n-1} v_i L_i(X).$$

*That is,  $\text{int}_H \circ \text{ev}_H = \text{id}_{\mathbb{F}[X]_{<n}}$  and  $\text{ev}_H \circ \text{int}_H = \text{id}_{\mathbb{F}^n}$ .*

*Proof.* Both maps are linear and act between spaces of the same finite dimension  $n$ , so verifying one composition would suffice (a one-sided inverse makes  $\text{ev}_H$  injective, hence surjective by rank-nullity, Theorem 8.19); we verify both. For  $v \in \mathbb{F}^n$ , the polynomial  $\text{int}_H(v) = \sum_i v_i L_i$  has value  $\sum_i v_i L_i(\omega^j) = v_j$  at  $\omega^j$  by Proposition 9.21, so  $\text{ev}_H(\text{int}_H(v)) = v$ . For  $f \in \mathbb{F}[X]_{<n}$ , the polynomial  $\text{int}_H(\text{ev}_H(f)) = \sum_i f(\omega^i) L_i$  is, by the uniqueness clause of Proposition 9.21, the unique degree- $< n$  polynomial taking the values  $f(\omega^i)$  on  $H$ —which is  $f$  itself. Hence the two maps are inverse; in particular  $\text{ev}_H$  is bijective and linear, i.e. an isomorphism (Definition 8.18).  $\square$

In coordinates,  $\text{ev}_H$  is given by a matrix, which identifies evaluation on  $H$  with the Fourier transform. Matrix notation is fixed as follows: an  $n \times n$  matrix over  $\mathbb{F}$  is a doubly indexed family  $A = (A_{ik})_{0 \leq i, k \leq n-1}$  of elements of  $\mathbb{F}$ , acting on a vector  $c \in \mathbb{F}^n$  by  $(Ac)_i = \sum_k A_{ik} c_k$ —the general form, written out through the standard basis (Example 8.15), of a linear map  $\mathbb{F}^n \rightarrow \mathbb{F}^n$ . The product of two matrices is defined so that  $(AB)c = A(Bc)$ , which forces  $(AB)_{ij} = \sum_k A_{ik} B_{kj}$ ; the identity matrix  $I$ , with entries  $\delta_{ik}$ , acts as the identity map; and  $A$  is invertible if some matrix  $A^{-1}$  satisfies  $A^{-1}A = AA^{-1} = I$ .

**Definition 9.28** (Vandermonde / DFT matrix). Let  $f(X) = \sum_{k=0}^{n-1} c_k X^k$  have coefficient vector  $c = (c_0, \dots, c_{n-1})$ . Then  $\text{ev}_H(f)$  has  $i$ -th entry  $f(\omega^i) = \sum_k c_k \omega^{ik}$ . Thus, relative to the monomial basis on  $\mathbb{F}[X]_{<n}$  and the standard basis on  $\mathbb{F}^n$ , the *Vandermonde matrix*

$$V = (\omega^{ik})_{0 \leq i, k \leq n-1} = \begin{pmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \omega & \omega^2 & \dots & \omega^{n-1} \\ 1 & \omega^2 & \omega^4 & \dots & \omega^{2(n-1)} \\ \vdots & & & & \vdots \\ 1 & \omega^{n-1} & \omega^{2(n-1)} & \dots & \omega^{(n-1)^2} \end{pmatrix}$$

—rows indexed by the evaluation point  $\omega^i$ , columns by the power  $k$ —represents  $\text{ev}_H$ , so that  $\text{ev}_H(f) = Vc$ . We call  $V$  the (discrete Fourier) transform matrix on  $H$ .

**Theorem 9.29** (Invertibility and the inverse transform). *The matrix  $V$  of Definition 9.28 is invertible, with*

$$(V^{-1})_{k,i} = \frac{1}{n} \omega^{-ik}.$$

Equivalently,  $V^{-1} = \frac{1}{n}\bar{V}$ , where  $\bar{V} = (\omega^{-ik})_{i,k}$  is the transform matrix built from  $\omega^{-1}$  (itself a primitive  $n$ -th root of unity). Consequently the coefficient and value representations are related by the mutually inverse formulas

$$c_k = \frac{1}{n} \sum_{i=0}^{n-1} f(\omega^i) \omega^{-ik}, \quad f(\omega^i) = \sum_{k=0}^{n-1} c_k \omega^{ik}.$$

*Proof.* First,  $\omega^{-1} = \omega^{n-1}$  has order  $n$  by Corollary 3.23 (with  $\gcd(n-1, n) = 1$ ), so  $\bar{V}$  is again a matrix of the shape in Definition 9.28. We show  $\bar{V}V = nI$ . The  $(k, k')$  entry of  $\bar{V}V$  is

$$\sum_{i=0}^{n-1} \omega^{-ik} \omega^{ik'} = \sum_{i=0}^{n-1} \omega^{i(k'-k)}.$$

By Proposition 9.16 this sum is  $n$  if  $n \mid (k' - k)$  and 0 otherwise; for  $k, k' \in \{0, \dots, n-1\}$  we have  $n \mid (k' - k) \iff k = k'$ . Hence  $\bar{V}V = nI$ , and the same computation with the roles of the two factors exchanged gives  $V\bar{V} = nI$ . Since  $n \neq 0$  in  $\mathbb{F}$  (Theorem 9.13), we may divide by  $n$ :  $V^{-1} = \frac{1}{n}\bar{V}$ . The component formulas write out the matrix identities  $\text{ev}_H(f) = Vc$  and  $c = V^{-1} \text{ev}_H(f)$ .  $\square$

*Remark 9.30.* Theorem 9.27 and Theorem 9.29 state one isomorphism in abstract and in matrix form. Since the inverse of  $V$  is, up to the scalar  $1/n$ , the same matrix with  $\omega$  replaced by  $\omega^{-1}$ , the fast inverse transform of Theorem 9.37 costs as much as the forward one.

## 9.6 The number-theoretic transform and convolution

The map  $\text{ev}_H$ , read as the matrix  $V$ , is the discrete Fourier transform over a finite field rather than over  $\mathbb{C}$ ; since it involves no analytic structure (no complex exponentials, no convergence), it is called the *number-theoretic transform*.

**Definition 9.31** (Number-theoretic transform). Let  $\omega$  be a primitive  $n$ -th root of unity in  $\mathbb{F}$ . The *number-theoretic transform* (NTT) sends the coefficient vector of a polynomial to its value vector on  $H$ : it is the evaluation map  $\text{ev}_H$  written in coordinates. Explicitly, the NTT of  $a = (a_0, \dots, a_{n-1}) \in \mathbb{F}^n$  is  $\hat{a} = (\hat{a}_0, \dots, \hat{a}_{n-1})$  with

$$\hat{a}_j = \sum_{k=0}^{n-1} a_k \omega^{jk}, \quad j = 0, \dots, n-1,$$

each entry being the value at  $\omega^j$  of the polynomial with coefficients  $a$ . The *inverse number-theoretic transform* is interpolation—the recovery of the coefficients from the values—and by Theorem 9.29 it too is a single explicit sum:

$$a_k = \frac{1}{n} \sum_{j=0}^{n-1} \hat{a}_j \omega^{-jk}.$$

The NTT has the convolution property of the classical DFT, the basis of fast polynomial multiplication. The coefficients of a product are a *convolution*: with  $f = \sum_i a_i X^i$  and  $g = \sum_j b_j X^j$ , the coefficient of  $X^k$  in  $fg$  is  $\sum_{i+j=k} a_i b_j$  (Definition 5.3). The *cyclic* variant folds indices modulo  $n$ : for  $a, b \in \mathbb{F}^n$ , the cyclic convolution  $a * b \in \mathbb{F}^n$  has entries

$$(a * b)_k = \sum_{\substack{0 \leq i, j \leq n-1 \\ i+j \equiv k \pmod{n}}} a_i b_j, \quad k = 0, \dots, n-1.$$

**Theorem 9.32** (Convolution theorem). *For  $f, g \in \mathbb{F}[X]_{<n}$  with  $\deg f + \deg g < n$ , let  $c = fg$  be their product (of degree  $< n$ ). Then for every  $h \in H$ ,*

$$c(h) = f(h) g(h),$$

*i.e. pointwise multiplication of value vectors corresponds to polynomial multiplication. Equivalently, writing  $a, b \in \mathbb{F}^n$  for the coefficient vectors of  $f, g$  (padded with zeros up to length  $n$ ),*

$$\widehat{a * b}_j = \hat{a}_j \hat{b}_j \quad (j = 0, \dots, n-1),$$

*so the NTT turns convolution into entrywise multiplication.*

*Proof.* The first identity is the statement that evaluation at  $h$  is a ring homomorphism  $\mathbb{F}[X] \rightarrow \mathbb{F}$  (Proposition 5.16):  $(fg)(h) = f(h) g(h)$ . Since  $\deg(fg) < n$ , interpolation recovers the product  $c = fg$  from its values on the  $n$ -point set  $H$  (Proposition 9.21), so the entrywise product of the value vectors uniquely determines  $c$ .

The convolution statement is proved in three steps: (i) multiplication in  $\mathbb{F}[X]/(X^n - 1)$  is cyclic convolution of coefficient vectors; (ii) evaluation on  $H$  is a ring isomorphism from that quotient onto  $\mathbb{F}^n$  with entrywise operations; (iii) the degree bound rules out wrap-around, so the cyclic product is the ordinary one. For (i), work in the quotient ring  $\mathbb{F}[X]/(X^n - 1)$  (Definition 4.19 and Proposition 4.20, applied to the ideal of multiples of  $X^n - 1$ , Definition 4.17). Every class has a unique representative of degree  $< n$ , by division with remainder (Theorem 5.12); and multiplying two such representatives and reducing replaces each  $X^k$  with  $k \geq n$  by  $X^{k-n}$ , since  $X^n \equiv 1$ . The coefficient of  $X^k$  in the reduced product of the representatives with coefficient vectors  $a$  and  $b$  is therefore  $\sum_{i+j \equiv k \pmod{n}} a_i b_j = (a * b)_k$ : multiplication in  $\mathbb{F}[X]/(X^n - 1)$  is cyclic convolution of coefficient vectors.

Next consider the map  $[p] \mapsto \text{ev}_H(p)$  from  $\mathbb{F}[X]/(X^n - 1)$  to  $\mathbb{F}^n$ , where  $\mathbb{F}^n$  is a commutative ring under entrywise addition and multiplication (the ring laws hold in each coordinate because they hold in  $\mathbb{F}$ ), with multiplicative identity  $(1, \dots, 1)$ . The map is well defined: two representatives of a class differ by a multiple of  $X^n - 1$ , which vanishes on  $H$  (Theorem 9.13). It is a ring homomorphism (Definition 4.21), because evaluation at each point of  $H$  is one (Proposition 5.16) and the operations on  $\mathbb{F}^n$  are entrywise. And it is bijective: classes correspond to their degree- $< n$  representatives, on which  $\text{ev}_H$  is a bijection by Theorem 9.27. Hence it is a ring isomorphism  $\mathbb{F}[X]/(X^n - 1) \xrightarrow{\sim} \mathbb{F}^n$ , and it sends the product of the classes of  $f$  and  $g$ —coefficient vector  $a * b$ —to the entrywise product of  $\hat{a}$  and  $\hat{b}$ . This is the displayed identity. Finally, when  $\deg f + \deg g < n$ , the ordinary product  $fg$  already has degree  $< n$ , so no folding occurs, the cyclic and ordinary products coincide, and the statement computes  $fg$  itself.  $\square$

**Remark 9.33** (Computational significance). Theorem 9.32 reduces multiplying two degree- $< n/2$  polynomials—a quadratic-cost operation by the schoolbook method—to: transform both (NTT), multiply the value vectors entrywise ( $n$  multiplications), and transform back (inverse NTT). If the NTT runs in  $O(n \log n)$  operations, the whole multiplication costs  $O(n \log n)$ . The fast Fourier transform, constructed below, meets that hypothesis.

**Example 9.34** (The convolution theorem in  $\mathbb{F}_{17}$ ). In  $\mathbb{F}_{17}$ , the element  $\omega = 4$  has  $\omega^2 = 16 = -1$ , so it is a primitive 4-th root of unity with domain  $H = \{1, 4, 16, 13\}$ . To multiply  $f = 1 + 2X$  and  $g = 3 + X$ , pad both coefficient vectors to length 4 and transform:  $\hat{a} = (3, 9, 16, 10)$  and  $\hat{b} = (4, 7, 2, 16)$ , the values of  $f$  and of  $g$  on  $H$ . Multiplying entrywise gives  $(12, 12, 15, 7)$ , the values of  $fg$  on  $H$ , and the inverse transform returns  $(3, 7, 2, 0)$ : the coefficients of  $fg = 3 + 7X + 2X^2$ , agreeing with the schoolbook product. At  $n = 4$  nothing is saved; the example exhibits the three-step procedure, which saves work once the transforms themselves cost  $O(n \log n)$  (§9.7).

## 9.7 The fast Fourier transform

Computing  $\hat{a} = Va$  directly from Definition 9.28 costs  $\Theta(n^2)$  field multiplications: each of the  $n$  outputs is a sum of  $n$  products. The fast Fourier transform (FFT)—here a number-theoretic FFT—computes the same vector in  $O(n \log n)$  operations by a divide-and-conquer recursion, provided  $n$  is a power of two or, more generally, smooth (Remark 9.10).

**Theorem 9.35** (Cooley–Tukey radix-2 step). *Suppose  $n = 2m$  and  $\omega$  is a primitive  $n$ -th root of unity in  $\mathbb{F}$ . Given  $f(X) = \sum_{k=0}^{n-1} a_k X^k$ , split its coefficients by parity into*

$$f_{\text{even}}(Y) = \sum_{k=0}^{m-1} a_{2k} Y^k, \quad f_{\text{odd}}(Y) = \sum_{k=0}^{m-1} a_{2k+1} Y^k,$$

*so that  $f(X) = f_{\text{even}}(X^2) + X f_{\text{odd}}(X^2)$ . Then  $\eta := \omega^2$  is a primitive  $m$ -th root of unity, and for  $j = 0, \dots, m-1$ ,*

$$f(\omega^j) = f_{\text{even}}(\eta^j) + \omega^j f_{\text{odd}}(\eta^j), \quad (7)$$

$$f(\omega^{j+m}) = f_{\text{even}}(\eta^j) - \omega^j f_{\text{odd}}(\eta^j). \quad (8)$$

*Thus the length- $n$  NTT of  $a$  reduces to two length- $m$  NTTs (of the even and odd coefficient subsequences, over the domain  $\langle \eta \rangle$ ) plus  $O(n)$  extra operations.*

*Proof.* The identity  $f(X) = f_{\text{even}}(X^2) + X f_{\text{odd}}(X^2)$  regroups the sum  $\sum_k a_k X^k$  into even-index and odd-index terms; in the odd terms one factor of  $X$  is pulled out, leaving powers  $X^{2k}$ . That  $\eta = \omega^2$  is a primitive  $m$ -th root of unity follows from Lemma 9.5 and the order-of-a-power formula (Corollary 3.23):  $\text{ord}(\omega^2) = n / \gcd(2, n) = n/2 = m$ , using  $2 \mid n$ .

It remains to evaluate. For (7), put  $X = \omega^j$ ; then  $X^2 = \omega^{2j} = \eta^j$ , so  $f(\omega^j) = f_{\text{even}}(\eta^j) + \omega^j f_{\text{odd}}(\eta^j)$ . For (8), put  $X = \omega^{j+m}$ ; then  $X^2 = \omega^{2j+2m} = \omega^{2j} \omega^n = \eta^j \cdot 1 = \eta^j$  (since  $2m = n$  and  $\omega^n = 1$ ), while the linear factor is  $\omega^{j+m} = \omega^j \omega^m = -\omega^j$ . Here  $\omega^m = -1$ , because  $\omega^m$  has order  $n / \gcd(m, n) = n/m = 2$  (Corollary 3.23), and  $-1$  is the unique element of order 2 in a field:  $x^2 = 1$

with  $x \neq 1$  forces  $(x - 1)(x + 1) = 0$  and so  $x = -1$ ; and  $-1 \neq 1$ , since  $\text{char } \mathbb{F} \nmid n$  (Theorem 9.13) with  $n$  even forces  $\text{char } \mathbb{F} \neq 2$ . Substituting,  $f(\omega^{j+m}) = f_{\text{even}}(\eta^j) - \omega^j f_{\text{odd}}(\eta^j)$ .  $\square$

**Remark 9.36** (The butterfly). Equations (7)–(8) constitute the *butterfly*: from the two subtransform outputs  $E_j = f_{\text{even}}(\eta^j)$  and  $O_j = f_{\text{odd}}(\eta^j)$  one forms the *twiddle*  $t_j = \omega^j O_j$  and reads off the two outputs  $E_j + t_j$  and  $E_j - t_j$ . In a diagram,  $E_j$  and  $O_j$  are two nodes on the left, each joined to both outputs on the right, the two  $E_j$ -edges with weight 1 and the two  $O_j$ -edges with weights  $+\omega^j$  and  $-\omega^j$ ; the pair of crossing edges forms the  $\times$  after which the operation is named. Each butterfly costs one multiplication (the twiddle) and two additions, and there are  $n/2$  of them per level.

**Theorem 9.37** ( $O(n \log n)$  cost of the FFT). *Let  $n = 2^t$  and let  $T(n)$  be the number of field additions and multiplications the radix-2 recursion of Theorem 9.35 uses to compute a length- $n$  NTT. Then*

$$T(n) = 2T(n/2) + cn \quad (n > 1), \quad T(1) = O(1),$$

*for a constant  $c$ , and therefore  $T(n) = O(n \log n)$ . The inverse NTT has the same cost, since it is the forward NTT with  $\omega$  replaced by  $\omega^{-1}$ , followed by scaling each entry by  $1/n$  (Theorem 9.29).*

*Proof.* The recursion is exactly Theorem 9.35: a length- $n$  transform calls two length- $n/2$  transforms (on the even and odd coefficient subsequences) and combines them with  $n/2$  butterflies of  $O(1)$  operations each, for the  $cn$  overhead. Unrolling, with  $n = 2^t$ ,

$$T(2^t) = 2T(2^{t-1}) + c2^t.$$

Divide by  $2^t$  and set  $S(t) = T(2^t)/2^t$ : then  $S(t) = S(t-1) + c$ , so  $S(t) = S(0) + ct = O(1) + ct$ , and  $T(2^t) = 2^t S(t) = O(2^t) + ct2^t = O(t2^t)$ . With  $t = \log_2 n$  this is  $T(n) = O(n \log n)$ . (Equivalently: the recursion tree has  $\log_2 n$  levels, each performing  $\Theta(n)$  total work across all subproblems at that level.) The inverse-transform claim is Theorem 9.29: the inverse is the same Cooley–Tukey recursion with  $\omega^{-1}$  in place of  $\omega$  (also a primitive  $n$ -th root of unity), plus a final pass of  $n$  divisions by  $n$ , all within  $O(n \log n)$ .  $\square$

**Remark 9.38** (Mixed radix and smoothness). Fast transforms apply when  $n$  is smooth and the field contains a primitive  $n$ -th root of unity, that is,  $n \mid q - 1$  (Theorem 9.7); for a power-of-two  $n$  dividing  $q - 1$ , every level of the recursion is a radix-2 step.

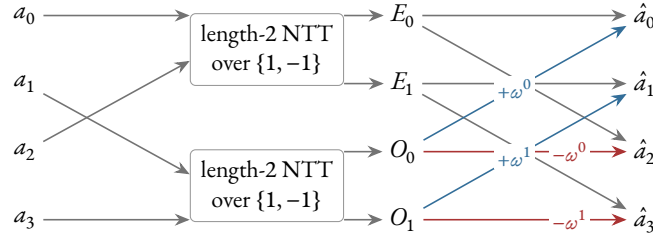
**Example 9.39** (A length-4 transform). Take  $n = 4$  with primitive 4-th root  $\omega$  (so  $\omega^2 = -1$  and  $\omega^4 = 1$ , and  $\eta = \omega^2 = -1$  is the primitive 2nd root). For  $f = a_0 + a_1X + a_2X^2 + a_3X^3$ , the two length-2 subtransforms over the domain  $\{1, -1\}$ —each itself a single butterfly, with twiddle  $\eta^0 = 1$ —are

$$E = (a_0 + a_2, a_0 - a_2), \quad O = (a_1 + a_3, a_1 - a_3),$$

and the butterflies give

$$\hat{a}_0 = E_0 + \omega^0 O_0, \quad \hat{a}_2 = E_0 - \omega^0 O_0, \quad \hat{a}_1 = E_1 + \omega^1 O_1, \quad \hat{a}_3 = E_1 - \omega^1 O_1.$$

(Outputs  $j$  and  $j + m$  are paired, as in (7)–(8).) This uses 2 subtransforms of 2 additions each plus 2 butterflies of 2 additions each ( $n/2 = 2$  per level, Remark 9.36)—together 2 twiddle multiplications and 8 additions—versus 16 coefficient-products and 12 additions for the direct  $4 \times 4$  matrix multiply; the difference grows with  $n$ .



**Figure 2.** The length-4 transform of Example 9.39 as a circuit. The inputs split by parity into two length-2 subtransforms, whose outputs  $E_0, E_1$  and  $O_0, O_1$  feed the two butterflies of Remark 9.36: each  $E_j$  is carried to outputs  $\hat{a}_j$  and  $\hat{a}_{j+2}$  with weight 1, each  $O_j$  with weights  $+\omega^j$  and  $-\omega^j$  (Eq. (7)–Eq. (8)); the crossing pair of weighted edges is the  $\times$  after which the butterfly is named.

## 9.8 The role of smooth multiplicative domains in polynomial proof systems

This subsection states how the preceding results are used in Halo 2, the proof system of Orchard, which encodes a computation as polynomial identities on the evaluation domain  $H = \mu_n$ : the  $n$ -step table described at the start of this section is indexed by the  $n$  points of  $H$ .

*Remark 9.40* (The roles played by  $H$ ,  $Z_H$ , the Lagrange basis, and the FFT).

- **Witnesses as evaluations on  $H$ .** A prover lays out the trace of a computation as the values of one or more polynomials on the  $n$  points of  $H$ . By Theorem 9.27 this is the same data as a polynomial of degree  $< n$ ; the prover may move freely between the value (Lagrange) representation, suited to stating constraints row by row, and the coefficient representation, suited to committing and to low-degree testing. The change of representation is the (inverse) NTT.
- **Constraints as divisibility by  $Z_H$ .** “Constraint polynomial  $C$  holds on every row” means  $C$  vanishes on all of  $H$ , which by Theorem 9.13 means exactly that  $Z_H = X^n - 1$  divides  $C$ . The prover demonstrates this by exhibiting a quotient  $t$  with  $C(X) = t(X)(X^n - 1)$ . The closed form  $Z_H(X) = X^n - 1$ , evaluated by one exponentiation, makes this check cheap at a random point. Proposition 9.14 extends the same technique to cosets  $\gamma H$ , used to separate quotient and remainder domains.
- **Lagrange selectors.** The Lagrange polynomials  $L_i$  of Definition 9.20 are *selectors*:  $L_i$  is 1 on row  $i$  and 0 elsewhere (Proposition 9.21), so boundary and instance constraints (e.g. “the first row equals the public input”) become  $L_0 \cdot (\dots)$ . The closed form and the barycentric formula (Theorem 9.22, Remark 9.23) let the verifier evaluate these at a random challenge in  $O(n)$  or even  $O(\log n)$  field operations rather than reconstructing polynomials.
- **Group structure for permutations and shifts.** Because  $H$  is a cyclic group (Definition 9.11), multiplication by  $\omega$  is the cyclic “next-row” shift  $\omega^i \mapsto \omega^{i+1}$ . Constraints relating a row to its neighbour become  $f(\omega X)$  versus  $f(X)$ ; permutation arguments, on which Halo 2’s copy constraints are built, rest on the action of the group on itself and on its cosets. An arbitrary set of  $n$  points has none of this structure.

- **Smoothness for speed.** Provers manipulate polynomials of size  $n$  (often after enlarging the domain by a small constant factor for low-degree testing). Transforms between representations, multiplications and coset evaluations are computed by NTTs, which by Theorem 9.37 cost  $O(n \log n)$  instead of  $O(n^2)$  (at the sizes used in practice, the difference between feasible and infeasible computation), provided  $n$  is smooth and  $\mathbb{F}_q$  contains a primitive  $n$ -th root of unity. By Theorem 9.7 that requires  $n \mid q - 1$ ; by Remark 9.10 this is why high-two-adicity primes are chosen for the scalar fields of the Pallas and Vesta curves (Section 10).

## 10 Elliptic curves

This section constructs elliptic-curve groups, the standard source of groups of roughly  $2^{255}$  elements for which no discrete-logarithm attack faster than the generic  $O(\sqrt{N})$  attacks of Remark 7.22 is known. The multiplicative groups  $\mathbb{F}_p^\times$  lack this property: the *index-calculus* algorithms exploit the factorisation of the integers behind the residues and run in subexponential time, so  $p$  must have *thousands* of bits to push the cheapest known attack beyond  $2^{128}$  operations. An *elliptic curve* is the solution set of a cubic equation in two variables together with one additional point, the “point at infinity”. The section proves that these points form an abelian group under a geometric rule (Theorem 10.16, associativity in sketch), derives explicit addition formulas, including complete ones valid for every input pair, bounds the group order by Hasse’s theorem (Theorem 10.22), and states the elliptic-curve discrete logarithm problem.

Throughout,  $K$  is a field whose characteristic (Definition 4.30) is neither 2 nor 3; in the applications  $K$  is a prime field  $\mathbb{F}_p$  with  $p$  large. The running example is the pair of *Pasta curves*

$$E_p : y^2 = x^3 + 5 \quad \text{over } \mathbb{F}_p, \quad E_q : y^2 = x^3 + 5 \quad \text{over } \mathbb{F}_q,$$

where  $p$  and  $q$  are the two 255-bit primes of Remark 6.23 and Remark 7.18,

$$\begin{aligned} p &= 2^{254} + 0x224698fc094cf91b992d30ed00000001, \\ q &= 2^{254} + 0x224698fc0994a8dd8c46eb2100000001. \end{aligned}$$

The primes and the curve equations are the deployed constants, specified in §5.4.9.6 (“Pallas and Vesta”) of the Zcash protocol specification. The curve  $E_p$  is called *Pallas* and  $E_q$  *Vesta*. By design they form a *cycle*: the number of  $\mathbb{F}_p$ -points of Pallas equals  $q$ , and the number of  $\mathbb{F}_q$ -points of Vesta equals  $p$ . A toy curve over  $\mathbb{F}_{11}$  is treated in parallel for hand computation.

### 10.1 Weierstrass equations

**Definition 10.1** (Short Weierstrass equation). Let  $K$  be a field of characteristic  $\neq 2, 3$ , and let  $a, b \in K$ . The *short Weierstrass equation* with coefficients  $a, b$  is the equation

$$y^2 = x^3 + ax + b. \tag{9}$$

The elements  $a$  and  $b$  are the *Weierstrass coefficients*.

Over a field of characteristic  $\neq 2, 3$ , an invertible affine change of variables reduces the most general cubic of this shape,  $y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$ , to the form (9): completing the square in  $y$  (possible since  $2 \neq 0$ ) removes the  $xy$  and  $y$  terms but changes the quadratic coefficient to  $a_2 + a_1^2/4$ ; translating with the substitution  $x = X - (a_2 + a_1^2/4)/3$  (possible since  $3 \neq 0$ ) removes the  $X^2$  term. Characteristics 2 and 3 are excluded for this reason: those cases require the longer Weierstrass form, and they do not arise for the Pasta curves, which are defined over large prime fields.

Cubics whose graph has a self-intersection or a cusp are excluded, because the group law defined below fails at such a singular point.

**Definition 10.2** (Discriminant and nonsingularity). The *discriminant* of the short Weierstrass equation (9) is

$$\Delta = -16(4a^3 + 27b^2) \in K. \quad (10)$$

The equation, and the cubic  $f(x) = x^3 + ax + b$ , is *nonsingular* (or *smooth*) if  $\Delta \neq 0$ , equivalently  $4a^3 + 27b^2 \neq 0$ . (The factor  $-16$  is a convention inherited from the general Weierstrass form; over the fields considered here it is nonzero and does not affect the condition.)

The following lemma expresses  $4a^3 + 27b^2$  through the roots of a *depressed cubic* (a cubic with no square term); the smoothness criterion is derived from it.

**Lemma 10.3** (Discriminant of a depressed cubic). Let  $f(x) = x^3 + ax + b = (x - r_1)(x - r_2)(x - r_3)$  with  $r_1, r_2, r_3$  in some field containing  $K$ . Then

$$\prod_{i < j} (r_i - r_j)^2 = -(4a^3 + 27b^2).$$

*Proof.* Expanding  $(x - r_1)(x - r_2)(x - r_3)$  and comparing coefficients with  $x^3 + ax + b$  gives Vieta's relations for the depressed cubic:

$$r_1 + r_2 + r_3 = 0, \quad r_1r_2 + r_1r_3 + r_2r_3 = a, \quad r_1r_2r_3 = -b.$$

Differentiating  $f(x) = \prod_i (x - r_i)$  by the product rule and evaluating at a root annihilates every term but one:  $f'(r_i) = \prod_{j \neq i} (r_i - r_j)$ . Multiplying the three evaluations pairs each factor  $(r_i - r_j)$  with  $(r_j - r_i)$ :

$$\prod_i f'(r_i) = \prod_{i < j} (r_i - r_j)(r_j - r_i) = (-1)^3 \prod_{i < j} (r_i - r_j)^2.$$

It remains to compute  $\prod_i f'(r_i) = \prod_i (3r_i^2 + a)$ . Expanding the product and collecting the elementary symmetric functions of the squares  $r_i^2$ ,

$$\prod_i (3r_i^2 + a) = 27 \prod_i r_i^2 + 9a \sum_{i < j} r_i^2 r_j^2 + 3a^2 \sum_i r_i^2 + a^3.$$

Vieta's relations evaluate each symmetric function: with  $e_1 = 0$ ,  $e_2 = a$ ,  $e_3 = -b$ ,

$$\sum_i r_i^2 = e_1^2 - 2e_2 = -2a, \quad \sum_{i < j} r_i^2 r_j^2 = e_2^2 - 2e_1 e_3 = a^2, \quad \prod_i r_i^2 = e_3^2 = b^2.$$

Substituting,  $\prod_i f'(r_i) = 27b^2 + 9a^3 - 6a^3 + a^3 = 4a^3 + 27b^2$ , and the two displays together give  $\prod_{i < j} (r_i - r_j)^2 = -(4a^3 + 27b^2)$ .  $\square$

**Proposition 10.4** (Equivalent conditions for smoothness). *Let  $f(x) = x^3 + ax + b \in K[x]$ . The following are equivalent.*

1. *The quantity  $4a^3 + 27b^2$  of Definition 10.2 is nonzero.*
2. *The cubic  $f$  has no repeated root in the algebraic closure  $\overline{K}$ —the smallest extension field of  $K$  in which every nonconstant polynomial over  $K$  factors into linear factors.*
3. *The polynomials  $f$  and  $f'$  are coprime in  $\overline{K}[x]$ .*
4. *There is no point  $(x_0, y_0) \in \overline{K}^2$  satisfying the curve equation  $y^2 = f(x)$  together with the two partial-derivative equations  $2y = 0$  and  $f'(x) = 0$ .*

*Proof.* Equivalence (1)  $\Leftrightarrow$  (2): over  $\overline{K}$  the monic cubic factors as  $f = (x - r_1)(x - r_2)(x - r_3)$ , and Lemma 10.3 gives  $\prod_{i < j} (r_i - r_j)^2 = -(4a^3 + 27b^2)$ . The left side vanishes exactly when two roots coincide, so  $f$  has a repeated root if and only if  $4a^3 + 27b^2 = 0$ .

Equivalence (2)  $\Leftrightarrow$  (3): this is the derivative test of Theorem 5.40, applied over  $\overline{K}$  where  $f$  splits: a repeated root of  $f$  is a common root of  $f$  and  $f'$ , hence a common factor  $(x - \alpha)$ , and conversely a nontrivial  $\gcd(f, f')$  has a root in  $\overline{K}$  which is then a repeated root of  $f$ .

Equivalence (4)  $\Leftrightarrow$  (2): a singular point of the affine curve  $y^2 = f(x)$  is a point where both partial derivatives of  $g(x, y) = y^2 - f(x)$  vanish, namely  $\partial g / \partial y = 2y = 0$  and  $\partial g / \partial x = -f'(x) = 0$ . Since  $\text{char}(K) \neq 2$ , the first equation forces  $y_0 = 0$ ; the curve equation then gives  $f(x_0) = 0$ , which combined with  $f'(x_0) = 0$  says  $x_0$  is a repeated root of  $f$  (Theorem 5.40(1)). Conversely a repeated root  $x_0$  of  $f$  yields the singular point  $(x_0, 0)$ .  $\square$

**Definition 10.5** (Elliptic curve and rational points). *An elliptic curve over  $K$  is a short Weierstrass equation (9) with nonzero discriminant,  $4a^3 + 27b^2 \neq 0$ . For any field extension  $L \supseteq K$ , the set of  $L$ -rational affine points is*

$$E^{\text{aff}}(L) = \{ (x, y) \in L \times L : y^2 = x^3 + ax + b \},$$

and the set of  $L$ -rational points is

$$E(L) = E^{\text{aff}}(L) \cup \{\mathcal{O}\},$$

where  $\mathcal{O}$  is a single formal symbol called the *point at infinity*. When  $L = K$  one writes  $E$  for the curve and  $E(K)$  for its rational points.

**Remark 10.6** (The Pasta curves are smooth). For the Pasta curves  $a = 0$  and  $b = 5$ , so  $4a^3 + 27b^2 = 27 \cdot 25 = 675 = 3^3 \cdot 5^2$ . The curve is therefore nonsingular whenever the characteristic does not divide 675, i.e. whenever the prime is neither 3 nor 5; the Pasta primes are far larger than 5, so the condition holds. More generally, any curve  $y^2 = x^3 + b$  with  $b \neq 0$  is nonsingular away from characteristics 2 and 3.

## 10.2 The point at infinity, geometrically

The point  $\mathcal{O}$  is the identity element of the group law defined below. To exhibit it as a point of the curve, embed the affine plane in the *projective plane*. A point of the projective plane  $\mathbb{P}^2(K)$  is an equivalence class  $[X : Y : Z]$  of nonzero triples  $(X, Y, Z) \in K^3 \setminus \{0\}$ , where  $(X, Y, Z) \sim (\lambda X, \lambda Y, \lambda Z)$  for every  $\lambda \in K^\times$ . The affine points  $(x, y)$  embed as  $[x : y : 1]$ ; the points with  $Z = 0$  form the *line at infinity*, the extra “directions”. Homogenising the Weierstrass equation (9) by setting  $x = X/Z, y = Y/Z$  and clearing denominators gives the *projective Weierstrass equation*

$$Y^2Z = X^3 + aXZ^2 + bZ^3. \quad (11)$$

Setting  $Z = 0$  in (11) forces  $X^3 = 0$ , hence  $X = 0$ , leaving the *single* projective point  $[0 : 1 : 0]$ . This is the point at infinity  $\mathcal{O}$ . It is the point at infinity in the vertical direction, and every vertical line  $x = \text{const}$  passes through it.

## 10.3 The chord-and-tangent group law: geometry

The geometric definition of addition rests on one fact: a line meets a cubic curve in exactly three points, counted with multiplicity, once the point at infinity is included.

**Definition 10.7** (The group law, geometric form). Let  $P, Q \in E(K)$ . Define  $P + Q$  as follows.

- If  $P = \mathcal{O}$ , set  $P + Q = Q$ ; if  $Q = \mathcal{O}$ , set  $P + Q = P$ .
- Otherwise  $P = (x_1, y_1)$  and  $Q = (x_2, y_2)$  are affine. Draw the line  $\ell$  through  $P$  and  $Q$ ; if  $P = Q$ , let  $\ell$  be the tangent line to the curve at  $P$ . The line  $\ell$  meets the curve in a third point  $R'$ , counted with multiplicity, where a vertical  $\ell$  has third point  $\mathcal{O}$ . Define  $P + Q$  to be the *reflection of  $R'$  across the  $x$ -axis*: if  $R' = (x_3, -y_3)$  then  $P + Q = (x_3, y_3)$ , and if  $R' = \mathcal{O}$  then  $P + Q = \mathcal{O}$ .

The reflection turns the rule that *the three intersection points of any line with the curve add up to  $\mathcal{O}$*  into a group law with identity  $\mathcal{O}$ : reflection across the  $x$ -axis sends an affine point  $(x, y)$  to  $(x, -y)$ , its additive inverse by the next subsection, and fixes  $\mathcal{O}$ . The rule is stated here as a heuristic; every use of it below is discharged by explicit computation with the formulas of §10.4, which obtain the third intersection point from Vieta’s relations, as in Lemma 10.3.

## 10.4 Explicit affine formulas

Fix an elliptic curve  $E : y^2 = x^3 + ax + b$  over  $K$ , and let  $P = (x_1, y_1)$  and  $Q = (x_2, y_2)$  denote affine points.

**Definition 10.8** (Negation). Define

$$-\mathcal{O} = \mathcal{O}, \quad -(x_1, y_1) = (x_1, -y_1).$$

The reflected point  $(x_1, -y_1)$  is again on the curve, since the curve equation depends on  $y$  only through  $y^2$ . Geometrically, the line through  $P$  and  $-P$  is the vertical line  $x = x_1$ , which meets the curve at  $P, -P$ , and  $\mathcal{O}$ .

**Construction 10.9** (Chord addition). Suppose  $P, Q \neq \mathcal{O}$  and  $x_1 \neq x_2$ . The line through  $P$  and  $Q$  is  $y = \lambda x + \mu$  with

$$\lambda = \frac{y_2 - y_1}{x_2 - x_1}, \quad \mu = y_1 - \lambda x_1.$$

Substituting the line into the curve equation,  $(\lambda x + \mu)^2 = x^3 + ax + b$ , and collecting terms gives

$$g(x) := x^3 - \lambda^2 x^2 + (a - 2\lambda\mu)x + (b - \mu^2) = 0.$$

Both  $x_1$  and  $x_2$  are roots of the monic cubic  $g$ : each point lies on the line and on the curve, so  $g(x_i) = f(x_i) - (\lambda x_i + \mu)^2 = f(x_i) - y_i^2 = 0$ . Since  $x_1 \neq x_2$ , factoring out  $(x - x_1)(x - x_2)$  (Theorem 5.17, applied twice) leaves a monic linear factor  $x - x_3$  with  $x_3 \in K$ : the  $x$ -coordinate of the third intersection point  $R'$ . By Vieta, the sum of the roots equals the negative of the  $x^2$ -coefficient—so  $x_1 + x_2 + x_3 = \lambda^2$ , which determines the third root without factoring. Hence

$$x_3 = \lambda^2 - x_1 - x_2, \quad y_3 = \lambda(x_1 - x_3) - y_1, \quad (12)$$

and  $P + Q = (x_3, y_3)$ . The  $y$ -formula already incorporates the reflection across the  $x$ -axis: the third intersection point is  $(x_3, \lambda x_3 + \mu)$ , and reflecting gives  $y_3 = -(\lambda x_3 + \mu) = -(\lambda x_3 + y_1 - \lambda x_1) = \lambda(x_1 - x_3) - y_1$ .

**Construction 10.10** (Tangent doubling). Suppose  $P = Q = (x_1, y_1)$  with  $y_1 \neq 0$ . Implicit differentiation of  $y^2 = x^3 + ax + b$  yields the tangent slope: from  $2y dy = (3x^2 + a) dx$ ,

$$\lambda = \frac{3x_1^2 + a}{2y_1}.$$

With  $\mu = y_1 - \lambda x_1$ , substituting  $y = \lambda x + \mu$  into the curve equation produces the same monic cubic  $g(x) = x^3 + ax + b - (\lambda x + \mu)^2$  as in Construction 10.9, and here  $x_1$  is a *double* root. The statement that “tangency counts the intersection twice” is verified directly:  $g(x_1) = f(x_1) - y_1^2 = 0$ , and

$$g'(x_1) = 3x_1^2 + a - 2\lambda(\lambda x_1 + \mu) = 3x_1^2 + a - 2\lambda y_1 = 0$$

by the choice of  $\lambda$ , so  $x_1$  is a repeated root of  $g$  (Theorem 5.40(1)). Factoring  $(x - x_1)^2$  out of  $g$  leaves a monic linear factor  $x - x_3$  with  $x_3 \in K$ , and Vieta now reads  $x_1 + x_1 + x_3 = \lambda^2$ , hence

$$x_3 = \lambda^2 - 2x_1, \quad y_3 = \lambda(x_1 - x_3) - y_1, \quad [2]P = (x_3, y_3). \quad (13)$$

The remaining cases are exactly those in which the line is vertical.

- If  $P = \mathcal{O}$ , then  $P + Q = Q$ ; symmetrically if  $Q = \mathcal{O}$ .
- If  $x_1 = x_2$  but  $y_1 = -y_2$  (so  $Q = -P$ , including the subcase  $y_1 = y_2 = 0$ ), the line is vertical, the third point is  $\mathcal{O}$ , and  $P + Q = \mathcal{O}$ .
- If  $P = Q$  and  $y_1 = 0$ , the tangent is vertical (the slope formula in Construction 10.10 has zero denominator), so  $[2]P = \mathcal{O}$ . These are exactly the points of order two, studied in Section 10.9.

Every case is covered: either an input is  $\mathcal{O}$ ; or both inputs are affine with  $x_1 \neq x_2$  (use (12)); or both are affine with  $x_1 = x_2$ , in which case either  $y_1 = -y_2$  (result  $\mathcal{O}$ ) or  $y_1 = y_2 \neq 0$  (use (13)).

**Proposition 10.11** (Closure). *For all  $P, Q \in E(K)$ , the point  $P + Q$  computed by the case analysis above lies in  $E(K)$ . In particular  $E(K)$  is closed under  $+$ .*

*Proof.* If either input is  $\mathcal{O}$ , the result is the other input, which lies in  $E(K)$  by hypothesis; if the result is  $\mathcal{O}$ , it lies in  $E(K)$  by definition. Otherwise  $(x_3, y_3)$  is produced by (12) or (13), and in both constructions  $x_3$  was exhibited as a root of the cubic  $g(x) = x^3 + ax + b - (\lambda x + \mu)^2$ —as the third root beside  $x_1, x_2$  in Construction 10.9, and beside the double root  $x_1$  in Construction 10.10. The equation  $g(x_3) = 0$  states precisely that the third intersection point  $(x_3, \lambda x_3 + \mu) = (x_3, -y_3)$  satisfies the curve equation; and since the curve equation is invariant under  $y \mapsto -y$ , the reflected point  $(x_3, y_3)$  is on the curve too, so  $P + Q \in E^{\text{aff}}(K) \subseteq E(K)$ . For rationality: all coordinates are rational expressions in  $x_1, x_2, y_1, y_2, a$  whose denominators are  $x_2 - x_1$  or  $2y_1$ , nonzero in the cases where each formula is invoked; so  $x_3, y_3 \in K$ .  $\square$

**Remark 10.12** (Complete versus incomplete addition). The chord formula (12) is *incomplete*: it fails whenever  $x_1 = x_2$ —the doubling and inverse cases—and an implementation must branch into the case analysis above. *Complete* addition formulas, valid for *all* input pairs including  $P = Q$  and  $P = -Q$ , exist for the curves used here, at the cost of more field operations per addition; Construction 10.13 exhibits the deployed set and Proposition 10.14 proves it complete. Later volumes depend on the distinction. Halo 2’s elliptic-curve gadgets provide complete addition and use incomplete gates only where their callers justify the exclusions. The Sinsemilla hash of Orchard deliberately uses the *incomplete* formulas nondeterministically: an exceptional case does not violate the constraints but leaves the affected output unconstrained (the prover may choose the intermediate slope freely), which the specification models by letting the hash return  $\perp$  and weakening the proved statement accordingly; its Merkle path validity check may pass on the resulting sentinel values. Security follows not from the proof system but from Theorem 5.4.4 of the Zcash protocol specification: any input reaching an exceptional case of the incomplete addition yields a nontrivial discrete-logarithm relation among the Sinsemilla generators, so under discrete-log hardness on Pallas (Remark 10.31) such inputs are infeasible to find, for honest and adversarial provers alike.

A circuit cannot branch. A proof system checks the group law by a fixed list of polynomial identities in the input, output, and auxiliary cells of a row, evaluated on every row alike, so each branch of the case analysis appears as a factor that vanishes exactly when the branch is not taken. The deployed gate uses five auxiliary cells, four of them “inverse-or-zero” hints that let a polynomial test a field element for zero.

**Construction 10.13** (Complete addition as deployed). Let  $E : y^2 = x^3 + b$  over  $\mathbb{F}_p$  with  $p > 3$ , and encode the point at infinity as the pair  $(0, 0)$ , so that every element of  $E(\mathbb{F}_p)$  is a pair of field elements. For  $z \in \mathbb{F}_p$  let  $\text{inv}_0(z)$  be  $z^{-1}$  if  $z \neq 0$  and 0 if  $z = 0$ . For encoded inputs  $P = (x_p, y_p)$

and  $Q = (x_q, y_q)$  the prover fills five auxiliary cells,

$$\alpha = \text{inv}_0(x_q - x_p), \quad \beta = \text{inv}_0(x_p), \quad \gamma = \text{inv}_0(x_q), \quad \delta = \begin{cases} \text{inv}_0(y_q + y_p) & \text{if } x_q = x_p, \\ 0 & \text{otherwise,} \end{cases}$$

$$\lambda = \begin{cases} (y_q - y_p) \alpha & \text{if } x_q \neq x_p, \\ 3x_p^2 (2y_p)^{-1} & \text{if } x_q = x_p \text{ and } y_p \neq 0, \\ 0 & \text{otherwise,} \end{cases}$$

and an output pair  $(x_r, y_r)$ : it is  $Q$  if  $x_p = 0$ ; else  $P$  if  $x_q = 0$ ; else  $(0, 0)$  if  $x_q = x_p$  and  $y_q = -y_p$ ; else  $x_r = \lambda^2 - x_p - x_q$  and  $y_r = \lambda(x_p - x_r) - y_p$ . The verifier accepts the row iff the twelve identities

$$(x_q - x_p)((x_q - x_p)\lambda - (y_q - y_p)) = 0, \quad (\text{C1})$$

$$(1 - (x_q - x_p)\alpha)(2y_p\lambda - 3x_p^2) = 0, \quad (\text{C2})$$

$$x_p x_q (x_q - x_p) (\lambda^2 - x_p - x_q - x_r) = 0, \quad (\text{C3a})$$

$$x_p x_q (x_q - x_p) (\lambda(x_p - x_r) - y_p - y_r) = 0, \quad (\text{C3b})$$

$$x_p x_q (y_q + y_p) (\lambda^2 - x_p - x_q - x_r) = 0, \quad (\text{C3c})$$

$$x_p x_q (y_q + y_p) (\lambda(x_p - x_r) - y_p - y_r) = 0, \quad (\text{C3d})$$

$$(1 - x_p \beta) (x_r - x_q) = 0, \quad (\text{C4a})$$

$$(1 - x_p \beta) (y_r - y_q) = 0, \quad (\text{C4b})$$

$$(1 - x_q \gamma) (x_r - x_p) = 0, \quad (\text{C5a})$$

$$(1 - x_q \gamma) (y_r - y_p) = 0, \quad (\text{C5b})$$

$$(1 - (x_q - x_p)\alpha - (y_q + y_p)\delta) x_r = 0, \quad (\text{C6a})$$

$$(1 - (x_q - x_p)\alpha - (y_q + y_p)\delta) y_r = 0 \quad (\text{C6b})$$

hold in  $\mathbb{F}_p$ . These are, in this order, the twelve constraints of the gate complete addition in the `halo2_gadgets` crate, each carrying there one further factor, a per-row cell equal to 1 on the rows where the gate applies and 0 elsewhere, which switches the list on or off; the crate assigns the cells as above, and the circuit field is the Pallas base field  $\mathbb{F}_p$ .

**Proposition 10.14** (The gate is complete and sound). *Suppose no affine point of  $E(\mathbb{F}_p)$  has  $x = 0$  and none has  $y = 0$ —for  $y^2 = x^3 + b$ , that  $b$  is a non-square and  $-b$  a non-cube in  $\mathbb{F}_p$ . Then for all  $P, Q \in E(\mathbb{F}_p)$ , encoded as in Construction 10.13:*

1. (completeness) *the prover's cells satisfy (C1)–(C6b), and the output pair encodes  $P + Q$ ;*
2. (soundness) *every  $(\lambda, \alpha, \beta, \gamma, \delta, x_r, y_r) \in \mathbb{F}_p^7$  satisfying (C1)–(C6b) has  $(x_r, y_r)$  equal to the encoding of  $P + Q$ .*

*This is what complete means for a gate: the twelve identities are satisfiable for every input pair—doublings, inverse pairs, and the identity included—and the output they admit is unique, so they*

are the graph of the total function  $+: E(\mathbb{F}_p) \times E(\mathbb{F}_p) \rightarrow E(\mathbb{F}_p)$ , with no case left to a branch outside the constraint system.

*Proof.* Under the hypothesis an encoded pair has  $x = 0$  iff it is  $\mathcal{O}$ , and two affine points with the same  $x$ -coordinate have  $y_q = \pm y_p$  with  $y_p \neq 0$ , so they are equal or inverse and not both. Every case uses one observation: for  $z \in \mathbb{F}_p$  and any cell value  $b$ , the factor  $1 - zb$  equals 1 when  $z = 0$ , whatever  $b$  is, while for  $z \neq 0$  the honest choice  $b = z^{-1}$  makes it 0. A constraint of the shape  $(1 - zb)t = 0$  therefore forces  $t = 0$  exactly when  $z = 0$  and is discharged by the honest hint otherwise. Five cases exhaust the input pairs.

*Case  $P = \mathcal{O}$ .* Then  $x_p = 0$ , so (C4a)–(C4b) force  $(x_r, y_r) = (x_q, y_q)$ , the encoding of  $\mathcal{O} + Q = Q$ ; that is soundness. The prover outputs  $Q$  by the first clause, and its cells satisfy the rest: (C1) reads  $x_q(x_q\lambda - y_q) = 0$  and holds for  $\lambda = y_q \text{ inv}_0(x_q)$ ; (C2) and (C3a)–(C3d) carry a factor  $x_p$  or  $2y_p\lambda - 3x_p^2$ , both zero; (C5a)–(C5b) are discharged by  $\gamma = x_q^{-1}$  if  $Q \neq \mathcal{O}$ , and read  $(x_r, y_r) = (0, 0) = (x_p, y_p)$  if  $Q = \mathcal{O}$ ; and (C6a)–(C6b) have first factor  $1 - x_q\alpha = 0$ , which is 0 for  $\alpha = x_q^{-1}$  when  $Q \neq \mathcal{O}$ , while for  $Q = \mathcal{O}$  the other factor  $x_r = y_r = 0$  vanishes.

*Case  $Q = \mathcal{O}, P \neq \mathcal{O}$ .* Then  $x_q = 0 \neq x_p$ , so (C5a)–(C5b) force  $(x_r, y_r) = (x_p, y_p)$ , the encoding of  $P$ : soundness. The prover outputs  $P$  by the second clause, with  $\alpha = (-x_p)^{-1}$  and  $\lambda = (-y_p)\alpha = y_p/x_p$ ; then (C1) reads  $(-x_p)(-x_p\lambda + y_p) = 0$  and holds; (C2) and (C6a)–(C6b) carry the factor  $1 - (x_q - x_p)\alpha = 0$ , the latter with  $\delta = 0$ ; (C3a)–(C3d) carry  $x_q = 0$ ; and (C4a)–(C4b) are discharged by  $\beta = x_p^{-1}$ .

*Case  $P, Q$  affine with  $x_q \neq x_p$ .* Constraint (C1) forces  $\lambda$  to be the chord slope  $(y_q - y_p)/(x_q - x_p)$ , and since  $x_p x_q (x_q - x_p) \neq 0$ , (C3a)–(C3b) force  $(x_r, y_r)$  to be the chord formula (12)—the encoding of  $P + Q$ , which is affine because  $x_p \neq x_q$ . That is soundness. The prover's  $\lambda$  is that slope and its output that formula, by the fourth clause; (C3c)–(C3d) contain the same two vanishing factors; (C2), (C4a)–(C4b), (C5a)–(C5b) are discharged by the honest  $\alpha, \beta, \gamma$ ; and (C6a)–(C6b) have first factor  $1 - 1 - 0 = 0$ , since  $\delta = 0$ .

*Case  $Q = -P, P$  affine.* Here  $x_q = x_p$  and  $y_q + y_p = 0$ , so both hint-bearing products in (C6a)–(C6b) are 0 whatever  $\alpha$  and  $\delta$  are, and the constraints force  $x_r = y_r = 0$ : the encoding of  $P + (-P) = \mathcal{O}$ , which is soundness. The prover outputs  $(0, 0)$  by the third clause; (C1) and (C3a)–(C3b) carry  $x_q - x_p = 0$ , and (C3c)–(C3d) carry  $y_q + y_p = 0$ ; (C2), whose first factor is now 1, needs  $2y_p\lambda = 3x_p^2$ , which the honest  $\lambda = 3x_p^2(2y_p)^{-1}$  supplies ( $y_p \neq 0$ ); and (C4a)–(C5b) are discharged by  $\beta = \gamma = x_p^{-1}$ .

*Case  $Q = P$  affine.* Again  $x_q = x_p$  fixes  $(x_q - x_p)\alpha = 0$ , so (C2) forces  $\lambda$  to be the tangent slope  $3x_p^2/(2y_p)$ , and since  $x_p x_q (y_q + y_p) = 2x_p^2 y_p \neq 0$ , (C3c)–(C3d) force  $(x_r, y_r)$  to be the doubling formula (13)—the encoding of  $[2]P$ , affine because  $y_p \neq 0$ . That is soundness. The prover's  $\lambda$  is that slope and its output that formula, by the fourth clause; (C1) and (C3a)–(C3b) carry  $x_q - x_p = 0$ ; (C4a)–(C5b) are discharged by  $\beta = \gamma = x_p^{-1}$ ; and (C6a)–(C6b) have first factor  $1 - 0 - 2y_p\delta = 0$  for the honest  $\delta = (2y_p)^{-1}$ .  $\square$

*Example 10.15* (The gate on the toy curve). Pallas satisfies both hypotheses of Proposition 10.14: 5 is a non-square in  $\mathbb{F}_p$  (Example 7.16) and  $-5$  is not a cube (Remark 10.26). The toy curve  $y^2 = x^3 + 5$  over  $\mathbb{F}_{11}$  of Example 10.38 satisfies neither— $5 = 4^2$  gives the points  $(0, \pm 4)$ , and  $(8, 0)$  has order 2—so on it the gate can be examined both on inputs satisfying the hypotheses and on

inputs violating them. For each input pair considered below, a script determines every output pair  $(x_r, y_r)$  that *some* assignment of the five auxiliary cells in  $\mathbb{F}_{11}$  makes the twelve identities accept—the search over  $\alpha, \beta, \gamma, \delta$  factored by which identities mention each:  $\beta$  only (C4),  $\gamma$  only (C5),  $\alpha$  and  $\delta$  only (C2) and (C6)—so soundness is checked against all provers, not only the honest one. On the nine points  $\mathcal{O}, (4, \pm 5), (5, \pm 3), (6, \pm 1), (10, \pm 2)$  with  $x, y \neq 0$ , all 81 ordered pairs—eight doublings, eight inverse pairs, seventeen involving  $\mathcal{O}$ —admit exactly one output, the encoding of the true sum. Three of them, worked: for  $[2](5, 3)$  the honest cells are  $\lambda = 3 \cdot 5^2 \cdot 6^{-1} = 9 \cdot 2 = 7$  and  $\delta = 6^{-1} = 2$ , giving  $(x_r, y_r) = (49 - 10, 7(5 - 6) - 3) = (6, 1)$ , the value of (13); for the inverse pair  $(5, 3) + (5, 8)$  the prover supplies the same  $\lambda = 7$  but  $\delta = \text{inv}_0(11) = 0$ , and (C6a)–(C6b) force  $(0, 0)$ ; for  $\mathcal{O} + (5, 3)$  the cells are  $\alpha = 5^{-1} = 9$ ,  $\lambda = 3 \cdot 9 = 5$ ,  $\beta = 0$ , and (C4a)–(C4b) force  $(5, 3)$ . Off the admissible set the gate fails as the proof predicts. With  $P = (0, 4)$  and  $Q = (5, 3)$  the identities admit only the output  $(5, 3)$ —the coordinate  $x_p = 0$  is read as  $\mathcal{O}$ —whereas  $P + Q = (10, 9)$  (Example 10.38); the pair  $(0, 4) + (0, 7)$  admits no output at all, (C4b) demanding  $y_r = 7$  and (C5b)  $y_r = 4$ ; and  $(8, 0) + (8, 0)$  admits none either, since (C2) reduces to  $3 \cdot 8^2 = 0$ , false in  $\mathbb{F}_{11}$ . On Pallas none of the three inputs exists, and the gate is complete.

## 10.5 Identity, inverses, commutativity, and associativity

**Theorem 10.16** (Group law). *Let  $E$  be an elliptic curve over a field  $K$ . The set  $E(K)$  with the chord-and-tangent operation  $+$  of Definition 10.7 is an abelian group with identity  $\mathcal{O}$ .*

Closure is Proposition 10.11. The next proposition establishes every remaining axiom except associativity.

**Proposition 10.17** (Identity, inverses, commutativity). *The operation  $+$  on  $E(K)$  has the following properties.*

1. *The point  $\mathcal{O}$  is a two-sided identity:  $P + \mathcal{O} = \mathcal{O} + P = P$  for all  $P$ .*
2. *Every  $P$  has the two-sided inverse  $-P$  of Definition 10.8:  $P + (-P) = \mathcal{O}$ .*
3. *The operation is commutative:  $P + Q = Q + P$ .*

*Proof.* Item (1) is the first clause of Definition 10.7. For item (2), let  $P = (x, y)$  be affine. If  $y \neq 0$ , the line through  $P$  and  $-P = (x, -y)$  is vertical, so its third point is  $\mathcal{O}$  and  $P + (-P) = \mathcal{O}$  by the vertical-line case; if  $y = 0$ , then  $P = -P$  and  $[2]P = \mathcal{O}$  directly by the vertical-tangent case; and  $\mathcal{O} + (-\mathcal{O}) = \mathcal{O} + \mathcal{O} = \mathcal{O}$ . For item (3), the chord through  $P$  and  $Q$  does not depend on the order in which the two points are named, and neither does the tangent in the doubling case; algebraically, the formulas (12) are symmetric under the swap  $(x_1, y_1) \leftrightarrow (x_2, y_2)$ —the slope  $\lambda$  is unchanged, and  $x_3 = \lambda^2 - x_1 - x_2$  is symmetric in  $x_1, x_2$ .  $\square$

The remaining axiom is *associativity*,  $(P + Q) + R = P + (Q + R)$ . A direct verification through the explicit formulas divides into many coordinate cases, each a lengthy rational-function identity; we sketch the conceptual proof instead.

*Proof of associativity (sketch).* The proof uses the classical *Cayley–Bacharach theorem* for plane cubics: if two cubics with no common component meet in nine points counted with intersection multiplicity, then any third cubic passing through eight of them with the required multiplicities also passes through the ninth. To prove  $(P + Q) + R = P + (Q + R)$ , one writes both sides via the chord-and-tangent construction and arranges the relevant intersections among the curve  $E$  and two degenerate cubics, each a product of three suitably chosen lines. Cayley–Bacharach forces the two candidate sums to coincide; its intersection-multiplicity formulation includes tangencies and coincident points. Replacing “reflect across the  $x$ -axis” by “the three points of any line sum to  $\mathcal{O}$ ” makes the bookkeeping uniform.

A complete, case-free treatment identifies  $E(K)$  with the degree-zero part  $\text{Pic}^0(E)$  of the divisor class group of the curve via the map  $P \mapsto [P] - [\mathcal{O}]$ , a bijection that intertwines the two operations; the group law on  $\text{Pic}^0(E)$  is manifestly associative, because it is inherited from addition of divisors modulo principal divisors. Either route yields a complete proof; both lie beyond the elementary scope of this volume, so we take associativity as established.  $\square$

*Remark 10.18* (The notation  $[n]P$ ). For  $n \in \mathbb{Z}$  and  $P \in E(K)$ , we write  $[n]P$  for the  $n$ -fold sum  $P + \dots + P$ , with  $[0]P = \mathcal{O}$  and  $[-n]P = -([n]P)$ . This makes the abelian group  $E(K)$  a  $\mathbb{Z}$ -module: integers act on points. The map  $n \mapsto [n]P$  is the *scalar multiplication* of Definition 10.20—not any coordinate-wise product.

## 10.6 Projective and Jacobian coordinates

The affine formulas (12) and (13) each require a field *inversion*, by  $x_2 - x_1$  or  $2y_1$ , which in a large prime field costs roughly as much as dozens of multiplications, whether computed by the extended Euclidean algorithm (Construction 6.1) or as  $z^{p-2}$  by Fermat (Construction 6.4). Over a long chain of group operations, as in scalar multiplication, it is cheaper to carry points in a redundant coordinate system whose group law uses only additions, subtractions, and multiplications, and to invert once at the end.

*Remark 10.19* (Inversion-free coordinate systems). Two redundant systems serve this purpose. In *(homogeneous) projective coordinates*, a triple  $[X : Y : Z]$  with  $Z \neq 0$  represents the affine point  $(X/Z, Y/Z)$  on the projective curve (11), with  $\mathcal{O} = [0 : 1 : 0]$ . In *Jacobian coordinates*, a triple  $(X : Y : Z)$  with  $Z \neq 0$  represents  $(X/Z^2, Y/Z^3)$ , under the equivalence  $(X, Y, Z) \sim (\lambda^2 X, \lambda^3 Y, \lambda Z)$ ; the curve equation becomes  $Y^2 = X^3 + aXZ^4 + bZ^6$ . In either system addition and doubling need no divisions, at the cost of more multiplications per operation; the final recovery of affine coordinates computes  $Z^{-1}$  once, then the needed powers of it by multiplication. Jacobian coordinates usually yield the faster doubling and are the standard choice in cryptographic libraries; the `pasta_curves` crate used by Halo 2 carries its curve points in them. For curves  $y^2 = x^3 + b$  with  $a = 0$ , the term  $aZ^4$  in the Jacobian doubling formulas vanishes, which lowers the cost of doubling; this is one reason the Pasta curves take  $a = 0$ .

## 10.7 Scalar multiplication and double-and-add

*Scalar multiplication* computes, for a point  $P$  and an integer  $n \geq 0$ ,

$$[n]P = \underbrace{P + P + \dots + P}_{n \text{ times}}.$$

Computing this by  $n - 1$  successive additions is infeasible when  $n$  has hundreds of bits. Double-and-add uses the binary expansion of  $n$  instead; it is the additive form of square-and-multiply (Construction 6.2), anticipated in Remark 6.3.

**Definition 10.20** (Double-and-add). If  $n = 0$ , return  $\mathcal{O}$ . Otherwise write  $n = \sum_{i=0}^{k-1} n_i 2^i$  with digits  $n_i \in \{0, 1\}$  and top digit  $n_{k-1} = 1$ . The *left-to-right double-and-add* algorithm computes  $[n]P$  as follows: set  $R \leftarrow P$ ; for  $i = k - 2$  down to 0, set  $R \leftarrow [2]R$ , and if  $n_i = 1$  set  $R \leftarrow R + P$ ; return  $R$ .

**Proposition 10.21.** For  $n \geq 1$ , *double-and-add* returns  $[n]P$  using exactly  $k - 1$  doublings and at most  $k - 1$  additions, where  $k = \lfloor \log_2 n \rfloor + 1$  is the bit length of  $n$ . For  $n = 0$ , it returns  $\mathcal{O}$  without a group operation.

*Proof.* The case  $n = 0$  is immediate. Let  $n \geq 1$ , and let  $R_j$  denote the value of  $R$  after bits  $n_{k-1}, \dots, n_j$  have been incorporated. We claim  $R_j = \lfloor [n/2^j] \rfloor P$ , by downward induction on  $j$ . At initialisation,  $R_{k-1} = P = \lfloor [n/2^{k-1}] \rfloor P$  because the top bit is 1. Assuming  $R_{j+1} = \lfloor [n/2^{j+1}] \rfloor P$ , the next iteration computes

$$[2]R_{j+1} + [n_j]P = [2\lfloor [n/2^{j+1}] \rfloor + n_j]P = \lfloor [n/2^j] \rfloor P,$$

where the last step is the digit identity  $2\lfloor [n/2^{j+1}] \rfloor + n_j = \lfloor [n/2^j] \rfloor$ : stripping the last bit of  $\lfloor [n/2^j] \rfloor$  and appending it again recovers the number. At  $j = 0$  the invariant reads  $R_0 = [n]P$ , which is what the algorithm returns. For the counts, the loop has exactly  $k - 1$  iterations, each performing one doubling and at most one addition.  $\square$

Scalar multiplication by an  $n$ -bit scalar therefore costs  $O(n)$  group operations— $O(n)$  field operations up to constant factors, or  $O(n \log^2 p)$  bit operations with schoolbook field arithmetic. Elliptic-curve cryptography rests on the contrast between this *forward* map and the apparent hardness of inverting it (§10.11).

## 10.8 The group structure of $E(\mathbb{F}_p)$

Let  $K = \mathbb{F}_p$  with  $p$  prime,  $p > 3$ . Each of the  $p$  values of  $x$  admits at most two values of  $y$  (Theorem 5.18, applied to  $y^2 - f(x)$  as a polynomial in  $y$ ), so with  $\mathcal{O}$  there are at most  $2p + 1$  points, and by Theorem 10.16 the set  $E(\mathbb{F}_p)$  is a *finite abelian group*. Hasse's theorem bounds its order.

**Theorem 10.22** (Hasse). Let  $E$  be an elliptic curve over  $\mathbb{F}_p$ . Write

$$\#E(\mathbb{F}_p) = p + 1 - t,$$

which defines the integer  $t$ , called the trace of Frobenius. Then

$$|t| \leq 2\sqrt{p};$$

equivalently,  $\#E(\mathbb{F}_p)$  lies in the Hasse interval  $[p + 1 - 2\sqrt{p}, p + 1 + 2\sqrt{p}]$ .

*Proof (sketch).* The count has an exact character-sum expression. For fixed  $x$ , the number of  $y \in \mathbb{F}_p$  with  $y^2 = f(x)$  is  $1 + \left(\frac{f(x)}{p}\right)$  when  $f(x) \neq 0$ —two solutions for a nonzero square, none for a non-square (Proposition 7.12)—and exactly  $1 = 1 + \left(\frac{0}{p}\right)$  when  $f(x) = 0$ , using the Legendre symbol of Definition 7.14 with the convention  $\left(\frac{0}{p}\right) = 0$ . Summing over  $x$ ,

$$\#E^{\text{aff}}(\mathbb{F}_p) = \sum_{x \in \mathbb{F}_p} \left(1 + \left(\frac{x^3 + ax + b}{p}\right)\right) = p + \sum_{x \in \mathbb{F}_p} \left(\frac{x^3 + ax + b}{p}\right),$$

so  $t = -\sum_x \left(\frac{x^3 + ax + b}{p}\right)$ . Heuristically, as  $x$  ranges over  $\mathbb{F}_p$  the value  $f(x)$  is a nonzero square about half the time (two points), a non-square about half the time (none), and zero occasionally (one point); the  $p$  summands  $\pm 1$  behave like a random walk, and Hasse's bound  $|t| \leq 2\sqrt{p}$  is exactly square-root cancellation in this character sum.

The conceptual proof realises  $t$  as the trace of the  $p$ -power *Frobenius endomorphism*  $\pi: (x, y) \mapsto (x^p, y^p)$  of the curve—the curve-level counterpart of the field automorphism of Definition 6.14. One shows that  $\pi$  satisfies the characteristic equation  $\pi^2 - [t]\pi + [p] = 0$  in the endomorphism ring of  $E$ , and that the associated quadratic form (the degree) is positive definite; the discriminant condition  $t^2 - 4p \leq 0$  for the complex roots of the characteristic polynomial then yields  $|t| \leq 2\sqrt{p}$ . The full argument requires the theory of isogenies and the Weil pairing, beyond the present scope.  $\square$

By Hasse's theorem the relative deviation of the order from  $p$  is of order  $p^{-1/2}$ , so the bound guarantees without a brute-force count that the group has about  $p$  elements, about 255 bits for the Pasta curves. The bound also underlies the point-counting algorithms (Schoof–Elkies–Atkin) that compute the *exact* order in polynomial time, by which curve designers certify their group orders.

Cryptographic use requires  $E(\mathbb{F}_p)$  to contain a large subgroup of *prime* order. The general structure theory of  $E(\mathbb{F}_p)$  is not needed in this series: the Pasta curves are designed so that  $\#E(\mathbb{F}_p)$  is itself prime, and a group of prime order is cyclic with no proper nontrivial subgroups (Corollary 3.34).

## 10.9 Torsion, the cofactor, and prime-order subgroups

**Definition 10.23** (Order of a point and torsion). The *order* of a point  $P \in E(\mathbb{F}_p)$  is its order as a group element (Definition 3.11): the least integer  $m \geq 1$  with  $[m]P = \mathcal{O}$ . It exists and divides  $\#E(\mathbb{F}_p)$ , by Lagrange's theorem (Theorem 3.29 via Corollary 3.30). For an integer  $m \geq 1$ , the  *$m$ -torsion subgroup* is

$$E[m] = \{P \in E(\overline{\mathbb{F}_p}) : [m]P = \mathcal{O}\},$$

the points over the algebraic closure killed by  $m$ ; its rational part is  $E(\mathbb{F}_p)[m] = E[m] \cap E(\mathbb{F}_p)$ .

**Definition 10.24** (Cofactor). Suppose  $\#E(\mathbb{F}_p) = h \cdot r$ , where  $r$  is the (large) prime to be used and  $h$  is the remaining factor. The integer  $h$  is the *cofactor*, and  $r$  is the order of the *prime-order subgroup*: a point  $P$  of order  $r$  generates a cyclic subgroup  $\langle P \rangle \cong \mathbb{Z}/r\mathbb{Z}$ , and cryptographic protocols operate in this subgroup.

A nontrivial cofactor  $b > 1$  is undesirable: it admits *small-subgroup* points, of order dividing  $b$ , which can leak information or break protocol assumptions—the small-subgroup attacks of Remark 7.25. Implementations therefore either *clear the cofactor*, multiplying incoming points by  $b$ , or use prime-order curves with  $b = 1$ . The Pasta curves have cofactor 1: the orders  $\#E_p(\mathbb{F}_p) = q$  and  $\#E_q(\mathbb{F}_q) = p$  are themselves prime, so  $E_p(\mathbb{F}_p) \cong \mathbb{Z}/q\mathbb{Z}$  and  $E_q(\mathbb{F}_q) \cong \mathbb{Z}/p\mathbb{Z}$ , and no cofactor clearing is needed.

**Proposition 10.25** (Two-torsion). *The affine points of order 2 in  $E(\mathbb{F}_p)$  are exactly the points  $(x_0, 0)$  with  $f(x_0) = x_0^3 + ax_0 + b = 0$ . Hence  $E(\mathbb{F}_p)$  has 0, 1, or 3 points of order 2 according as  $f$  has 0, 1, or 3 roots in  $\mathbb{F}_p$ ; and  $\#E(\mathbb{F}_p)$  is even if and only if  $f$  has a root in  $\mathbb{F}_p$ .*

*Proof.* A point  $P = (x, y)$  has order 2 iff  $P = -P$  and  $P \neq \mathcal{O}$ , i.e.  $y = -y$ , i.e.  $y = 0$  (since  $\text{char}(\mathbb{F}_p) \neq 2$ ); the curve equation then forces  $f(x) = 0$ . Conversely each root  $x_0$  of  $f$  in  $\mathbb{F}_p$  gives the order-2 point  $(x_0, 0)$ .

Next, the number  $N$  of roots of  $f$  in  $\mathbb{F}_p$  can only be 0, 1, or 3: two roots force a third. Suppose  $f$  has distinct roots  $r_1 \neq r_2$  in  $\mathbb{F}_p$ . The factor theorem (Theorem 5.17) gives  $f = (x - r_1)b$ ; evaluating at  $r_2$  yields  $0 = (r_2 - r_1)b(r_2)$ , so  $b(r_2) = 0$ , and a second application factors the monic cubic completely,  $f = (x - r_1)(x - r_2)(x - r_3)$  with  $r_3 \in \mathbb{F}_p$ —explicitly  $r_3 = -r_1 - r_2$  by Vieta, the depressed cubic having no  $x^2$  term. Nonsingularity says  $f$  has no repeated root even over  $\overline{\mathbb{F}_p}$  (Proposition 10.4(2)), so  $r_3 \notin \{r_1, r_2\}$ , and  $f$  has exactly three roots (Theorem 5.18 caps the count). Hence  $N \in \{0, 1, 3\}$ , matching the trichotomy of the statement. Together with  $\mathcal{O}$  the order-2 points form the rational 2-torsion subgroup  $E(\mathbb{F}_p)[2]$ , which is accordingly trivial,  $\mathbb{Z}/2\mathbb{Z}$ , or  $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ .

For the parity claim, consider negation  $P \mapsto -P$ , an *involution*—a self-inverse map—of the finite set  $E(\mathbb{F}_p)$ . Its fixed points are exactly  $\mathcal{O}$  and the points  $(x_0, 0)$ ; every other point pairs off with its distinct negative into a 2-element orbit. Counting the set by orbits,

$$\#E(\mathbb{F}_p) \equiv 1 + N \pmod{2},$$

where  $N \in \{0, 1, 3\}$  is the number of roots of  $f$  in  $\mathbb{F}_p$ . Hence  $\#E(\mathbb{F}_p)$  is even iff  $N$  is odd, i.e. iff  $N \geq 1$ , i.e. iff  $f$  has a root in  $\mathbb{F}_p$ . (In general, an involution of a finite set determines the cardinality of the set modulo 2 from its fixed points alone.)  $\square$

*Remark 10.26* (No two-torsion on the Pasta curves). For  $y^2 = x^3 + 5$  over  $\mathbb{F}_p$ , the curve has a point of order 2 iff  $x^3 + 5 = 0$  has a solution, i.e. iff  $-5$  is a cube in  $\mathbb{F}_p$ . For the Pasta primes the order  $\#E_p(\mathbb{F}_p) = q$  is odd (it is prime), so there are *no* rational 2-torsion points; consistently,  $-5$  is not a cube in  $\mathbb{F}_p$ .

## 10.10 Base fields, scalar fields, and the Pasta cycle

An elliptic-curve computation involves two prime fields. Every operation of the group law on  $E(\mathbb{F}_p)$ —the slope  $\lambda = (y_2 - y_1)(x_2 - x_1)^{-1}$ , the squarings, and the inversions needed to add and to double—is arithmetic in  $\mathbb{F}_p$ , the *base field* (or coordinate field), performed with the prime-field algorithms of Section 6.1. Keys and signatures reside in the subgroup of large prime order  $r$  (Definition 10.24), and a scalar multiplication  $P \mapsto [s]P$  on a point of order  $r$  depends only on  $s \bmod r$ ,

since  $[r]P = \mathcal{O}$  (Proposition 3.12); scalars are therefore residues modulo  $r$ , and since  $r$  is prime, they form a *field* (Corollary 4.26).

**Definition 10.27** (Base field and scalar field). For an elliptic curve used in cryptography, with a prime-order subgroup of order  $r$ :

- the *base field*  $\mathbb{F}_p$  is the field over which the curve is defined and in which point coordinates lie;
- the *scalar field*  $\mathbb{F}_r = \mathbb{Z}/r\mathbb{Z}$  is the field of scalars by which one multiplies points, i.e. the integers modulo the subgroup order  $r$ .

Both are prime fields, but in general  $p \neq r$ .

*Remark 10.28* (Two fields at once, and the Pasta cycle). Scalar arithmetic runs in  $\mathbb{F}_r$  and coordinate arithmetic in  $\mathbb{F}_p$ . Halo 2’s Pallas and Vesta curves (the “Pasta” pair) form a *cycle of curves*: the base field of one is the scalar field of the other and vice versa,

$$\#E_p(\mathbb{F}_p) = q, \quad \#E_q(\mathbb{F}_q) = p,$$

so the scalar field of Pallas is  $\mathbb{F}_q$ , the base field of Vesta, and symmetrically (§5.4.9.6 of the Zcash protocol specification). This lets a proof system express the verifier of one curve’s arithmetic *natively* in the field of the other, which enables the recursive proof composition of Halo 2. Both primes are chosen  $\equiv 1 \pmod{2^{32}}$ , so by Corollary 6.22 each field contains a multiplicative subgroup of order  $2^{32}$ , furnishing the high-order roots of unity that the number-theoretic transform of Section 9 needs for fast polynomial multiplication.

## 10.11 The elliptic-curve discrete logarithm problem

The discrete logarithm problem of Definition 7.21, stated there for an abstract cyclic group, reads as follows for elliptic curves in additive notation.

**Definition 10.29** (ECDLP). Let  $P \in E(\mathbb{F}_p)$  be a point of large prime order  $r$ , generating  $G = \langle P \rangle$ . The *elliptic-curve discrete logarithm problem* (ECDLP) is: given  $P$  and a point  $Q \in G$ , find the unique integer  $n \in \{0, 1, \dots, r-1\}$  with  $Q = [n]P$ . One writes  $n = \log_p Q$ .

The map  $n \mapsto [n]P$  is a group isomorphism  $\mathbb{Z}/r\mathbb{Z} \rightarrow G$  (Theorem 3.43), so a discrete logarithm always exists and is unique; the difficulty of the problem is purely *computational*. The forward map is fast by double-and-add (Proposition 10.21); on a well-chosen curve the best known algorithms to invert it are the two generic attacks, which transfer verbatim from Remark 7.22 to the additive setting.

**Proposition 10.30** (Square-root attacks). *Baby-step giant-step solves the ECDLP in a group of prime order  $r$  in  $O(\sqrt{r})$  group operations and  $O(\sqrt{r})$  space. Pollard’s rho method matches the  $O(\sqrt{r})$  bound in expectation with only  $O(1)$  space, under a modelling assumption on its pseudo-random walk that is stated explicitly in the proof and is unproved for every concrete walk.*

*Proof.* For baby-step giant-step, set  $m = \lceil \sqrt{r} \rceil$  and write the unknown logarithm as  $n = im + j$  with  $0 \leq i, j < m$ ; since  $m^2 \geq r$ , every  $n \in \{0, \dots, r-1\}$  has such a decomposition. Tabulate the *baby steps*  $[j]P$  for  $j = 0, \dots, m-1$  (a table of  $m$  points, built with  $m-1$  additions), compute  $S = [m]P$  once, and walk the *giant steps*  $Q - [i]S$  for  $i = 0, 1, 2, \dots$ , testing each against the table. At  $i = \lfloor n/m \rfloor$  the walk hits  $Q - [im]P = [n - im]P = [j]P$ , a table entry; the collision reveals  $i$  and  $j$ , hence  $n = im + j$ . Both the table and the walk cost  $O(m) = O(\sqrt{r})$  group operations, and the table holds  $O(\sqrt{r})$  points.

Pollard's rho eliminates the table, at the price of an explicitly heuristic analysis. The algorithm takes a pseudo-random walk on  $G$  whose every position is maintained in the form  $[a]P + [b]Q$  with known coefficients  $a, b$ , started from such a combination with random coefficients; the walk's next step depends only on the current point, so once the walk revisits a point it cycles from then on. A repeat is therefore detected with  $O(1)$  memory: run a second copy of the walk at double speed, and the two copies meet at a common point inside the cycle within a constant factor of the first repeat time. *Modelling assumption:* the successive positions of the walk behave like independent uniform samples from  $G$ . No proof of this is known for any concrete walk—the assumption is what “well-designed” means, and experiment supports it—and the rest of the argument is conditional on it. Under the assumption, the first  $k$  positions are pairwise distinct with probability at most  $e^{-k(k-1)/(2r)}$  by the birthday bound (a forward reference: Proposition 11.18 belongs to the closing section), so the number  $T$  of steps to the first repeat satisfies  $\Pr[T > k] \leq e^{-k(k-1)/(2r)}$ . The tail-sum formula (Lemma 11.33) converts the tail bound into an expectation bound: with  $m = \lceil \sqrt{r} \rceil$  and  $r \geq 4$ ,

$$\mathbb{E}[T] = \sum_{k \geq 1} \Pr[T \geq k] \leq m \sum_{j \geq 0} \Pr[T > jm] \leq m \left( 1 + \sum_{j \geq 1} e^{-j^2/4} \right) = O(\sqrt{r}),$$

where the middle step groups the indices  $k$  into blocks of length  $m$ , bounding each block by its first term, and the last uses  $jm(jm-1) \geq j^2 m(m-1) \geq j^2(r - \sqrt{r}) \geq j^2 r/2$  for  $j \geq 1$ , so that  $\Pr[T > jm] \leq e^{-j^2/4} \leq e^{-j/4}$ , a convergent geometric tail. A repeat presents one group element with two representations,  $[a]P + [b]Q = [a']P + [b']Q$ . If  $b' \not\equiv b \pmod{r}$ , this rearranges to  $[a - a']P = [b' - b]Q = [(b' - b)n]P$ , whence

$$n \equiv (a - a')(b' - b)^{-1} \pmod{r},$$

the inverse existing because  $r$  is prime. If instead  $b' \equiv b$ , then  $[a]P = [a']P$  forces  $a \equiv a'$ : the two representations coincide, the collision reveals nothing, and the walk is restarted with fresh random starting coefficients; that such degenerate collisions are rare is a further part of the heuristic, borne out in practice. Both algorithms run in time exponential in the bit length  $\log_2 r$ .  $\square$

*Remark 10.31* (Best known attacks on well-chosen curves). For well-chosen curves, no unrestricted classical attack faster than  $O(\sqrt{r})$  is known; this is not an unconditional lower bound for all classical algorithms. In the classical generic-group model, however, Shoup's matching lower bound is proved (Remark 7.22). For  $r \approx 2^{254}$ —the 255-bit Pasta orders—this is about  $2^{127}$  operations, infeasible. Curves with special structure can be weaker, and the known weak classes are avoided by design. *Anomalous* curves, those with  $\#E(\mathbb{F}_p) = p$ , admit a polynomial-time attack through the formal group. The MOV/Frey–Rück attack transfers the ECDLP into a finite-field

discrete logarithm when the *embedding degree*—the least  $k \geq 1$  with  $r \mid p^k - 1$ , i.e. the degree (Example 8.4; the dimension of the extension over its base) of the smallest extension of  $\mathbb{F}_p$  whose multiplicative group contains a subgroup of order  $r$ —is small, where index calculus applies. Cryptographically secure curves, the Pasta curves included, are chosen with a large prime subgroup order, a large embedding degree, and  $\#E(\mathbb{F}_p) \neq p$ , which excludes every known subexponential attack.

*Remark 10.32* (The index-calculus contrast). In  $\mathbb{F}_p^\times$  the index-calculus method solves the discrete logarithm in subexponential time  $L_p[1/3]$ —shorthand for  $2^{\Theta(\lambda^{1/3}(\log \lambda)^{2/3})}$  operations in the bit length  $\lambda = \log_2 p$ , a notation defined precisely in Definition 11.60 of the closing section—exploiting the integer arithmetic behind the residues (Remark 7.22); those groups must therefore be large. On a well-chosen elliptic curve no index calculus is known—the points offer no analogue of “factoring into small primes”—so the generic  $O(\sqrt{r})$  bound stands, and much smaller groups suffice: about 256 bits for 128-bit security, versus thousands of bits for  $\mathbb{F}_p^\times$ . Smaller groups mean faster arithmetic and shorter keys, which is why elliptic curves dominate modern protocols, including Halo 2.

## 10.12 The $j$ -invariant and the GLV endomorphism

The  $j$ -invariant classifies curves up to isomorphism; when it vanishes, as it does for the Pasta curves by design, the curve has an additional endomorphism.

**Definition 10.33** ( $j$ -invariant). The  $j$ -invariant of the curve  $y^2 = x^3 + ax + b$  (with  $4a^3 + 27b^2 \neq 0$ ) is

$$j = 1728 \frac{4a^3}{4a^3 + 27b^2} \in K,$$

where  $1728 = 12^3$  is a normalising constant of historical origin.

**Proposition 10.34.** Two elliptic curves over  $\bar{K}$  are isomorphic over  $\bar{K}$  if and only if they have the same  $j$ -invariant. The admissible coordinate changes preserving short Weierstrass form are  $(x, y) \mapsto (u^2x, u^3y)$  for  $u \in \bar{K}^\times$ , under which  $(a, b) \mapsto (u^4a, u^6b)$ ; the combination  $j$  is invariant.

*Proof (sketch).* A direct computation shows that the only isomorphisms between short Weierstrass curves fixing  $\mathcal{O}$  are the scalings  $(x, y) \mapsto (u^2x, u^3y)$ . Substituting  $x = u^{-2}x'$ ,  $y = u^{-3}y'$  into  $y^2 = x^3 + ax + b$  and clearing  $u^6$  gives  $y'^2 = x'^3 + u^4ax' + u^6b$ , which is the stated action on coefficients. Under it, numerator and denominator of  $4a^3/(4a^3 + 27b^2)$  both scale by  $u^{12}$ , so  $j$  is unchanged. Conversely, given two curves with equal  $j$ -invariants one solves for a suitable  $u$  transforming one coefficient pair into the other, treating separately the cases  $j = 0$  (both  $a$  vanish) and  $j = 1728$  (both  $b$  vanish).  $\square$

*Remark 10.35* (The Pasta curves have  $j = 0$ ). For  $y^2 = x^3 + b$  one has  $a = 0$ , so  $j = 0$ ; both Pasta curves have  $j = 0$ . Curves with  $j = 0$  admit an additional automorphism of order 3, which underlies the efficient endomorphism described next. (In the other exceptional case,  $b = 0$  gives

$j = 1728$  and an order-4 automorphism.) The Pasta design chooses  $j = 0$  to make the Gallant–Lambert–Vanstone (GLV) speedup available, which roughly halves the doubling count in scalar multiplication.

**Proposition 10.36** (The GLV endomorphism for  $y^2 = x^3 + b$ ). *Suppose  $\mathbb{F}_p$  contains a primitive cube root of unity  $\omega$ , i.e.  $p \equiv 1 \pmod{3}$  (Corollary 6.22), so  $\omega \in \mathbb{F}_p$  satisfies  $\omega^2 + \omega + 1 = 0$ . On  $E : y^2 = x^3 + b$ , the map*

$$\phi(x, y) = (\omega x, y), \quad \phi(\mathcal{O}) = \mathcal{O},$$

*is a group endomorphism of  $E(\mathbb{F}_p)$ . Let  $G = \langle P \rangle$  be a subgroup of prime order  $r$  with  $r \equiv 1 \pmod{3}$  and  $\phi(G) \subseteq G$ . Then  $\phi$  acts on  $G$  as multiplication by a scalar: there is  $\lambda \in \mathbb{Z}/r\mathbb{Z}$  with  $\lambda^2 + \lambda + 1 \equiv 0 \pmod{r}$  and*

$$\phi(Q) = [\lambda] Q \quad \text{for all } Q \in G.$$

*For the Pasta curves the hypothesis  $\phi(G) \subseteq G$  is automatic, since  $E(\mathbb{F}_p)$  itself is the prime-order group  $G$ ; and  $r \equiv 1 \pmod{3}$  holds for both orders.*

*Proof.* The map lands on the curve. If  $y^2 = x^3 + b$  then  $y^2 = (\omega x)^3 + b$ , because  $\omega^3 = 1$ ; so  $(\omega x, y) \in E(\mathbb{F}_p)$ .

*The map is a homomorphism.* We check  $\phi(P_1 + P_2) = \phi(P_1) + \phi(P_2)$  against the case analysis of §10.4. The  $\mathcal{O}$  cases are immediate, and  $\phi(-P) = -\phi(P)$  since negation touches only  $y$ ; in particular the inverse case  $\phi(P) + \phi(-P) = \mathcal{O} = \phi(\mathcal{O})$  is respected, and  $\phi$  preserves the case split, since it preserves equality of  $x$ -coordinates. For a chord  $(x_1 \neq x_2)$ , the slope transforms as

$$\lambda' = \frac{y_2 - y_1}{\omega x_2 - \omega x_1} = \omega^{-1} \lambda = \omega^2 \lambda,$$

using  $\omega^{-1} = \omega^2$ . Then, by (12) and  $\omega^4 = \omega$ ,

$$x'_3 = \lambda'^2 - \omega x_1 - \omega x_2 = \omega^4 \lambda^2 - \omega x_1 - \omega x_2 = \omega (\lambda^2 - x_1 - x_2) = \omega x_3,$$

$$y'_3 = \lambda'(\omega x_1 - x'_3) - y_1 = \omega^2 \lambda \cdot \omega(x_1 - x_3) - y_1 = \lambda(x_1 - x_3) - y_1 = y_3,$$

so  $\phi(P_1) + \phi(P_2) = (\omega x_3, y_3) = \phi(P_1 + P_2)$ . The tangent case is identical, with  $\lambda' = 3(\omega x_1)^2 / (2y_1) = \omega^2 \lambda$  in (13).

*The relation  $\phi^2 + \phi + \text{id} = 0$ .* Let  $P = (x, y)$  be affine; we show  $P + \phi(P) + \phi^2(P) = \mathcal{O}$  directly from the formulas of §10.4.

First let  $x \neq 0$ . The points  $P = (x, y)$ ,  $\phi(P) = (\omega x, y)$ ,  $\phi^2(P) = (\omega^2 x, y)$  share their  $y$ -coordinate and have pairwise distinct  $x$ -coordinates, since  $1, \omega, \omega^2$  are distinct and  $x \neq 0$ . The chord formula (12) applies to  $P + \phi(P)$ , with slope  $\lambda = (y - y) / (\omega x - x) = 0$ :

$$x_3 = 0 - x - \omega x = -(1 + \omega)x = \omega^2 x, \quad y_3 = 0 \cdot (x - x_3) - y = -y,$$

using  $1 + \omega + \omega^2 = 0$ . Hence  $P + \phi(P) = (\omega^2 x, -y) = -\phi^2(P)$  (Definition 10.8), and  $P + \phi(P) + \phi^2(P) = \mathcal{O}$ .

If instead  $x = 0$ , the three points coincide:  $\phi(P) = \phi^2(P) = P$ . Here  $y \neq 0$ , since  $x = y = 0$  on the curve would force  $b = 0$ , which nonsingularity  $27b^2 \neq 0$  forbids; so the doubling formula (13) applies, with  $a = 0$ :

$$\lambda = \frac{3 \cdot 0^2 + 0}{2y} = 0, \quad x_3 = 0 - 2 \cdot 0 = 0, \quad y_3 = 0 \cdot (0 - 0) - y = -y,$$

so  $[2]P = (0, -y) = -P$ , hence  $P + \phi(P) + \phi^2(P) = [3]P = \mathcal{O}$  in this case too. (The three points  $P, \phi(P), \phi^2(P)$  are exactly the intersections of the horizontal line through  $P$  with the curve, the roots of  $X^3 = y^2 - b$  being  $x, \omega x, \omega^2 x$ ; the computation above proves the rule “three points of a line sum to  $\mathcal{O}$ ” for this line, including the triple intersection at  $x = 0$ .)

With  $\phi(\mathcal{O}) = \mathcal{O}$  the identity holds on all of  $E(\mathbb{F}_p)$ ; as an identity of endomorphisms,  $\phi^2 + \phi + \text{id} = 0$ . Moreover  $\phi^3 = \text{id}$ , since  $\omega^3 = 1$ ; and whenever  $\#E(\mathbb{F}_p) > 3$ —as always at cryptographic sizes, the Pasta curves included— $\phi$  has order exactly 3, because at most two affine points,  $(0, \pm y_0)$  with  $y_0^2 = b$ , share the coordinate  $x = 0$ , so some point of  $E(\mathbb{F}_p)$  has  $x \neq 0$  and  $\phi$  moves it. The proviso is necessary:  $y^2 = x^3 + 4$  over  $\mathbb{F}_7$  has only the three points  $\mathcal{O}, (0, 2), (0, 5)$ , and there  $\phi = \text{id}$ .

*Action on  $G$  as a scalar.* The subgroup  $G \cong \mathbb{Z}/r\mathbb{Z}$  is cyclic with generator  $P$ , and  $\phi(G) \subseteq G$  by hypothesis, so  $\phi(P) = [\lambda]P$  for some  $\lambda \in \mathbb{Z}/r\mathbb{Z}$ ; then for any  $Q = [m]P \in G$ , the homomorphism property gives  $\phi(Q) = [m]\phi(P) = [m\lambda]P = [\lambda]Q$ . (This is the statement  $\text{Hom}(\mathbb{Z}/r\mathbb{Z}, \mathbb{Z}/r\mathbb{Z}) \cong \mathbb{Z}/r\mathbb{Z}$ : an endomorphism of a cyclic group is determined by, and equal to multiplication by, the image of a generator.) Applying the endomorphism relation to  $P$ ,

$$[\lambda^2 + \lambda + 1]P = \mathcal{O} \implies \lambda^2 + \lambda + 1 \equiv 0 \pmod{r},$$

since  $P$  has order  $r$ . Consistently, this congruence is solvable exactly when  $\mathbb{F}_r$  contains a primitive cube root of unity, i.e. when  $r \equiv 1 \pmod{3}$  (Corollary 6.22 again)—the hypothesis of the statement.  $\square$

*Remark 10.37* (How GLV halves the doubling count). Computing  $\phi$  costs a single field multiplication  $x \mapsto \omega x$ , negligible against a group operation. A scalar  $n \in \mathbb{Z}/r\mathbb{Z}$  is decomposed as

$$[n]P = [n_1]P + [n_2]\phi(P), \quad n \equiv n_1 + n_2\lambda \pmod{r},$$

where  $n_1, n_2$  can each be taken about  $\frac{1}{2} \log_2 r$  bits long; such a pair is found by lattice reduction in the lattice of pairs  $(c_1, c_2)$  with  $c_1 + c_2\lambda \equiv 0 \pmod{r}$ . The two half-length scalar multiplications are then run *simultaneously*, sharing their doublings (Straus–Shamir’s trick): one double-and-add loop of half the length, adding  $P, \phi(P)$ , or both according to the bits of  $n_1$  and  $n_2$ . The result costs about half the doublings of a single full-length double-and-add; this saving is one reason the Pasta curves take the form  $y^2 = x^3 + b$  over primes  $p \equiv 1 \pmod{3}$ .

### 10.13 A worked toy example

*Example 10.38* (The curve  $y^2 = x^3 + 5$  over  $\mathbb{F}_{11}$ ). Take  $p = 11$ : then  $p > 3$ , and  $4 \cdot 0 + 27 \cdot 5^2 = 675 \equiv 4 \not\equiv 0 \pmod{11}$ , so the curve  $E : y^2 = x^3 + 5$  is nonsingular over  $\mathbb{F}_{11}$  (Definition 10.5). Squaring the residues  $0, 1, \dots, 10$  yields the squares  $\{0, 1, 3, 4, 5, 9\}$ , so the nonzero

quadratic residues are  $\{1, 3, 4, 5, 9\}$ . Tabulating  $f(x) = x^3 + 5 \pmod{11}$  and reading off the  $y$ -values:

| $x$ | $x^3 \pmod{11}$ | $f(x) = x^3 + 5$ | $y$ with $y^2 = f(x)$           |
|-----|-----------------|------------------|---------------------------------|
| 0   | 0               | 5                | $\pm 4$ ( $4^2 = 16 \equiv 5$ ) |
| 1   | 1               | 6                | none ( $6 \notin \text{QR}$ )   |
| 2   | 8               | 2                | none                            |
| 3   | 5               | 10               | none                            |
| 4   | 9               | 3                | $\pm 5$ ( $5^2 = 25 \equiv 3$ ) |
| 5   | 4               | 9                | $\pm 3$                         |
| 6   | 7               | 1                | $\pm 1$                         |
| 7   | 2               | 7                | none                            |
| 8   | 6               | 0                | 0                               |
| 9   | 3               | 8                | none                            |
| 10  | 10              | 4                | $\pm 2$                         |

Counting: the five values  $x \in \{0, 4, 5, 6, 10\}$  contribute two points each (10 points),  $x = 8$  contributes the single point  $(8, 0)$ , and  $\mathcal{O}$  is the remaining point:

$$\#E(\mathbb{F}_{11}) = 10 + 1 + 1 = 12.$$

Hasse's bound holds:  $p + 1 = 12$ , so the trace is  $t = 0$ , and  $|t| = 0 \leq 2\sqrt{11} \approx 6.63$ . The point  $(8, 0)$  has order 2, consistent with Proposition 10.25 and the even order 12; here  $-5 \equiv 6$  is a cube, namely  $8^3 = 512 \equiv 6$ .

*An addition.* Let  $P = (0, 4)$  and  $Q = (5, 3)$ . Since  $x_1 = 0 \neq 5 = x_2$ , the chord formula (12) applies:

$$\lambda = \frac{3 - 4}{5 - 0} = (-1) \cdot 5^{-1} = (-1) \cdot 9 = -9 \equiv 2 \pmod{11},$$

using  $5^{-1} = 9$  (check:  $5 \cdot 9 = 45 \equiv 1$ ). Then

$$x_3 = \lambda^2 - x_1 - x_2 = 4 - 0 - 5 = -1 \equiv 10, \quad y_3 = \lambda(x_1 - x_3) - y_1 = 2(0 - 10) - 4 = -24 \equiv 9,$$

so  $P + Q = (10, 9)$ . The result is on the curve:  $9^2 = 81 \equiv 4$  and  $10^3 + 5 = 1005 \equiv 4 \pmod{11}$ , as required.

*A doubling.* Take  $P = (0, 4)$ . With  $a = 0$ , formula (13) gives

$$\lambda = \frac{3x_1^2 + a}{2y_1} = \frac{0}{8} = 0, \quad x_3 = \lambda^2 - 2x_1 = 0, \quad y_3 = \lambda(x_1 - x_3) - y_1 = -4 \equiv 7,$$

so  $[2](0, 4) = (0, 7) = -(0, 4)$ . Hence  $[3](0, 4) = \mathcal{O}$ : the point  $(0, 4)$  has order 3.

*The GLV endomorphism.* Since  $11 \equiv 2 \pmod{3}$ , the field  $\mathbb{F}_{11}$  contains *no* primitive cube root of unity (Corollary 6.22 requires  $3 \mid p - 1$ ), so the GLV map is not available over  $\mathbb{F}_{11}$ . The Pasta primes do satisfy  $p \equiv q \equiv 1 \pmod{3}$ , so GLV applies there. For a small GLV-friendly illustration take instead  $p = 13 \equiv 1 \pmod{3}$ , where  $\omega = 3$  satisfies  $3^2 + 3 + 1 = 13 \equiv 0 \pmod{13}$ ; then  $\phi(x, y) = (3x, y)$  is a nontrivial endomorphism of  $y^2 = x^3 + b$  over  $\mathbb{F}_{13}$ .

### 10.14 Pallas and Vesta assembled

*Example 10.39* (The Pasta curves). For the running example  $y^2 = x^3 + 5$ , with  $p$  and  $q$  the 255-bit primes of the section opening:

- *Pallas*, over  $\mathbb{F}_p$ : the order is  $\#E_p(\mathbb{F}_p) = q$ , a prime, so the group is cyclic,  $E_p(\mathbb{F}_p) \cong \mathbb{Z}/q\mathbb{Z}$  (Corollary 3.34), with cofactor  $h = 1$  (Definition 10.24). The trace is  $t = p + 1 - q$ , an 87-bit (negative) integer, far inside the Hasse bound  $2\sqrt{p} \approx 2^{128}$  of Theorem 10.22.
- *Vesta*, over  $\mathbb{F}_q$ : the order is  $\#E_q(\mathbb{F}_q) = p$ , a prime, so  $E_q(\mathbb{F}_q) \cong \mathbb{Z}/p\mathbb{Z}$ , again with cofactor 1.
- Both curves have  $j = 0$  (Remark 10.35), and both primes satisfy  $p \equiv q \equiv 1 \pmod{3}$ , so each curve admits the order-3 endomorphism  $\phi(x, y) = (\omega x, y)$  of Proposition 10.36 and supports the GLV speedup of Remark 10.37.
- Both fields have two-adicity 32 in the sense of Remark 9.10:  $2^{32}$  divides  $p - 1$  and  $q - 1$ , so each field carries the multiplicative subgroups of order  $2^{32}$  asserted in Remark 10.28, the evaluation domains of the fast polynomial arithmetic of Section 9.
- The cycle property  $\#E_p(\mathbb{F}_p) = q$  and  $\#E_q(\mathbb{F}_q) = p$  means the scalar field of each curve is the base field of the other (Definition 10.27): the base field of Pallas is  $\mathbb{F}_p$  and its scalar field is  $\mathbb{F}_q$ , matching the base field of Vesta, and symmetrically. This lets recursive proof systems chain proofs across the two curves without expensive non-native field arithmetic (Remark 10.28). The cycle is deployed: the `pasta_curves` crate defines the scalar field of Pallas to be  $\mathbb{F}_q$  and that of Vesta to be  $\mathbb{F}_p$ , as specified in §5.4.9.6 of the Zcash protocol specification.

The discrete logarithm problem in  $E_p(\mathbb{F}_p)$  and  $E_q(\mathbb{F}_q)$  is believed to require about  $2^{127}$  operations— $\sqrt{r}$  for the 255-bit orders  $r \approx 2^{254}$  (Proposition 10.30)—with no faster attack known (Remark 10.31), slightly below the 128-bit security level targeted by Halo 2 and Zcash Orchard.

Each Pasta curve is thus a cyclic group of prime 255-bit order (Theorem 10.22 and Corollary 3.34), with a fast forward map (Proposition 10.21, accelerated by Remark 10.37) whose inversion, the ECDLP, resists every known attack except the generic  $O(\sqrt{r})$  attacks of Proposition 10.30. Later volumes build on this pair.

## 11 Probability, asymptotics, and computation

This section fixes the language of security claims: probability, asymptotic notation, and efficient computation. It proves the union and birthday bounds, the properties of statistical distance, the bias bound for modular reduction, Wald’s identity, Chernoff bounds for the arrival counts of exponential clocks, and the closure of negligible functions under addition and multiplication by polynomials. The birthday bound sets hash output lengths, and the bias bound justifies the sampling of Orchard scalars; later volumes build their block-discovery and extraction analyses on the waiting-time and clock results. A signature scheme advertised as offering “128-bit security” cannot exclude forgery: a forger that guesses the secret key, a finite string, succeeds with positive

probability, so the claim bounds the success probability of every attacker within stated resources; §11.11 states what it asserts. Probability spaces are finite until §11.7; finite spaces suffice for keys, bounded-length messages, group elements, and the coins of an algorithm with a fixed running-time bound. The prerequisites are finite sets, sums, and elementary calculus: limits, the exponential function, and, for the density-defined laws of §11.7, the improper Riemann integral.

## 11.1 Finite probability spaces and events

**Definition 11.1** (Finite probability space). A *finite probability space* is a pair  $(\Omega, \Pr)$  where  $\Omega$  is a nonempty finite set, called the *sample space*, and  $\Pr: \Omega \rightarrow [0, 1]$  is a function, called the *probability mass function*, satisfying the normalisation condition

$$\sum_{\omega \in \Omega} \Pr[\omega] = 1.$$

An element  $\omega \in \Omega$  is an *outcome* (or *elementary event*).

**Definition 11.2** (Event). An *event* is a subset  $A \subseteq \Omega$ . Its *probability* is

$$\Pr[A] = \sum_{\omega \in A} \Pr[\omega],$$

with the convention  $\Pr[\emptyset] = 0$  (an empty sum). The event  $A$  *occurs* on outcome  $\omega$  if  $\omega \in A$ .

Because  $\Omega$  is finite, every subset is an event. On an uncountable sample space the general theory specifies a family of admissible events together with its probability assignment; Remark 11.31 marks the one place this series requires measure theory. The following facts are used without further citation.

**Proposition 11.3** (Basic axioms). For any finite probability space  $(\Omega, \Pr)$  and events  $A, B \subseteq \Omega$ :

1.  $0 \leq \Pr[A] \leq 1$ ,  $\Pr[\emptyset] = 0$ , and  $\Pr[\Omega] = 1$ .
2. (Complement.)  $\Pr[\Omega \setminus A] = 1 - \Pr[A]$ .
3. (Monotonicity.) If  $A \subseteq B$  then  $\Pr[A] \leq \Pr[B]$ .
4. (Inclusion–exclusion for two events.)  $\Pr[A \cup B] = \Pr[A] + \Pr[B] - \Pr[A \cap B]$ .
5. (Finite additivity.) If  $A_1, \dots, A_n$  are pairwise disjoint then  $\Pr[\bigcup_{i=1}^n A_i] = \sum_{i=1}^n \Pr[A_i]$ .

*Proof.* Every claim follows by manipulating the defining sums. For (1), each summand  $\Pr[\omega]$  lies in  $[0, 1]$  and the total is 1 by normalisation; an event's probability is a partial sum of nonnegative terms, hence lies in  $[0, 1]$ . For (2), the sets  $A$  and  $\Omega \setminus A$  partition  $\Omega$ , so  $\Pr[A] + \Pr[\Omega \setminus A] = \sum_{\omega \in \Omega} \Pr[\omega] = 1$ . For (3), splitting the sum over  $B$  gives  $\Pr[B] = \sum_{\omega \in A} \Pr[\omega] + \sum_{\omega \in B \setminus A} \Pr[\omega] \geq \Pr[A]$ , the second term being a sum of nonnegative numbers. For (5), disjointness means each  $\omega$  in the union lies in exactly one  $A_i$ , so the double sum  $\sum_i \sum_{\omega \in A_i} \Pr[\omega]$  counts each  $\omega \in \bigcup_i A_i$  exactly

once. For (4), write  $A \cup B = A \sqcup (B \setminus A)$  (disjoint), so by (5)  $\Pr[A \cup B] = \Pr[A] + \Pr[B \setminus A]$ ; also  $B = (A \cap B) \sqcup (B \setminus A)$ , giving  $\Pr[B \setminus A] = \Pr[B] - \Pr[A \cap B]$ . Substituting yields the claim.  $\square$

The distribution used most often in cryptography is the uniform distribution.

**Definition 11.4** (Uniform distribution). Let  $S$  be a nonempty finite set. The *uniform distribution* on  $S$  is the probability space  $(S, \Pr)$  with  $\Pr[\omega] = 1/|S|$  for every  $\omega \in S$ . The notation  $x \leftarrow S$  (or  $x \xleftarrow{\$} S$ ) means that  $x$  is sampled according to the uniform distribution on  $S$ . Under the uniform distribution, for any event  $A \subseteq S$ ,

$$\Pr[A] = \frac{|A|}{|S|}.$$

## 11.2 Conditional probability and independence

Conditioning updates the probability of one event on the information that another has occurred.

**Definition 11.5** (Conditional probability). Let  $A, B$  be events with  $\Pr[B] > 0$ . The *conditional probability of  $A$  given  $B$*  is

$$\Pr[A \mid B] = \frac{\Pr[A \cap B]}{\Pr[B]}.$$

When  $\Pr[B] = 0$  the quantity is left undefined.

Conditioning on  $B$  restricts the sample space to  $B$  and renormalises so that  $B$  has probability 1: the map  $\omega \mapsto \Pr[\{\omega\} \mid B]$  is itself a probability mass function on  $B$ , and  $\Pr[A \mid B]$  is the probability it assigns to  $A \cap B$ .

**Proposition 11.6** (Chain rule and total probability). Let  $A_1, \dots, A_n$  be events with  $\Pr[A_1 \cap \dots \cap A_{n-1}] > 0$ . Then

$$\Pr[A_1 \cap \dots \cap A_n] = \Pr[A_1] \Pr[A_2 \mid A_1] \Pr[A_3 \mid A_1 \cap A_2] \dots \Pr[A_n \mid A_1 \cap \dots \cap A_{n-1}].$$

Moreover, if  $B_1, \dots, B_k$  partition  $\Omega$  (pairwise disjoint with union  $\Omega$ ) and each  $\Pr[B_i] > 0$ , then for any event  $A$ ,

$$\Pr[A] = \sum_{i=1}^k \Pr[A \mid B_i] \Pr[B_i].$$

*Proof.* The chain rule follows by telescoping: each factor  $\Pr[A_j \mid A_1 \cap \dots \cap A_{j-1}]$  equals  $\Pr[A_1 \cap \dots \cap A_j] / \Pr[A_1 \cap \dots \cap A_{j-1}]$  by Definition 11.5, and the product collapses to  $\Pr[A_1 \cap \dots \cap A_n] / 1$ . (Monotonicity ensures that all the intermediate intersections have positive probability, so every factor is defined.) For total probability, the events  $A \cap B_i$  are pairwise disjoint with union  $A$  (since the  $B_i$  partition  $\Omega$ ), so finite additivity and  $\Pr[A \cap B_i] = \Pr[A \mid B_i] \Pr[B_i]$  give  $\Pr[A] = \sum_i \Pr[A \cap B_i] = \sum_i \Pr[A \mid B_i] \Pr[B_i]$ .  $\square$

Independence is the formal statement that one event carries no information about another.

**Definition 11.7** (Independence of events). Two events  $A, B$  are *independent* if

$$\Pr[A \cap B] = \Pr[A] \Pr[B].$$

A family  $\{A_i\}_{i \in I}$  of events is *mutually independent* if for every finite subset  $J \subseteq I$ ,

$$\Pr\left[\bigcap_{i \in J} A_i\right] = \prod_{i \in J} \Pr[A_i].$$

*Remark 11.8.* When  $\Pr[B] > 0$ , independence of  $A$  and  $B$  is equivalent to  $\Pr[A \mid B] = \Pr[A]$ . Mutual independence is strictly stronger than pairwise independence. The standard counterexample consists of two fair coin tosses, with  $A$  “the first is heads”,  $B$  “the second is heads”, and  $C$  “the two coins differ”. Each pair is independent, yet  $\Pr[A \cap B \cap C] = 0 \neq \frac{1}{8} = \Pr[A] \Pr[B] \Pr[C]$ , so the three are not mutually independent.

### 11.3 Random variables and expectation

**Definition 11.9** (Random variable). Let  $(\Omega, \Pr)$  be a finite probability space and  $T$  any set. A ( $T$ -valued) *random variable* is a function  $X: \Omega \rightarrow T$ . When  $T \subseteq \mathbb{R}$ , we call  $X$  *real-valued*. For  $t \in T$ ,

$$\Pr[X = t] = \Pr[\{\omega \in \Omega : X(\omega) = t\}],$$

and more generally  $\Pr[X \in B]$  denotes the probability that  $X$  takes a value in  $B \subseteq T$ . The function  $t \mapsto \Pr[X = t]$  on the finite image of  $X$  is the *distribution* (or *law*) of  $X$ .

**Definition 11.10** (Independent random variables). Random variables  $X_1, \dots, X_n$  (with values in sets  $T_1, \dots, T_n$ ) are *mutually independent* if for all  $t_1 \in T_1, \dots, t_n \in T_n$ ,

$$\Pr[X_1 = t_1, \dots, X_n = t_n] = \prod_{i=1}^n \Pr[X_i = t_i].$$

**Definition 11.11** (Expectation). Let  $X: \Omega \rightarrow \mathbb{R}$  be a real-valued random variable on a finite space. Its *expectation* (or *mean*) is

$$\mathbb{E}[X] = \sum_{\omega \in \Omega} X(\omega) \Pr[\omega] = \sum_{t \in \text{im}(X)} t \cdot \Pr[X = t],$$

where  $\text{im}(X)$  denotes the (finite) image of  $X$ . The two expressions agree by grouping outcomes according to their  $X$ -value.

**Definition 11.12** (Indicator). For an event  $A$ , its *indicator random variable*  $\mathbf{1}_A: \Omega \rightarrow \{0, 1\}$  takes the value  $\mathbf{1}_A(\omega) = 1$  if  $\omega \in A$  and 0 otherwise.

**Proposition 11.13** (Indicator expectation). *For any event  $A$ ,  $\mathbb{E}[\mathbf{1}_A] = \Pr[A]$ .*

*Proof.* By Definition 11.11,  $\mathbb{E}[\mathbf{1}_A] = \sum_{\omega} \mathbf{1}_A(\omega) \Pr[\omega] = \sum_{\omega \in A} \Pr[\omega] = \Pr[A]$ , since the indicator vanishes off  $A$ .  $\square$

**Theorem 11.14** (Linearity of expectation). *Let  $X, Y: \Omega \rightarrow \mathbb{R}$  be random variables on a finite space and  $a, b \in \mathbb{R}$ . Then*

$$\mathbb{E}[aX + bY] = a \mathbb{E}[X] + b \mathbb{E}[Y].$$

*More generally, for random variables  $X_1, \dots, X_n$  and scalars  $a_1, \dots, a_n$ ,*

$$\mathbb{E}\left[\sum_{i=1}^n a_i X_i\right] = \sum_{i=1}^n a_i \mathbb{E}[X_i].$$

*This holds whether or not the  $X_i$  are independent.*

*Proof.* Directly from the definition,

$$\begin{aligned} \mathbb{E}[aX + bY] &= \sum_{\omega \in \Omega} (aX(\omega) + bY(\omega)) \Pr[\omega] \\ &= a \sum_{\omega} X(\omega) \Pr[\omega] + b \sum_{\omega} Y(\omega) \Pr[\omega] \\ &= a \mathbb{E}[X] + b \mathbb{E}[Y], \end{aligned}$$

splitting the finite sum and extracting the constants. The  $n$ -term statement follows by induction on  $n$ .  $\square$

The following consequence underlies every averaging (“heavy-row”) argument in the series.

**Proposition 11.15** (Markov’s inequality). *Let  $X \geq 0$  be a real-valued random variable and  $a > 0$ . Then  $\Pr[X \geq a] \leq \mathbb{E}[X]/a$ .*

*Proof.* Every outcome contributes nonnegatively to  $\mathbb{E}[X]$ , and each outcome with  $X(\omega) \geq a$  contributes at least  $a \Pr[\omega]$ ; hence  $\mathbb{E}[X] \geq a \Pr[X \geq a]$ .  $\square$

## 11.4 The union bound and a birthday calculation

This subsection proves the union bound, the inequality used most often in cryptographic proofs; it requires no independence. It gives the upper half of the *birthday bound*, which governs how soon collisions appear under uniform sampling: among roughly  $\sqrt{N}$  uniform samples from a set of size  $N$ , two are likely to coincide. The birthday bound sets the output lengths of cryptographic hash functions.

**Theorem 11.16** (Union bound / Boole’s inequality). *For any events  $A_1, \dots, A_n$  in a finite probability space,*

$$\Pr\left[\bigcup_{i=1}^n A_i\right] \leq \sum_{i=1}^n \Pr[A_i].$$

*Proof.* Disjointify the union into “first occurrence” events:  $B_1 = A_1$  and  $B_i = A_i \setminus (A_1 \cup \dots \cup A_{i-1})$  for  $i \geq 2$ . The  $B_i$  are pairwise disjoint,  $B_i \subseteq A_i$ , and  $\bigcup_i B_i = \bigcup_i A_i$ . By finite additivity and monotonicity (Proposition 11.3),

$$\Pr\left[\bigcup_i A_i\right] = \Pr\left[\bigcup_i B_i\right] = \sum_i \Pr[B_i] \leq \sum_i \Pr[A_i]. \quad \square$$

The “first occurrence” disjointification recurs, in a telescoping form for increasing chains, in the proof of the continuity theorem of §11.7. Taking complements in the union bound gives the corresponding inequality for conjunctions.

**Corollary 11.17** (Union bound for the good event). *If each event  $A_i$  holds except with probability at most  $\varepsilon_i$  (that is,  $\Pr[\Omega \setminus A_i] \leq \varepsilon_i$ ), then all of them hold simultaneously except with probability at most  $\sum_i \varepsilon_i$ :*

$$\Pr\left[\bigcap_{i=1}^n A_i\right] \geq 1 - \sum_{i=1}^n \varepsilon_i.$$

*Proof.* By De Morgan,  $\Omega \setminus \bigcap_i A_i = \bigcup_i (\Omega \setminus A_i)$ . Apply the union bound to the complements and take the complement of both sides.  $\square$

**Proposition 11.18** (Birthday bound). *Sample  $x_1, \dots, x_k$  independently and uniformly from a set of size  $N$ , and let  $C$  be the event that some two of them collide, i.e.  $x_i = x_j$  for some  $i < j$ . Then*

$$\Pr[C] \leq \binom{k}{2} \frac{1}{N} = \frac{k(k-1)}{2N},$$

and, on the other side,

$$\Pr[C] = 1 - \prod_{i=0}^{k-1} \left(1 - \frac{i}{N}\right) \geq 1 - e^{-k(k-1)/(2N)}.$$

In particular a collision becomes likely once  $k = \Theta(\sqrt{N})$ .

*Proof. Upper bound.* For  $i < j$  let  $A_{ij}$  be the event  $x_i = x_j$ . Since  $x_i$  and  $x_j$  are independent and uniform,  $\Pr[A_{ij}] = \sum_v \Pr[x_i = v] \Pr[x_j = v] = N \cdot (1/N)^2 = 1/N$ . Now  $C = \bigcup_{i < j} A_{ij}$ , a union of  $\binom{k}{2}$  events, one per two-element subset of the indices (Lemma 6.12), so the union bound (Theorem 11.16) gives  $\Pr[C] \leq \binom{k}{2}/N$ .

*Lower bound.* The complement  $\bar{C}$  is the event that all  $k$  samples are distinct. If  $k > N$ , the pigeonhole principle forces a collision, so  $\Pr[\bar{C}] = 0$ ; the product also vanishes, through its factor  $1 - N/N$ , so the stated equality holds and the exponential bound is trivial. Let then  $k \leq N$ , and for  $0 \leq i \leq k$  let  $D_i$  be the event that the first  $i$  samples are pairwise distinct ( $D_0 = D_1 = \Omega$ , and  $D_k = \bar{C}$ ). We show  $\Pr[D_i] = \prod_{j=0}^{i-1} (1 - j/N)$  by induction on  $i \leq k$ ; every factor is positive, since  $j \leq k-1 \leq N-1$ , so in particular each  $D_i$  has positive probability and conditioning on it is legitimate (Definition 11.5). For the step, the events pinning the first  $i$  samples to specific distinct values partition  $D_i$ , each with probability  $N^{-i} > 0$  by independence and uniformity; given

any one of them, the next sample avoids the  $i$  values taken with probability  $1 - i/N$ , again by independence and uniformity. Averaging over the cells (total probability, Proposition 11.6) gives  $\Pr[D_{i+1} \mid D_i] = 1 - i/N$ , and  $\Pr[D_{i+1}] = \Pr[D_{i+1} \mid D_i] \Pr[D_i]$  completes the induction. Taking  $i = k$  yields the stated equality, in which every factor is nonnegative, so the inequality  $1 - x \leq e^{-x}$  applies factor by factor. (For  $0 \leq x < 1$  this inequality follows from Bernoulli's inequality,  $(1 - x/n)^n \geq 1 - x$  for  $n > x$ , by letting  $n \rightarrow \infty$  in the limit definition of  $e^{-x}$ ; for  $x \geq 1$  the left side is nonpositive.) Thus

$$\Pr[\overline{C}] = \prod_{i=0}^{k-1} \left(1 - \frac{i}{N}\right) \leq \prod_{i=0}^{k-1} e^{-i/N} = \exp\left(-\frac{1}{N} \sum_{i=0}^{k-1} i\right) = e^{-k(k-1)/(2N)},$$

summing the exponents via  $\sum_{i=0}^{k-1} i = k(k-1)/2$ . Taking complements gives  $\Pr[C] = 1 - \Pr[\overline{C}] \geq 1 - e^{-k(k-1)/(2N)}$ .  $\square$

*Example 11.19.* With  $N = 365$  and  $k = 23$ , the classical birthday problem, the exact product gives  $\Pr[C] = 0.5073$ , a probability above one half, and the proposition brackets it as  $0.5000 \leq \Pr[C] \leq 0.6932$ . The threshold appears at small scale as well: with  $N = 16$ , the fifth sample, the first beyond  $\sqrt{16} = 4$ , raises the collision probability above one half,  $\Pr[C] = 0.5001$  at  $k = 5$ .

*Remark 11.20.* The threshold  $k \approx \sqrt{N}$  is the reason an  $n$ -bit hash function (so  $N = 2^n$ ) offers only about  $2^{n/2}$  collision resistance: an adversary expects a collision after roughly  $2^{n/2}$  random evaluations — for  $N = 2^{256}$ , that is  $\sqrt{2^{256}} = 2^{128}$ . Output lengths are therefore chosen twice the desired security level. The same square-root phenomenon recurs in random-walk discrete-log attacks (Pollard's rho, Proposition 10.30), so the curves used in Halo 2 and Orchard are sized so that  $\sqrt{N}$  itself lies far beyond feasible computation.

## 11.5 Statistical distance

Statistical distance measures how far apart two distributions are, such as a real key-generation procedure and an idealised uniform one; it formalises “no statistical test can distinguish them well”.

**Definition 11.21** (Statistical distance). Let  $P$  and  $Q$  be probability distributions on the same finite set  $S$  (i.e. probability mass functions  $P, Q: S \rightarrow [0, 1]$ ). Their *statistical distance* (or *total variation distance*) is

$$\Delta(P, Q) = \frac{1}{2} \sum_{s \in S} |P(s) - Q(s)|.$$

For random variables  $X, Y$  with values in  $S$ ,  $\Delta(X, Y)$  denotes the statistical distance of their distributions.

The factor  $\frac{1}{2}$  makes  $\Delta$  range over  $[0, 1]$  and makes it equal to the maximum advantage of a distinguisher, as part (3) of the next theorem establishes.

**Theorem 11.22** (Properties of statistical distance). Let  $P, Q, R$  be distributions on a finite set  $S$ . Then:

1.  $0 \leq \Delta(P, Q) \leq 1$ ; and  $\Delta(P, Q) = 0$  if and only if  $P = Q$ .
2. (Metric.)  $\Delta$  is symmetric and satisfies the triangle inequality  $\Delta(P, R) \leq \Delta(P, Q) + \Delta(Q, R)$ .
3. (Variational characterisation.)  $\Delta(P, Q) = \max_{A \subseteq S} (P(A) - Q(A)) = \max_{A \subseteq S} |P(A) - Q(A)|$ , where  $P(A) = \sum_{s \in A} P(s)$ .
4. (Data processing / post-processing.) For any (possibly randomised) function  $f$  applied to a sample,  $\Delta(f(P), f(Q)) \leq \Delta(P, Q)$ . In particular no algorithm can increase statistical distance.

*Proof.* (1) Nonnegativity holds termwise. For the upper bound, split  $S$  into  $S^+ = \{s : P(s) \geq Q(s)\}$  and  $S^- = S \setminus S^+$ . Since  $\sum_s P(s) = \sum_s Q(s) = 1$ , we have  $\sum_s (P(s) - Q(s)) = 0$ , so  $\sum_{s \in S^+} (P(s) - Q(s)) = \sum_{s \in S^-} (Q(s) - P(s)) =: D \geq 0$ . Hence  $\Delta(P, Q) = \frac{1}{2} (\sum_{S^+} (P - Q) + \sum_{S^-} (Q - P)) = D$ . As  $D = \sum_{S^+} (P(s) - Q(s)) \leq \sum_{S^+} P(s) \leq 1$ , we get  $\Delta \leq 1$ . If  $\Delta = 0$  then every  $|P(s) - Q(s)| = 0$ , so  $P = Q$ ; the converse is immediate.

(2) Symmetry holds termwise. For the triangle inequality,  $|P(s) - R(s)| \leq |P(s) - Q(s)| + |Q(s) - R(s)|$  termwise by the triangle inequality on  $\mathbb{R}$ ; summing over  $s$  and halving gives the claim.

(3) With  $D$  as above, take  $A = S^+$ : then  $P(A) - Q(A) = \sum_{S^+} (P - Q) = D = \Delta(P, Q)$ , so the maximum is at least  $\Delta$ . Conversely, for any  $A \subseteq S$ ,  $P(A) - Q(A) = \sum_{s \in A} (P(s) - Q(s)) \leq \sum_{s \in A \cap S^+} (P(s) - Q(s)) \leq D = \Delta$ , since dropping the negative terms only increases the sum and extending from  $A \cap S^+$  to all of  $S^+$  increases it further. Thus the maximum over  $A$  of  $P(A) - Q(A)$  is exactly  $\Delta$ ; replacing  $A$  by its complement swaps the sign, giving the same value for  $\max_A |P(A) - Q(A)|$ .

(4) First take  $f$  deterministic,  $f: S \rightarrow U$ . For  $u \in U$ ,  $f(P)(u) = P(f^{-1}(u))$ . Then

$$\Delta(f(P), f(Q)) = \frac{1}{2} \sum_u |P(f^{-1}(u)) - Q(f^{-1}(u))| \leq \frac{1}{2} \sum_u \sum_{s \in f^{-1}(u)} |P(s) - Q(s)| = \Delta(P, Q),$$

using  $|\sum_s a_s| \leq \sum_s |a_s|$  on each fibre, and that the fibres  $f^{-1}(u)$  partition  $S$ . A randomised  $f$  is a deterministic function of the sample  $s$  together with independent fresh coins  $r$ ; apply the deterministic case to the joint variable  $(s, r)$ , noting that the two joint distributions  $P \times U_r$  and  $Q \times U_r$  have the same statistical distance  $\Delta(P, Q)$ , the shared coin distribution contributing nothing.  $\square$

*Remark 11.23* (Distinguishing interpretation). Part (3) states that  $\Delta(P, Q)$  is exactly the best advantage of any test — even a computationally unbounded one — that, given a single sample, must guess whether it came from  $P$  or  $Q$ : the optimal test outputs “ $P$ ” on the set  $A = S^+$ , and its advantage  $\Pr_P[A] - \Pr_Q[A]$  equals  $\Delta(P, Q)$ . Part (4) makes statistical distance compose in proofs: if  $\Delta(P, Q) \leq \varepsilon$ , then  $P$  and  $Q$  are interchangeable in any analysis at the cost of an additive  $\varepsilon$  in every probability. When  $\varepsilon$  is negligible (Definition 11.61 below), we call  $P$  and  $Q$  *statistically indistinguishable*.

## 11.6 Uniform sampling and the bias of modular reduction

The most tractable source of randomness is a stream of independent uniform bits, whereas cryptography requires uniform elements of other finite sets: a scalar in  $\mathbb{Z}/q\mathbb{Z}$  (Section 2), a field element of  $\mathbb{F}_p$  (Section 6), a point in a group. If the target set  $S$  has  $|S| = 2^\ell$ , reading  $\ell$  uniform bits as a number in  $\{0, \dots, 2^\ell - 1\}$  and indexing into  $S$  is exact. When  $|S|$  is not a power of two — most importantly when  $S = \mathbb{Z}/q\mathbb{Z}$  for a prime  $q$ , the scalar field of an elliptic curve (Section 10) — a common method is *reduce-modulo*: sample a long uniform integer and reduce it mod  $q$ . Some residues then receive one more preimage than others; the next proposition bounds the resulting bias.

**Proposition 11.24** (Bias of modular reduction). *Let  $q \geq 2$  be an integer and let  $K = 2^L$  for a nonnegative integer  $L$ . Sample  $U$  uniformly from  $\{0, 1, \dots, K - 1\}$  (equivalently, read  $L$  uniform bits as an integer), and set  $R = U \bmod q \in \{0, 1, \dots, q - 1\}$ . Let  $\mathcal{U}_q$  denote the uniform distribution on  $\{0, \dots, q - 1\}$ . Then the statistical distance of  $R$  from uniform satisfies*

$$\Delta(R, \mathcal{U}_q) \leq \frac{q}{K} = \frac{q}{2^L}.$$

Consequently, if  $q < 2^n$  and one takes  $L = n + s$  bits, then  $\Delta(R, \mathcal{U}_q) < 2^{-s}$ .

*Proof.* Write  $K = mq + t$  with  $m = \lfloor K/q \rfloor$  and remainder  $t = K \bmod q$ , so  $0 \leq t < q$ . For a residue  $r \in \{0, \dots, q - 1\}$ , the number of integers in  $\{0, \dots, K - 1\}$  congruent to  $r$  modulo  $q$  is  $m + 1$  if  $r < t$  and  $m$  if  $r \geq t$ . Hence

$$\Pr[R = r] = \begin{cases} (m + 1)/K, & r < t, \\ m/K, & r \geq t. \end{cases}$$

Comparing with the target probability  $1/q$  and using  $m/K \leq 1/q \leq (m + 1)/K$  (since  $m \leq K/q \leq (m + 1)q$ ),

$$\left| \Pr[R = r] - \frac{1}{q} \right| \leq \frac{m + 1}{K} - \frac{m}{K} = \frac{1}{K} \quad \text{for every } r.$$

Therefore

$$\Delta(R, \mathcal{U}_q) = \frac{1}{2} \sum_{r=0}^{q-1} \left| \Pr[R = r] - \frac{1}{q} \right| \leq \frac{1}{2} \cdot q \cdot \frac{1}{K} = \frac{q}{2K} \leq \frac{q}{K}.$$

(The computation yields the sharper bound  $q/(2K)$ ;  $q/K$  is the usual stated form.) For the final claim,  $q < 2^n$  and  $K = 2^{n+s}$  give  $q/K < 2^n/2^{n+s} = 2^{-s}$ .  $\square$

*Example 11.25.* Let  $L = 6$  uniform bits be reduced modulo  $q = 5$ . Then  $K = 64 = 12 \cdot 5 + 4$ , so the four residues  $r < 4$  receive 13 preimages each and the residue  $r = 4$  receives 12; the statistical distance from uniform is exactly  $1/80$ , within the guarantee  $2^{-3}$  of the proposition (as  $q < 2^3$  and  $s = 6 - 3 = 3$ ).

*Remark 11.26* (Sampling scalars in practice). By the proposition, a scalar modulo a  $\sim 256$ -bit prime  $q$  with statistical distance below  $2^{-128}$  from uniform is obtained by reducing  $256 + 128 = 384$  uniform bits modulo  $q$ . The bias  $2^{-s}$ , with the slack  $s$  set to the security parameter, is negligible (Definition 11.61), so the reduce-modulo sample is statistically indistinguishable from a

perfectly uniform scalar, and by the data-processing inequality (Theorem 11.22(4)) substituting one for the other anywhere costs at most  $2^{-128}$  per use. An alternative, *rejection sampling*, draws  $\lceil \log_2 q \rceil$  bits, accepts if the result is  $< q$ , and otherwise retries; it gives *exactly* uniform output at the cost of a variable but (with overwhelming probability) small number of retries. For this series the negligible bias of reduce-modulo is acceptable and simpler to analyse. The Orchard specification employs this reduce-modulo sampling when deriving scalars from hash outputs: its ToScalar reduces the full 512-bit PRFexpand output modulo the 255-bit group order, a slack of 257 bits, well beyond the 128-bit slack computed above (Zcash protocol specification §4.2.3 and §5.4.2).

## 11.7 Beyond finite spaces: countable additivity, limits, and densities

Two uses require more than a finite sample space. A process observed until an event occurs, such as an unbounded retry loop or a random walk run until it first hits a barrier, has no finite horizon, even when it terminates with probability one; a waiting time measured in continuous units has no finite sample space. This subsection adds countable additivity, continuity along monotone events, and laws defined by densities, and states in Remark 11.31 the measure-theoretic results it assumes.

**Definition 11.27** (Countable probability space). A *countable probability space* is a pair  $(\Omega, \Pr)$  in which  $\Omega$  is a countable set and  $\Pr: \Omega \rightarrow [0, 1]$  satisfies  $\sum_{\omega \in \Omega} \Pr[\omega] = 1$ ; the sum is independent of the enumeration order because its terms are nonnegative. Events are arbitrary subsets  $A \subseteq \Omega$ , with  $\Pr[A] = \sum_{\omega \in A} \Pr[\omega]$ , and Definitions 11.2–11.11 carry over with “finite” read as “countable” (a random variable’s image may now be countably infinite), with one exception: Definition 11.4 does not — equal weights cannot sum to 1 on a countably infinite set, so the uniform distribution remains a strictly finite notion. Expectation additionally requires its defining sum to converge absolutely.

**Proposition 11.28** (Countable additivity). Let  $(\Omega, \Pr)$  be a countable probability space and  $A_1, A_2, \dots$  pairwise disjoint events. Then

$$\Pr\left[\bigcup_{n \geq 1} A_n\right] = \sum_{n \geq 1} \Pr[A_n],$$

and in particular the basic axioms of Proposition 11.3 hold unchanged.

*Proof.* Every outcome of the union lies in exactly one  $A_n$ , so both sides sum the same nonnegative terms  $\Pr[\omega]$ , grouped differently; a series of nonnegative terms may be summed in any order, or in groups, without changing its value.  $\square$

**Theorem 11.29** (Continuity along monotone events). Let  $\Pr$  be a probability assignment, on any sample space, defined for a family of events closed under complement and countable union, and countably additive on it. If  $A_1 \subseteq A_2 \subseteq \dots$  is an increasing chain of events, then

$$\Pr\left[\bigcup_{n \geq 1} A_n\right] = \lim_{n \rightarrow \infty} \Pr[A_n],$$

and dually  $\Pr[\bigcap_n B_n] = \lim_n \Pr[B_n]$  for a decreasing chain  $B_1 \supseteq B_2 \supseteq \dots$ . On a countable space the hypothesis holds automatically, by Proposition 11.28.

*Proof.* Disjointify by telescoping: set  $D_1 = A_1$  and  $D_n = A_n \setminus A_{n-1}$  for  $n \geq 2$ . The  $D_n$  are pairwise disjoint,  $\bigcup_{k \leq n} D_k = A_n$ , and  $\bigcup_n D_n = \bigcup_n A_n$ . Countable additivity gives

$$\Pr\left[\bigcup_n A_n\right] = \sum_{n \geq 1} \Pr[D_n] = \lim_{n \rightarrow \infty} \sum_{k=1}^n \Pr[D_k] = \lim_{n \rightarrow \infty} \Pr[A_n],$$

the final step by finite additivity. The decreasing case follows by complementation.  $\square$

**Definition 11.30** (Infinite sequence of independent trials). Fix a finite outcome set  $T$  and a distribution  $p$  on  $T$ . An *infinite sequence of independent trials* with law  $p$  is a sequence of random variables  $X_1, X_2, \dots$  such that every prefix  $(X_1, \dots, X_n)$  consists of  $n$  mutually independent  $p$ -distributed trials in the sense of Definition 11.10 — equivalently, the prefix's law is that of the finite probability space  $T^n$  carrying the product weights  $\Pr[(t_1, \dots, t_n)] = p(t_1) \cdots p(t_n)$ . An event is *finitely determined* if membership in it is decided by some fixed finite prefix; its probability is the finite-space value. An event of the form “some prefix satisfies  $P$ ” is the union of an increasing chain of finitely determined events, and Theorem 11.29 assigns it the limit of the prefix probabilities.

*Remark 11.31* (Assumed measure-theoretic results). Definition 11.30 presumes that a single, countably additive probability assignment on infinite sequences exists that restricts correctly to every finite prefix at once. That existence statement is the Kolmogorov extension theorem, and its proof belongs to measure theory, which lies off this series' critical path: the space of infinite sequences over at least two possible outcomes is uncountable, so it escapes Definition 11.27, and the extension specifies the admissible events. This series therefore takes it, with its countable additivity and hence Theorem 11.29, as an assumption rather than a theorem. On infinite-sequence spaces, only finitely determined events and their monotone limits appear in this series, and for those the values are determined by the finite theory together with continuity; the density-defined laws of Definition 11.32 rest on the analogous assumption, their interval and joint-event probabilities being taken to cohere as the stated integrals. The independent families of Definition 11.40 extend that assumption in two ways: joint events of several density-defined quantities are taken to be iterated integrals, in whichever order of integration is convenient — the exchange of order for nonnegative integrands is the Fubini–Tonelli theorem, likewise measure-theoretic — and the gap sequences of Definition 11.44 are infinite families, of which only finite prefixes enter any event this series uses, except through countable unions over the values of an arrival count —  $N(t)$  of Definition 11.44, or the count  $X$  of Theorem 11.51, each event “count =  $x$ ” being a joint event of finitely many gaps. For any set  $B$  of values, the event “count  $\in B$ ” — among them the event that the count is finite, and the tail events of Theorem 11.49 — is taken to have probability  $\sum_{x \in B} \Pr[\text{count} = x]$ , by the countable additivity assumed above.

**Definition 11.32** (Law defined by a density). A real-valued random quantity  $X$  has *density*  $f: \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$  if  $\int_{-\infty}^{\infty} f(t) dt = 1$  and

$$\Pr[a \leq X \leq b] = \int_a^b f(t) dt \quad \text{for all } a \leq b;$$

improper Riemann integrals suffice throughout this series. The *distribution function* is  $F(t) = \Pr[X \leq t] = \int_{-\infty}^t f$ , so  $\Pr[X > t] = 1 - F(t)$ , and single points carry probability zero. The expectation is  $E[X] = \int_{-\infty}^{\infty} t f(t) dt$  when this integral converges absolutely. Two density-defined quantities  $X, Y$  are *independent* if  $\Pr[X \in I, Y \in J] = \Pr[X \in I] \Pr[Y \in J]$  for all intervals  $I, J$ ; probabilities of joint events are then taken to factor through iterated integrals of the product  $f_X(s) f_Y(t)$ .

**11.7.1 Waiting times: tail sums, geometric trials, and Wald's identity.** The results of this unit need only countable expectation and finitely determined events. For a variable with values in  $\mathbb{N} \cup \{\infty\}$ , the defining sum of the expectation has nonnegative terms, and the value  $+\infty$  is permitted when it diverges.

**Lemma 11.33** (Tail-sum formula). *Let  $T$  be a random variable with values in  $\mathbb{N} \cup \{\infty\}$  and  $\Pr[T = \infty] = 0$ . Then  $E[T] = \sum_{k \geq 1} \Pr[T \geq k]$ , either side being infinite exactly when the other is.*

*Proof.* Expanding  $\Pr[T \geq k] = \sum_{m \geq k} \Pr[T = m]$  and regrouping the doubly indexed nonnegative series, each term  $\Pr[T = m]$  is counted once for every  $k \leq m$ , that is,  $m$  times; nonnegative series may be regrouped freely.  $\square$

**Proposition 11.34** (Geometric waiting time). *In an infinite sequence of independent trials, each succeeding with probability  $\varepsilon > 0$ , let  $T$  be the index of the first success. Then  $\Pr[T \geq k] = (1 - \varepsilon)^{k-1}$  and  $E[T] = 1/\varepsilon$ .*

*Proof.* The event  $T \geq k$  says the first  $k - 1$  trials all fail — a finitely determined event of probability  $(1 - \varepsilon)^{k-1}$ ; in particular  $\Pr[T = \infty] = \lim_k (1 - \varepsilon)^{k-1} = 0$  by continuity (Theorem 11.29). The tail-sum formula then gives  $E[T] = \sum_{k \geq 1} (1 - \varepsilon)^{k-1} = 1/\varepsilon$ , a geometric series.  $\square$

**Proposition 11.35** (Wald's identity). *Let  $X_1, X_2, \dots$  be an infinite sequence of independent trials with common law, each  $X_i \geq 0$  with finite mean, and let  $T$  be a stopping rule: a random index such that, for every  $k$ , whether  $T = k$  is decided by the first  $k$  trials alone. If  $E[T] < \infty$ , then*

$$E\left[\sum_{i=1}^T X_i\right] = E[T] \cdot E[X_1].$$

*Proof.* Write  $S = \sum_{i=1}^T X_i$  and  $S_k = \sum_{i=1}^k X_i$ , so that  $S = S_k$  on the event  $T = k$ . The hypothesis  $E[T] < \infty$  forces  $\Pr[T = \infty] = 0$ , by the convention preceding Lemma 11.33. The variable  $S$  takes countably many values — on  $\{T = k\}$  it agrees with  $S_k$ , whose values run over a finite set, the trials having finitely many outcomes — and each event  $\{S = s\}$  is the disjoint countable union  $\bigcup_k \{T = k, S_k = s\}$  of finitely determined events, so its probability is assigned by countable additivity (Remark 11.31). By  $E[S]$  we mean  $\sum_s s \Pr[S = s]$ , the expectation of this countably supported distribution, a sum of nonnegative terms.

Expanding each  $\Pr[S = s]$  and regrouping the doubly indexed nonnegative series,

$$\mathbb{E}[S] = \sum_s s \sum_{k \geq 1} \Pr[T = k, S_k = s] = \sum_{k \geq 1} \mathbb{E}[S_k \mathbf{1}_{T=k}],$$

each inner expectation living on the finite prefix space of the first  $k$  trials (Definition 11.30), where linearity (Theorem 11.14) expands it as  $\sum_{i=1}^k \mathbb{E}[X_i \mathbf{1}_{T=k}]$ . Regrouping the nonnegative double series once more,

$$\mathbb{E}[S] = \sum_{i \geq 1} \sum_{k \geq i} \mathbb{E}[X_i \mathbf{1}_{T=k}].$$

Fix  $i$  and let  $n \geq i$ . The events  $T = k$  (for  $k \leq n$ ),  $T > n$ , and  $T \geq i$  are all decided by the first  $n$  trials, and on that prefix space the pointwise identity  $\mathbf{1}_{T \geq i} = \sum_{k=i}^n \mathbf{1}_{T=k} + \mathbf{1}_{T > n}$  holds, so linearity gives

$$\sum_{k=i}^n \mathbb{E}[X_i \mathbf{1}_{T=k}] = \mathbb{E}[X_i \mathbf{1}_{T \geq i}] - \mathbb{E}[X_i \mathbf{1}_{T > n}].$$

The trials take finitely many values, so  $X_i \leq M$  for a constant  $M$ , and  $0 \leq \mathbb{E}[X_i \mathbf{1}_{T > n}] \leq M \Pr[T > n] \rightarrow 0$  as  $n \rightarrow \infty$ : the terms  $\Pr[T \geq k]$  of the convergent series  $\mathbb{E}[T]$  tend to zero. Hence  $\sum_{k \geq i} \mathbb{E}[X_i \mathbf{1}_{T=k}] = \mathbb{E}[X_i \mathbf{1}_{T \geq i}]$ .

The remaining step is a factorisation. The event  $T \geq i$  — the negation of  $T \leq i - 1$  — is decided by the first  $i - 1$  trials, so its indicator is a function  $g(t_1, \dots, t_{i-1})$  of the first  $i - 1$  trial values, and on the prefix space of the first  $i$  trials the expectation splits along the product weights of Definition 11.30: with  $p$  the common law,

$$\mathbb{E}[X_i \mathbf{1}_{T \geq i}] = \sum_{t_1, \dots, t_i} g(t_1, \dots, t_{i-1}) t_i p(t_1) \dots p(t_i) = \Pr[T \geq i] \cdot \mathbb{E}[X_i],$$

the double sum factoring into the sum over  $(t_1, \dots, t_{i-1})$ , which totals  $\Pr[T \geq i]$ , times  $\sum_{t_i} t_i p(t_i) = \mathbb{E}[X_i]$ . Assembling the pieces and applying the tail-sum formula (Lemma 11.33),

$$\mathbb{E}[S] = \sum_{i \geq 1} \mathbb{E}[X_i] \Pr[T \geq i] = \mathbb{E}[X_1] \cdot \mathbb{E}[T]. \quad \square$$

**11.7.2 Exponential clocks, Poisson counts, and the Chernoff method.** Block discovery is modelled by *clocks*: each party's next block arrives after a random delay whose remaining law does not depend on how long the delay has already run, and two parties race. A light client that samples a chain needs a tail bound on the probability that an adversary has found  $cL$  blocks while the honest chain found  $L$ , and a wallet that spaces its network queries needs a delay law with a prescribed mean and no memory. This unit constructs these clocks and proves the tail bounds from density-defined laws (Definition 11.32) and Markov's inequality with a free parameter.

**Definition 11.36** (Exponential law). A random quantity  $X$  is *exponential with rate*  $\lambda > 0$ , written  $X \sim \text{Exp}(\lambda)$ , if it has the density  $f(t) = \lambda e^{-\lambda t}$  for  $t \geq 0$  and  $f(t) = 0$  for  $t < 0$ .

**Lemma 11.37** (Tail, mean, and memorylessness). Let  $X \sim \text{Exp}(\lambda)$ . Then  $\Pr[X > t] = e^{-\lambda t}$  for all  $t \geq 0$ ,  $\mathbb{E}[X] = 1/\lambda$ , and

$$\Pr[X > s + t \mid X > s] = \Pr[X > t] \quad \text{for all } s, t \geq 0,$$

conditional probability being the ratio of Definition 11.5: a wait without an arrival leaves the law of the remaining wait unchanged. Conversely, a nonnegative quantity with  $\Pr[X > t] = e^{-\lambda t}$  for all  $t \geq 0$  is exponential with rate  $\lambda$ .

*Proof.* The density integrates to 1, and  $\Pr[X > t] = 1 - \int_0^t \lambda e^{-\lambda s} ds = e^{-\lambda t}$ . Integrating by parts,

$$\mathbb{E}[X] = \int_0^\infty t \lambda e^{-\lambda t} dt = [-te^{-\lambda t}]_0^\infty + \int_0^\infty e^{-\lambda t} dt = \frac{1}{\lambda}.$$

Since  $X > s + t$  implies  $X > s$ , the conditional probability is  $e^{-\lambda(s+t)} / e^{-\lambda s} = e^{-\lambda t}$ . For the converse, the events  $X > a$  and  $X > b$  ( $a \leq b$ ) differ by  $a < X \leq b$ , so  $\Pr[a < X \leq b] = e^{-\lambda a} - e^{-\lambda b} = \int_a^b \lambda e^{-\lambda t} dt$ ; and the point  $a$  carries probability at most  $e^{-\lambda(a-\delta)} - e^{-\lambda a}$  for every  $\delta > 0$ , hence zero, so  $\Pr[a \leq X \leq b]$  is the same integral.  $\square$

*Remark 11.38* (Exponential clocks as limits of geometric waiting times). The exponential law is the continuous limit of the geometric waiting time (Proposition 11.34). If trials succeed with a small probability  $\varepsilon$  and each occupies time  $\varepsilon/\lambda$ , the first success arrives at time  $\varepsilon T/\lambda$ , and for integer-valued  $T$

$$\Pr[\varepsilon T/\lambda > t] = \Pr[T \geq \lfloor \lambda t/\varepsilon \rfloor + 1] = (1 - \varepsilon)^{\lfloor \lambda t/\varepsilon \rfloor} \longrightarrow e^{-\lambda t} \quad (\varepsilon \rightarrow 0),$$

because  $(1 - \varepsilon)^{1/\varepsilon} \rightarrow e^{-1}$  and the floor alters the exponent by less than one. Whether a proof of work satisfies this model, with each candidate independent of the last, depends on the trials; the volumes on consensus treat that question.

*Example 11.39* (A jittered schedule). A wallet that must poll each of  $n$  addresses about once a day waits  $\Delta \sim \text{Exp}(1/T)$  between polls, with  $T = 86400/n$  seconds:  $\mathbb{E}[\Delta] = T$  by Lemma 11.37, one expected poll per address per day. For  $n = 3$ ,  $T = 28\,800$  seconds, eight hours; the gap exceeds a full day with probability  $e^{-3} = 0.0498$ , and exceeds twice its mean with probability  $e^{-2} = 0.1353$  whatever  $T$  is. By memorylessness, an observer who has seen no poll for some time obtains no information about the time of the next one.

Clock races and clock restarts both involve joint events of several density-defined quantities, which Definition 11.32 provides only for pairs.

**Definition 11.40** (Independent family of density-defined quantities). Density-defined quantities  $X_1, \dots, X_n$  with densities  $f_1, \dots, f_n$  form an *independent family* if  $\Pr[X_1 \in I_1, \dots, X_n \in I_n] = \prod_i \Pr[X_i \in I_i]$  for all intervals  $I_1, \dots, I_n$ . As in Definition 11.32, the probability of a joint event  $(X_1, \dots, X_n) \in R$ , for a region  $R \subseteq \mathbb{R}^n$  cut out by finitely many inequalities, is then taken to be the iterated integral of  $f_1(s_1) \dots f_n(s_n)$  over  $R$ , in whichever order of integration is convenient (Remark 11.31). In particular a quantity that is a function of one block of the family is independent of a quantity that is a function of a disjoint block: integrate the two blocks separately. A sequence  $X_1, X_2, \dots$  is an independent family if every finite prefix is one, in the manner of Definition 11.30.

**Lemma 11.41** (Integrating out one quantity). *Let  $U$  be a density-defined quantity, with density  $f_U$ , that is a function of the first  $k$  members of an independent family  $X_1, \dots, X_n$ , and write  $W = (X_{k+1}, \dots, X_n)$ . Let  $R$  be a region in the  $(U, W)$ -coordinates whose section  $\{s : (s, w) \in R\}$  is an interval for every fixed value  $w$  of  $W$ . Then*

$$\Pr[(U, W) \in R] = \int_{-\infty}^{\infty} f_U(s) \Pr[W \in R_s] ds, \quad R_s = \{w : (s, w) \in R\},$$

*the inner probability being a joint event of  $X_{k+1}, \dots, X_n$  alone.*

*Proof.* Integrate the product density over the first  $k$  coordinates first, holding  $w$  fixed: the result is the probability that  $U$  lies in the interval  $\{s : (s, w) \in R\}$ , which by Definition 11.32 equals  $\int f_U(s) \mathbf{1}[(s, w) \in R] ds$ . What remains is the integral of  $f_{k+1} \dots f_n$  against this function of  $w$ ; exchanging the order and integrating over  $w$  first leaves, for each  $s$ , the inner probability  $\Pr[W \in R_s]$ .  $\square$

**Lemma 11.42** (Density of a sum). *Let  $U, V$  be independent density-defined quantities, non-negative (their densities vanish on the negative axis). Then  $U + V$  is density-defined, with density the convolution  $(f_U * f_V)(x) = \int_0^x f_U(u) f_V(x - u) du$  for  $x \geq 0$ .*

*Proof.* For  $0 \leq a \leq b$ , the event  $a \leq U + V \leq b$  is the region  $\{(u, v) : u, v \geq 0, a \leq u + v \leq b\}$ , so

$$\Pr[a \leq U + V \leq b] = \int_0^\infty f_U(u) \int_{\max(a-u, 0)}^{\max(b-u, 0)} f_V(v) dv du = \int_0^\infty f_U(u) \int_{\max(a, u)}^{\max(b, u)} f_V(x - u) dx du,$$

substituting  $x = u + v$  in the inner integral. Exchanging the order, the region becomes  $a \leq x \leq b$ ,  $0 \leq u \leq x$ , and the right-hand side is  $\int_a^b (f_U * f_V)(x) dx$ . With  $a = 0$  and  $b = \infty$  the left-hand side is 1, so  $f_U * f_V$  is a density.  $\square$

**Lemma 11.43** (Two-clock race). *Let  $X \sim \text{Exp}(\lambda)$  and  $Y \sim \text{Exp}(\mu)$  be independent. Then, for every  $u \geq 0$ ,*

$$\Pr[X < Y, X \leq u] = \frac{\lambda}{\lambda + \mu} (1 - e^{-(\lambda + \mu)u}),$$

*symmetrically with the roles of  $(X, \lambda)$  and  $(Y, \mu)$  exchanged, and  $\Pr[X = Y] = 0$ . Consequently  $\Pr[X < Y] = \lambda/(\lambda + \mu)$ ; the winning time  $\min(X, Y)$  is exponential with rate  $\lambda + \mu$ ; and the winner is independent of the winning time,*

$$\Pr[X < Y, \min(X, Y) \leq u] = \Pr[X < Y] \cdot \Pr[\min(X, Y) \leq u] \quad \text{for all } u \geq 0.$$

*Proof.* The event is the region  $0 \leq s \leq u, t > s$  in the  $(X, Y)$ -plane. Integrating over  $t$  first (Definition 11.40),

$$\Pr[X < Y, X \leq u] = \int_0^u \lambda e^{-\lambda s} \Pr[Y > s] ds = \int_0^u \lambda e^{-(\lambda + \mu)s} ds = \frac{\lambda}{\lambda + \mu} (1 - e^{-(\lambda + \mu)u}).$$

The diagonal  $s = t$  has inner integral zero, so  $\Pr[X = Y] = 0$ , and  $u = \infty$  gives  $\Pr[X < Y]$ . On the event  $X < Y$  the winning time is  $X$ , so the displayed formula is  $\Pr[X < Y, \min(X, Y) \leq u]$ ; adding

its mirror for  $Y < X$  gives  $\Pr[\min(X, Y) \leq u] = 1 - e^{-(\lambda+\mu)u}$ , whence  $\min(X, Y)$  is exponential with rate  $\lambda + \mu$  by the characterisation in Lemma 11.37, and the formula is exactly the product  $\Pr[X < Y] \cdot \Pr[\min(X, Y) \leq u]$ . Differencing at two values of  $u$  extends the factorisation from half-lines to intervals.  $\square$

**Definition 11.44** (Exponential clock process). An *exponential clock of rate  $\lambda$*  — a *Poisson process*, by Theorem 11.47 — is an independent family  $G_1, G_2, \dots$  of  $\text{Exp}(\lambda)$  quantities, the *gaps*. Its *arrival times* are the partial sums  $S_n = G_1 + \dots + G_n$ , with  $S_0 = 0$ , and its *arrival count* up to time  $t \geq 0$  is  $N(t) = \sup\{n : S_n \leq t\}$ , the number of arrivals in  $[0, t]$ , a value in  $\mathbb{N} \cup \{\infty\}$  that Theorem 11.47 shows finite with probability one. The event  $N(t) = n$  is  $S_n \leq t < S_{n+1}$ , a joint event of the first  $n + 1$  gaps.

**Lemma 11.45** (Arrival-time density). *The  $n$ -th arrival time  $S_n$  of an exponential clock of rate  $\lambda$  has density  $f_n(s) = \lambda^n s^{n-1} e^{-\lambda s} / (n-1)!$  for  $s \geq 0$ .*

*Proof.* Induction on  $n$ :  $f_1$  is the exponential density, and  $S_{n+1} = S_n + G_{n+1}$  with  $S_n$  a function of the first  $n$  gaps, hence independent of  $G_{n+1}$  (Definition 11.40), so Lemma 11.42 gives

$$f_{n+1}(s) = \int_0^s \frac{\lambda^n u^{n-1} e^{-\lambda u}}{(n-1)!} \lambda e^{-\lambda(s-u)} du = \frac{\lambda^{n+1} e^{-\lambda s}}{(n-1)!} \int_0^s u^{n-1} du = \frac{\lambda^{n+1} s^n e^{-\lambda s}}{n!}. \quad \square$$

**Definition 11.46** (Poisson law). A random variable  $K$  with values in  $\mathbb{N}$  is *Poisson with mean  $\nu > 0$*  if  $\Pr[K = k] = e^{-\nu} \nu^k / k!$  for every  $k \geq 0$ . The weights sum to 1 by the exponential series  $e^\nu = \sum_{k \geq 0} \nu^k / k!$ , and  $\mathbb{E}[K] = \sum_{k \geq 1} k e^{-\nu} \nu^k / k! = \nu e^{-\nu} \sum_{k \geq 1} \nu^{k-1} / (k-1)! = \nu$ , so the parameter is the mean.

An arrival count is a function of the gaps of a clock family, so it lives on no finite or countable space, and the expectation of Definition 11.11 does not literally apply to it. The following convention covers  $N(t)$  above and the count  $X$  of Theorem 11.51 below. Let  $X$  be a variable with values in  $\mathbb{N} \cup \{\infty\}$  that is a function of the gaps, each event  $X = x$  with  $x$  finite being a joint event of finitely many gaps, with the probability of Definition 11.40. Events  $X \in B$ , for sets  $B$  of finite values, have the probabilities  $\sum_{x \in B} \Pr[X = x]$  (Remark 11.31), and when  $X$  is finite with probability 1 its *mean* is defined by the law,

$$\mathbb{E}[X] = \sum_{x \geq 0} x \Pr[X = x],$$

a sum of nonnegative terms, permitted to be  $+\infty$  as in the convention preceding Lemma 11.33. On a finite or countable space this sum is the second expression of Definition 11.11, the expectation regrouped by value (Definition 11.27), so the two definitions agree whenever both apply.

**Theorem 11.47** (Poisson arrival counts). *For an exponential clock of rate  $\lambda$  and every  $t \geq 0$ , the arrival count  $N(t)$  is Poisson with mean  $\lambda t$ :*

$$\Pr[N(t) = n] = e^{-\lambda t} \frac{(\lambda t)^n}{n!}, \quad n = 0, 1, 2, \dots$$

*In particular  $N(t)$  is finite with probability 1 and  $\mathbb{E}[N(t)] = \lambda t$ : a rate- $\lambda$  clock delivers  $\lambda$  arrivals per unit time on average.*

*Proof.* For  $n = 0$  the event is  $G_1 > t$ , of probability  $e^{-\lambda t}$ . For  $n \geq 1$  the event  $S_n \leq t < S_n + G_{n+1}$  is, in the  $(S_n, G_{n+1})$ -coordinates, the region  $s \leq t, g > t - s$ , whose section at fixed  $g$  is the interval  $(t - g, t]$ ; the quantity  $S_n$  is a function of the first  $n$  gaps with the density of Lemma 11.45, so Lemma 11.41 gives

$$\Pr[N(t) = n] = \int_0^t \frac{\lambda^n s^{n-1} e^{-\lambda s}}{(n-1)!} \Pr[G_{n+1} > t - s] ds = \frac{\lambda^n e^{-\lambda t}}{(n-1)!} \int_0^t s^{n-1} ds = e^{-\lambda t} \frac{(\lambda t)^n}{n!}.$$

The event  $N(t) < \infty$  is the disjoint union of the events  $N(t) = n$ , so its probability is the sum of the weights (Remark 11.31), which is 1 (Definition 11.46); and the mean — the sum over the law, by the convention above — is the one computed in Definition 11.46.  $\square$

Markov's inequality applied to  $X$  itself gives a bound decaying only like  $1/a$ ; applied to  $e^{tX}$ , it gives a bound decaying exponentially, with the free parameter  $t$  optimised for the threshold. This is the Chernoff method.

**Definition 11.48** (Moment generating function). For a real-valued random variable  $X$  with finite or countable image — one on a finite or countable space, or an arrival count of a clock family in the sense of the convention preceding Theorem 11.47, restricted to its finite values (the value  $\infty$  has probability zero there), whose law is countably additive over its values:  $\Pr[X \in B] = \sum_{x \in B} \Pr[X = x]$  for every set  $B$  of values (Remark 11.31) — and a real  $t$ , the *moment generating function* is

$$M_X(t) = \sum_x e^{tx} \Pr[X = x] \in (0, \infty],$$

the sum running over the finite values of  $X$ ; its terms are positive, so it either converges or is  $+\infty$ . On a countable space it is  $\mathbb{E}[e^{tX}]$ , the expectation regrouped by value, and  $M_X(0) = \sum_x \Pr[X = x] = 1$ .

**Theorem 11.49** (The Chernoff method). *Let  $X$  be as above, a real, and  $t > 0$  with  $M_X(t) < \infty$ . Then*

$$\Pr[X \geq a] \leq e^{-ta} M_X(t),$$

*and hence  $\Pr[X \geq a] \leq \inf_t e^{-ta} M_X(t)$ , the infimum over the  $t > 0$  at which  $M_X$  is finite.*

*Proof.* The events  $X = x$  are disjoint, so  $\Pr[X \geq a] = \sum_{x \geq a} \Pr[X = x]$  by countable additivity (Proposition 11.28, or the clause of Definition 11.48 for a clock family). Since  $t > 0$ , each term with  $x \geq a$  satisfies  $\Pr[X = x] \leq e^{t(x-a)} \Pr[X = x]$ , and summing gives  $\Pr[X \geq a] \leq e^{-ta} \sum_x e^{tx} \Pr[X = x]$ , the terms with  $x < a$  being nonnegative. This is Markov's inequality (Proposition 11.15) applied to  $e^{tX}$ , written on the law of  $X$ .  $\square$

**Corollary 11.50** (Chernoff bound for a Poisson variable). *Let  $K$  be Poisson with mean  $\nu$  and let  $a > \nu$ . Then  $M_K(t) = \exp(\nu(e^t - 1))$  for every real  $t$ , and*

$$\Pr[K \geq a] \leq e^{a-\nu} \left(\frac{\nu}{a}\right)^a.$$

*In particular, for  $\nu = \mu L$  and  $a = cL$  with  $c > \mu$ , the bound reads  $e^{L(c-\mu)} (c/\mu)^{-cL}$ .*

*Proof.* Expanding,  $M_K(t) = \sum_k e^{tk} e^{-\nu} \nu^k / k! = e^{-\nu} \sum_k (\nu e^t)^k / k! = e^{-\nu} e^{\nu e^t}$ . The exponent  $\phi(t) = -ta + \nu(e^t - 1)$  of the Chernoff bound has  $\phi'(t) = -a + \nu e^t$  and  $\phi''(t) = \nu e^t > 0$ , so it is minimised at  $e^t = a/\nu$ , a positive  $t$  exactly when  $a > \nu$ ; there  $\phi = -a \log(a/\nu) + a - \nu$ , with  $\log$  the natural logarithm — base-free log is natural throughout, base two being written  $\log_2$  — and exponentiating gives the bound. Substituting  $\nu = \mu L$  and  $a = cL$  gives  $e^{cL-\mu L}(\mu/c)^{cL}$ .  $\square$

**Theorem 11.51** (Negative-binomial arrival count). *Let two exponential clocks be independent — their gaps together form one independent family — with rates  $\lambda$ , the honest clock, and  $\mu$ , the adversary clock, and let  $X$  be the number of adversary arrivals strictly before the  $L$ -th honest arrival, for a fixed  $L \geq 1$ . Then  $X$  is finite with probability 1, with the negative binomial law*

$$\Pr[X = m] = \binom{m+L-1}{m} p^L q^m, \quad p = \frac{\lambda}{\lambda + \mu}, \quad q = \frac{\mu}{\lambda + \mu}, \quad m = 0, 1, 2, \dots,$$

of mean  $Lq/p = L\mu/\lambda$ , and, for every real  $t$  with  $\mu e^t < \lambda + \mu$ ,

$$M_X(t) = \left( \frac{\lambda}{\lambda + \mu - \mu e^t} \right)^L;$$

with the honest rate normalised to 1 this is  $M_X(t) = (1/(1 + \mu - \mu e^t))^L$ . Only the ratio  $\mu/\lambda$  enters.

*Proof.* Write  $S_L$  for the  $L$ -th honest arrival time, a function of the honest gaps with the density  $f_L$  of Lemma 11.45, and  $N_A$  for the adversary's arrival count. The event  $X = m$  differs from  $N_A(S_L) = m$  only on the events  $S_L = A_1 + \dots + A_m$  and  $S_L = A_1 + \dots + A_{m+1}$ , with  $A_i$  the adversary gaps; each of these, in the coordinates  $(S_L, A_1, \dots, A_{m+1})$ , has a single point as its section at fixed adversary gaps, hence probability zero (Definition 11.32), so  $\Pr[X = m] = \Pr[N_A(S_L) = m]$ . The latter event is the region  $A_1 + \dots + A_m \leq S_L < A_1 + \dots + A_{m+1}$ , whose section at fixed adversary gaps is an interval. By Lemma 11.41 and Theorem 11.47,

$$\begin{aligned} \Pr[X = m] &= \int_0^\infty f_L(s) \Pr[N_A(s) = m] ds = \frac{\lambda^L \mu^m}{(L-1)! m!} \int_0^\infty s^{m+L-1} e^{-(\lambda+\mu)s} ds \\ &= \frac{\lambda^L \mu^m}{(L-1)! m!} \cdot \frac{(m+L-1)!}{(\lambda + \mu)^{m+L}}, \end{aligned}$$

using  $\int_0^\infty s^k e^{-\alpha s} ds = k!/\alpha^{k+1}$  for  $\alpha > 0$ , which  $k$  integrations by parts establish. Rearranging the factorials gives the binomial coefficient and the powers of  $p$  and  $q$ .

For the generating function, set  $x = qe^t$ , so that  $0 \leq x < 1$  exactly when  $\mu e^t < \lambda + \mu$ . Then  $M_X(t) = p^L \sum_{m \geq 0} \binom{m+L-1}{m} x^m$ , and we claim the series is  $(1-x)^{-L}$ , by induction on  $L$ . For  $L = 1$  it is the geometric series. For the step, write  $T_L(n) = \sum_{m=0}^n \binom{m+L-1}{m} x^m$  for the partial sums. Pascal's rule  $\binom{m+L}{m} = \binom{m+L-1}{m-1} + \binom{m+L-1}{m}$  — sort the  $m$ -subsets of  $m+L$  items, which the coefficients count (Lemma 6.12), by whether they contain a distinguished item — gives, on subtracting  $x T_{L+1}(n)$  from  $T_{L+1}(n)$  term by term,

$$(1-x) T_{L+1}(n) = T_L(n) - \binom{n+L}{n} x^{n+1}.$$

Hence  $T_{L+1}(n) \leq T_L(n)/(1-x) \leq (1-x)^{-L-1}$  by the induction hypothesis: the partial sums  $T_{L+1}(n)$  increase and are bounded, so the series converges, and its terms  $\binom{n+L}{n}x^n$  tend to zero. Letting  $n \rightarrow \infty$  in the display gives  $(1-x) \sum_{m \geq 0} \binom{m+L}{m}x^m = (1-x)^{-L}$ , the claim for  $L+1$ . Hence  $M_X(t) = (p/(1-qe^t))^L$ , which is the displayed form after multiplying numerator and denominator by  $\lambda + \mu$ . At  $t = 0$  this gives  $\sum_m \Pr[X = m] = 1$ ; the event  $X < \infty$  is the disjoint union of the events  $X = m$ , so its probability is this sum (Remark 11.31), and  $X$  is finite with probability 1. For the mean — the sum over the law, by the convention preceding Theorem 11.47 —  $m \binom{m+L-1}{m} = L \binom{m+L-1}{m-1}$ , so  $E[X] = Lq p^L \sum_{k \geq 0} \binom{k+L}{k} q^k = Lq p^L (1-q)^{-L-1} = Lq/p$  by the same series with  $L+1$  in place of  $L$ .  $\square$

*Remark 11.52* (The coin-sequence form). The same law arises without clocks. In an infinite sequence of independent trials (Definition 11.30) succeeding with probability  $p$ , the number of failures strictly before the  $L$ -th success equals  $m$  exactly when the  $(m+L)$ -th trial succeeds and, among the first  $m+L-1$ , precisely  $m$  fail:  $\binom{m+L-1}{m}$  orders, one per choice of the  $m$  failing positions (Lemma 6.12), each of probability  $p^{L-1}q^m \cdot p$ . The two-clock race and memorylessness (Lemma 11.43 and Lemma 11.37) identify the clock model's attribution sequence — which clock fires next — with such a coin sequence of success probability  $\lambda/(\lambda + \mu)$ ; later volumes carry out that identification.

**Theorem 11.53** (Chernoff bound for the arrival count). *In the setting of Theorem 11.51 with honest rate 1 and adversary rate  $\mu$ , for every  $c > \mu$ ,*

$$\Pr[X \geq cL] \leq \left[ \left( \frac{1+c}{1+\mu} \right)^{1+c} \left( \frac{\mu}{c} \right)^c \right]^L,$$

*the Chernoff bound optimised at  $e^t = c(1+\mu)/(\mu(1+c))$ . The base in brackets is strictly less than 1, so the bound decays exponentially in  $L$ .*

*Proof.* The count  $X$  is finite with probability one with values in  $\mathbb{N}$  (Theorem 11.51) and has a law countably additive over its values (Remark 11.31), so Theorem 11.49 applies: with Theorem 11.51,  $\Pr[X \geq cL] \leq e^{-tcL} M_X(t) = e^{L\phi(t)}$  with  $\phi(t) = -tc - \log(1+\mu - \mu e^t)$ , for  $0 < t < \log((1+\mu)/\mu)$ . On that interval

$$\phi'(t) = -c + \frac{\mu e^t}{1+\mu - \mu e^t}, \quad \phi''(t) = \frac{\mu e^t(1+\mu)}{(1+\mu - \mu e^t)^2} > 0,$$

so  $\phi'$  is strictly increasing (positive derivative) and vanishes at most once, namely where  $\mu e^t(1+\mu) = c(1+\mu)$ , that is, at  $e^{t^*} = c(1+\mu)/(\mu(1+c))$ ; this lies in the interval because  $c > \mu$  gives  $e^{t^*} > 1$  and  $c/(1+c) < 1$  gives  $e^{t^*} < (1+\mu)/\mu$ . Thus  $\phi$  decreases before  $t^*$  and increases after it, and  $\phi(t^*)$  is its minimum on the interval. At  $t^*$ ,  $1+\mu - \mu e^{t^*} = (1+\mu)/(1+c)$ , so  $e^{\phi(t^*)} = (e^{t^*})^{-c} (1+c)/(1+\mu)$ , which is the bracketed base. Finally  $\phi(0) = 0$  and  $\phi$  decreases on  $[0, t^*]$ , since  $\phi'(0) = \mu - c < 0$ , so  $\phi(t^*) < 0$ : the base is below 1.  $\square$

*Example 11.54* (The Poisson expression is not a bound for the arrival count). With the honest rate 1, the arrival count  $X$  has mean  $\mu L$ , the same as the Poisson variable of Corollary 11.50 with  $\nu = \mu L$ , which counts adversary arrivals in a window of fixed length  $L$ ; the two laws differ because

the window here ends at the random time  $S_L$ . Substituting one for the other is an error. At  $\mu = 1/2$  and  $c = 3/4$ , the exact tail  $\Pr[X \geq cL]$  (summing Theorem 11.51), the bound of Theorem 11.53, and the Poisson expression  $e^{L(c-\mu)}(c/\mu)^{-cL}$  are

| $L$ | exact tail           | clock bound          | Poisson expression    |
|-----|----------------------|----------------------|-----------------------|
| 40  | $5.05 \cdot 10^{-2}$ | $2.53 \cdot 10^{-1}$ | $1.15 \cdot 10^{-1}$  |
| 200 | $1.09 \cdot 10^{-4}$ | $1.04 \cdot 10^{-3}$ | $2.00 \cdot 10^{-5}$  |
| 400 | $8.23 \cdot 10^{-8}$ | $1.08 \cdot 10^{-6}$ | $4.00 \cdot 10^{-10}$ |

At  $L = 200$  and  $L = 400$  the Poisson expression lies *below* the exact tail, so it is not an upper bound for the arrival count; its base per unit of  $L$  is 0.9473, against 0.9662 for the valid bound.

Later volumes build block-discovery and extraction analyses on these results: block clocks are the exponential clocks above, their races and arrival counts the lemmas of this unit, block-attribution sequences are infinite sequences of independent trials, extractor running times are the waiting times of the previous unit (the geometric waiting time and Wald's identity), and the limit arguments instantiate Theorem 11.29.

## 11.8 Asymptotic notation

Efficiency and hardness are stated by comparing how functions of an integer parameter grow as that parameter tends to infinity, in the Bachmann–Landau notation. Throughout, functions map  $\mathbb{N}$  (or a cofinite subset of it) to the nonnegative reals  $\mathbb{R}_{\geq 0}$ , and “for sufficiently large  $n$ ” means “for all  $n$  beyond some threshold  $n_0$ ”.

**Definition 11.55** (Big-O, Omega, Theta, little-o, little-omega). Let  $f, g: \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$  with  $g$  eventually positive.

- $f = O(g)$  (read “ $f$  is big-O of  $g$ ”, an asymptotic upper bound) if there exist constants  $c > 0$  and  $n_0$  with  $f(n) \leq c g(n)$  for all  $n \geq n_0$ .
- $f = \Omega(g)$  (asymptotic lower bound) if there exist  $c > 0$  and  $n_0$  with  $f(n) \geq c g(n)$  for all  $n \geq n_0$ ; equivalently,  $f = \Omega(g)$  if and only if  $g = O(f)$ .
- $f = \Theta(g)$  (tight bound) if  $f = O(g)$  and  $f = \Omega(g)$ ; i.e. there are  $c_1, c_2 > 0$  and  $n_0$  with  $c_1 g(n) \leq f(n) \leq c_2 g(n)$  for  $n \geq n_0$ .
- $f = o(g)$  (read “little-o”, strictly smaller order) if for every  $c > 0$  there is  $n_0$  with  $f(n) \leq c g(n)$  for all  $n \geq n_0$ ; equivalently  $\lim_{n \rightarrow \infty} f(n)/g(n) = 0$ .
- $f = \omega(g)$  (strictly larger order) if  $g = o(f)$ ; equivalently  $\lim_{n \rightarrow \infty} f(n)/g(n) = \infty$ .

*Remark 11.56.* The “=” in “ $f = O(g)$ ” is traditional but denotes set membership:  $O(g)$  is the class of all functions bounded above by a constant multiple of  $g$ , and  $f = O(g)$  means  $f \in O(g)$ . Such equations are not read right to left. We follow the customary abuse of notation.

*Example 11.57.* The polynomial  $3n^2 + 10n + 7$  is  $\Theta(n^2)$ : for  $n \geq 1$  one has  $3n^2 \leq 3n^2 + 10n + 7 \leq 20n^2$ , with equality on the right at  $n = 1$ , so the constant 20 is exact there. Also  $n^2 = o(n^3)$ , since

$n^2/n^3 = 1/n \rightarrow 0$ ;  $\log_2 n = o(n)$ ;  $n^{100} = o(2^n)$  (any polynomial is little-o of any exponential with base  $> 1$ ); and  $n! = \omega(2^n)$ . A constant factor does not change the order:  $2^n = o(2^{n+1})$  is *false*, but  $2^n = \Theta(2^{n+1})$  is true, since  $2^{n+1} = 2 \cdot 2^n$  is within a constant factor.

**Proposition 11.58** (Useful closure facts). *Let  $f_1 = O(g_1)$  and  $f_2 = O(g_2)$ . Then  $f_1 + f_2 = O(\max(g_1, g_2)) = O(g_1 + g_2)$  and  $f_1 f_2 = O(g_1 g_2)$ . The same closure statements hold with  $O$  replaced uniformly by  $\Omega$  or  $\Theta$ . Moreover  $O(\cdot)$  is transitive:  $f = O(g)$  and  $g = O(h)$  imply  $f = O(h)$ .*

*Proof.* Choose witnesses  $(c_1, n_1)$  and  $(c_2, n_2)$  for the two hypotheses and let  $n_0 = \max(n_1, n_2)$ . For  $n \geq n_0$ ,  $f_1(n) + f_2(n) \leq c_1 g_1(n) + c_2 g_2(n) \leq (c_1 + c_2) \max(g_1(n), g_2(n))$ , and  $\max(g_1, g_2) \leq g_1 + g_2 \leq 2 \max(g_1, g_2)$ , so the two right-hand forms agree up to a constant; this proves the sum rule. For products,  $f_1(n) f_2(n) \leq c_1 c_2 g_1(n) g_2(n)$ . Transitivity:  $f \leq c g$  and  $g \leq c' h$  eventually give  $f \leq cc' h$  eventually. The  $\Omega$  statements follow by the equivalence  $f = \Omega(g) \Leftrightarrow g = O(f)$ , and  $\Theta$  by combining the two.  $\square$

## 11.9 Polynomial, exponential, and negligible functions

Fix a distinguished variable  $\lambda \in \mathbb{N}$ , the *security parameter* (§11.11); informally, larger  $\lambda$  means more security and larger keys. Resources (running time, key length) and success probabilities are functions of  $\lambda$ . This subsection classifies growth rates in  $\lambda$  and proves (Proposition 11.63) that the *negligible* functions, those shrinking faster than the reciprocal of every polynomial, are closed under the operations that security proofs perform, addition and multiplication by polynomials; this is why “negligible” formalises “ignorable”.

**Definition 11.59** (Polynomial and super-polynomial). A function  $f: \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$  is *polynomially bounded*, written  $f = \text{poly}(\lambda)$ , if there is a constant  $c$  (and a threshold) with  $f(\lambda) \leq \lambda^c + c$  for all large  $\lambda$ ; equivalently,  $f(\lambda) = O(\lambda^c)$  for some constant  $c$ . It is *super-polynomial* if it grows faster than every polynomial, i.e.  $f(\lambda) = \omega(\lambda^c)$  for every constant  $c$  (equivalently,  $\lambda^c = o(f(\lambda))$  for all  $c$ ).

**Definition 11.60** (Exponential and sub-exponential). A function  $f$  is *exponential* if  $f(\lambda) = 2^{\Omega(\lambda)}$ ; the prototypical case is  $f(\lambda) = 2^{c\lambda}$  for a constant  $c > 0$ . A super-polynomial function with  $f(\lambda) = 2^{o(\lambda)}$  is *sub-exponential*; examples are  $\lambda^{\log_2 \lambda} = 2^{(\log_2 \lambda)^2}$  and the index-calculus running time  $L_p[1/3] = 2^{\Theta(\lambda^{1/3}(\log \lambda)^{2/3})}$  in the bit length  $\lambda = \log_2 p$  (Remark 7.22). Every exponential function is super-polynomial; the converse fails by these examples.

**Definition 11.61** (Negligible function). A function  $\nu: \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$  is *negligible*, written  $\nu = \text{negl}(\lambda)$ , if for every positive constant  $c$  there exists  $\lambda_0$  such that

$$\nu(\lambda) < \frac{1}{\lambda^c} \quad \text{for all } \lambda \geq \lambda_0.$$

Equivalently,  $\nu(\lambda) = o(\lambda^{-c})$  for every constant  $c > 0$ ; equivalently,  $\lambda^c \nu(\lambda) \rightarrow 0$  as  $\lambda \rightarrow \infty$  for every constant  $c$ . A function is *non-negligible* if it is not negligible, i.e.  $f(\lambda) \geq \lambda^{-c}$  for infinitely many  $\lambda$ , for some constant  $c$ ; it is *noticeable* if there exist  $c$  and  $\lambda_0$  with  $f(\lambda) \geq \lambda^{-c}$  for all  $\lambda \geq \lambda_0$  (noticeable implies non-negligible, but not conversely); it is *overwhelming* if  $1 - f$  is negligible.

*Example 11.62.* The functions  $2^{-\lambda}$ ,  $2^{-\sqrt{\lambda}}$ , and  $\lambda^{-\log \lambda}$  are negligible. For  $2^{-\lambda}$ : given any  $c$ , the product  $2^{-\lambda} \lambda^c = \lambda^c / 2^\lambda$  tends to 0 (a polynomial over an exponential), meeting the definition. By contrast,  $1/\lambda^{100}$  is *not* negligible — taking  $c = 101$  shows  $1/\lambda^{100} > 1/\lambda^{101}$  for all  $\lambda \geq 2$  — it is noticeable, even though it tends to 0. Tending to zero therefore does not suffice.

**Proposition 11.63** (Closure properties of negligible functions). *Let  $\nu, \mu$  be negligible and let  $p$  be a polynomially bounded function. Then:*

1.  $\nu + \mu$  is negligible (the class is closed under addition, hence under any fixed finite sum).
2.  $p \cdot \nu$  is negligible (a polynomial times a negligible function is negligible).
3. If  $f(\lambda) \leq \nu(\lambda)$  for all large  $\lambda$ , then  $f$  is negligible (domination by a negligible function).
4. Summing  $p(\lambda)$  functions, each bounded by a single negligible function  $\nu$ , yields a negligible function; in particular a union bound over polynomially many negligible events is negligible.

*Proof.* The proof rests on an exponent shift: to obtain a bound below  $\lambda^{-c}$ , negligibility is invoked at a strictly larger exponent, so that the surplus absorbs the combining factor. The same shift underlies the hybrid and union-bound steps of security proofs.

(1) Fix  $c$ . Applying the definition at the exponent  $c + 1$ , there is a threshold beyond which  $\nu(\lambda), \mu(\lambda) < \lambda^{-(c+1)} \leq \frac{1}{2} \lambda^{-c}$  (the latter once  $\lambda \geq 2$ ); adding the two bounds gives  $\nu(\lambda) + \mu(\lambda) < \lambda^{-c}$  for all large  $\lambda$ .

(2) Suppose  $p(\lambda) \leq \lambda^d$  for large  $\lambda$  (and some constant  $d$ ). Fix  $c$ . By negligibility applied at the exponent  $c + d$ ,  $\nu(\lambda) < \lambda^{-(c+d)}$  for large  $\lambda$ , whence  $p(\lambda)\nu(\lambda) < \lambda^d \lambda^{-(c+d)} = \lambda^{-c}$  for large  $\lambda$ . Hence  $p\nu$  is negligible.

(3) Immediate: if  $f \leq \nu$  eventually and  $\nu < \lambda^{-c}$  eventually, then  $f < \lambda^{-c}$  eventually, for every  $c$ .

(4) Write the sum  $\Sigma(\lambda) = \sum_{i=1}^{p(\lambda)} \nu_i(\lambda)$ , where each  $\nu_i \leq \nu$  by hypothesis (in security proofs all the  $\nu_i$  arise from one bound). Then  $\Sigma(\lambda) \leq p(\lambda) \nu(\lambda)$ , which is negligible by (2); conclude by (3).  $\square$

*Remark 11.64* (Why negligible is the threshold of security). Suppose an adversary wins a game with negligible probability  $\nu(\lambda)$ . If it repeats the attack  $p(\lambda)$  times (polynomially many, the most an efficient adversary can afford), the probability that some repetition succeeds is, by the union bound, at most  $p(\lambda)\nu(\lambda)$  — still negligible by Proposition 11.63(2). Thus negligible success probability remains negligible under any efficient amplification, and may therefore be ignored. Conversely, a noticeable success probability  $\geq \lambda^{-c}$  can be boosted to a constant by  $\lambda^c$  repetitions. The negligible/non-negligible split formalises the phrase “the scheme is secure except with negligible probability”: security asserts a negligible advantage, and an attack means a non-negligible one. (A merely non-negligible — not noticeable — success probability is boosted to a constant only on the infinitely many  $\lambda$  where the inverse-polynomial bound holds, which already suffices to contradict a security definition.)

### 11.10 Algorithms, running time, and PPT

This subsection makes “efficient algorithm” and “infeasible” precise in the standard model: algorithms are Turing machines (equivalently, up to polynomial factors, programs in any reasonable model of computation), inputs and outputs are finite binary strings, and the measured resource is the number of elementary steps as a function of input length. Write  $\{0, 1\}^*$  for the set of all finite binary strings and  $|x|$  for the length of  $x \in \{0, 1\}^*$ .

**Definition 11.65** (Deterministic polynomial-time algorithm). An algorithm  $M$  runs in (*deterministic*) *polynomial time* if there is a polynomial  $p$  such that, on every input  $x \in \{0, 1\}^*$ ,  $M$  halts within at most  $p(|x|)$  steps and produces an output. The class of decision problems solvable by such machines is  $P$ .

Key generation and sampling require randomness, which a probabilistic algorithm reads from an auxiliary tape of independent uniform random bits, its “coins”.

**Definition 11.66** (Probabilistic algorithm and PPT). A *probabilistic* (or *randomised*) algorithm  $M$  is a Turing machine with an additional read-only *random tape* initialised with independent uniformly random bits  $r \in \{0, 1\}^{\mathbb{N}}$ . For a fixed input  $x$ , the output  $M(x)$  is a random variable over the choice of  $r$ ; write  $M(x; r)$  to display the coins explicitly, and then  $M(x; \cdot)$  is a deterministic function. The algorithm runs in *probabilistic polynomial time* (it is *PPT*) if there is a polynomial  $p$  such that, for every input  $x$  and every choice of coins  $r$ ,  $M$  halts within  $p(|x|)$  steps. (Bounding the time for every  $r$ , not merely in expectation, is the standard “strict” notion and ensures  $M$  ever reads only  $p(|x|)$  coins.)

*Remark 11.67* (Non-uniformity and circuits). There are two standard ways to formalise an efficient adversary. The *uniform* model is a single PPT machine that works for all input lengths, as above. The *non-uniform* model allows a separate algorithm (equivalently, a Boolean circuit) of size  $\text{poly}(\lambda)$  for each  $\lambda$ , which may be hard-wired with an arbitrary polynomial-length “advice” string depending on  $\lambda$ . Non-uniform adversaries are at least as powerful as uniform ones and are often more convenient in reduction proofs, because the advice can encode a “best” attack. The asymptotic theory of this section is identical for both models; security definitions in the Halo 2/Orchard setting typically state security against non-uniform PPT adversaries, the more conservative choice.

The hardness notion below rests on the Cobham–Edmonds thesis: “efficiently computable” means “computable in polynomial time”. Feasibility and infeasibility are opposite regimes but not logical negations: intermediate success probabilities can fall into neither class.

**Definition 11.68** (Computational infeasibility). A computational task parameterised by the security parameter  $\lambda$  is (*computationally*) *feasible* if there is a PPT algorithm solving it with overwhelming success probability. It is *infeasible* if for every PPT algorithm  $\mathcal{A}$  the probability that  $\mathcal{A}$  solves it is negligible in  $\lambda$ :

$$\Pr[\mathcal{A} \text{ solves the task on parameter } \lambda] = \text{negl}(\lambda),$$

where the probability is over the task’s randomness and  $\mathcal{A}$ ’s coins. Equivalently: no efficient algorithm succeeds with non-negligible probability.

*Remark 11.69* (Reading a hardness assumption). Cryptographic assumptions are infeasibility statements. For instance, the discrete-logarithm assumption in a family of groups  $\{G_\lambda\}$  of order  $q(\lambda)$  asserts: for every PPT  $\mathcal{A}$ ,

$$\Pr[\mathcal{A}(g, g^x) = x] = \text{negl}(\lambda),$$

the probability taken over a uniformly chosen generator  $g$  of  $G_\lambda$ , a uniform  $x \leftarrow \mathbb{Z}/q\mathbb{Z}$ , and  $\mathcal{A}$ 's coins. The security of Halo 2 and Orchard rests on such asymptotic infeasibility assumptions; choosing a parameter size does not prove them. Separately, concrete analysis asks how much work the best known attacks need at one fixed size. Generic discrete-log attacks take  $\Theta(\sqrt{q})$  group operations by the birthday/rho phenomenon of Remark 11.20; absent a faster structure-specific attack, an order near  $2^{2\lambda}$  therefore targets about  $\lambda$  bits of generic-group security. The Pasta orders lie slightly above  $2^{254}$ , giving roughly  $2^{127}$  generic work. The asymptotic language permits the terms “negligible” and “polynomial”; concrete estimates remain conditional on the attacks presently known.

### 11.11 The security parameter

This subsection fixes the parameter in which negligibility, efficiency, and infeasibility are measured, and states what the “128-bit security” claim of the section opening asserts.

**Definition 11.70** (Security parameter). The *security parameter* is a positive integer  $\lambda \in \mathbb{N}$  supplied to all algorithms of a cryptographic scheme, conventionally in unary as the string  $1^\lambda$  (so that “polynomial in the input length” coincides with “polynomial in  $\lambda$ ”). It governs simultaneously:

- the *sizes* of objects — key lengths and encodings, each  $\text{poly}(\lambda)$ ; in a generic discrete-log instantiation one commonly chooses a group order near  $2^{2\lambda}$ ;
- the *running time* of honest algorithms, which must be  $\text{poly}(\lambda)$ ; and
- the *security level*: every PPT adversary's advantage must be  $\text{negl}(\lambda)$ .

*Remark 11.71* (Unary encoding of the security parameter). Passing  $1^\lambda$  (a string of  $\lambda$  ones, of length  $\lambda$ ) rather than the integer  $\lambda$  in binary (length  $\log_2 \lambda$ ) is a deliberate convention: it forces “polynomial time in the input length” to mean polynomial in  $\lambda$  itself, so an honest key generator may take time  $\text{poly}(\lambda)$  to produce  $\lambda$ -bit keys. In a scheme satisfying the security definition, increasing  $\lambda$  raises honest cost only polynomially while every PPT adversary's success remains negligible; in common concrete families, the best known attack cost grows exponentially. Asymptotic security is the statement that this gap holds for all sufficiently large  $\lambda$ ; the practitioner then fixes a single value (commonly  $\lambda = 128$ , targeting  $2^{128}$ -operation attacks) and checks the concrete estimates against that target: the Pallas/Vesta group orders, slightly above  $2^{254}$ , give roughly  $2^{127}$  generic group operations (Section 10; Zcash protocol specification §5.4.9.6), hash outputs of  $\geq 256$  bits per the birthday bound of Proposition 11.18, and sampling slack of at least 128 extra bits per Proposition 11.24 (the deployed ToScalar of Remark 11.26 reduces 512 bits, a slack of 257). Every “negligible” in the remainder of the series is implicitly a function of this  $\lambda$ .

The “128-bit security” claim of the section opening has two separate layers. Asymptotic infeasibility (Definition 11.68) requires the success probability of every probabilistic polynomial-time forger, uniform or hard-wired with advice (Remark 11.67), to be negligible in  $\lambda$ , a bound that no polynomial number of repetitions amplifies to a noticeable probability (Remark 11.64); it holds conditionally on the stated cryptographic assumptions and does not exclude the positive success probability of a forger that guesses a key. A concrete “128-bit” estimate states that the best known applicable attack at a fixed parameter size costs on the order of  $2^{128}$  operations; it is an attack estimate, not a theorem about every possible algorithm.